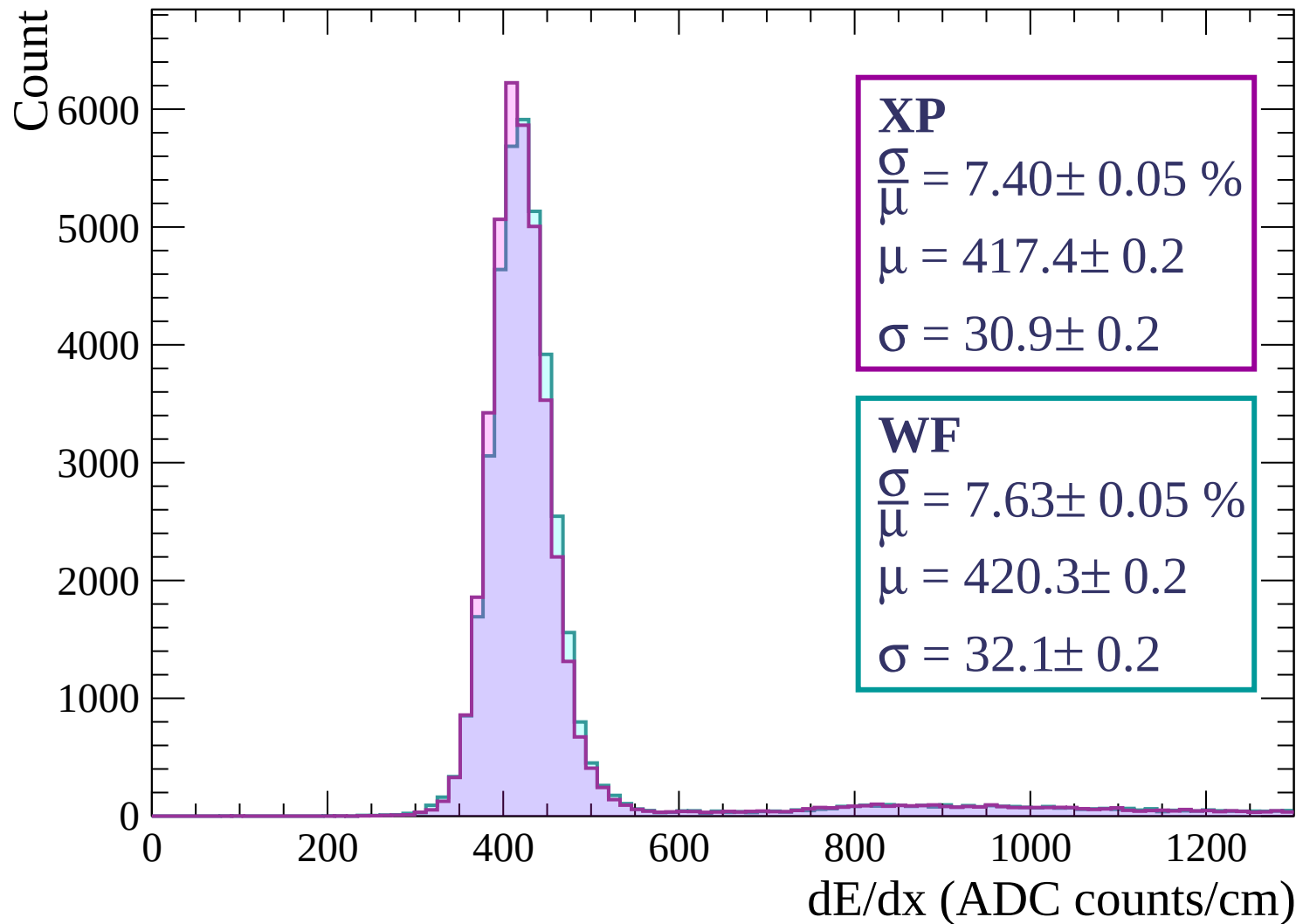
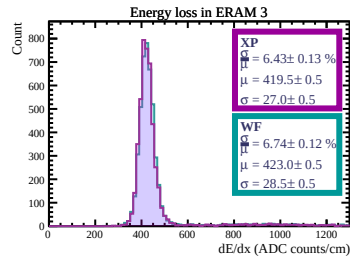
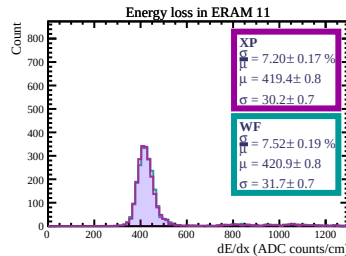
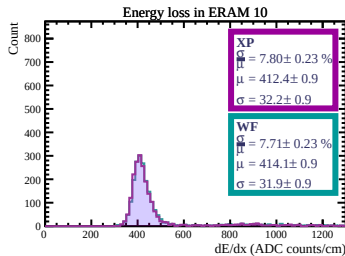
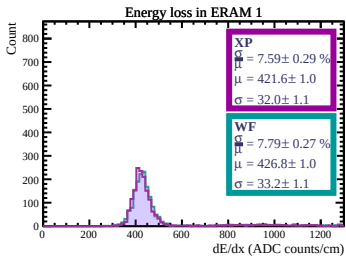
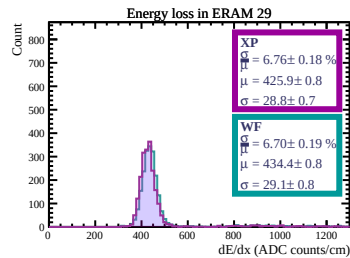
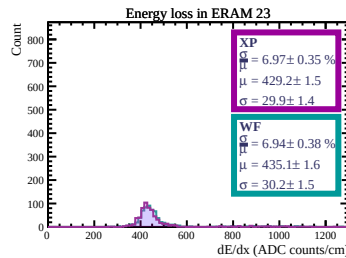
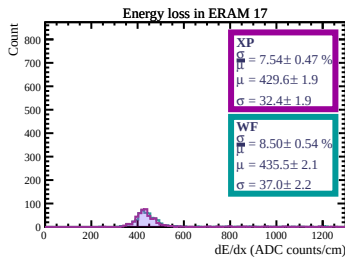
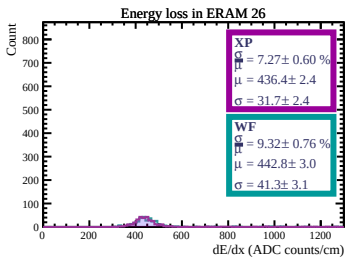
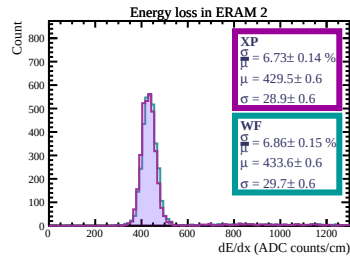
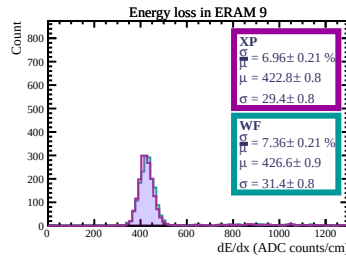
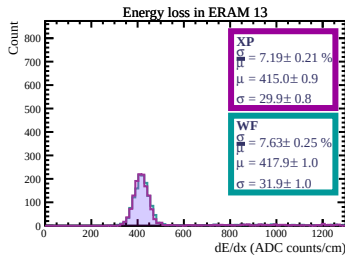
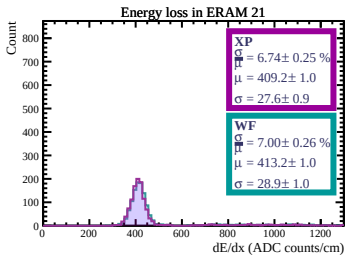
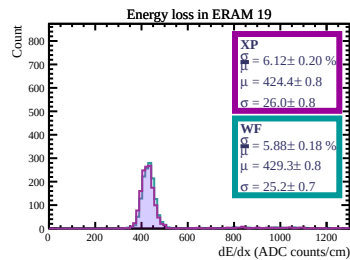
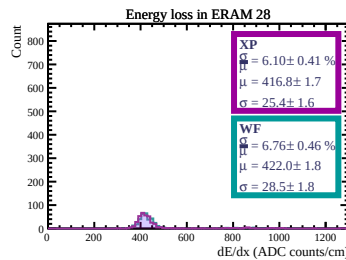
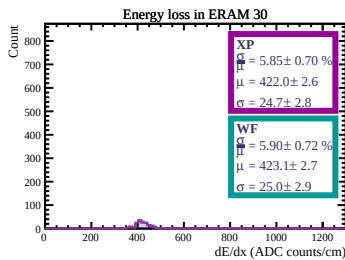
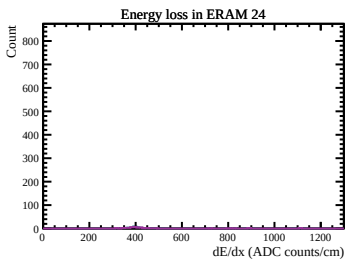
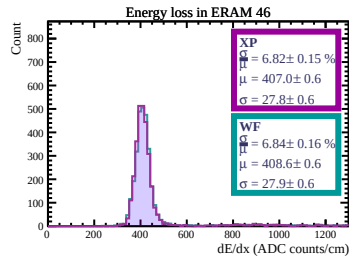
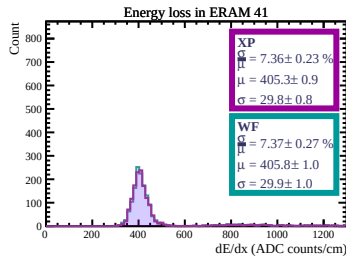
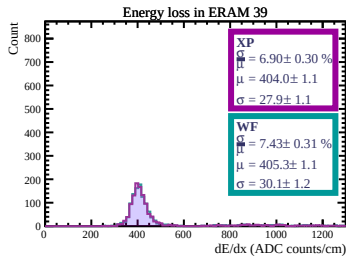
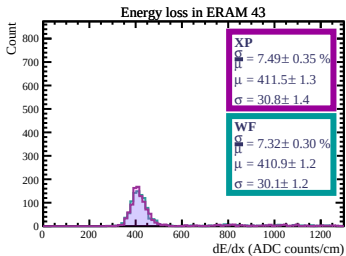
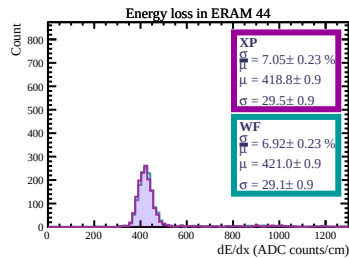
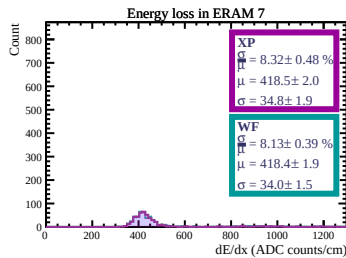
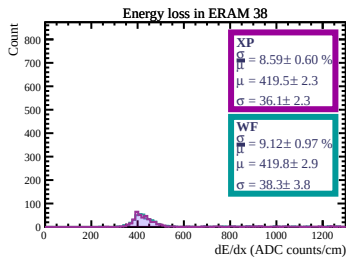
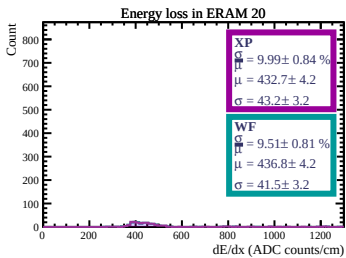
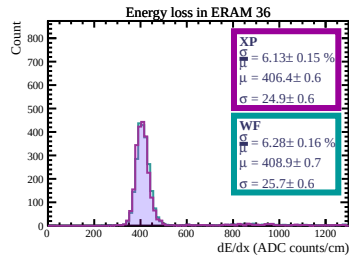
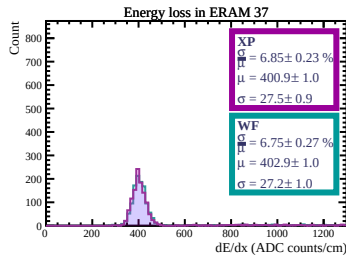
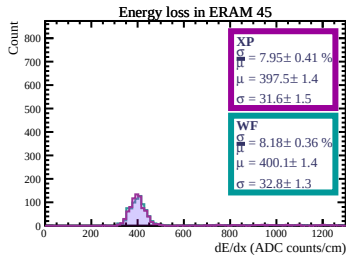
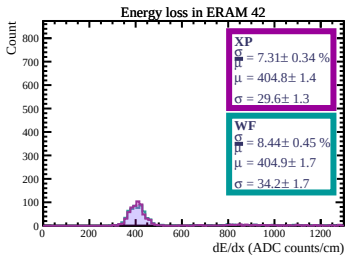
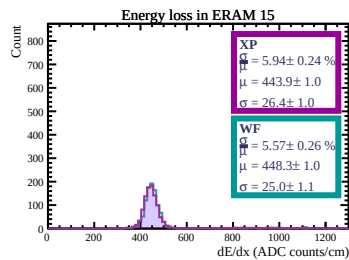
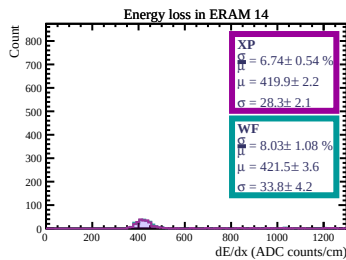
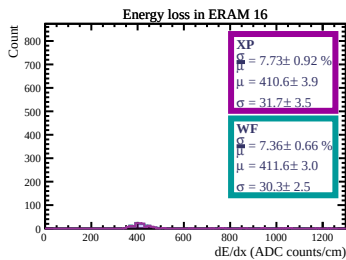
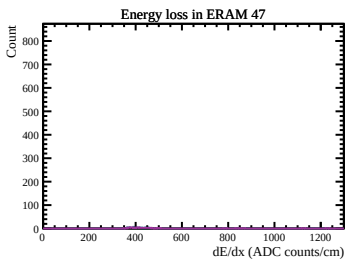
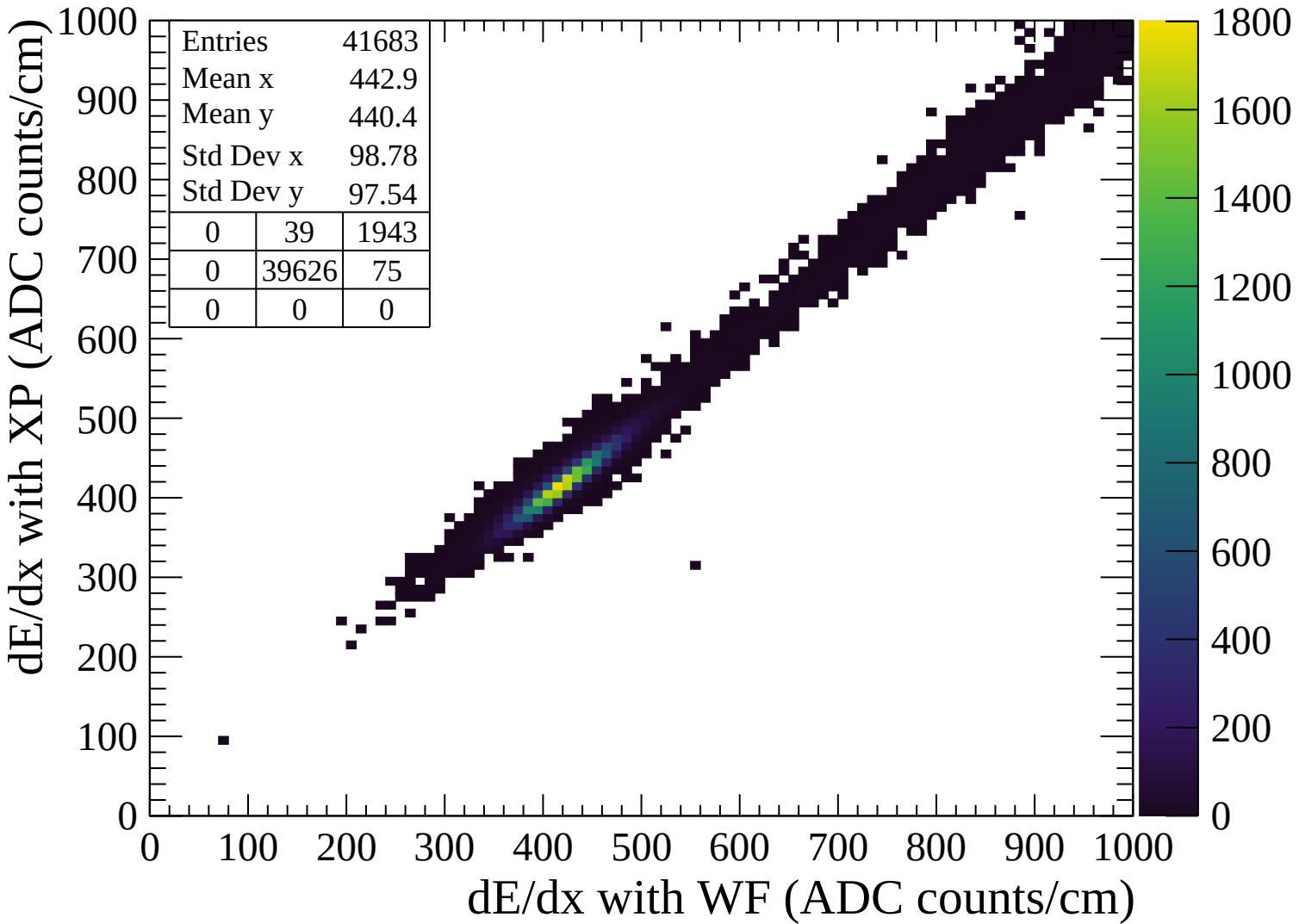
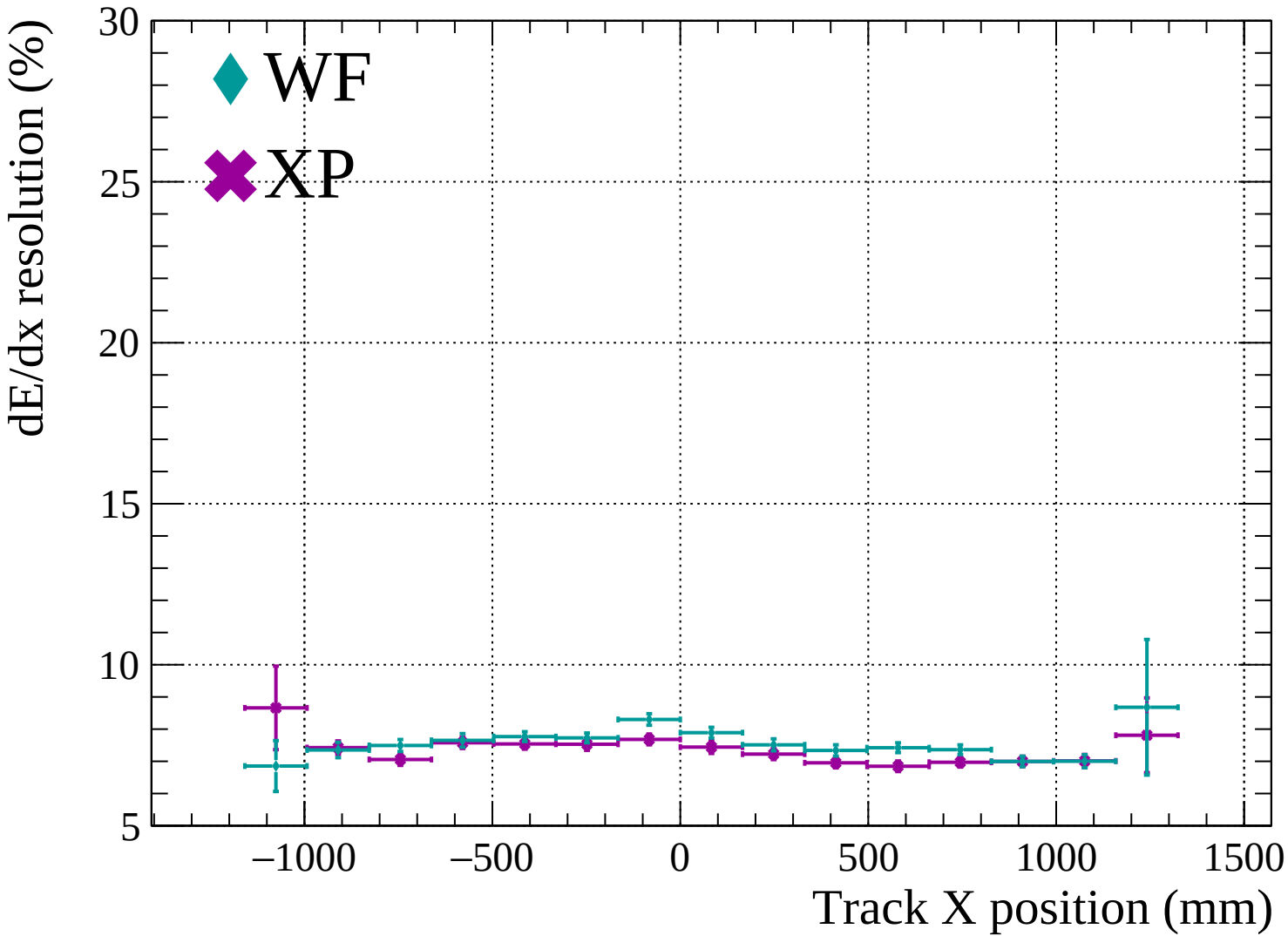


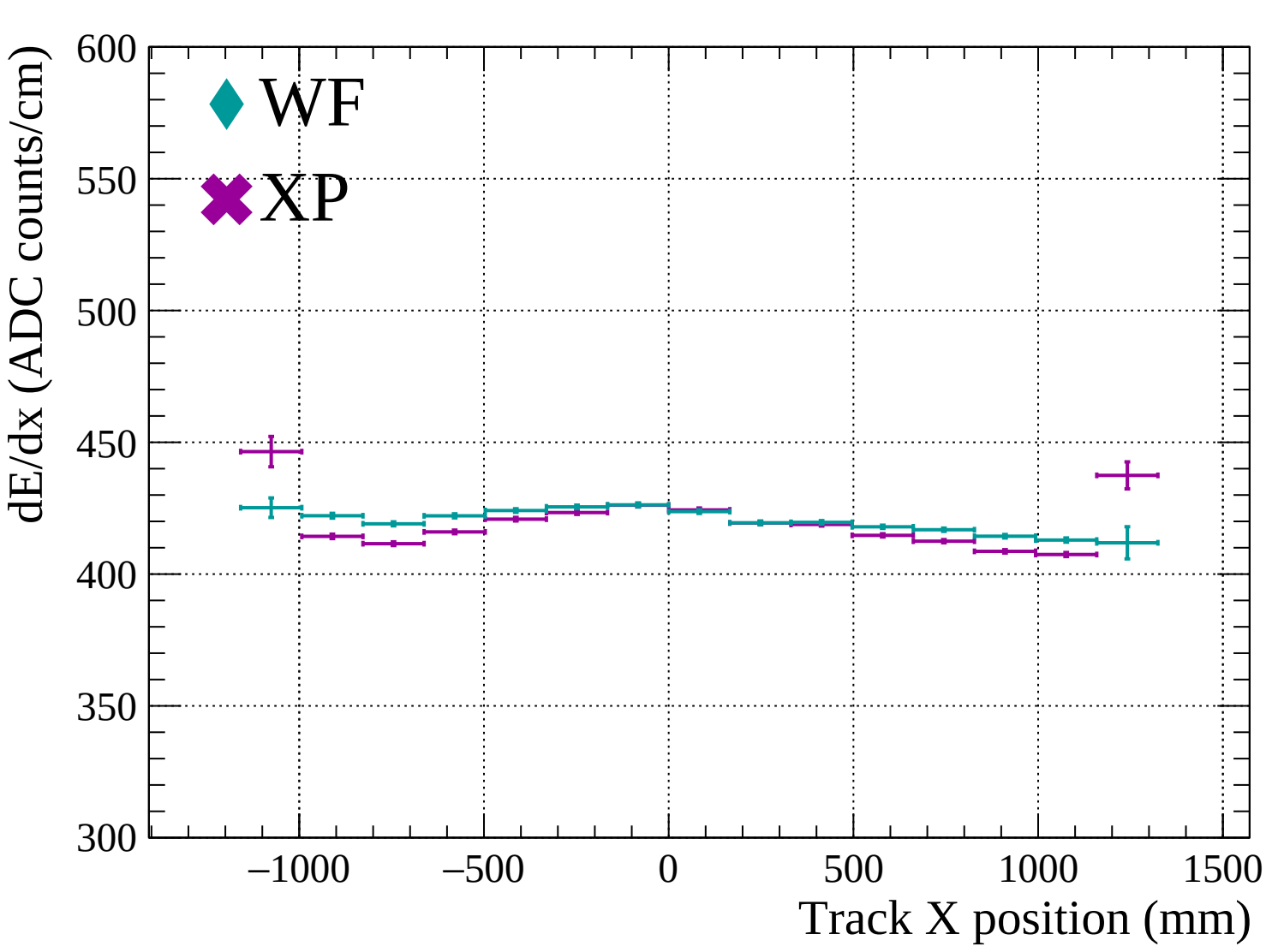


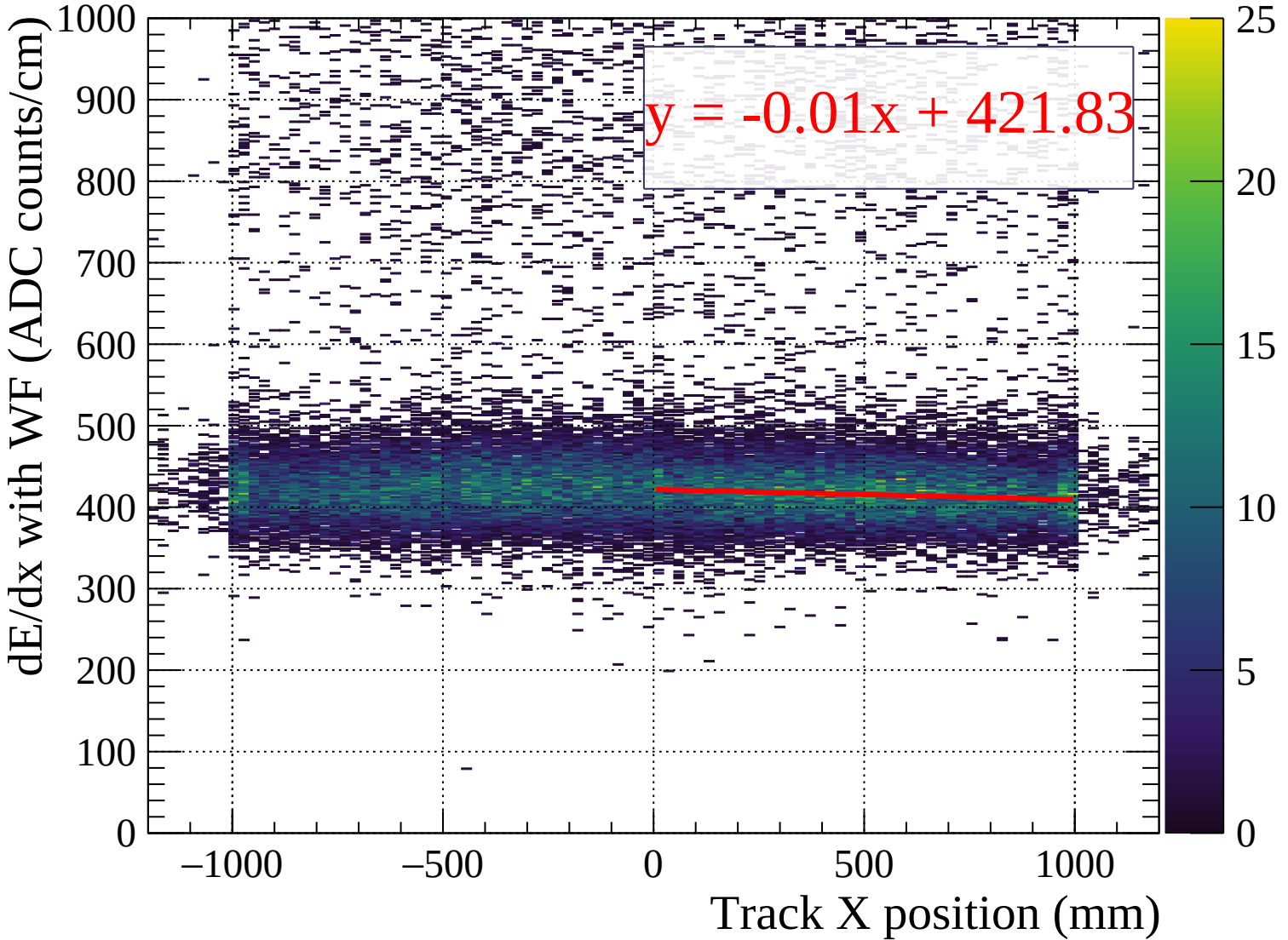


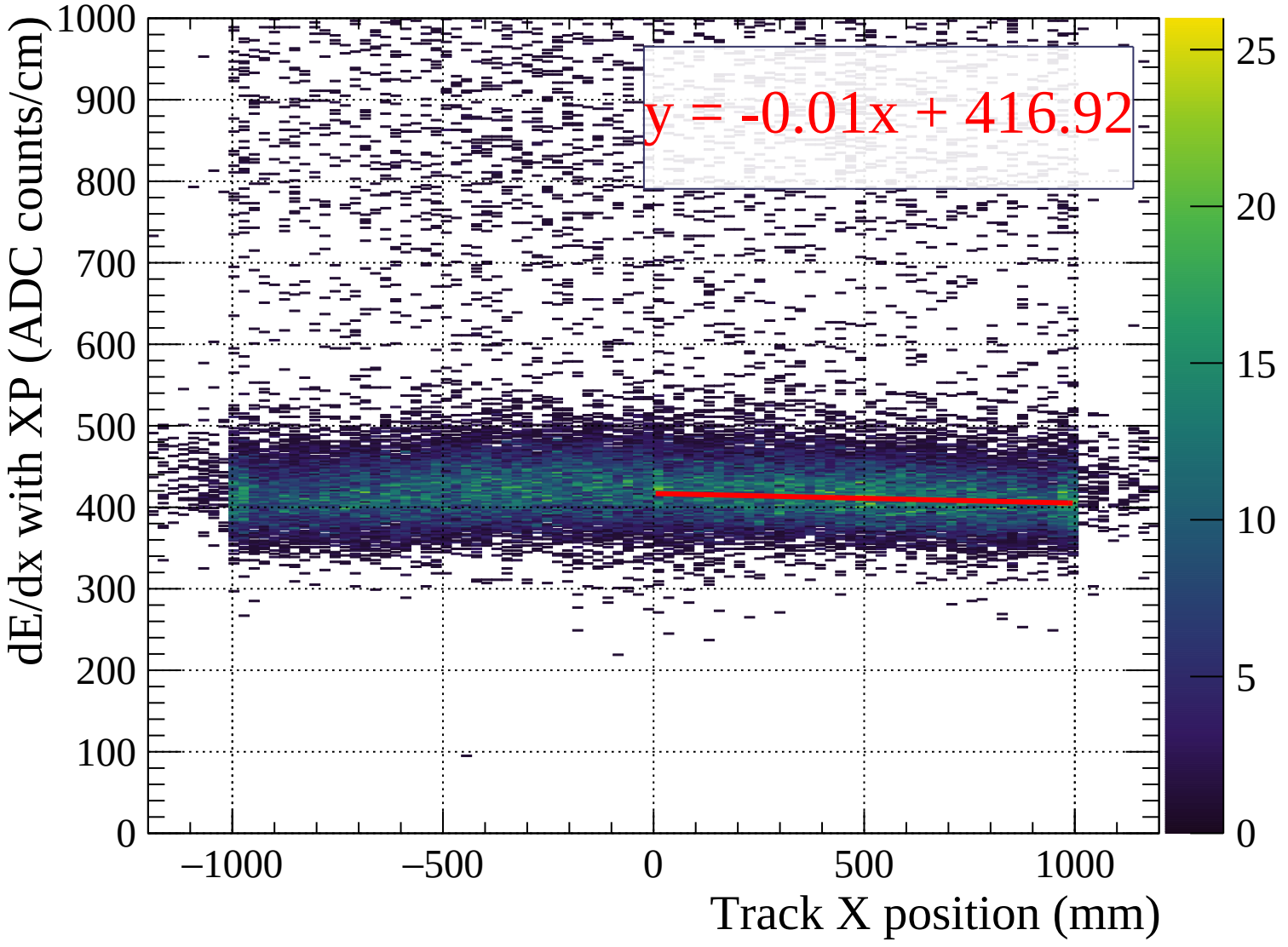


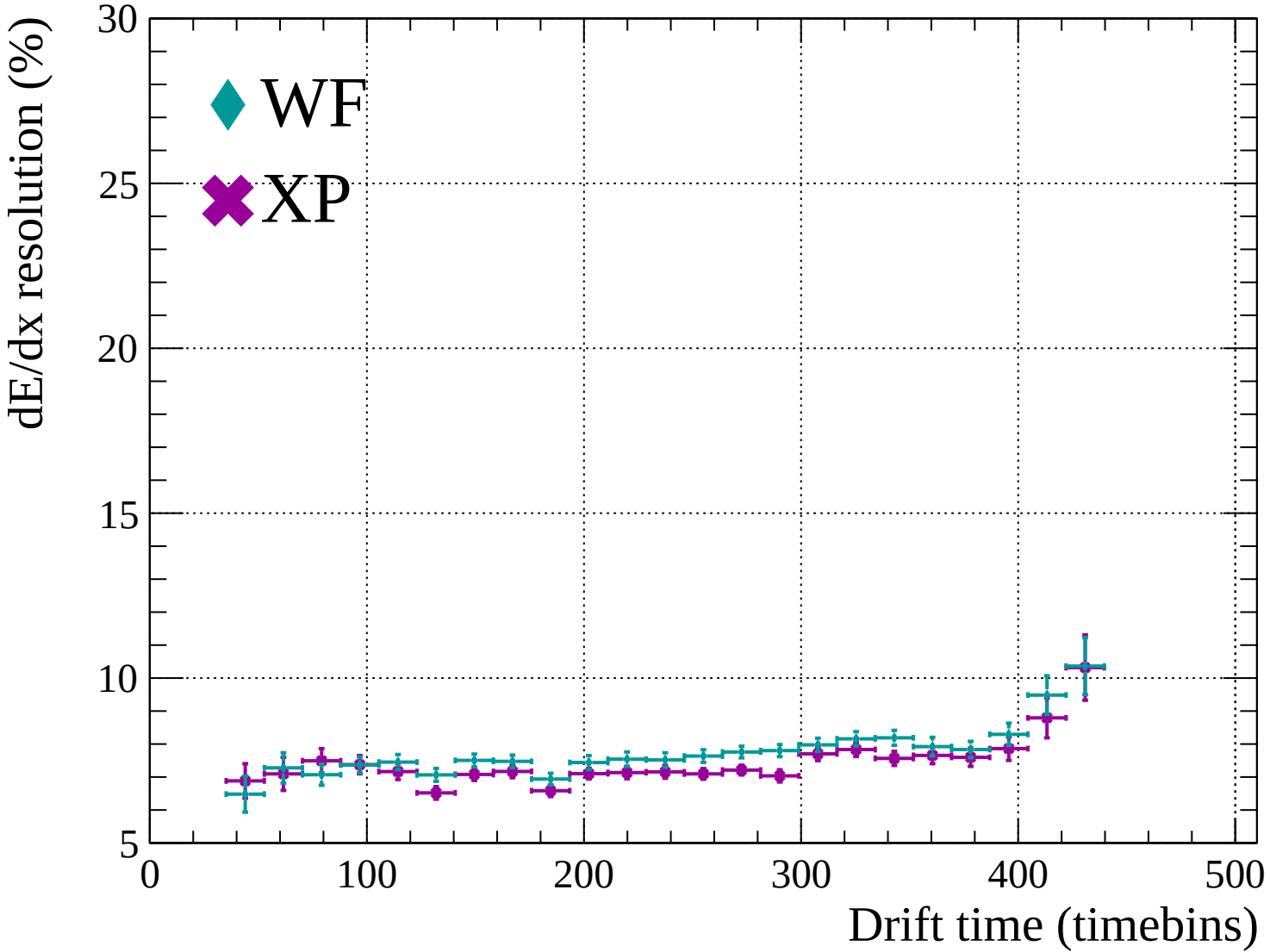


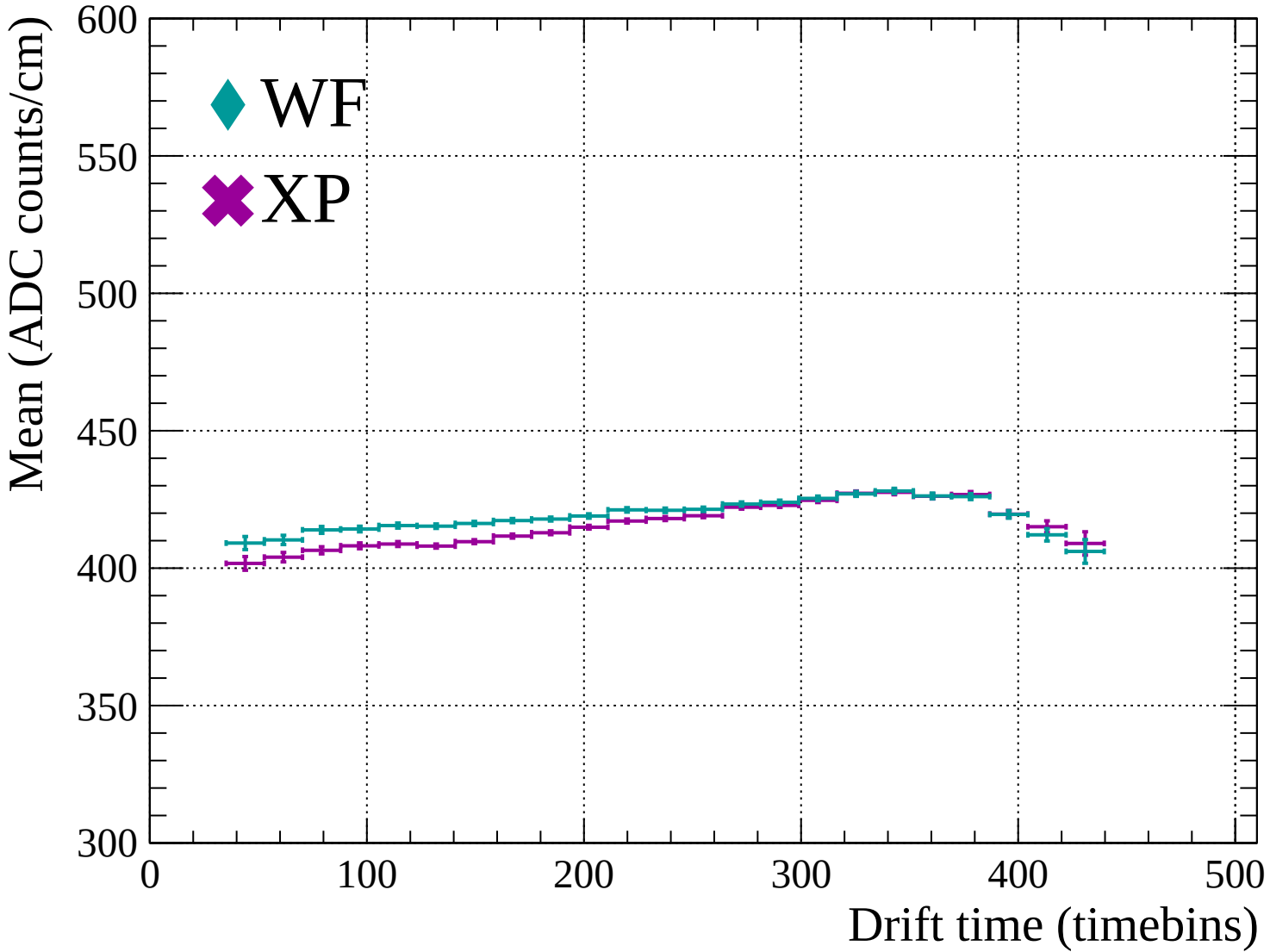


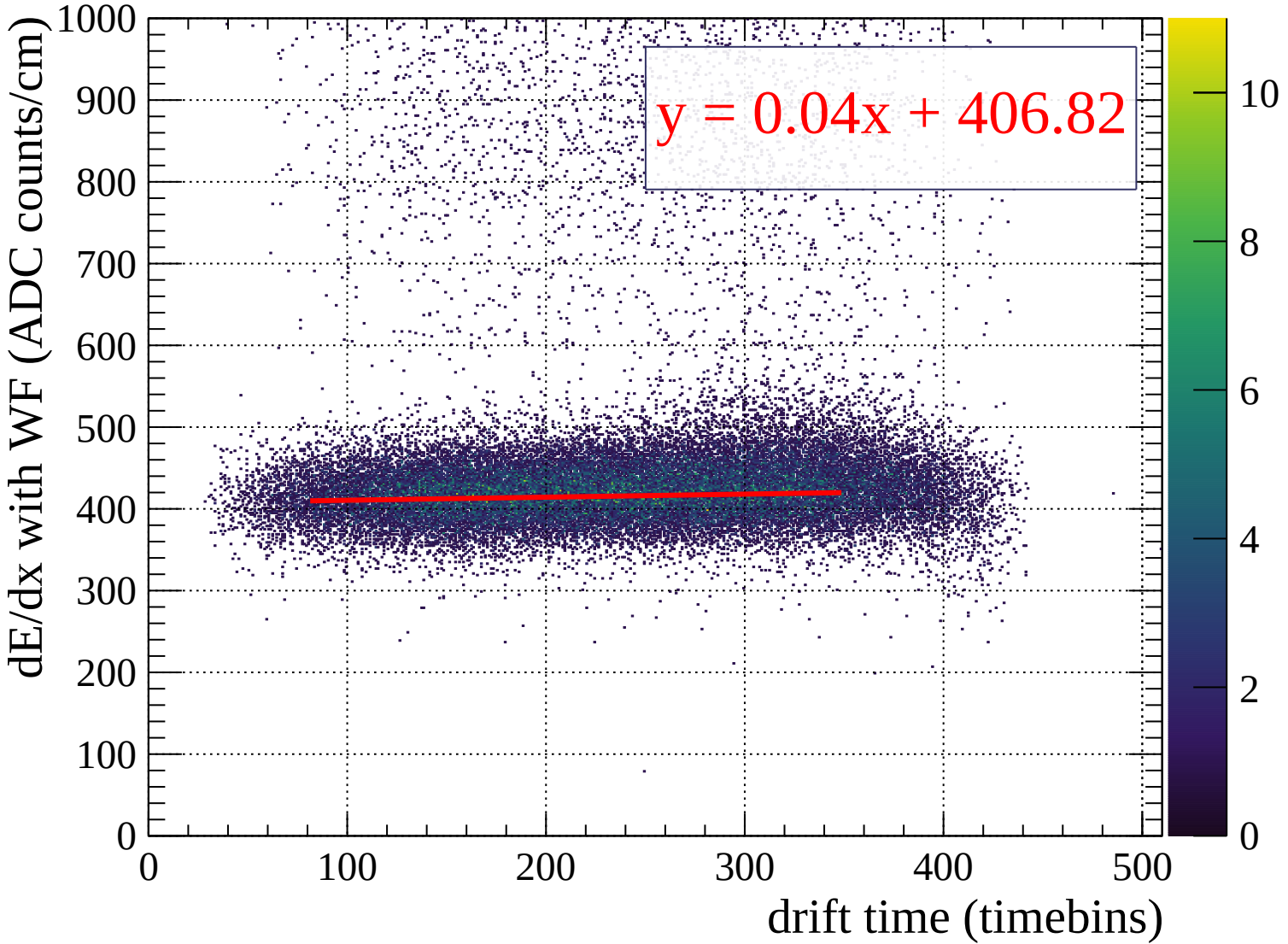


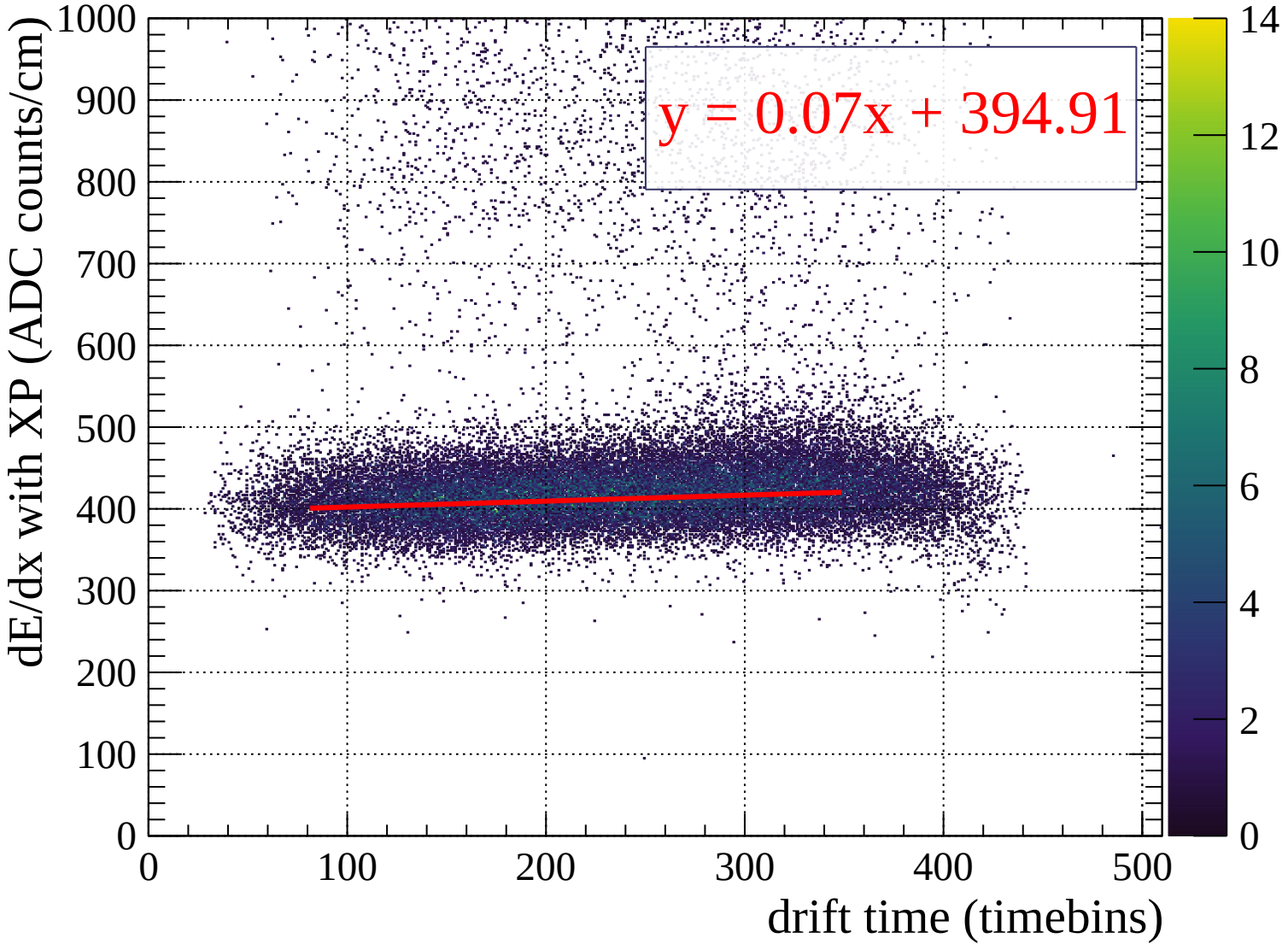


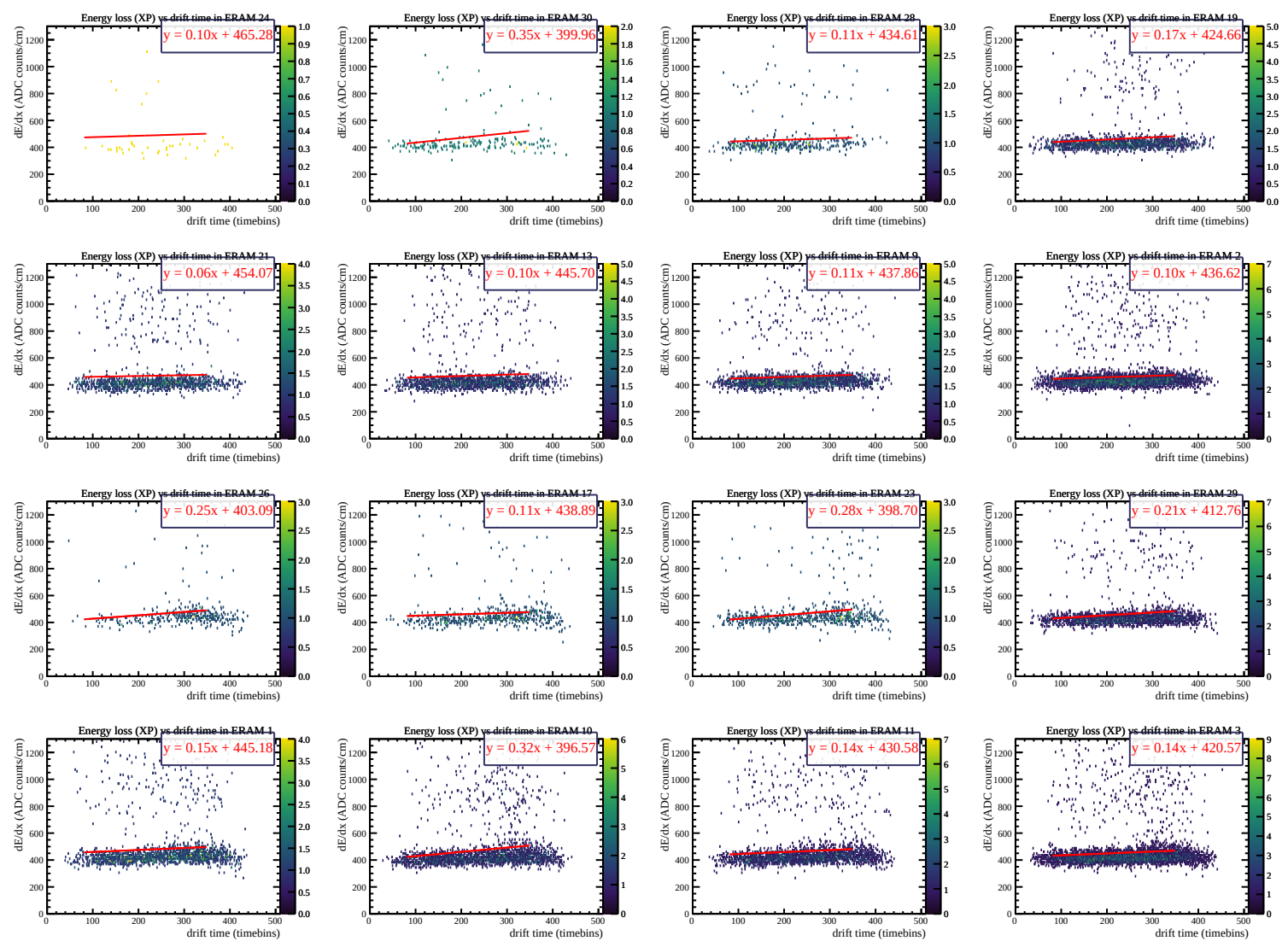


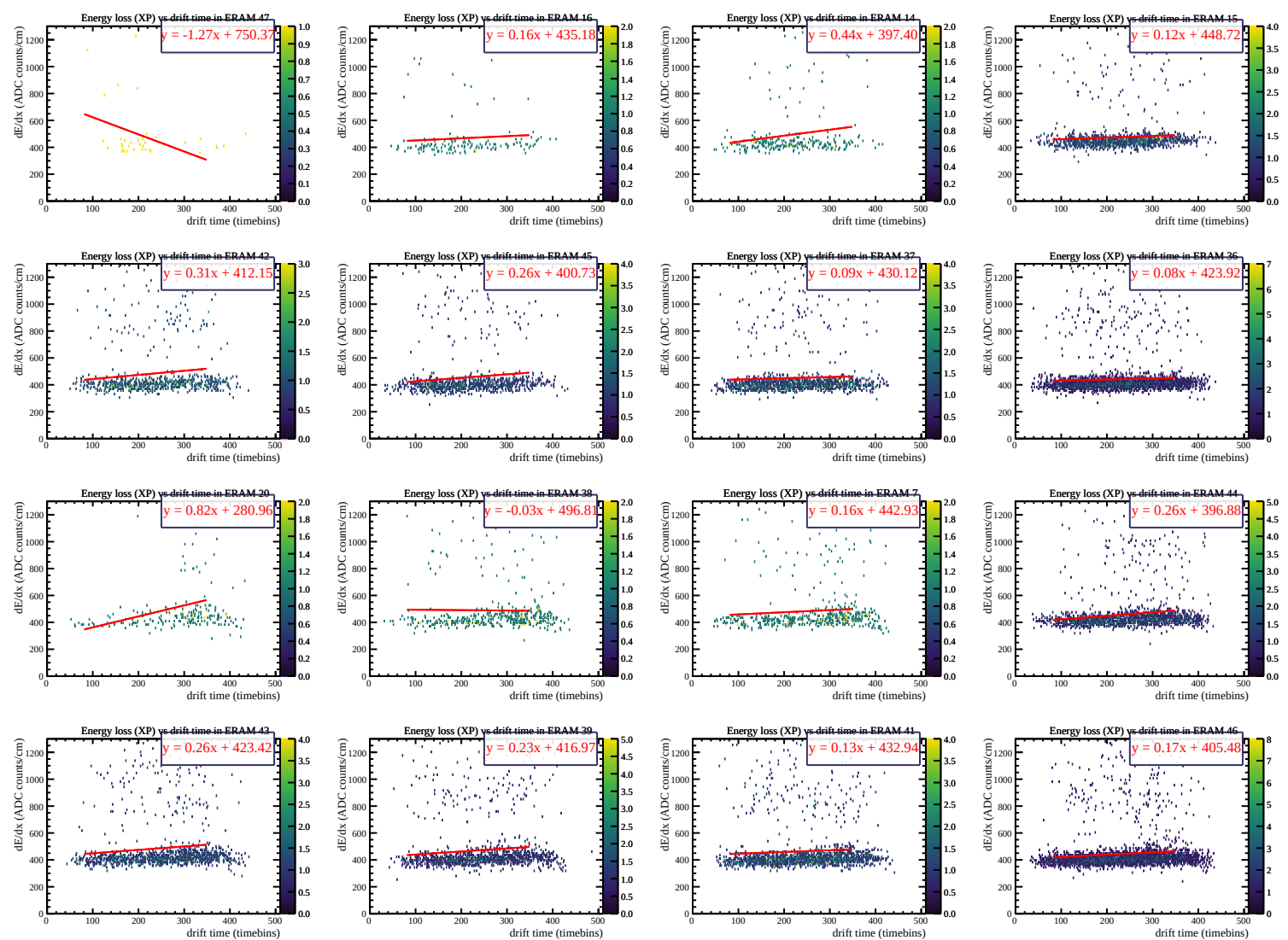


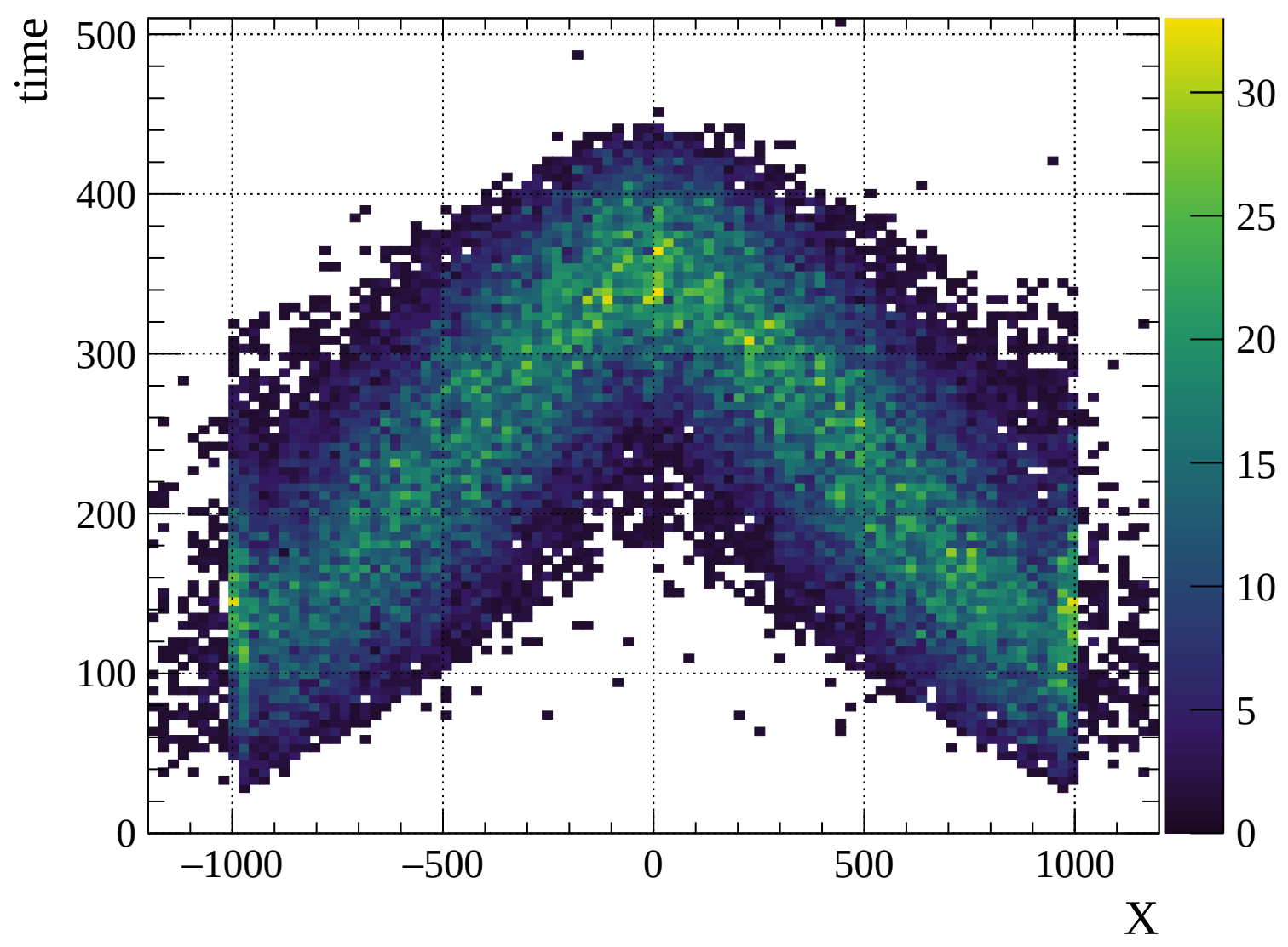


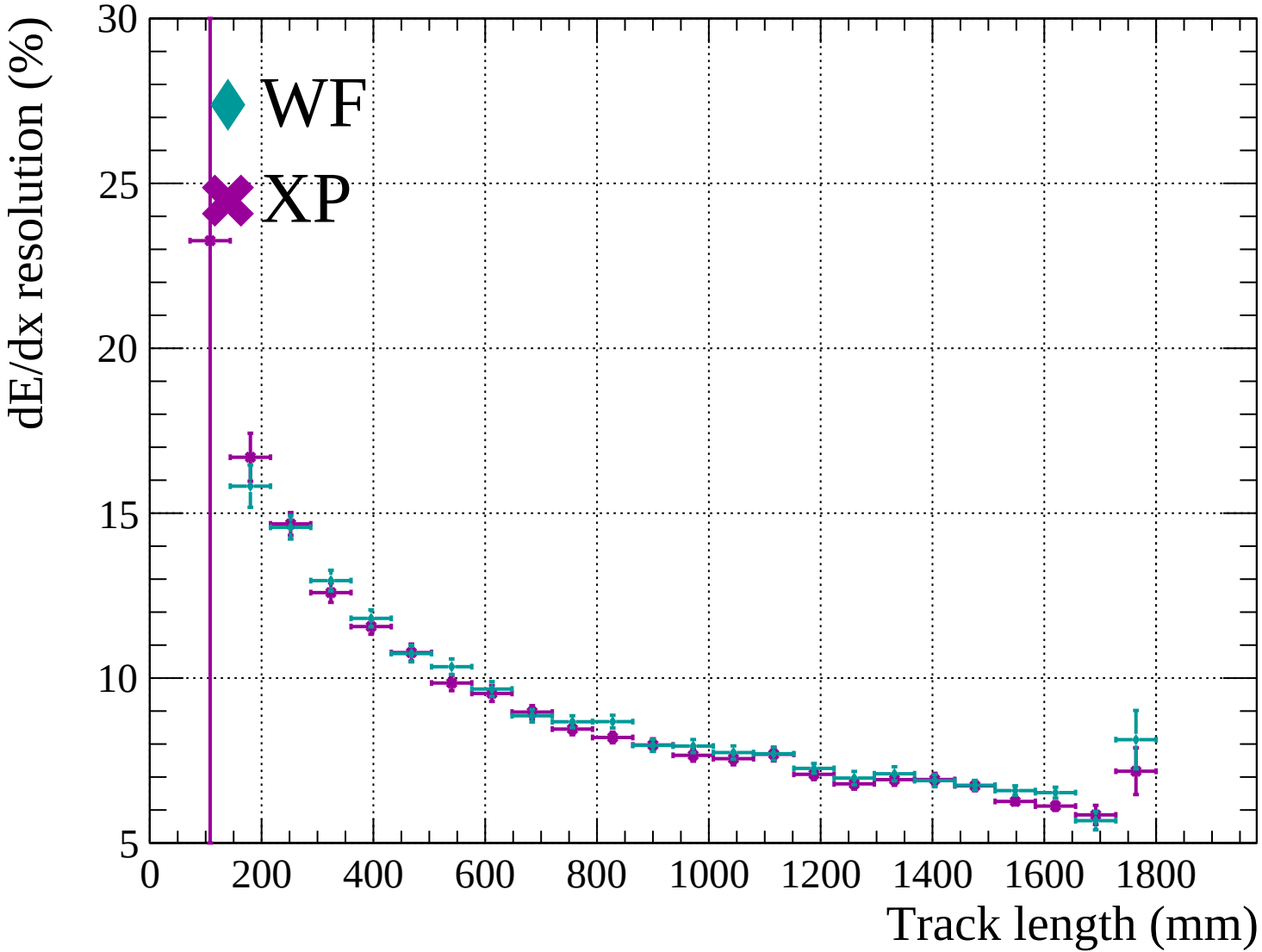


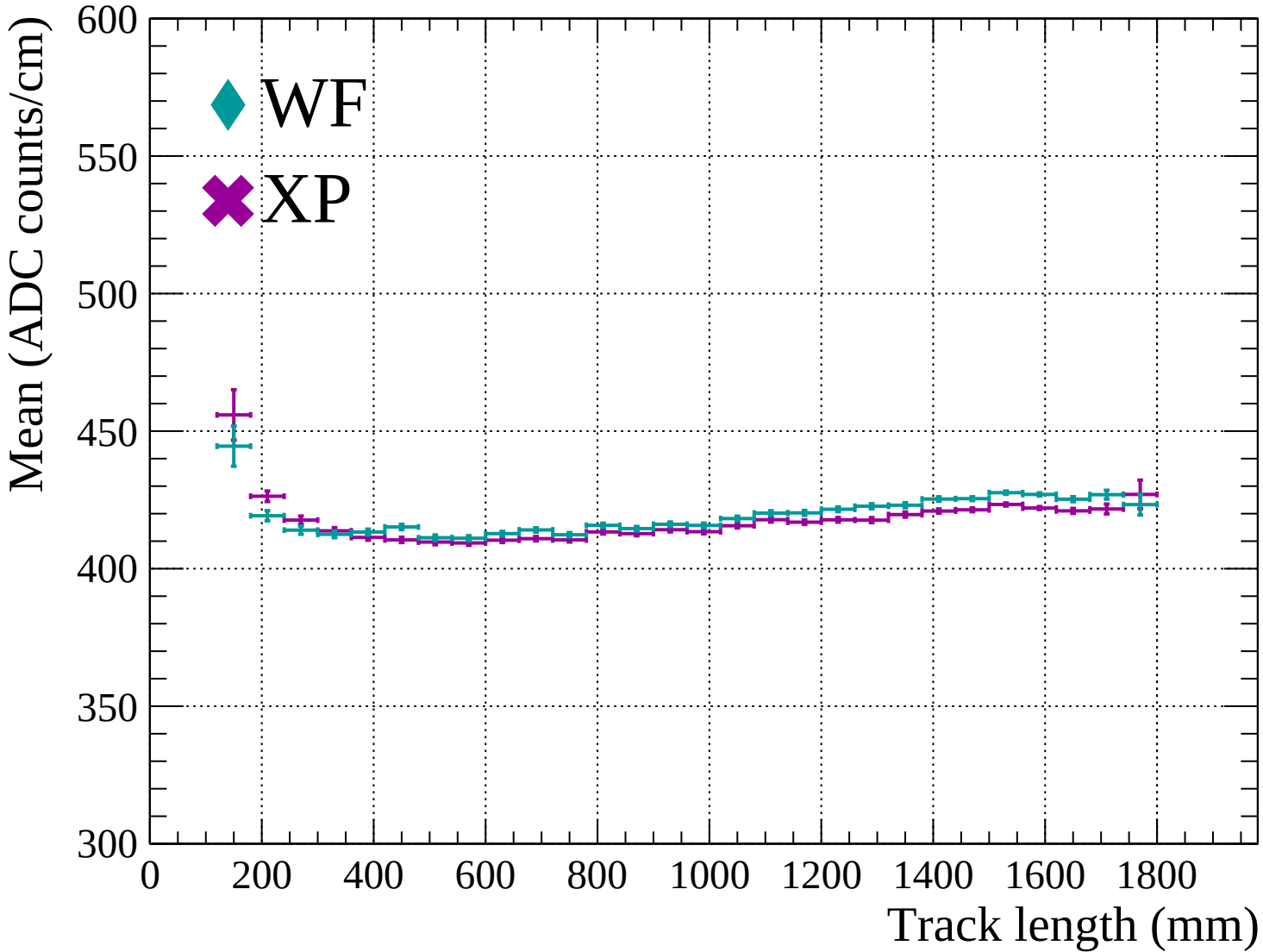


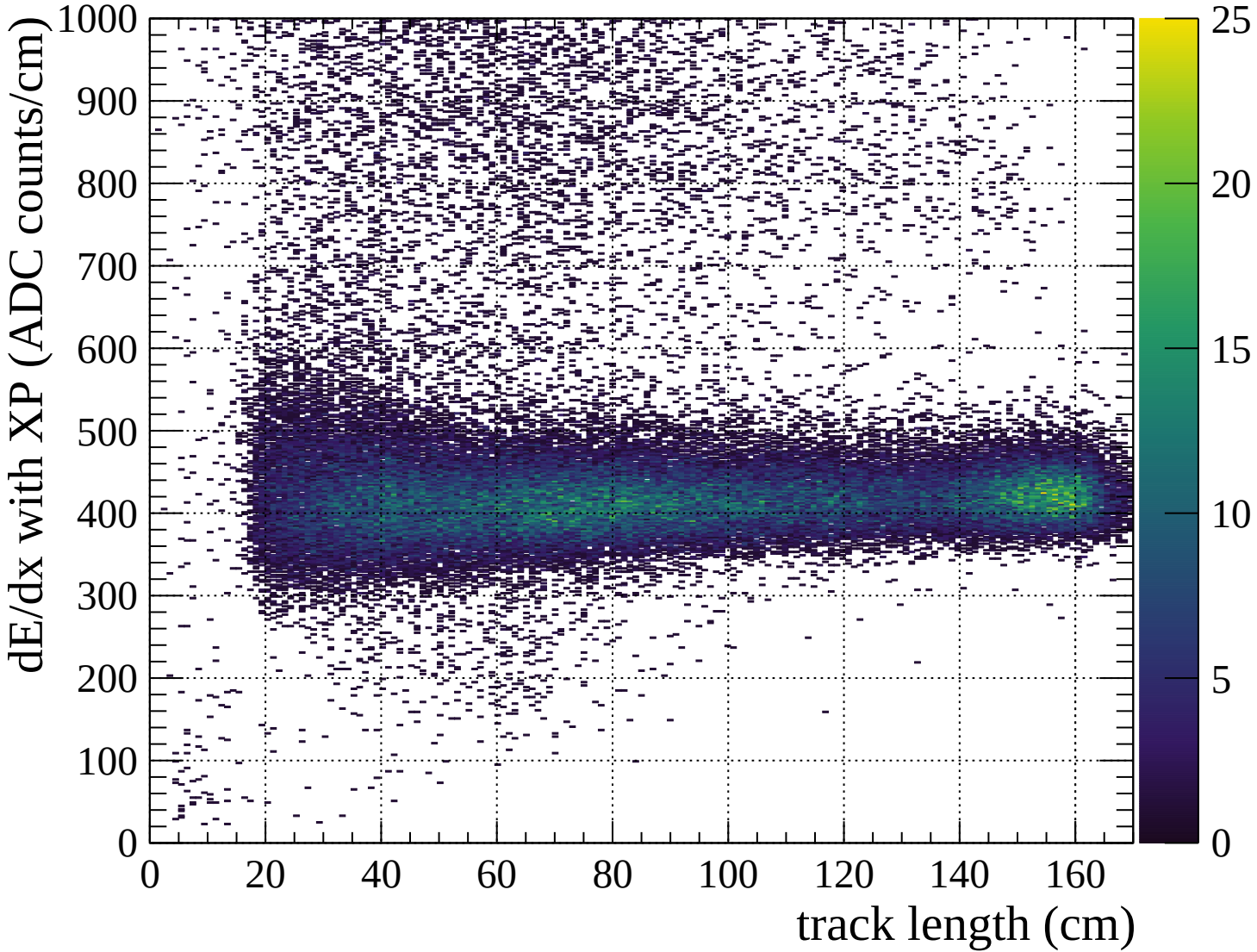


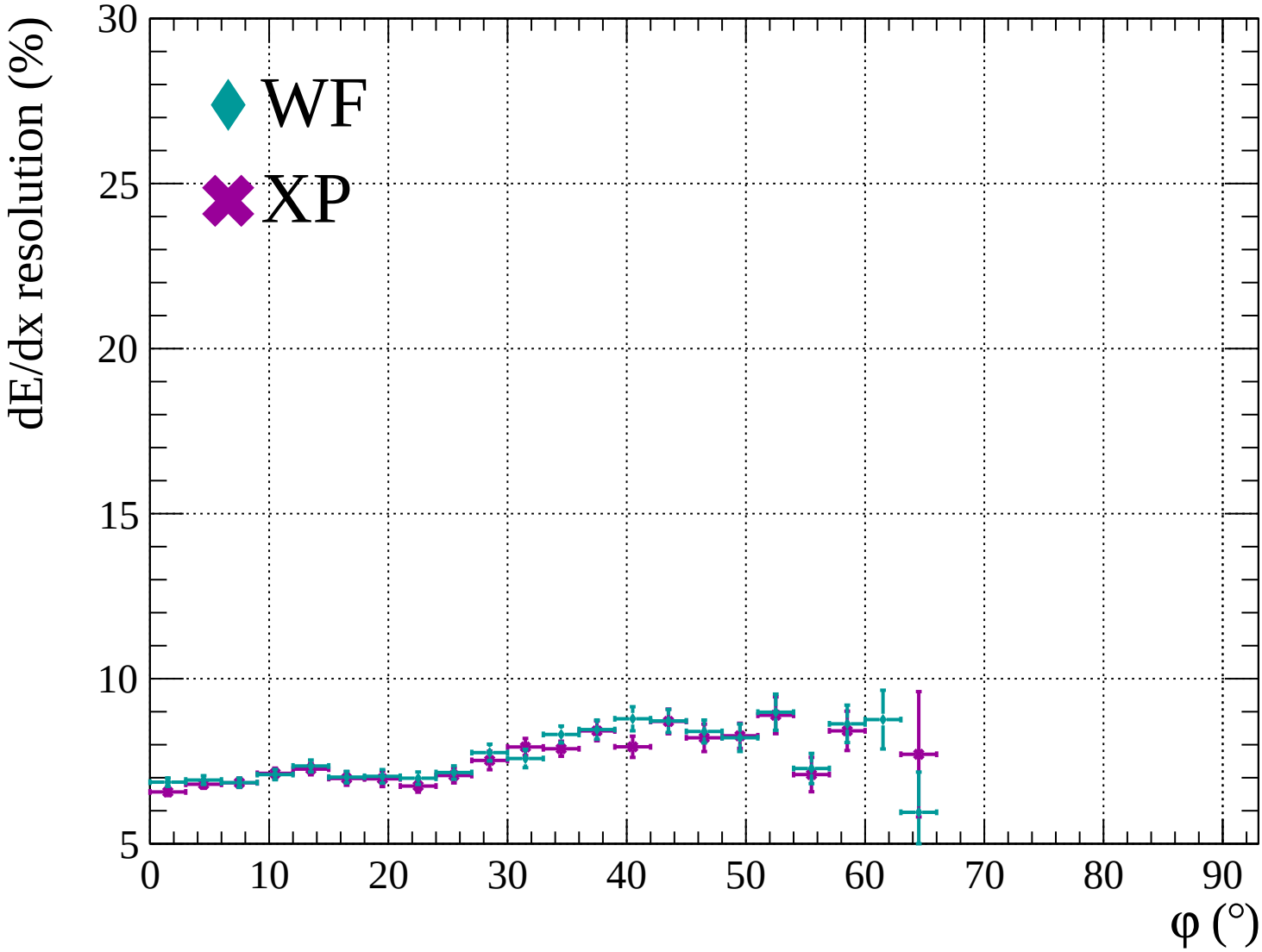


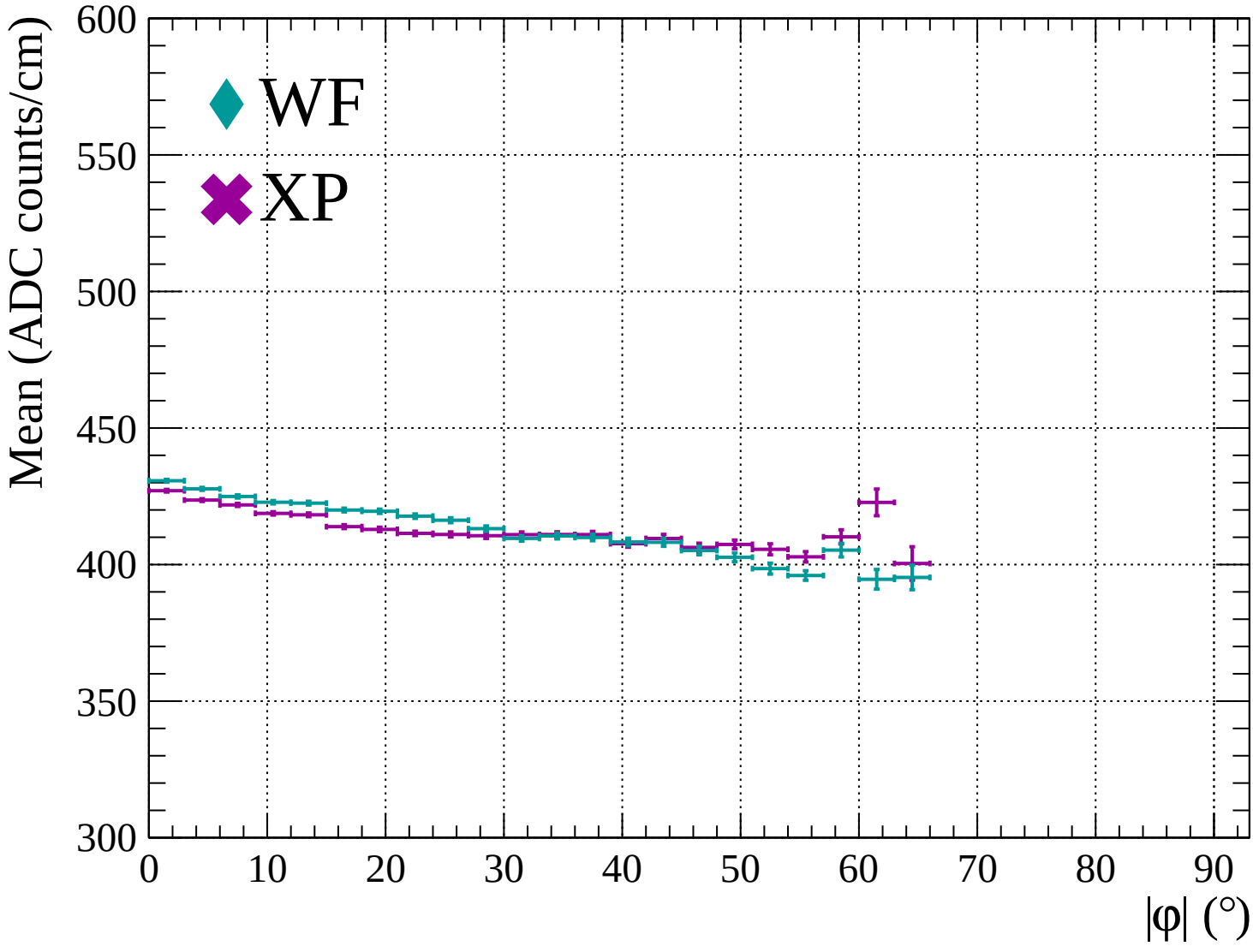


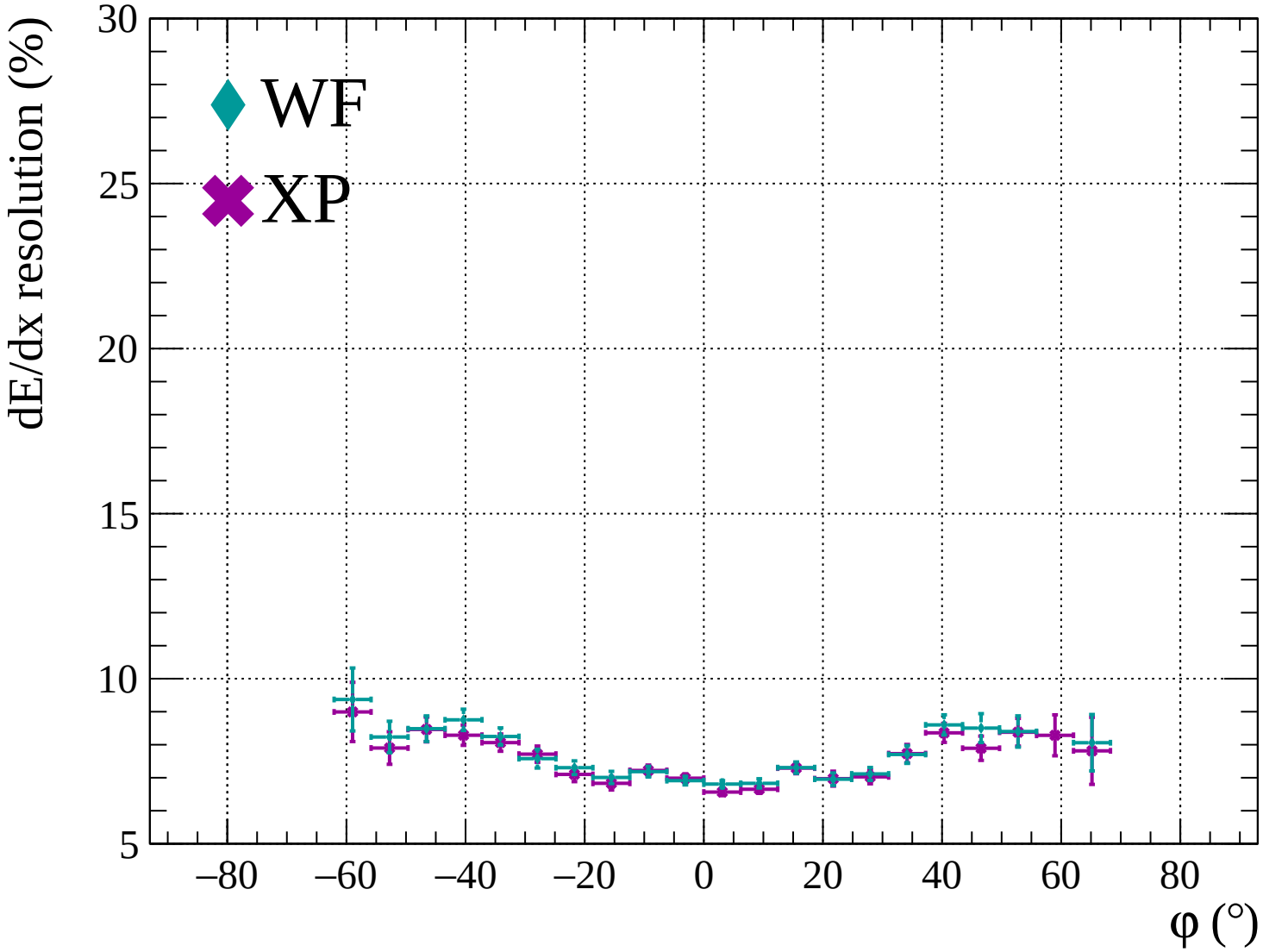


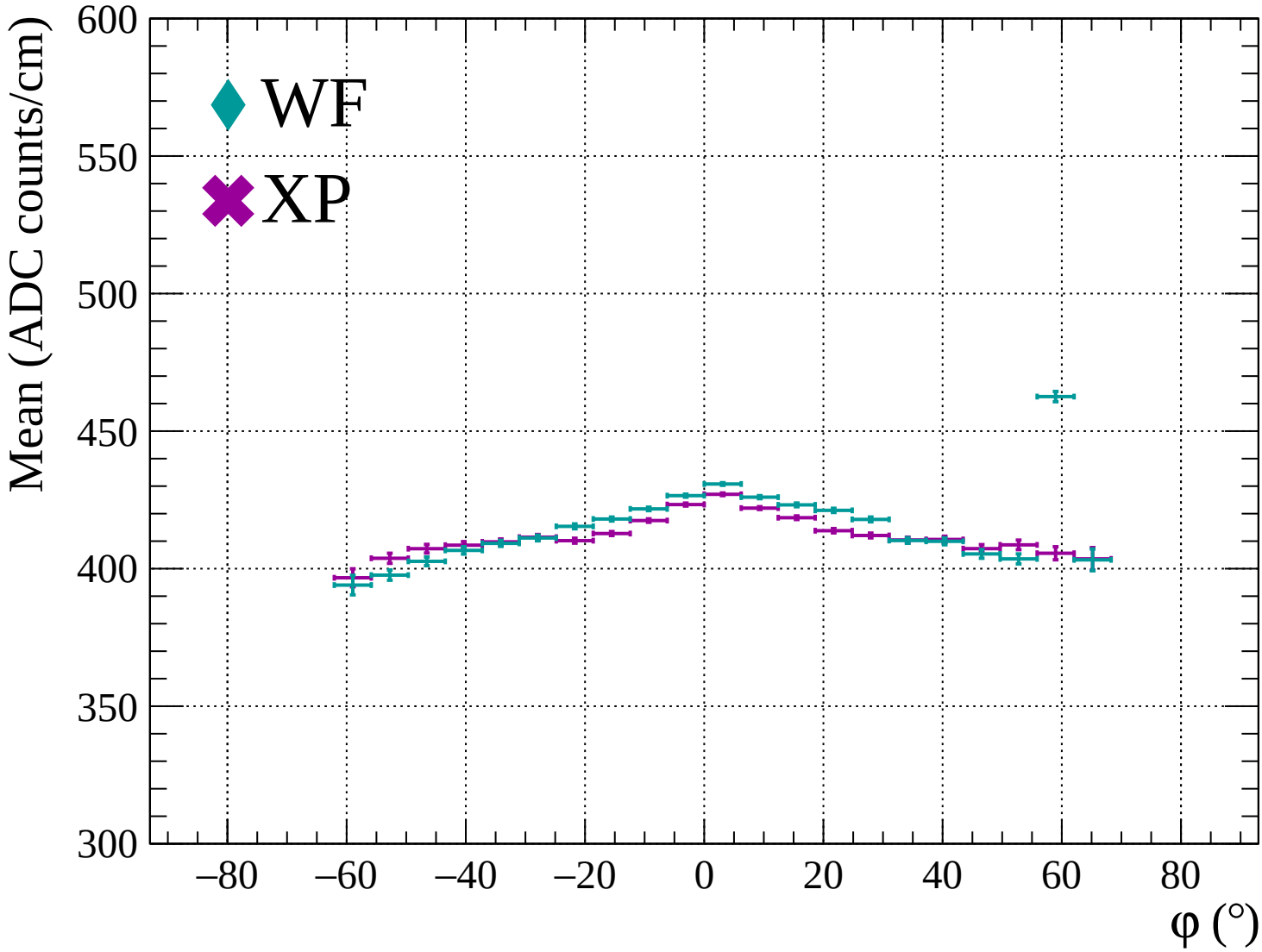


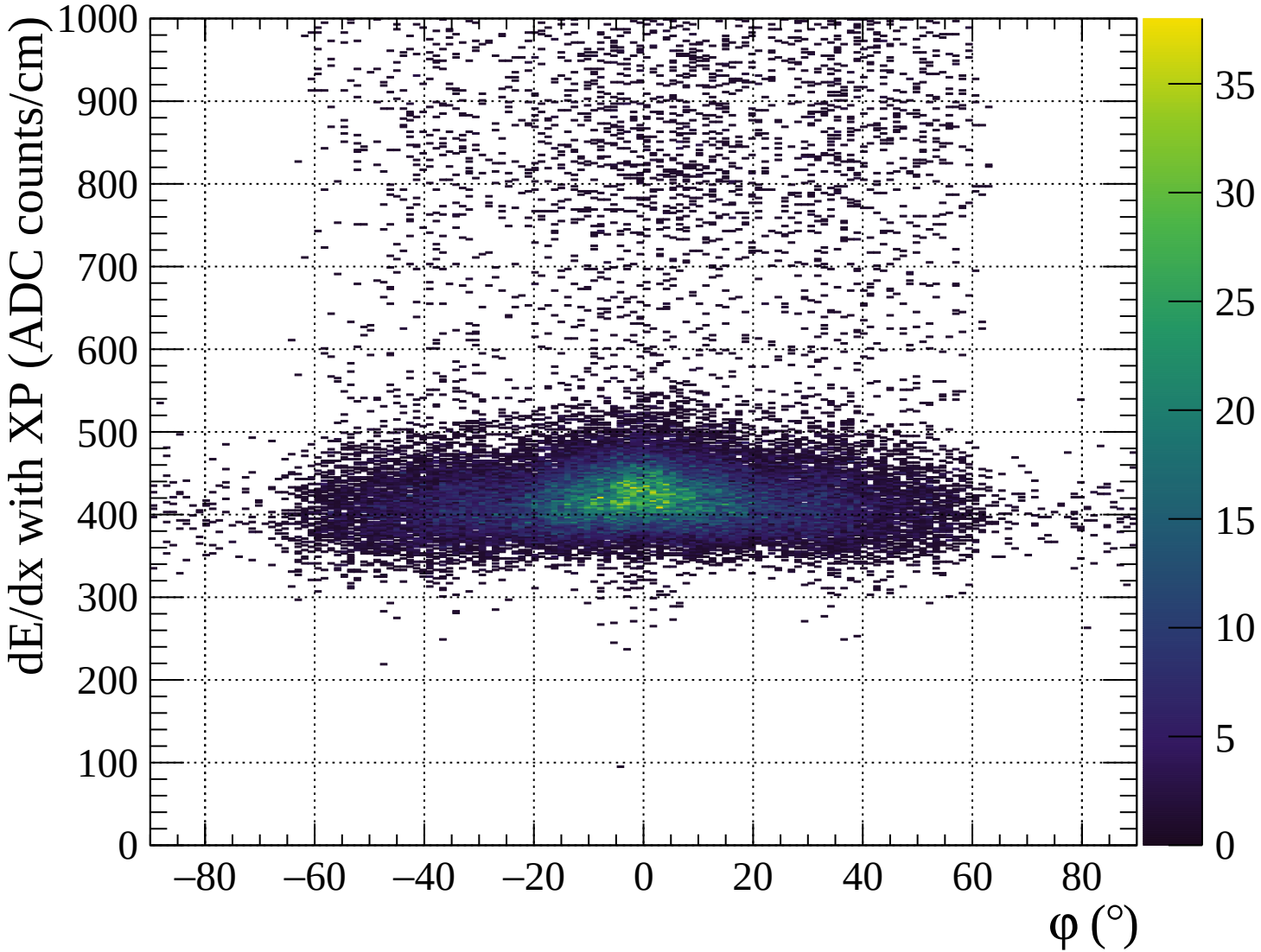


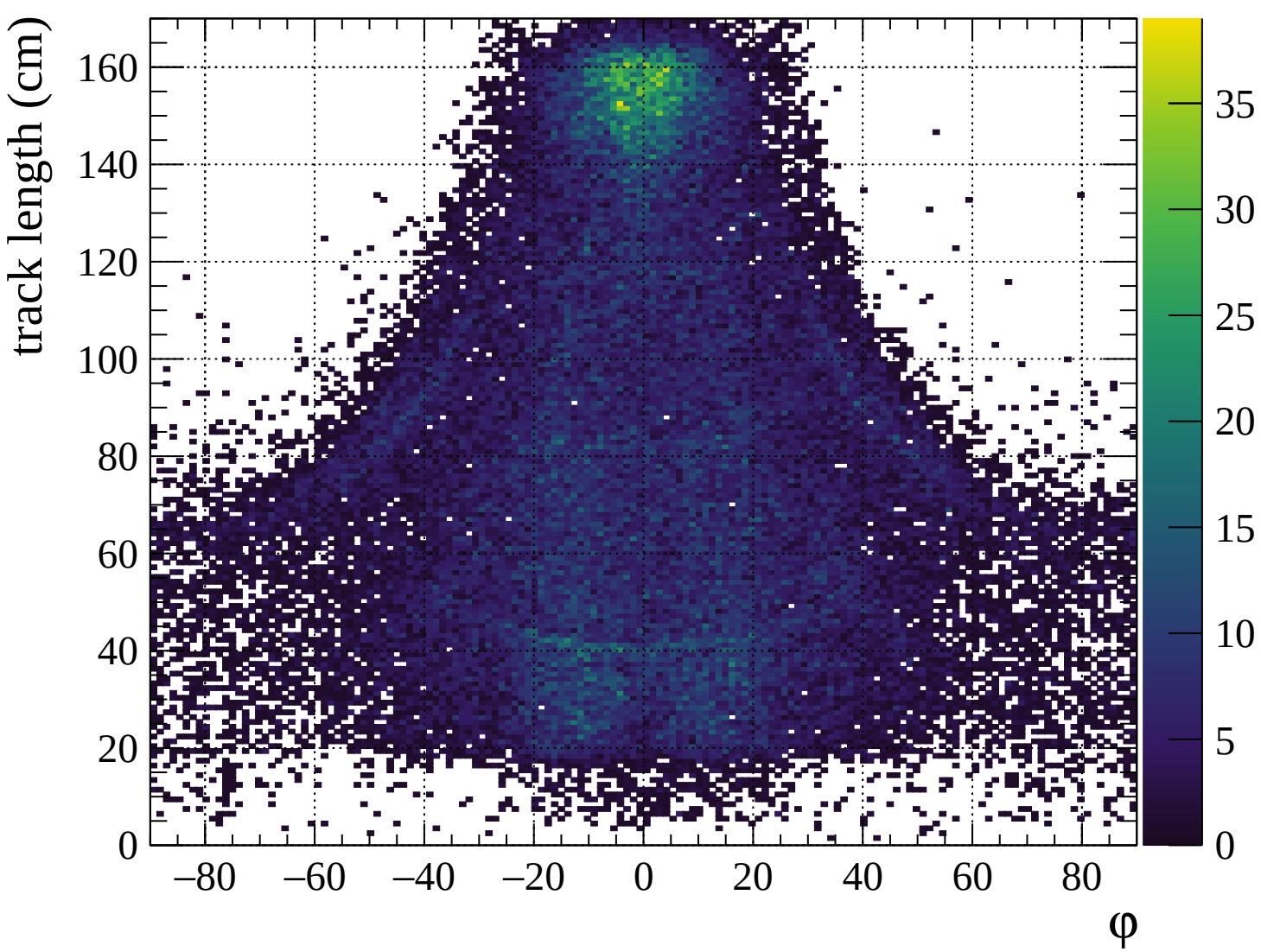


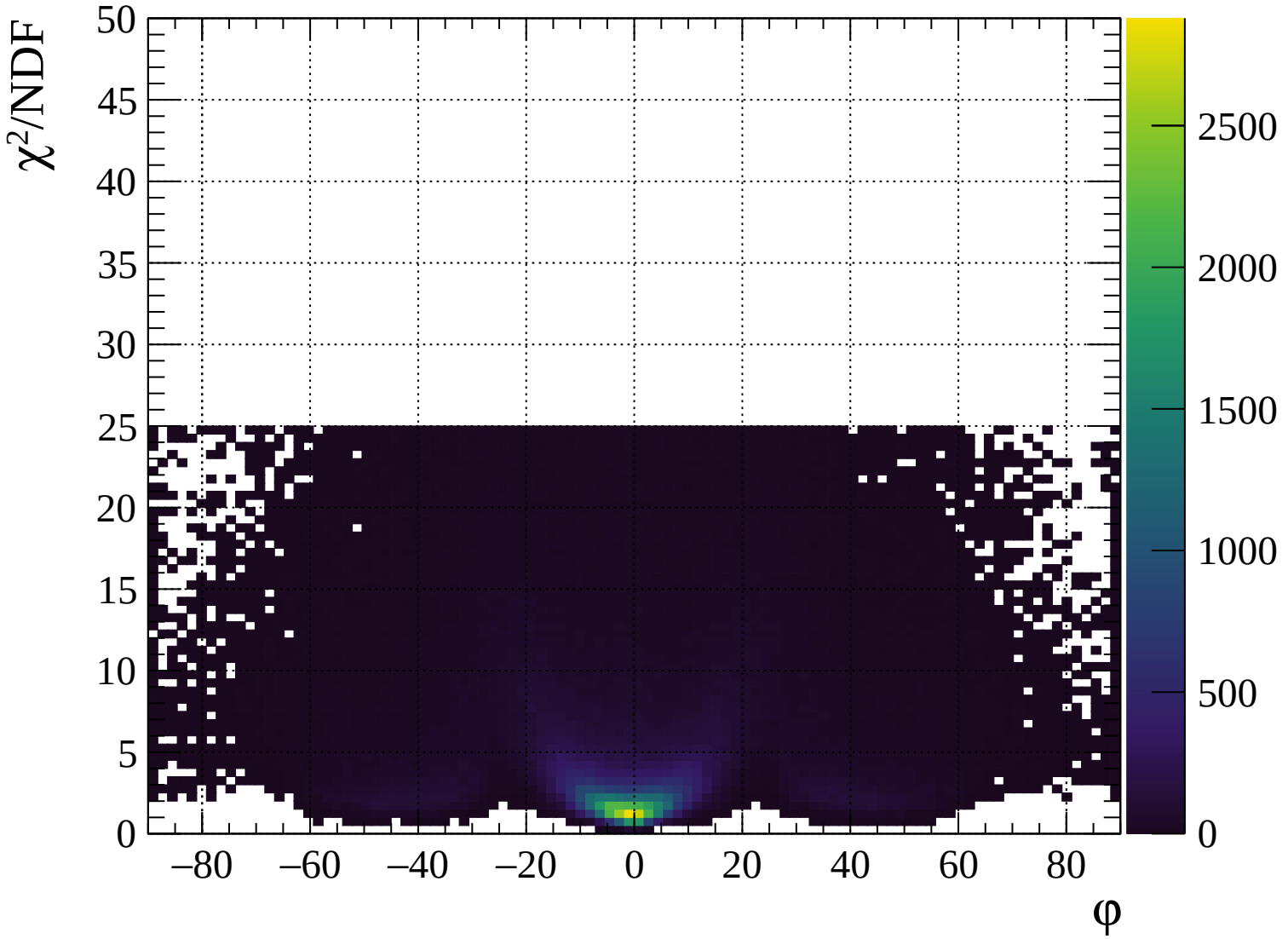


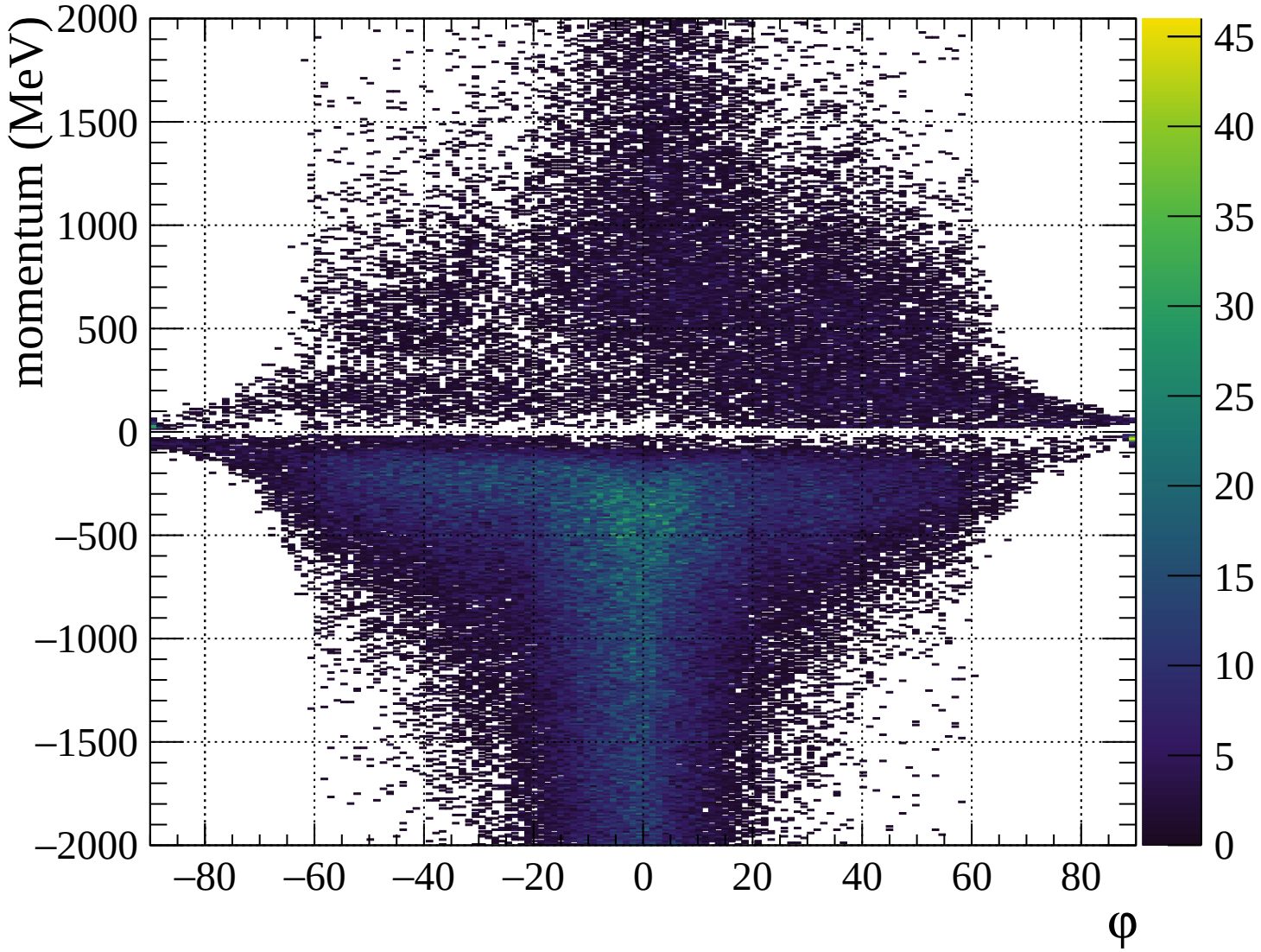


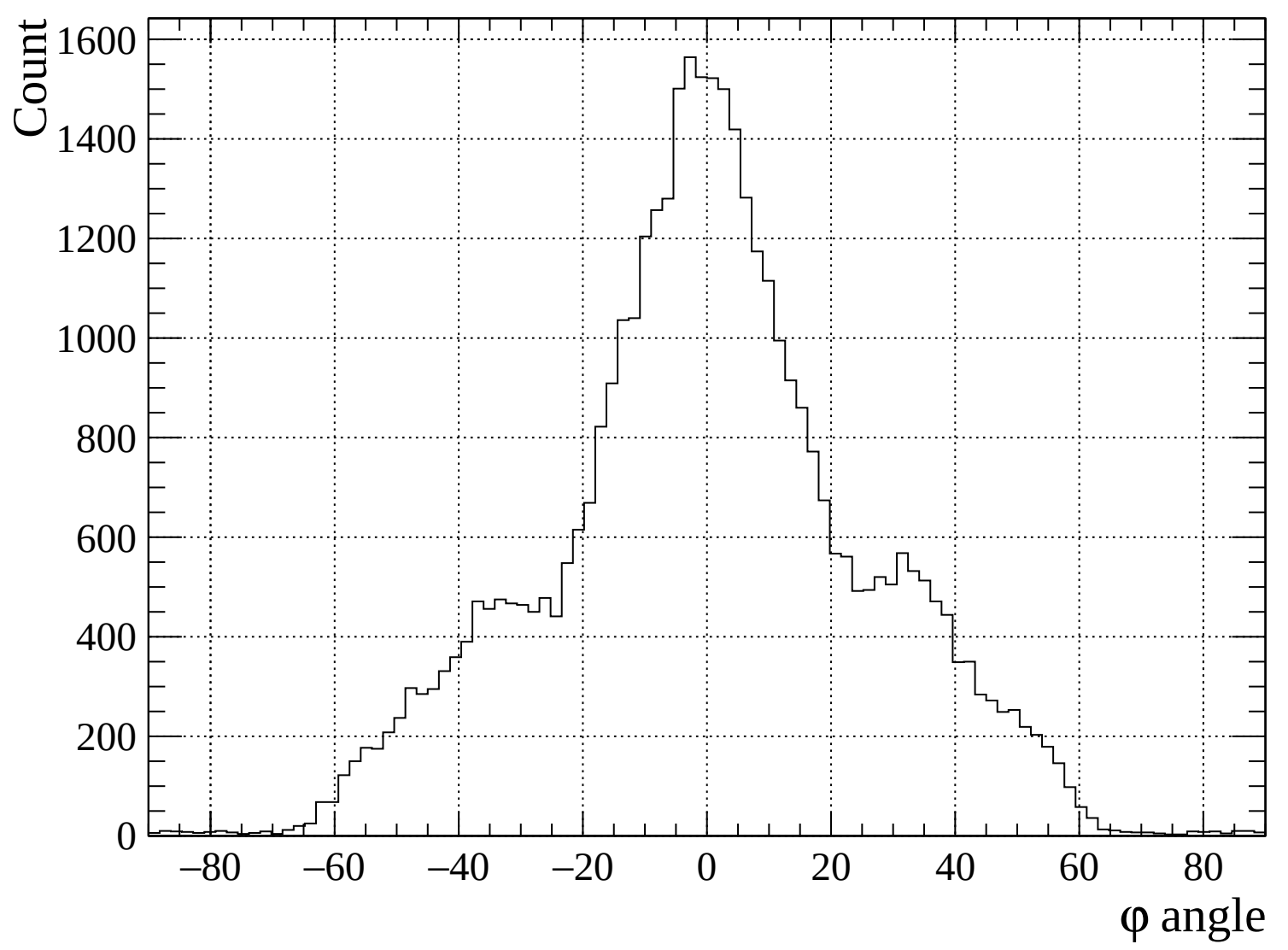


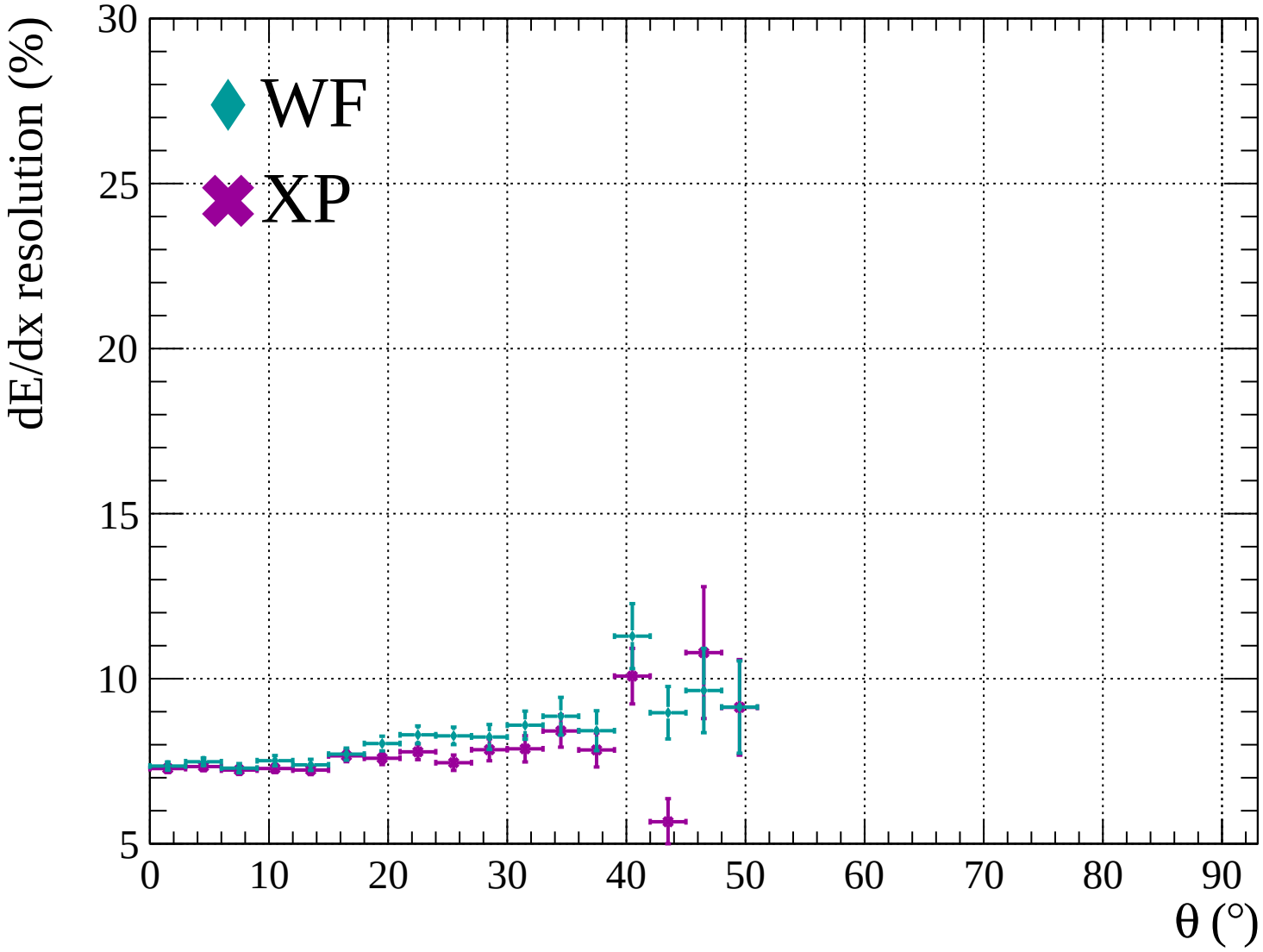


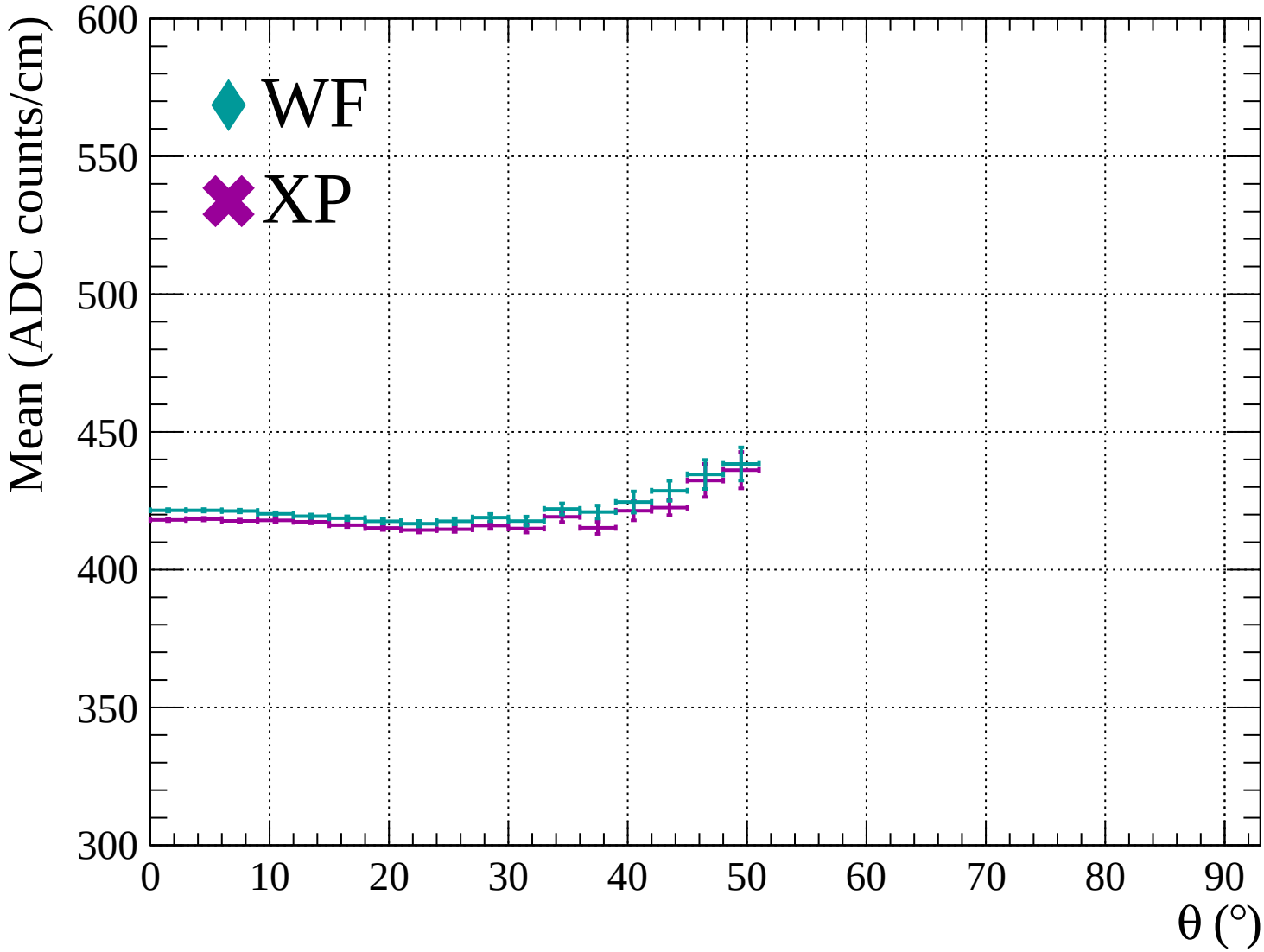


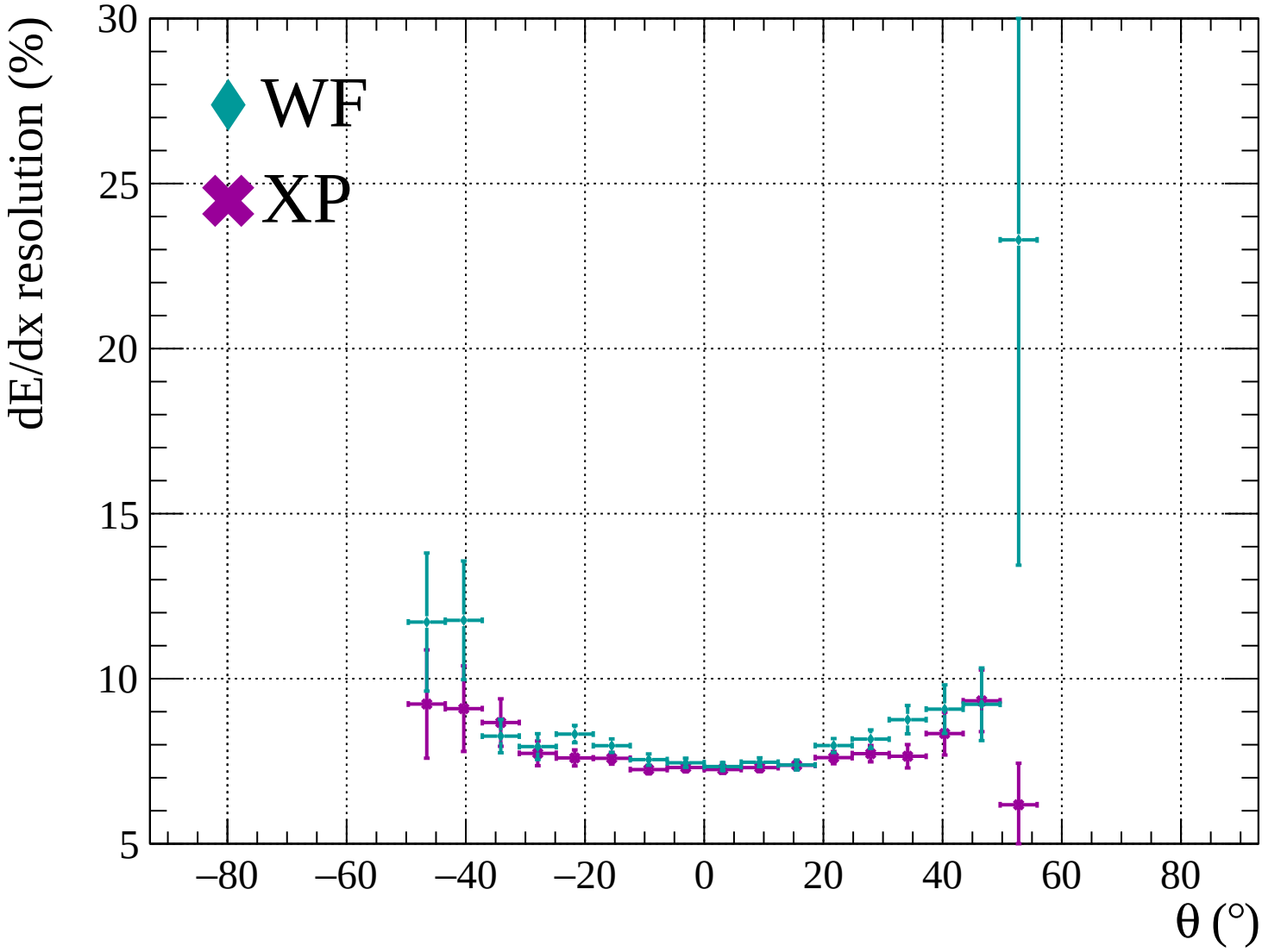


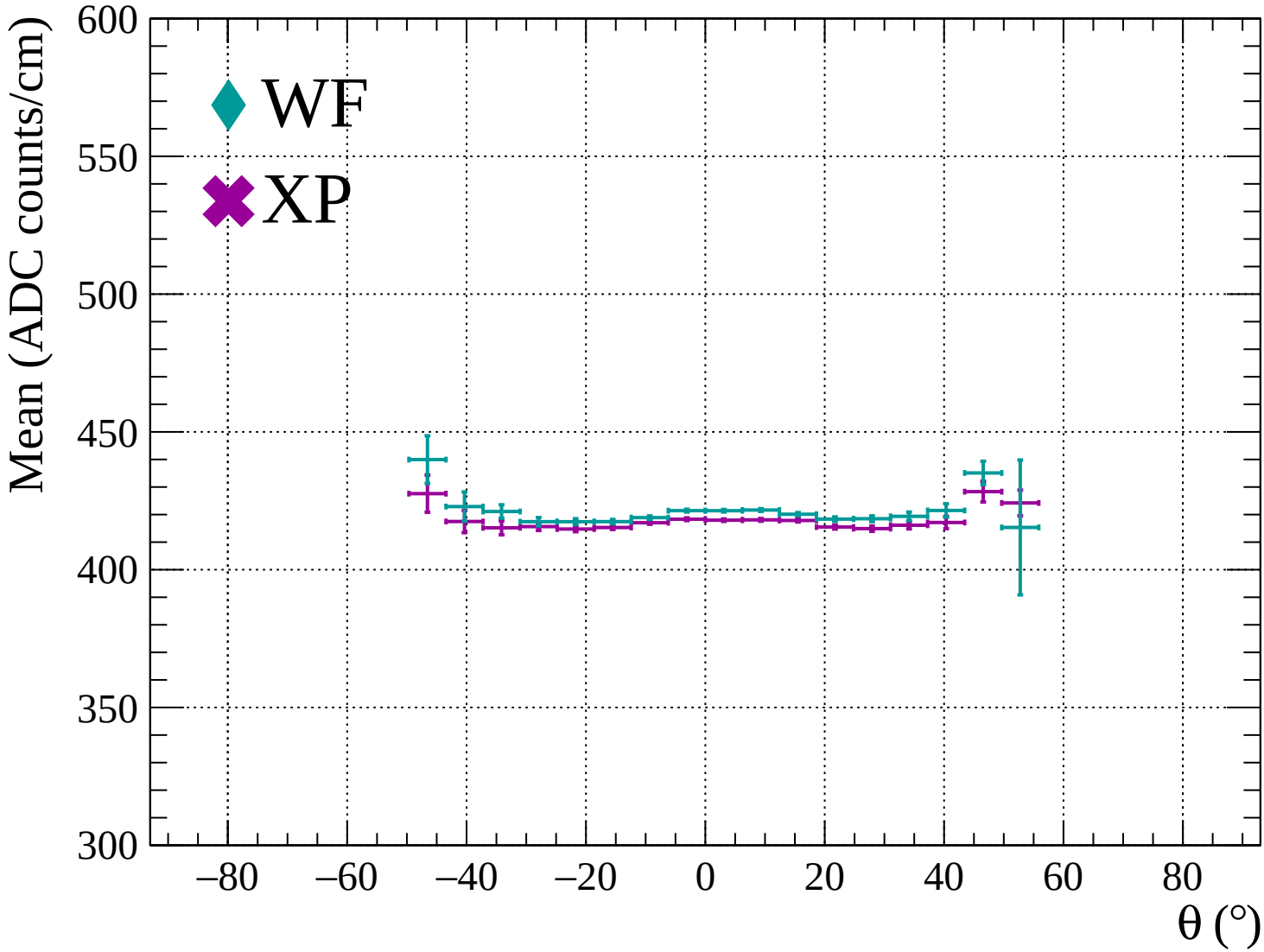


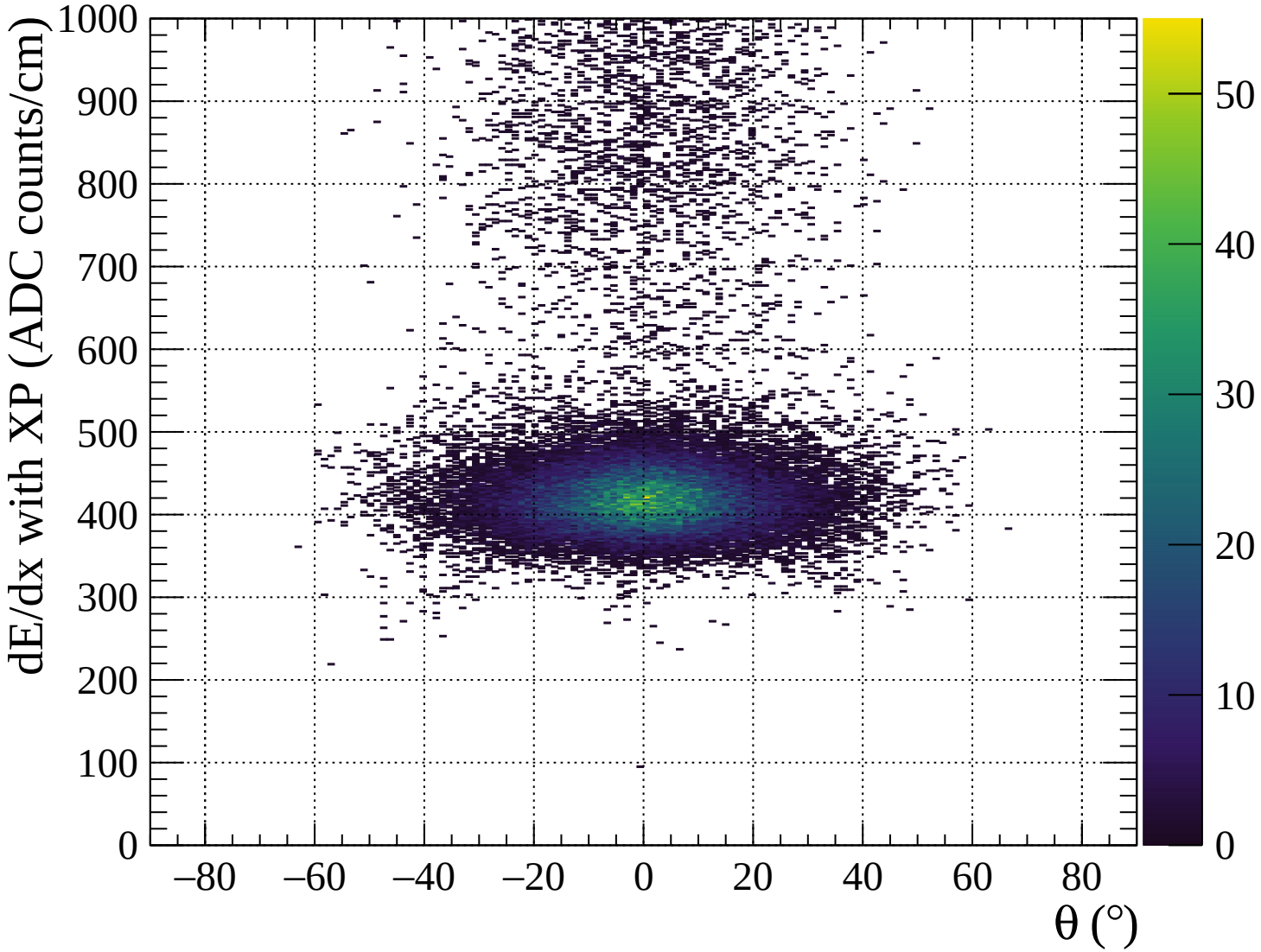


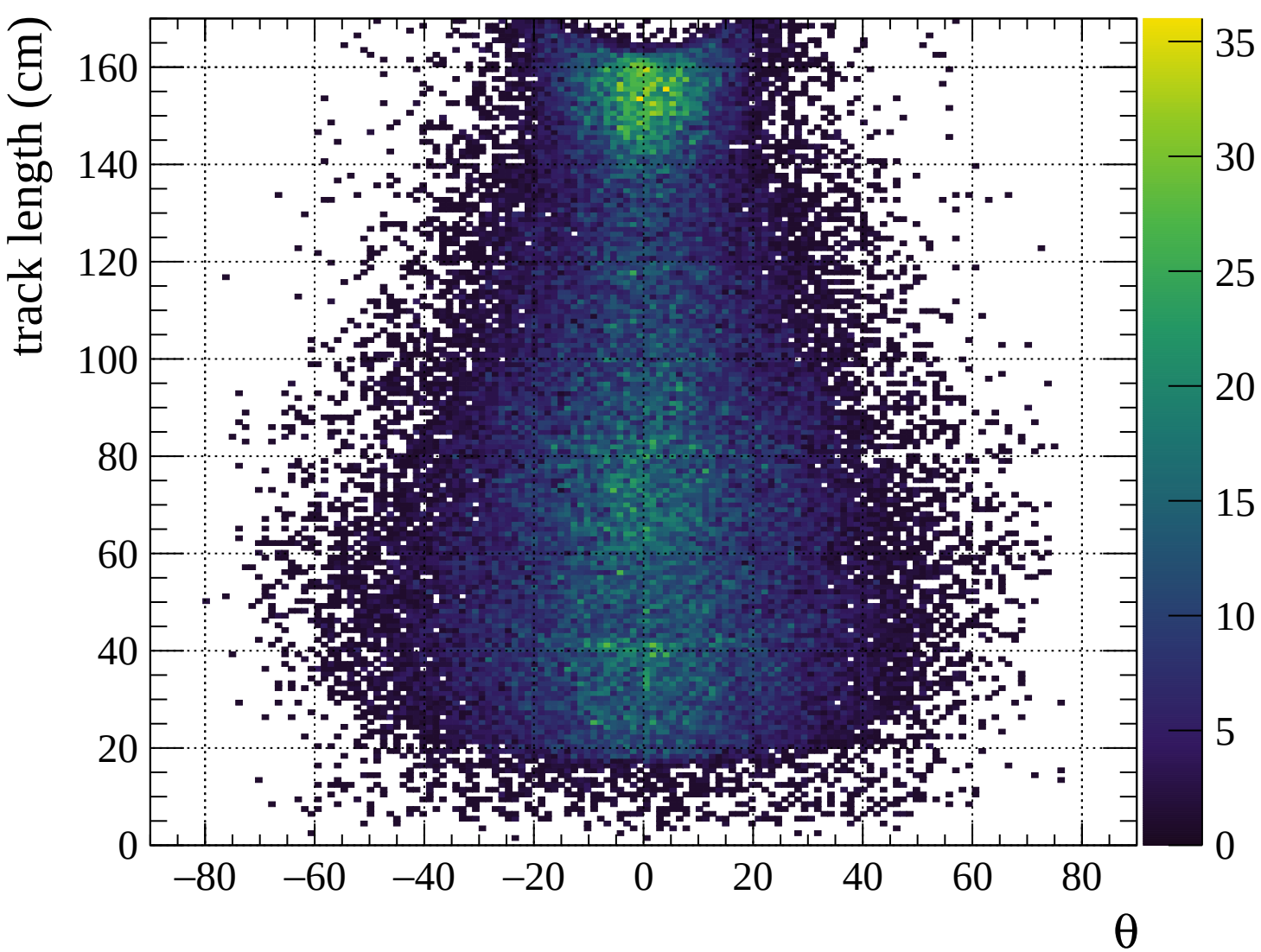


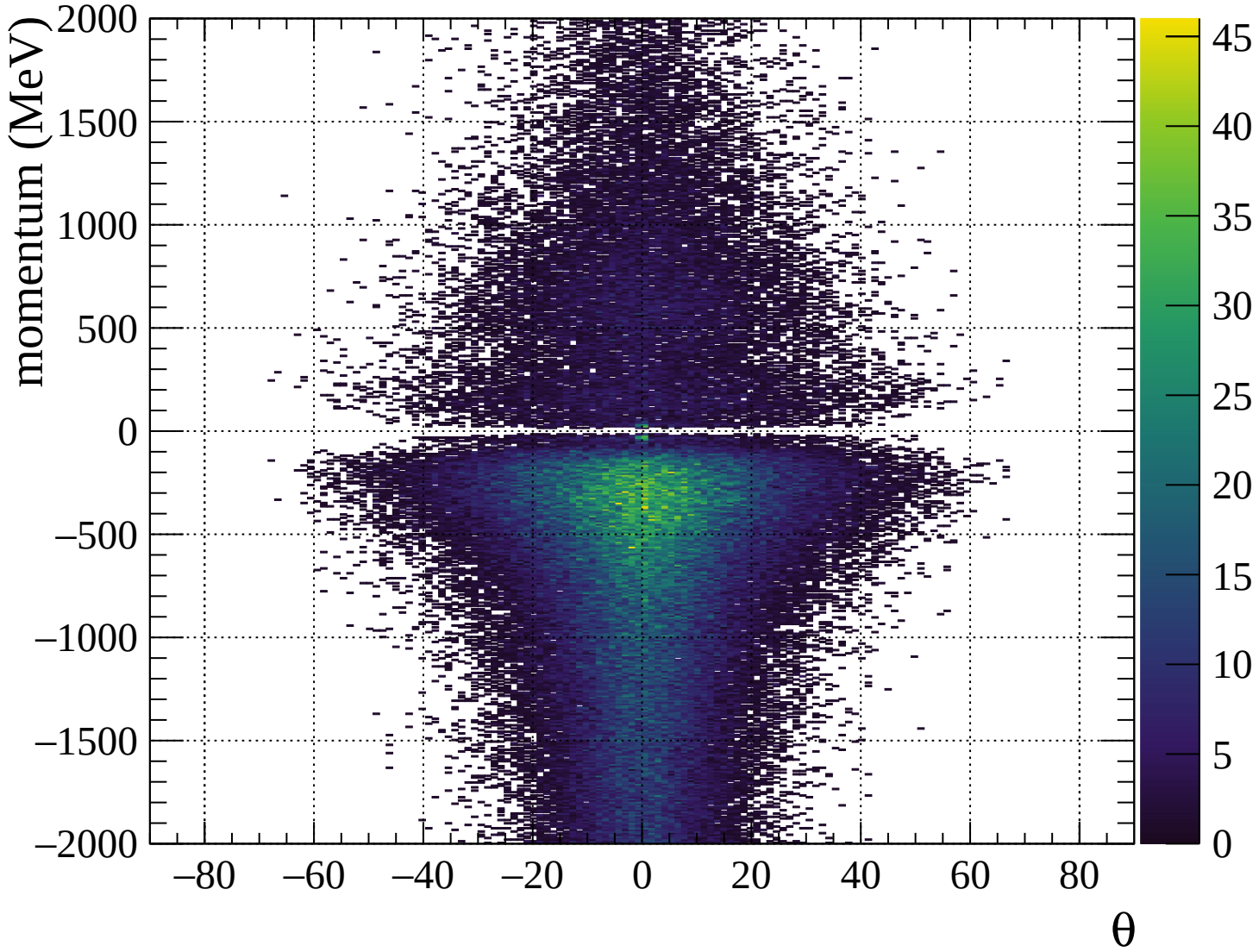


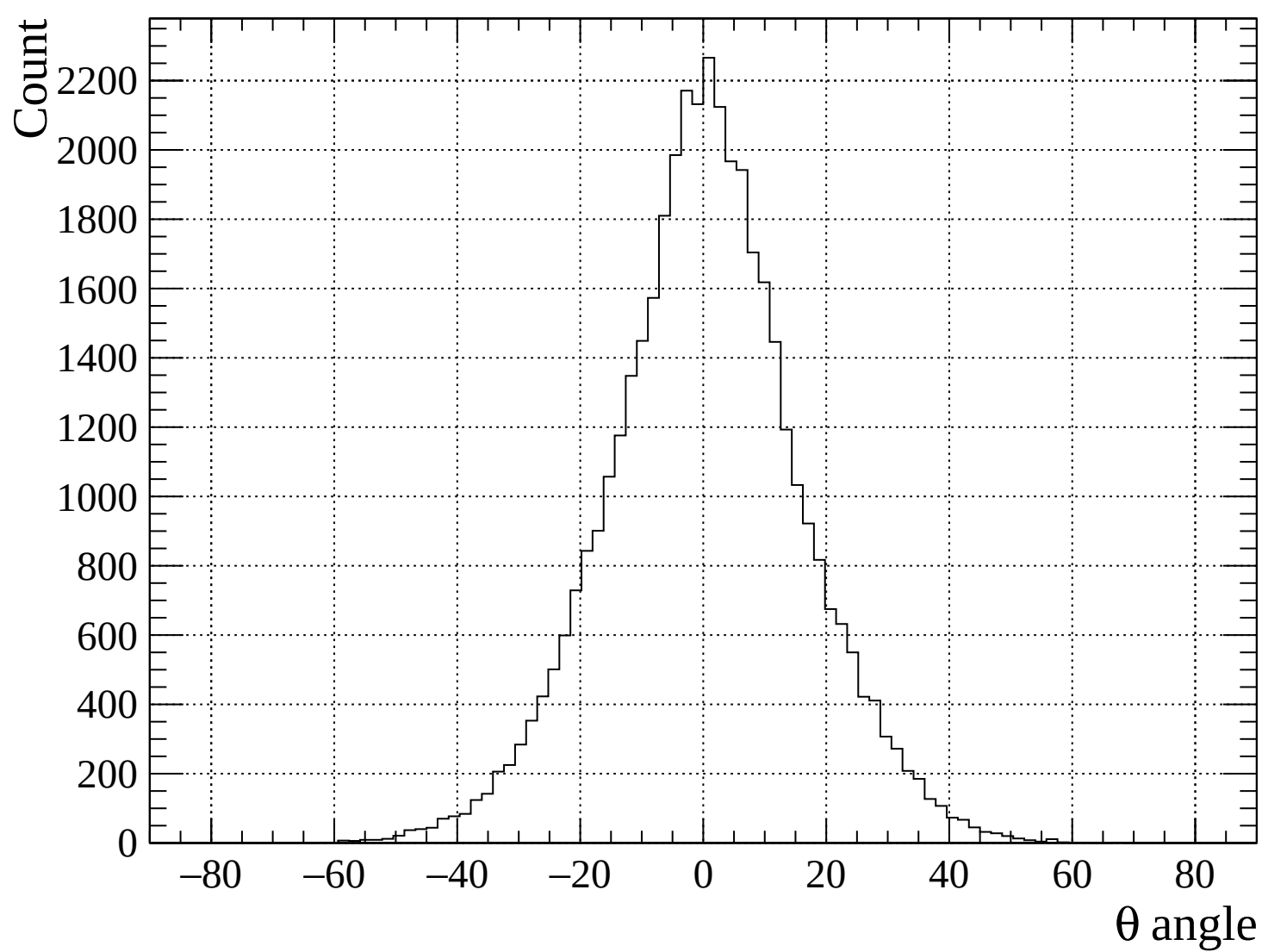


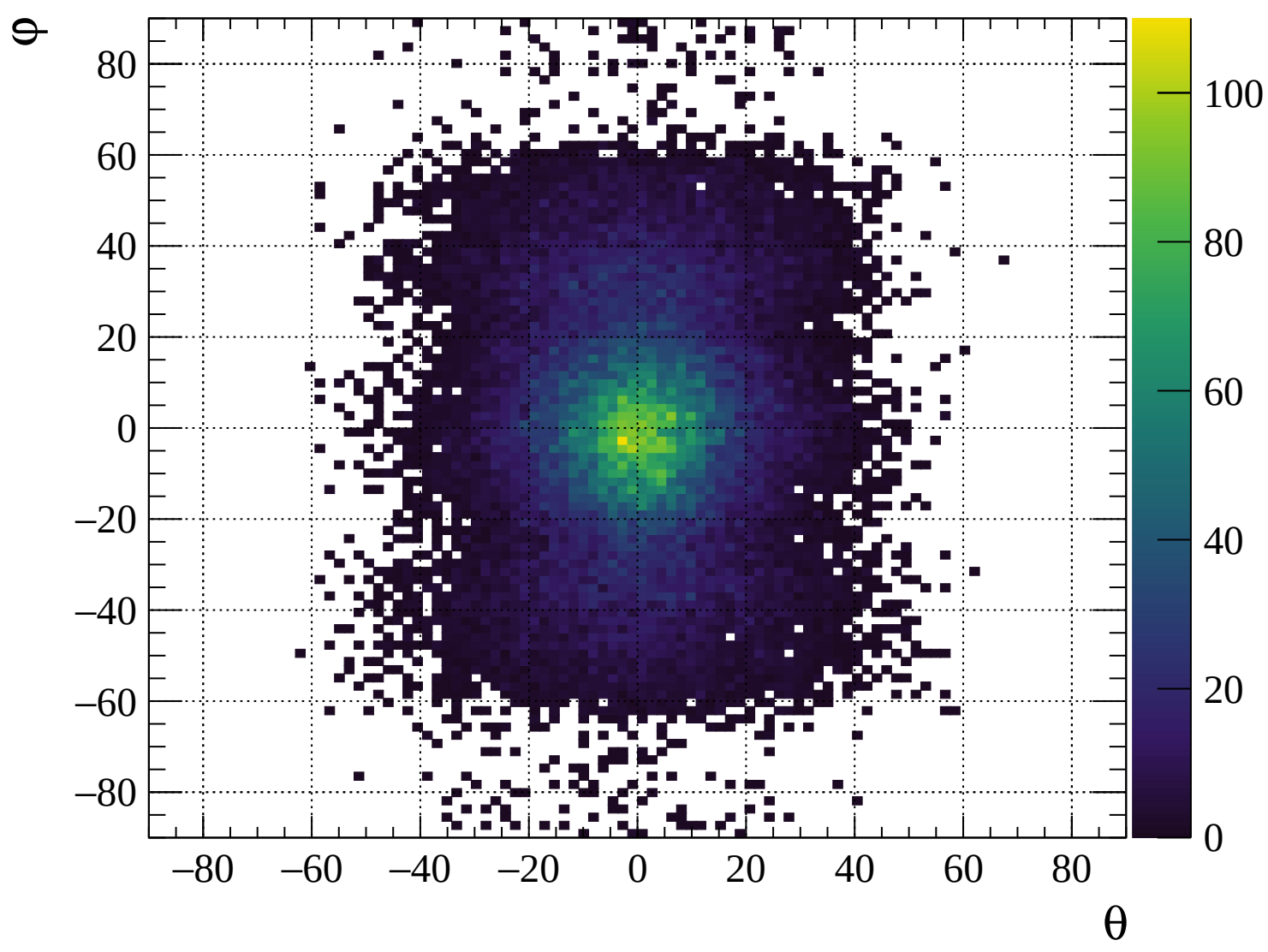


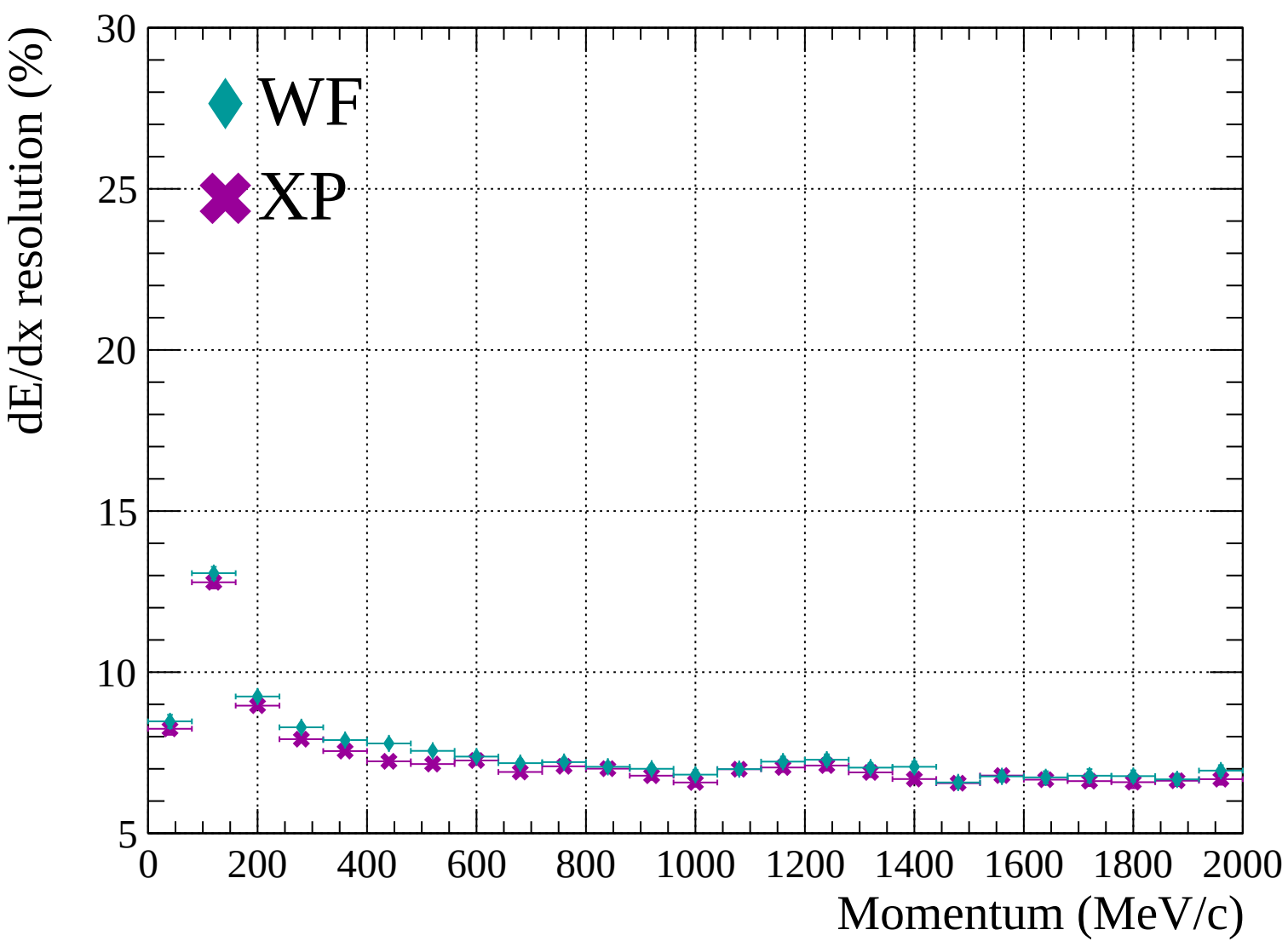


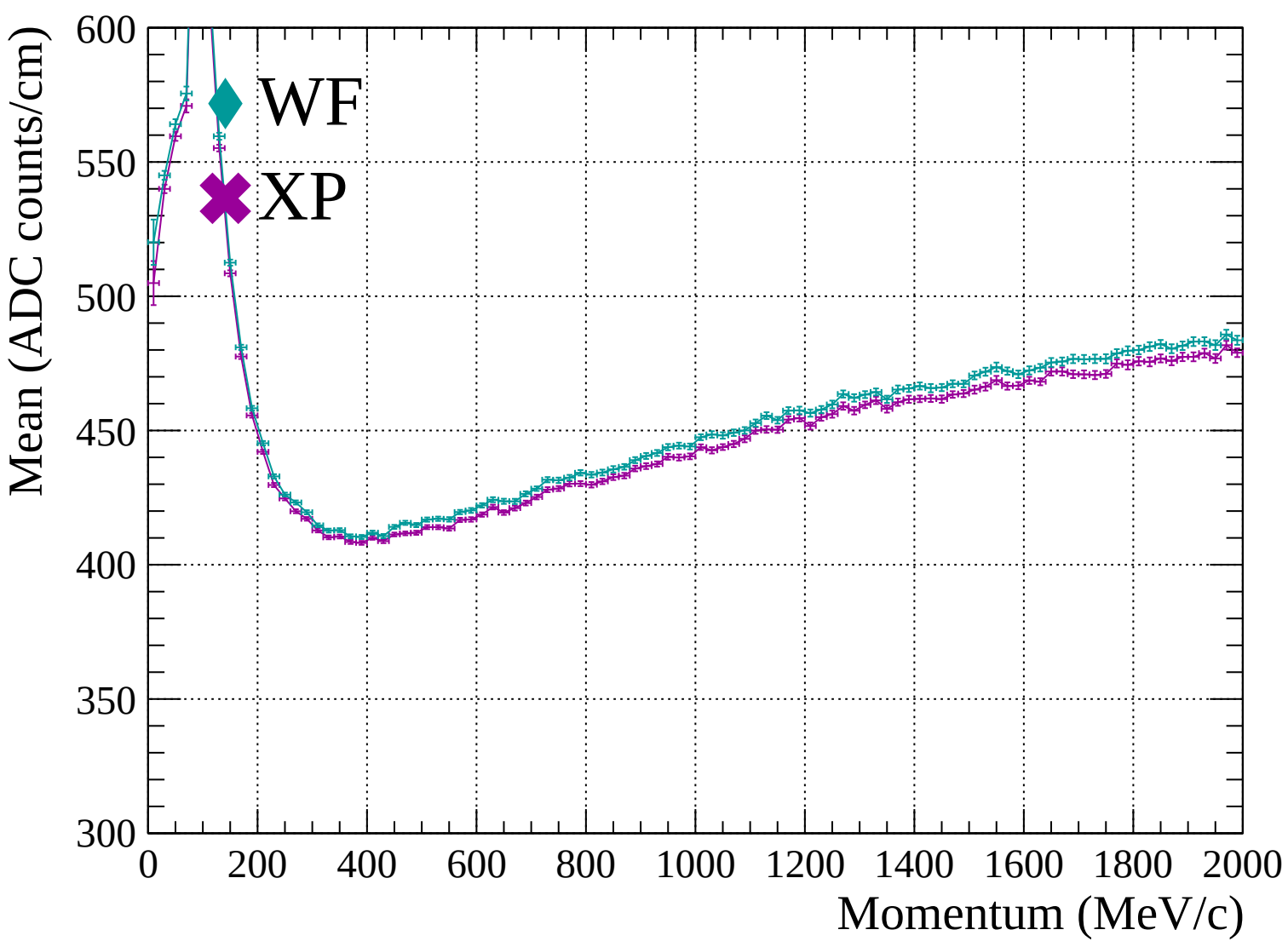


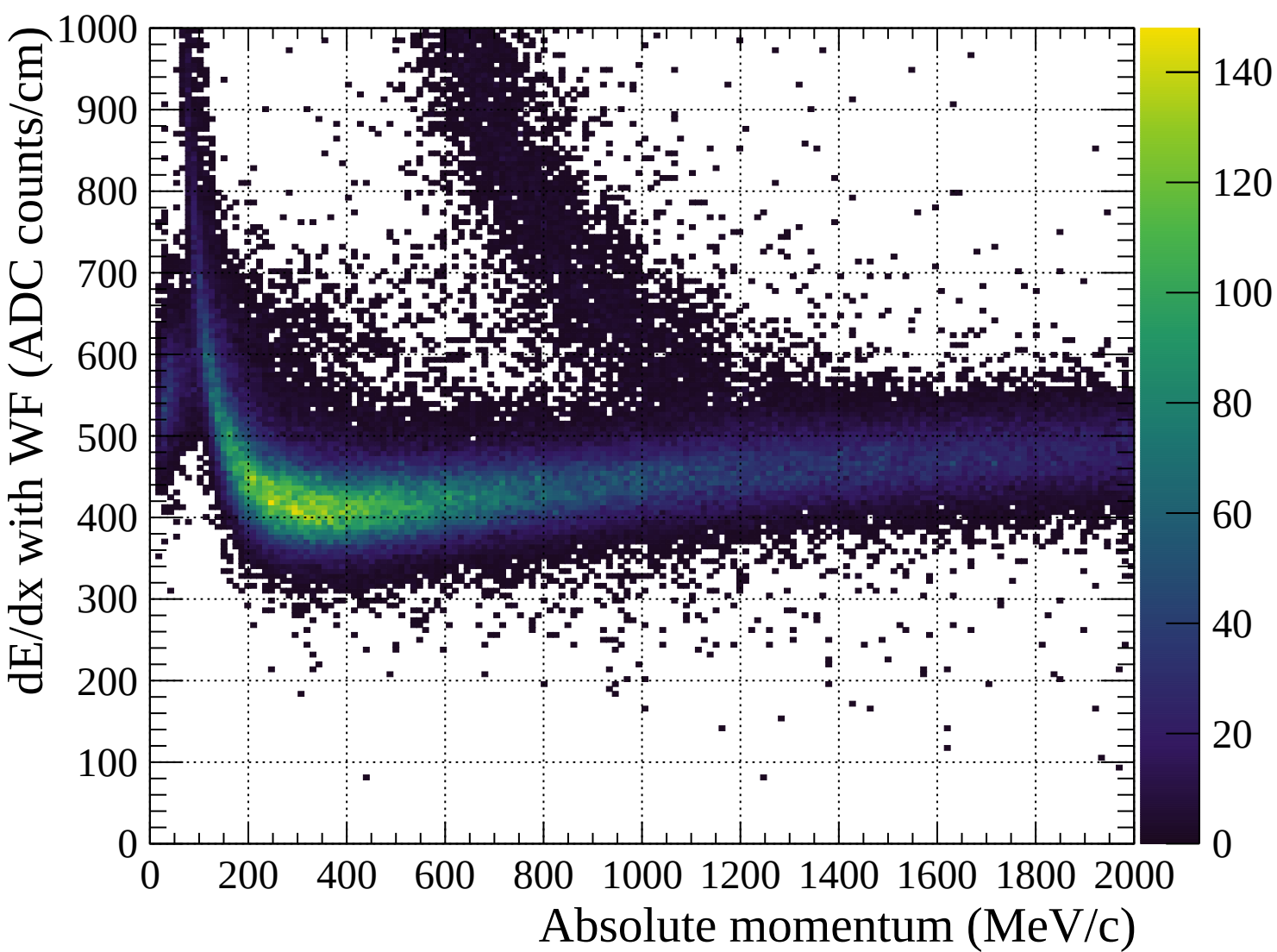


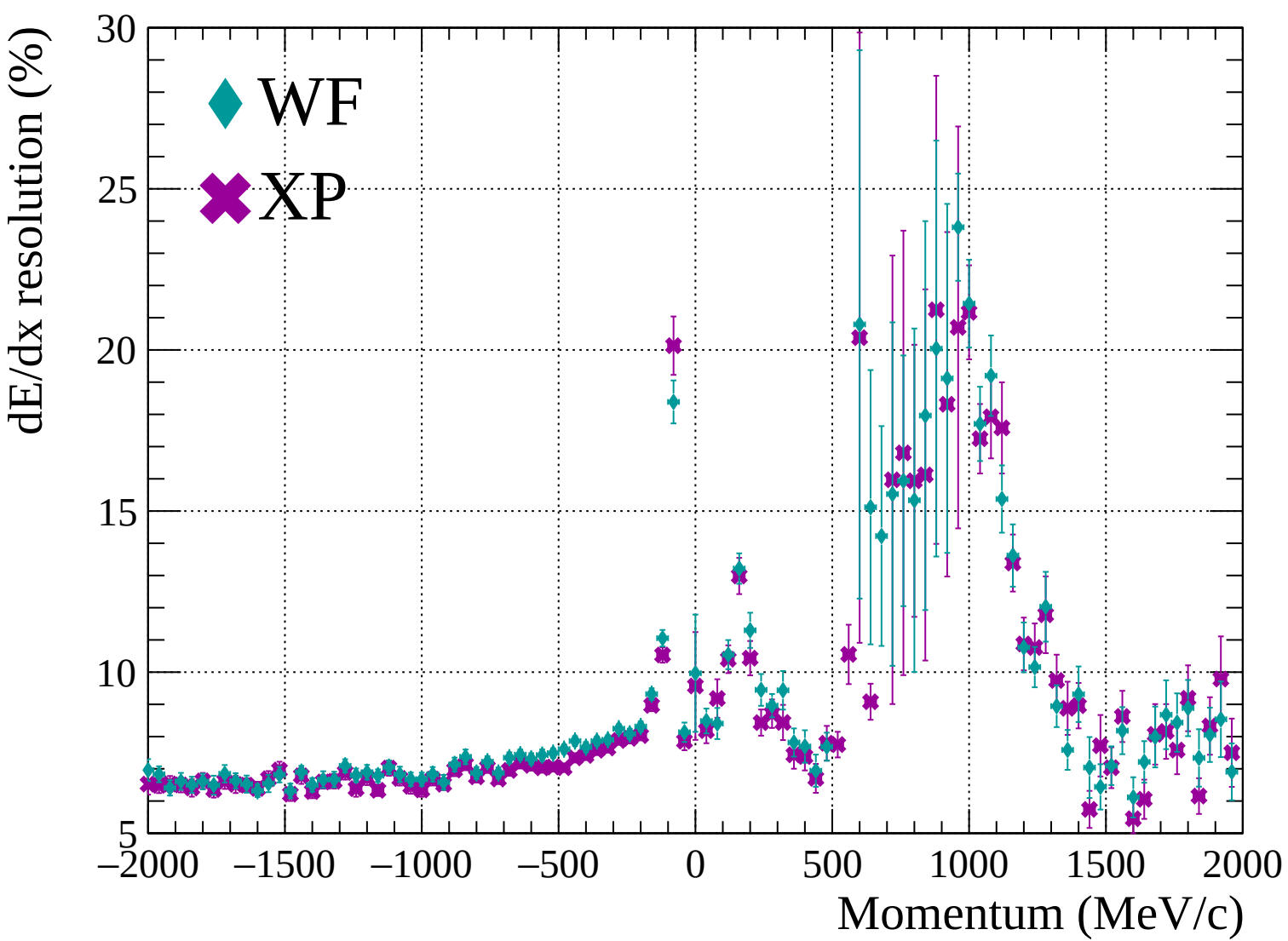


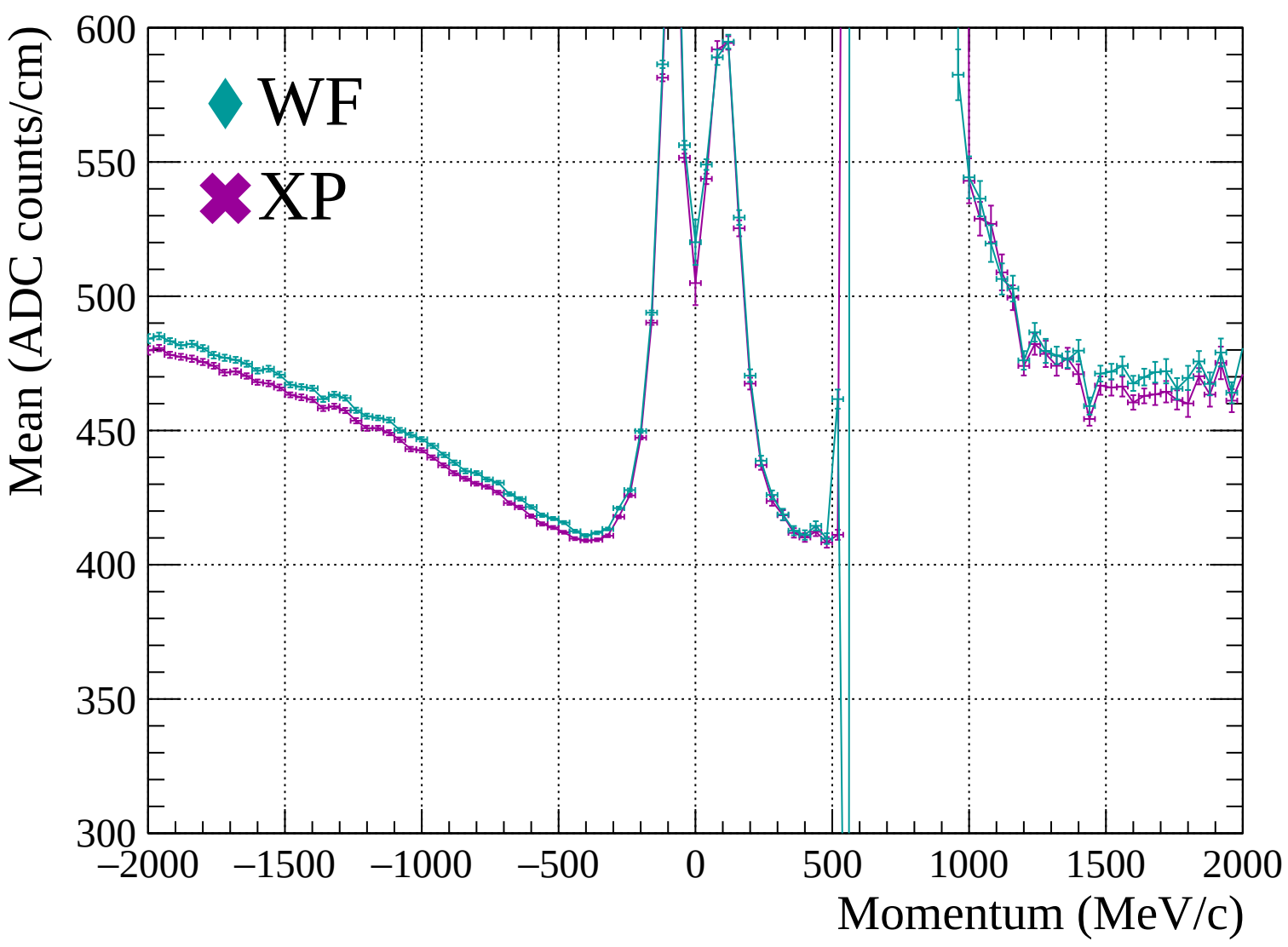


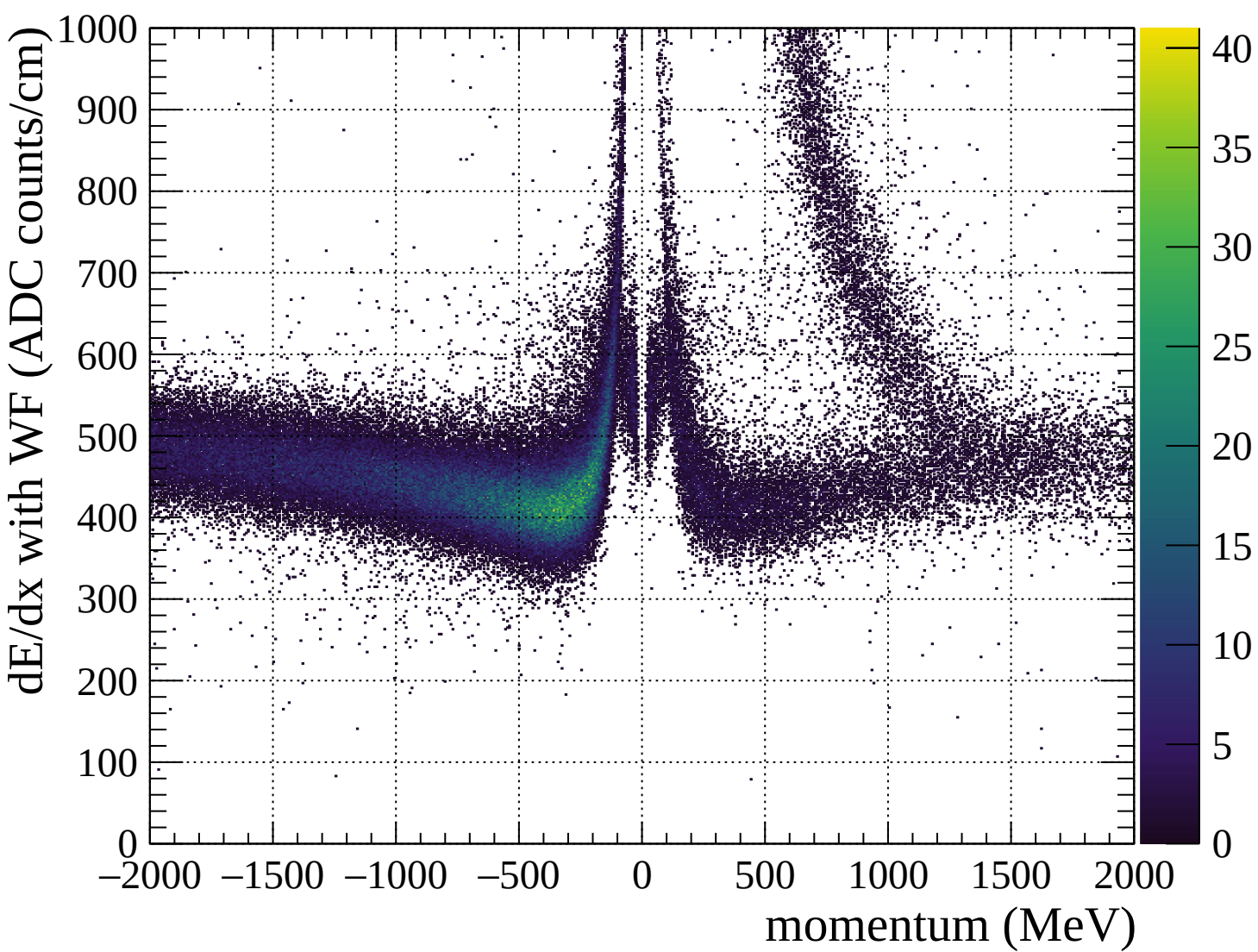


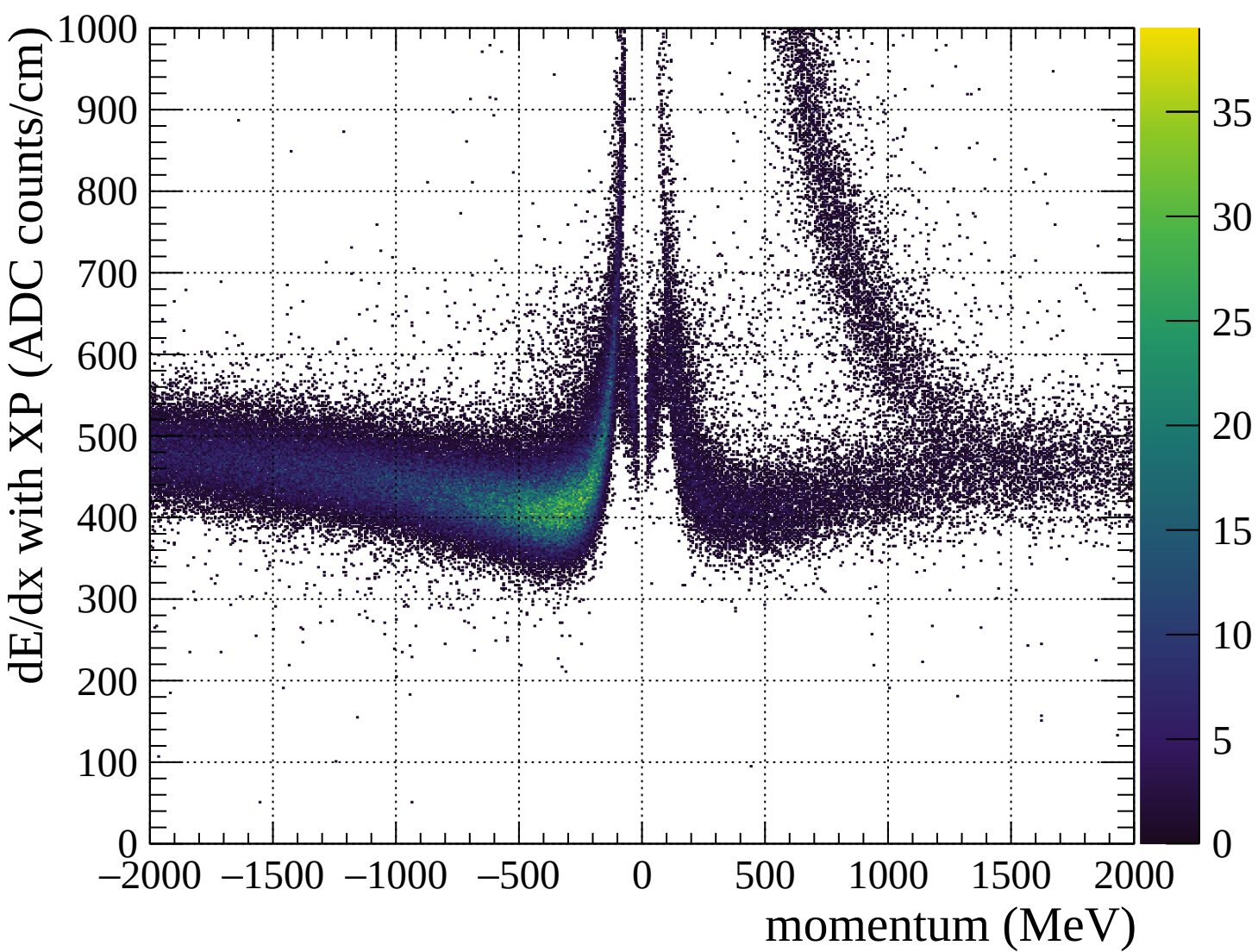


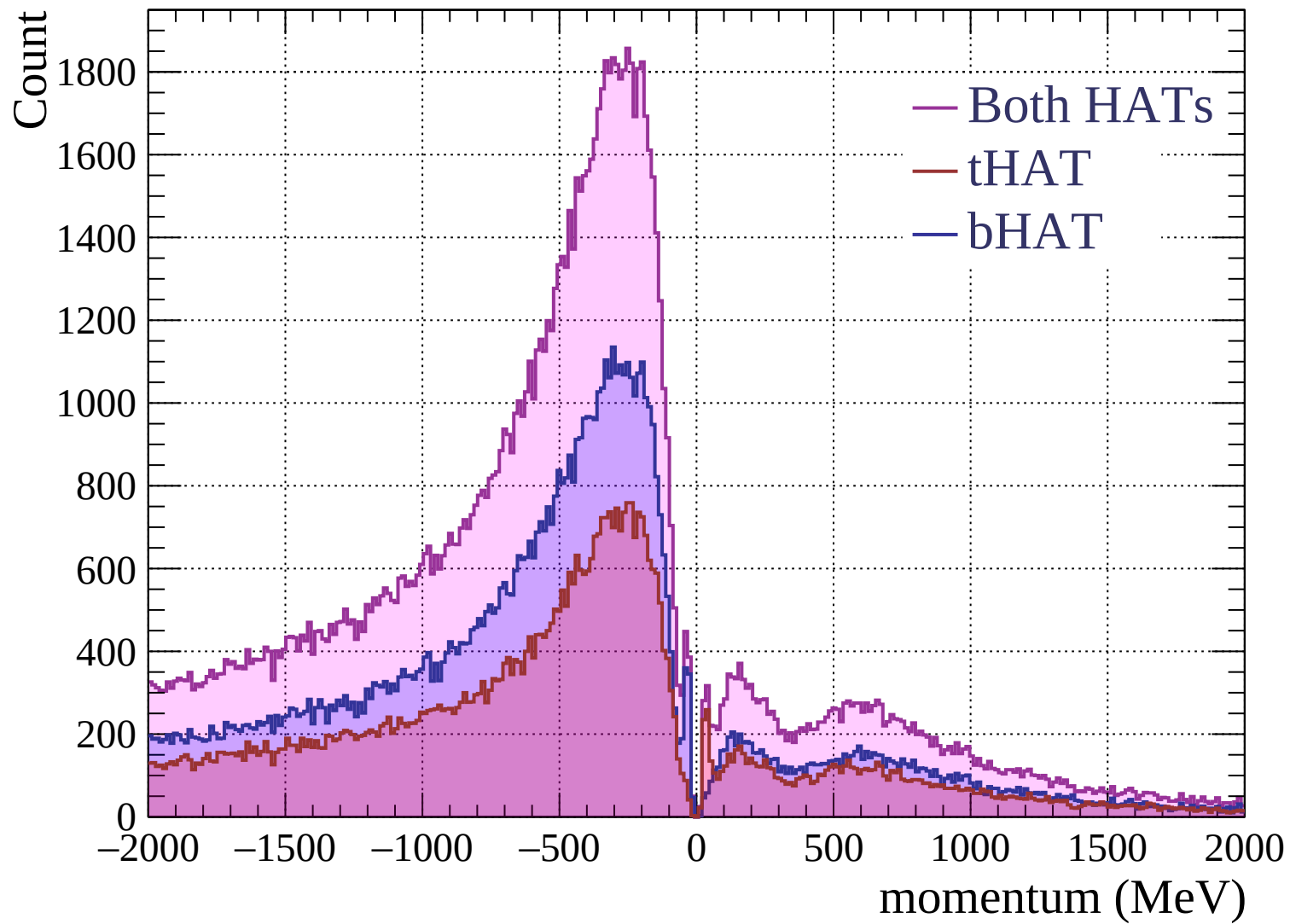


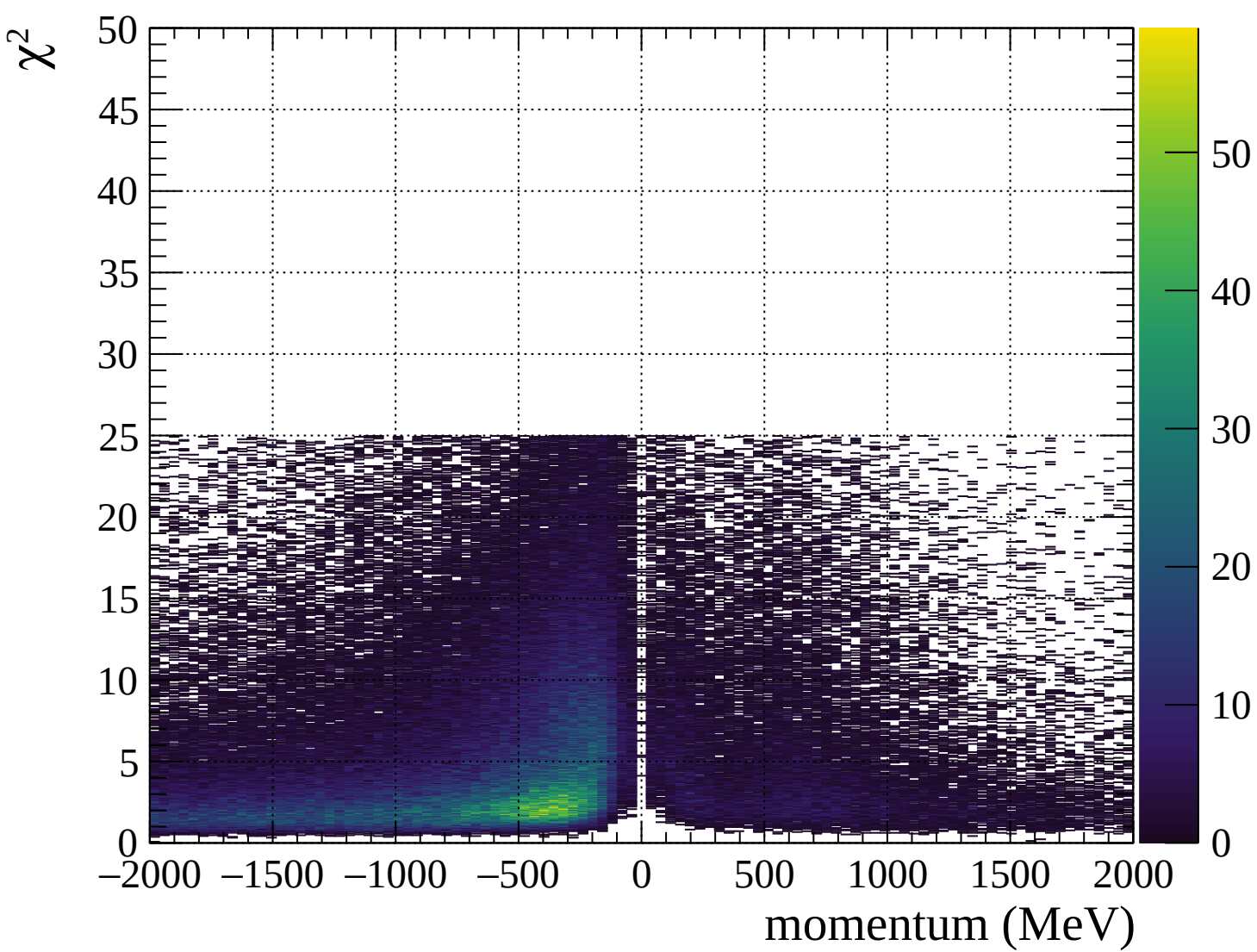




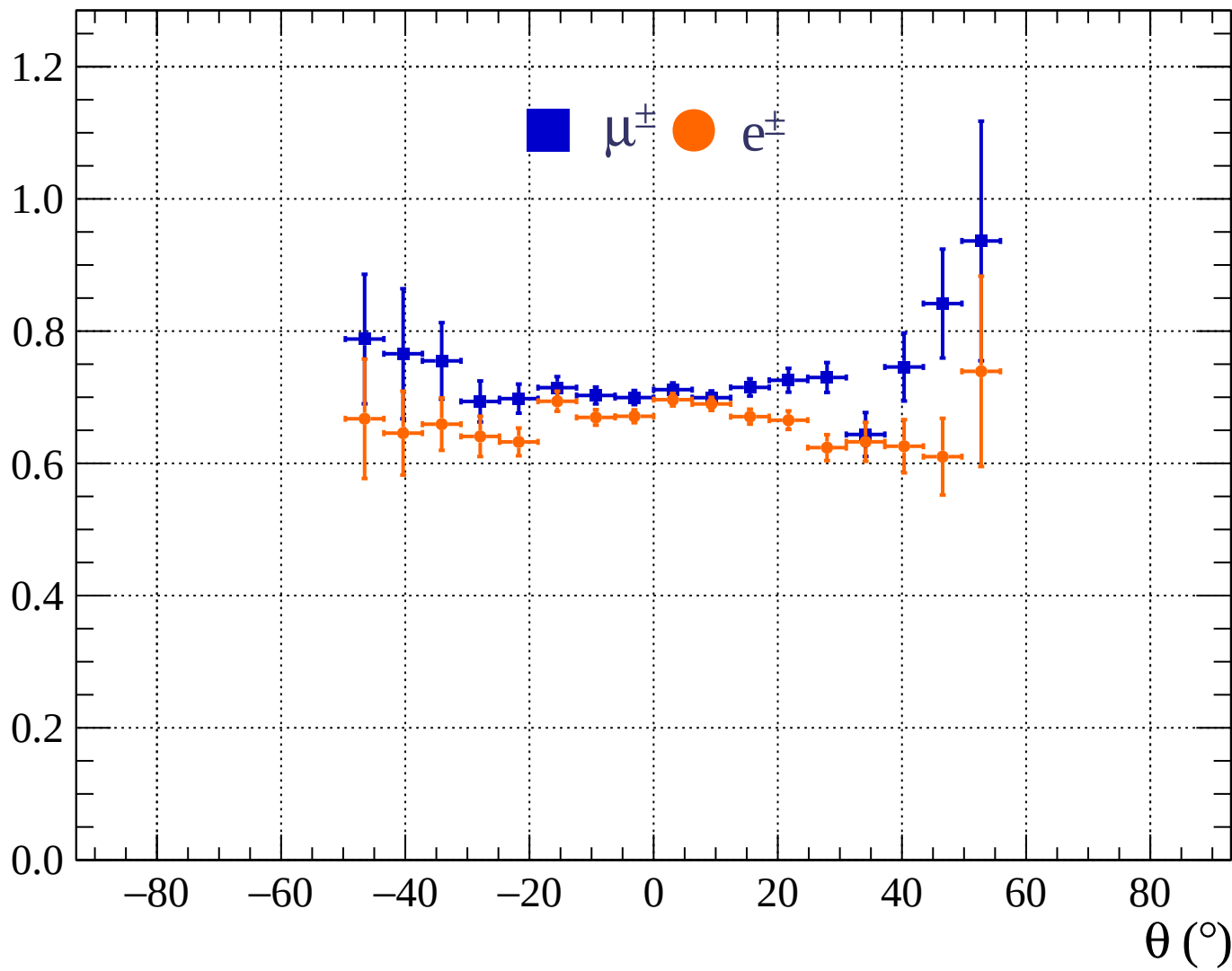




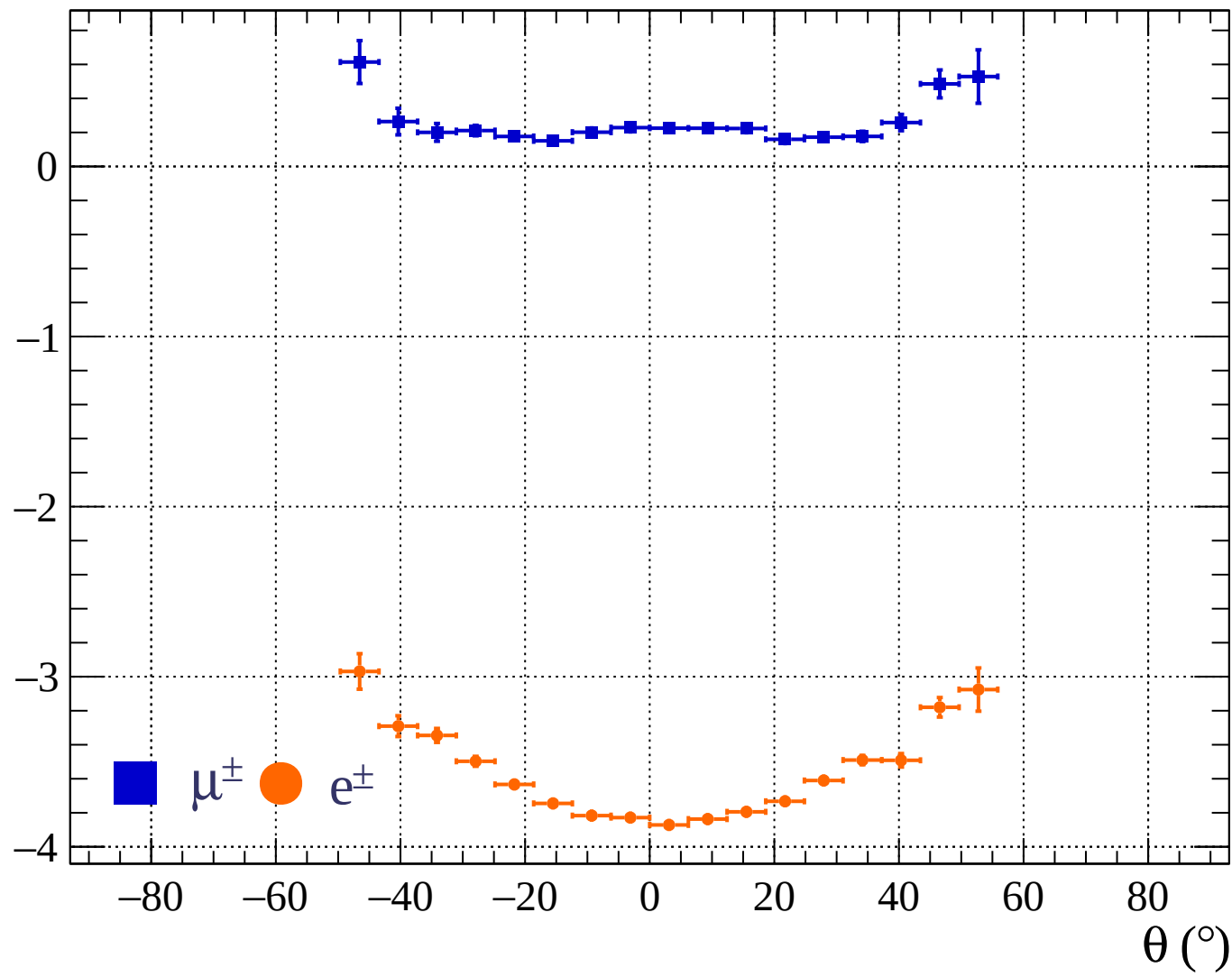




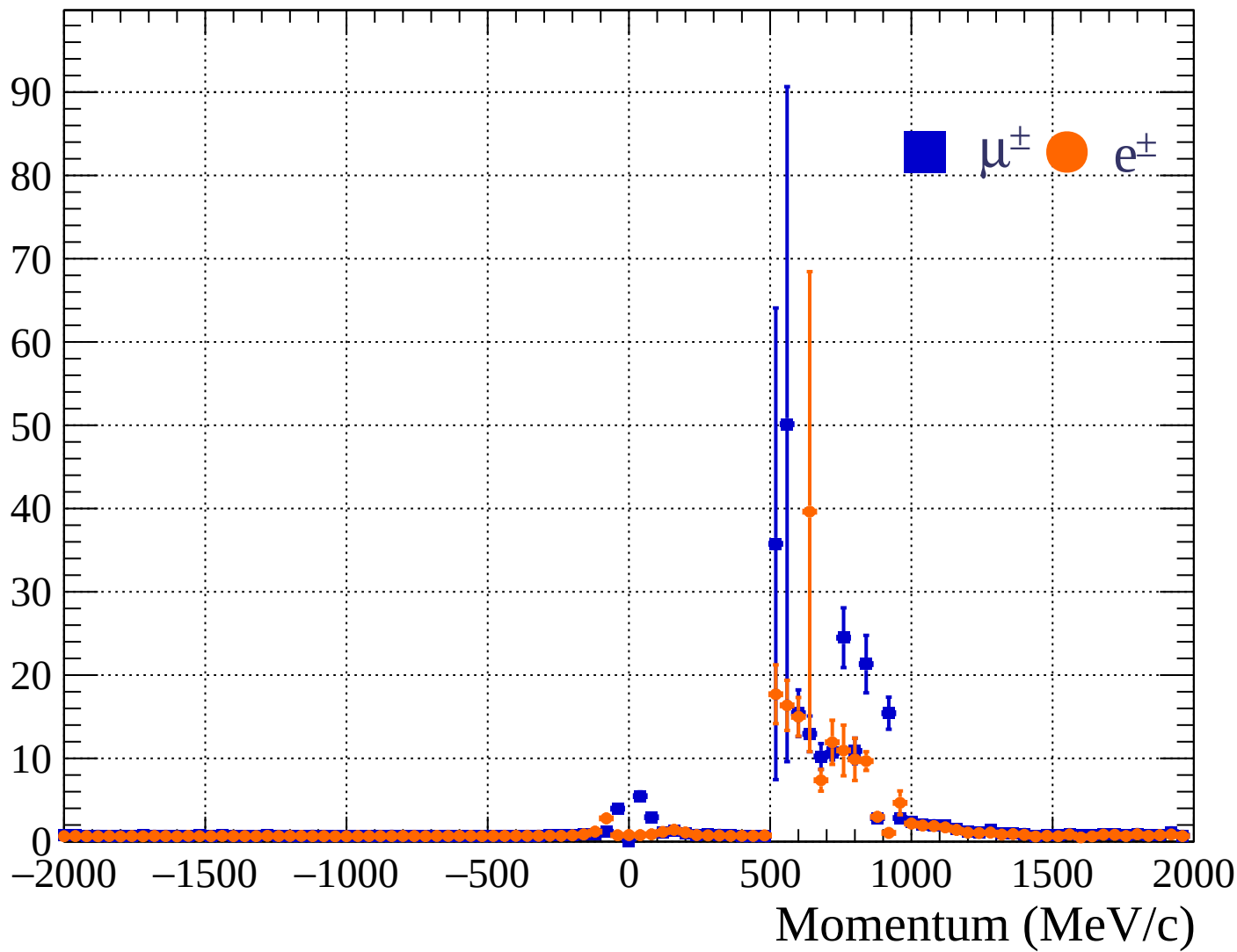
Pulls standard deviation

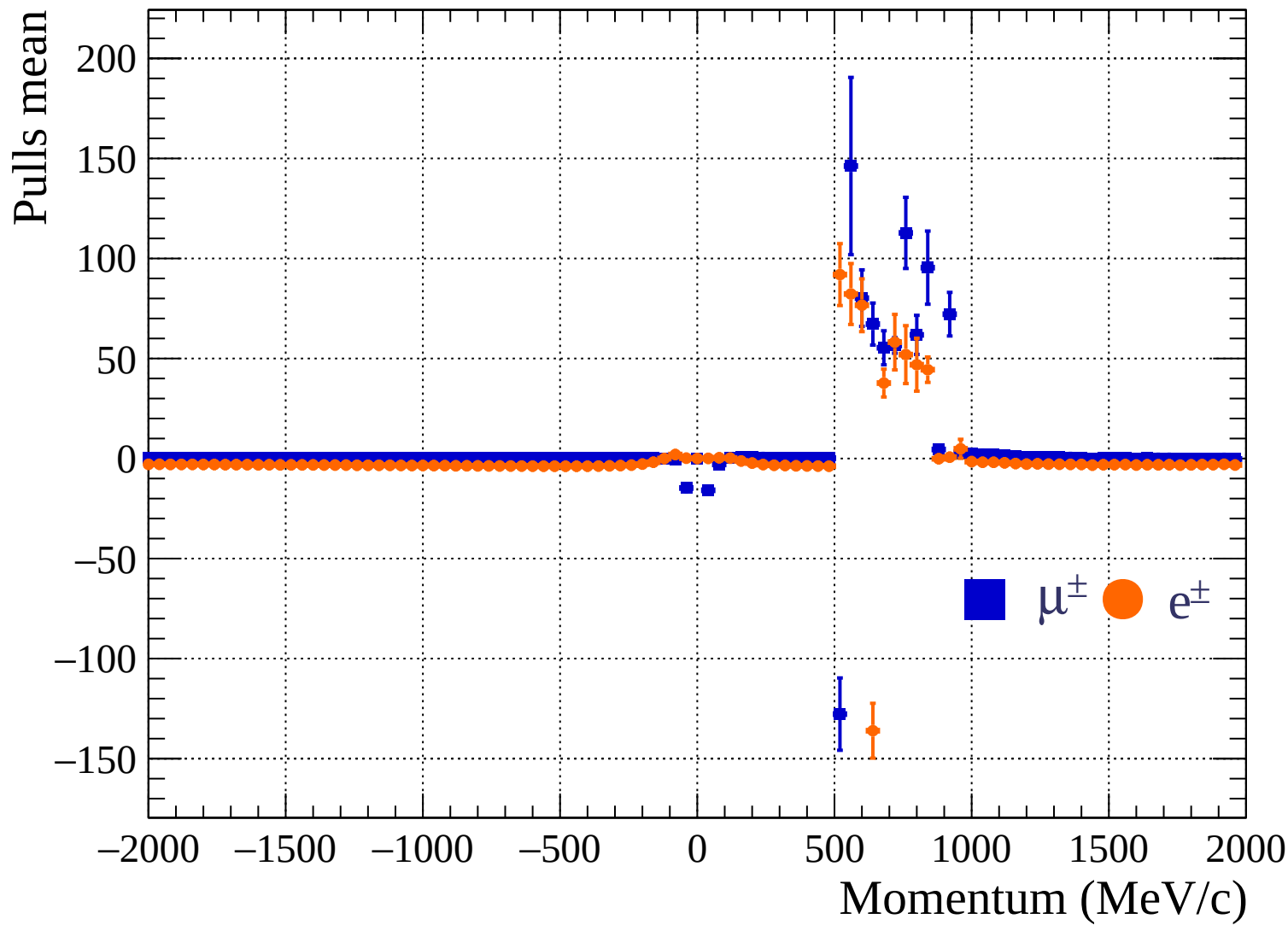


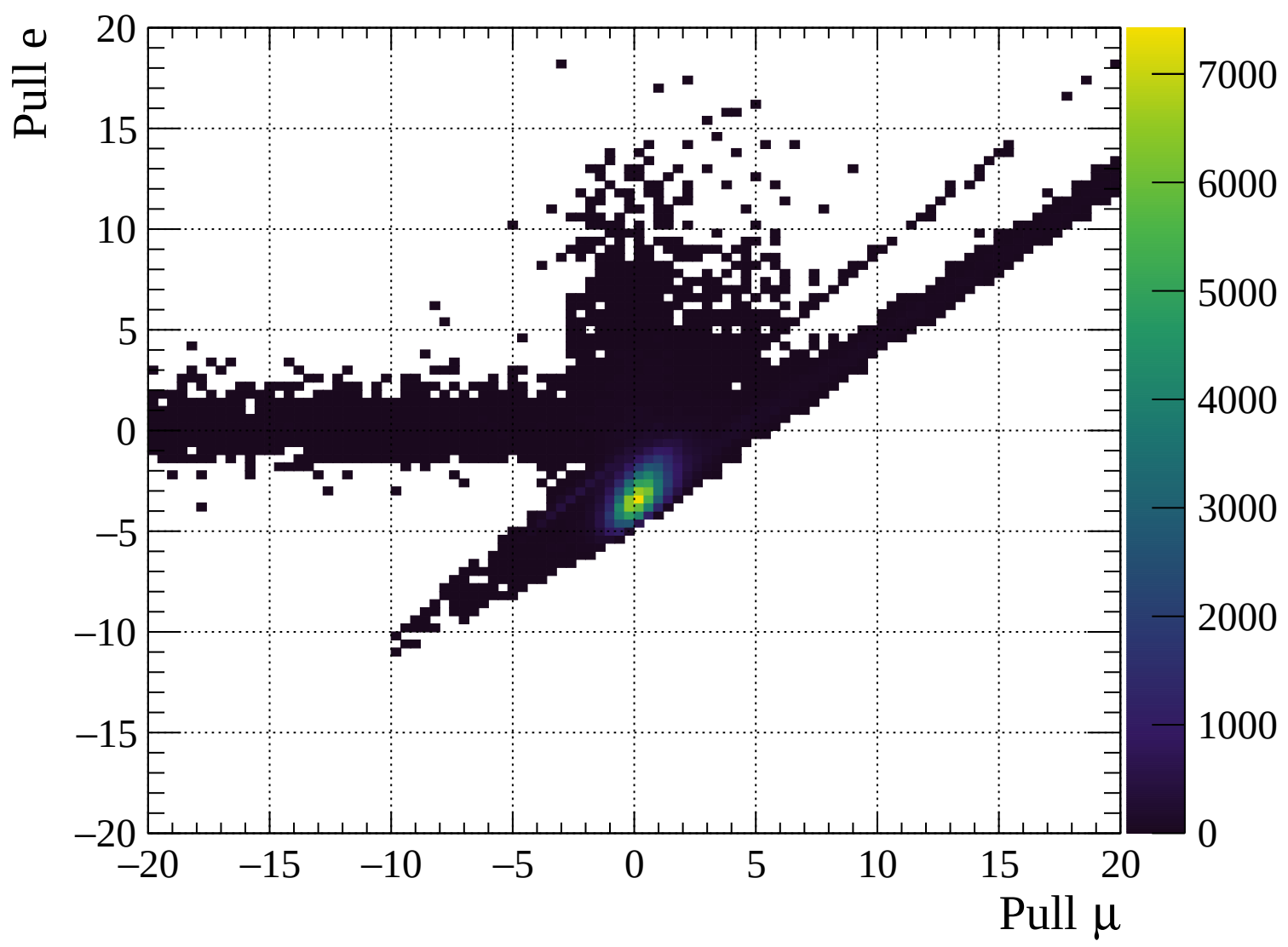
Pulls mean

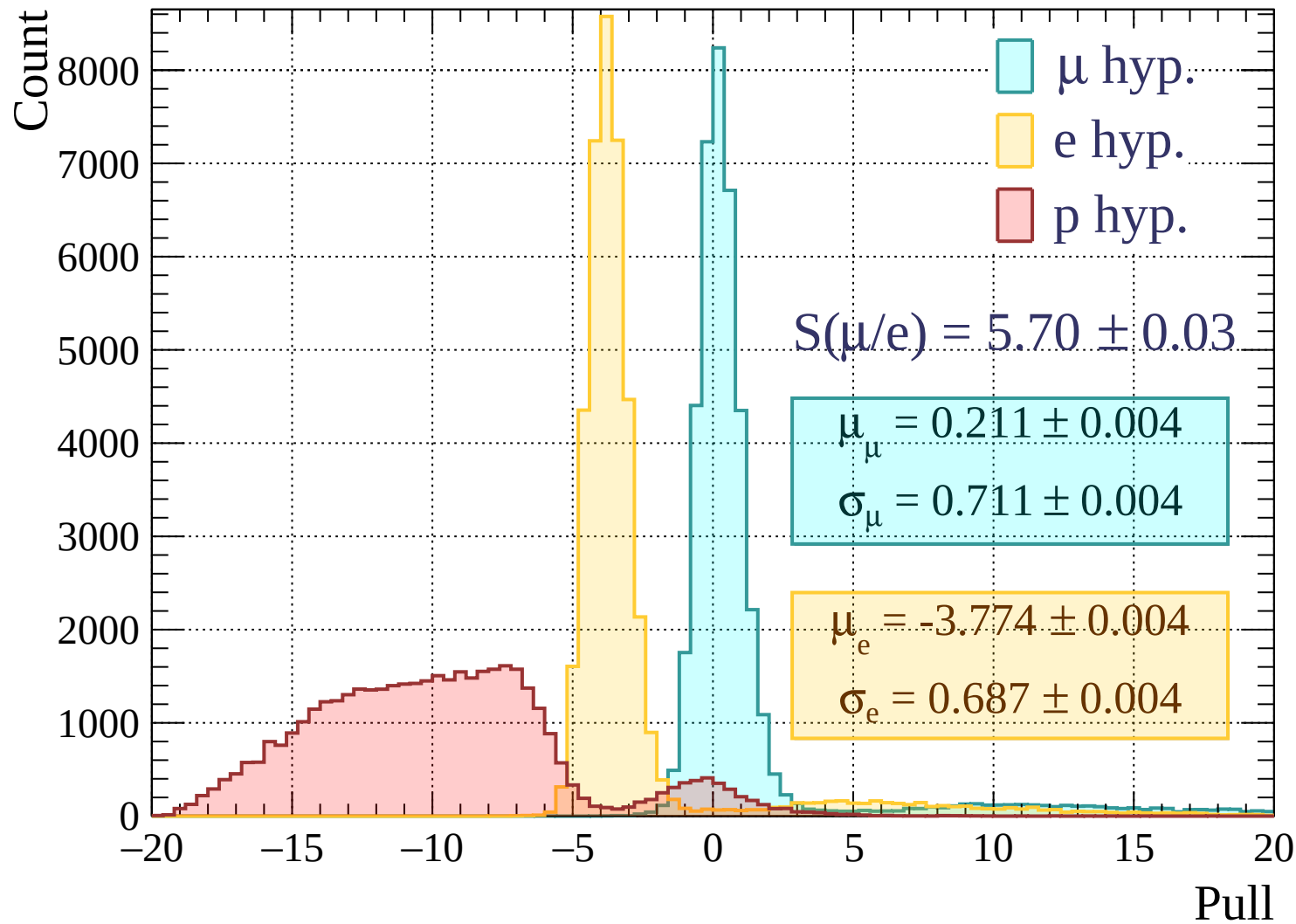


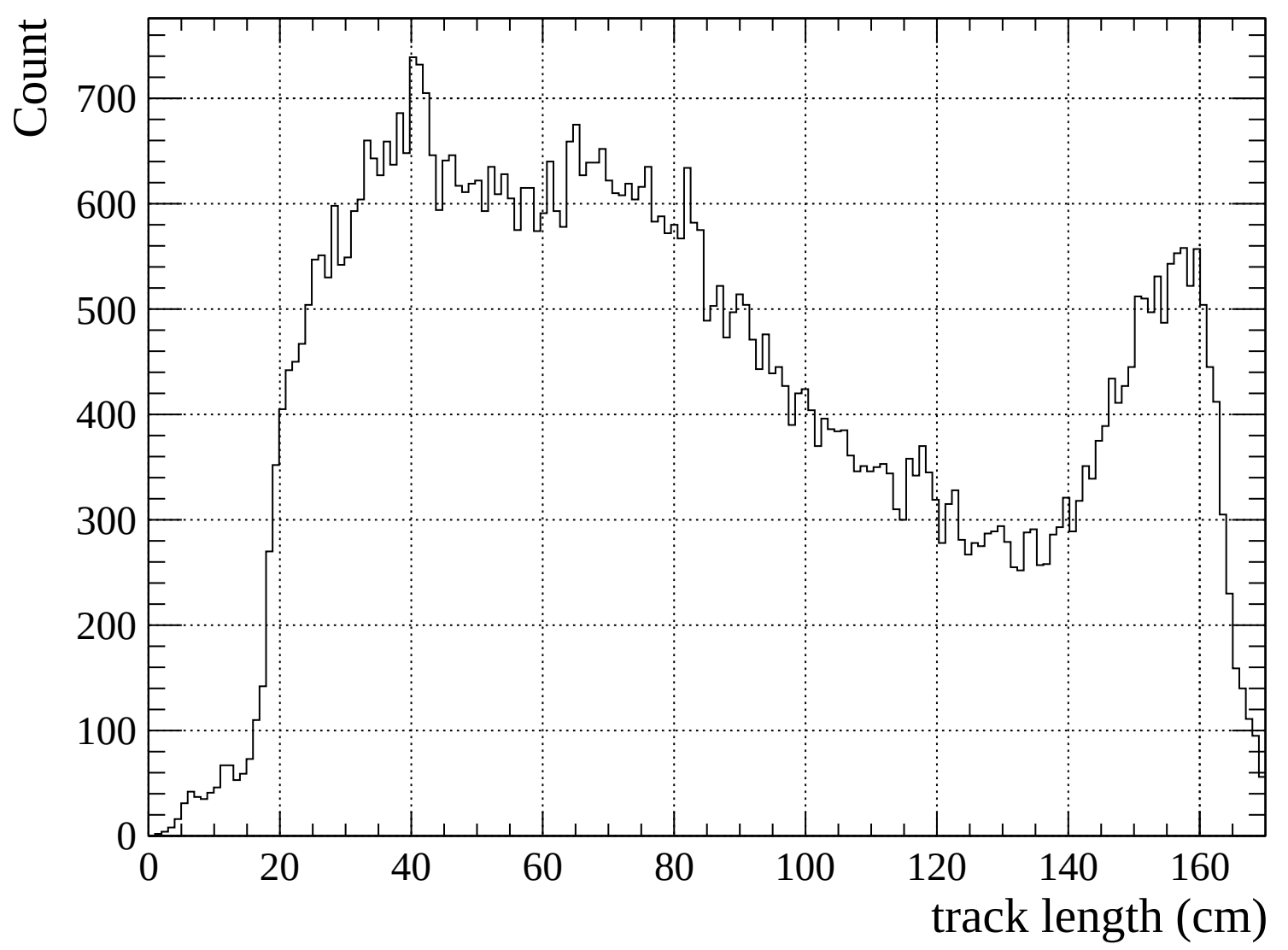
Pulls standard deviation

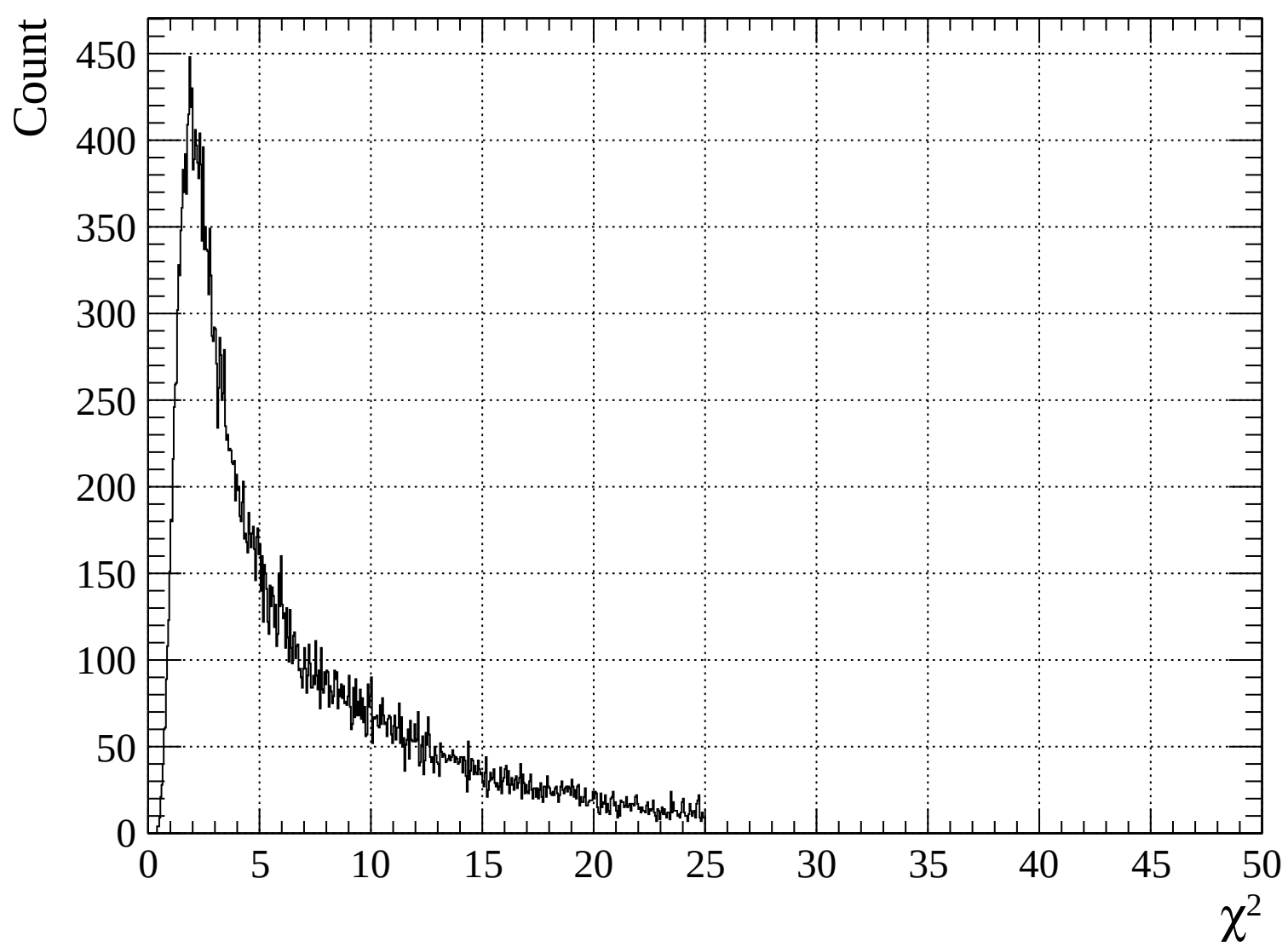


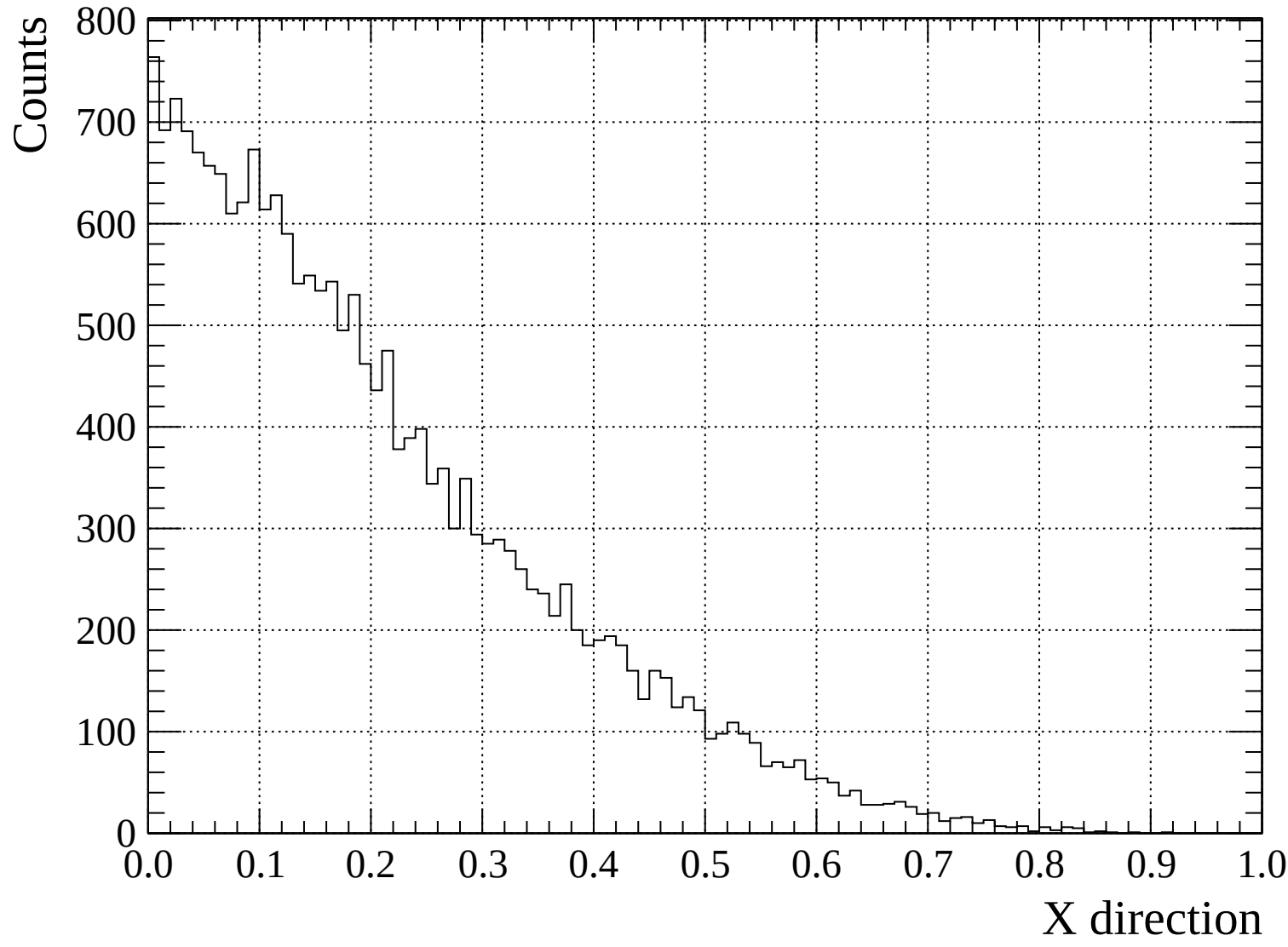


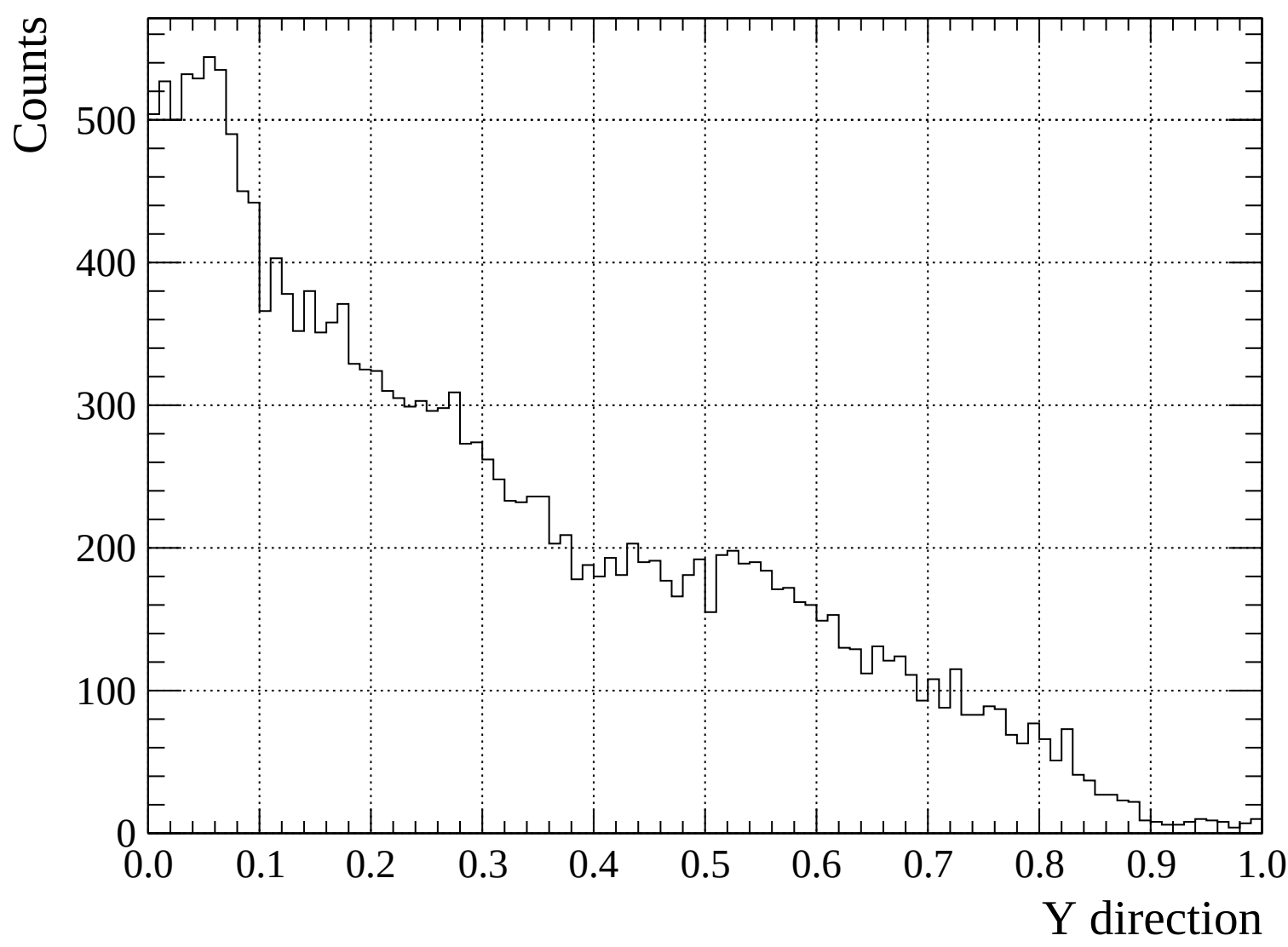


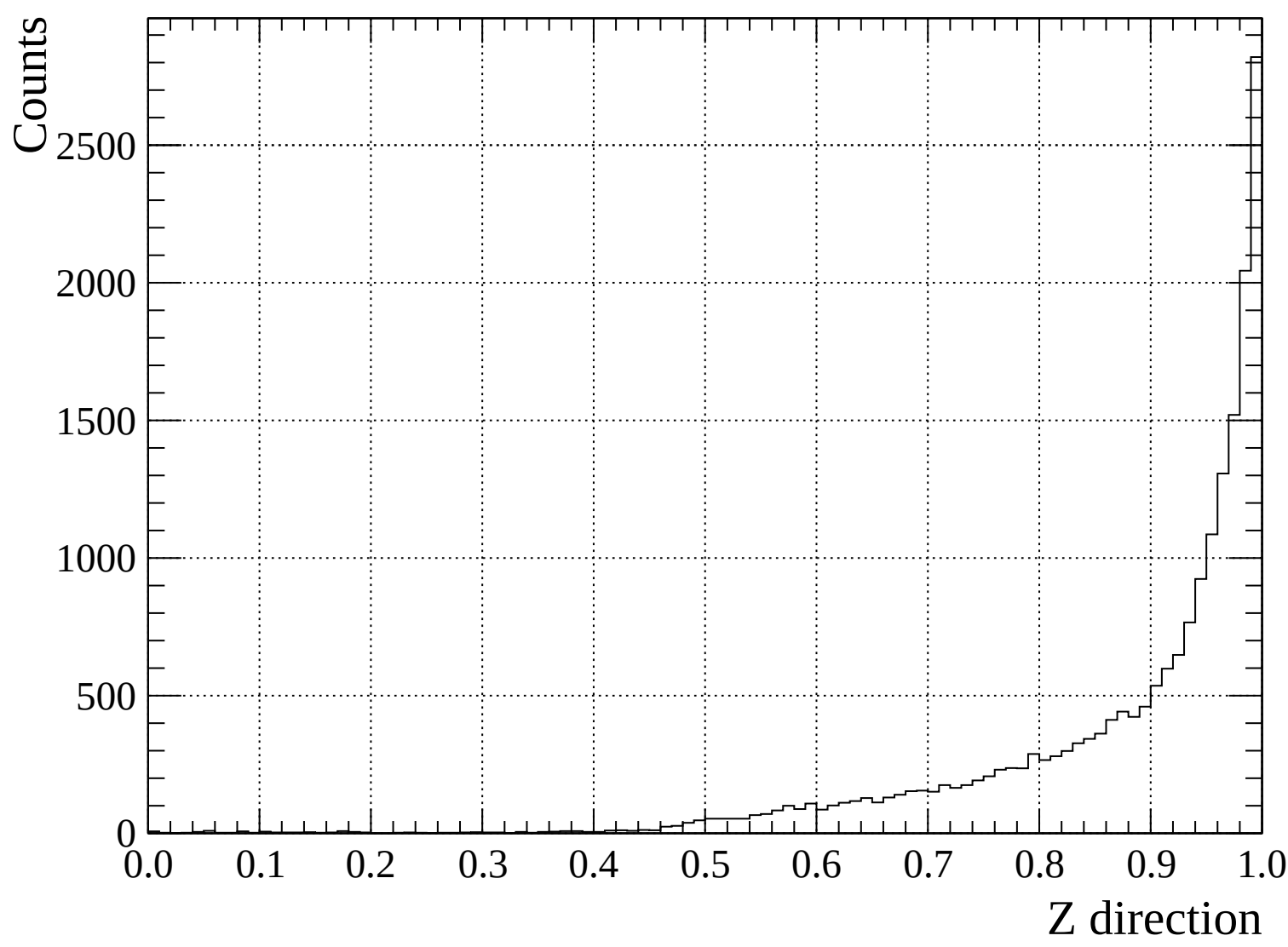


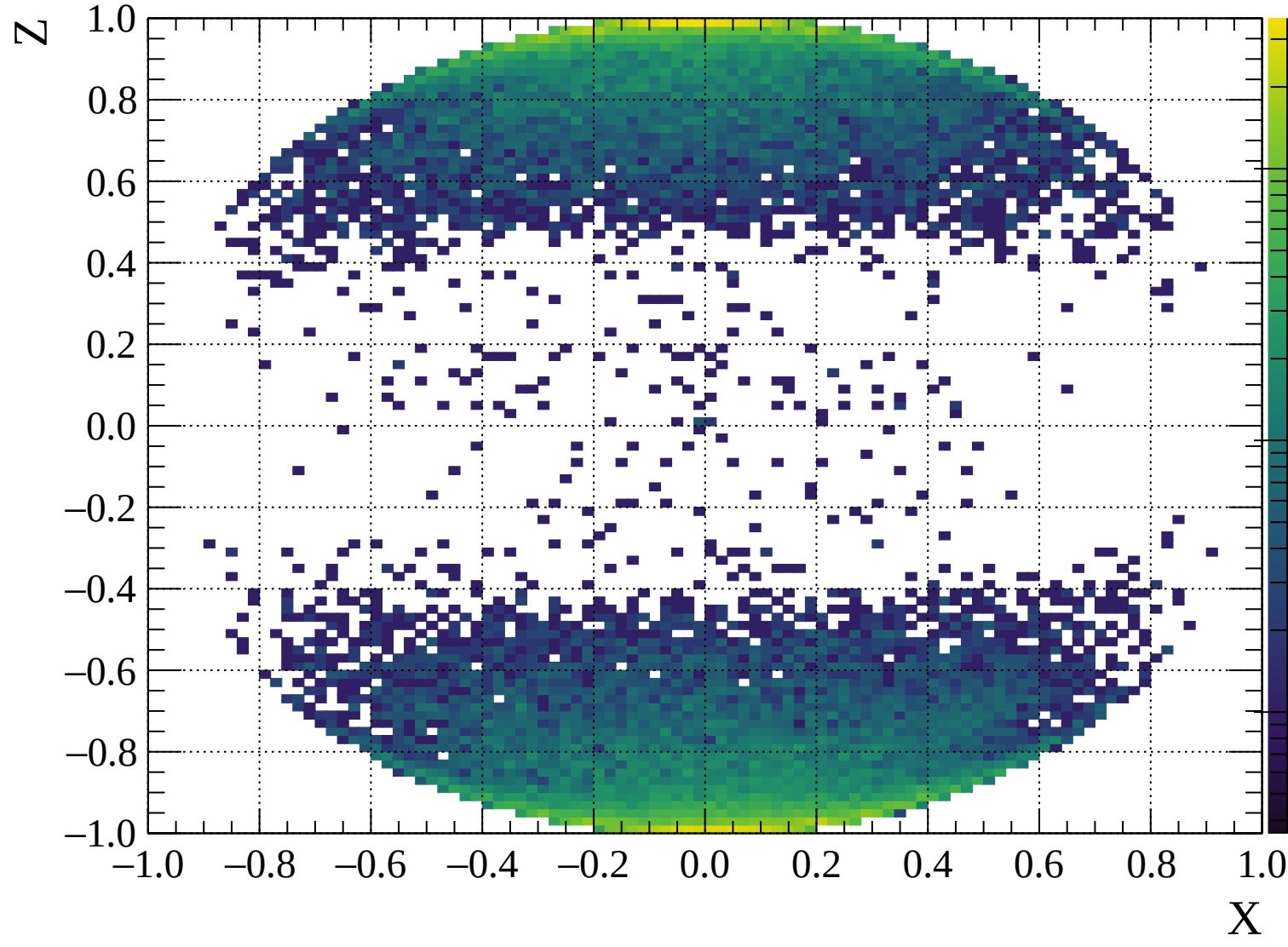


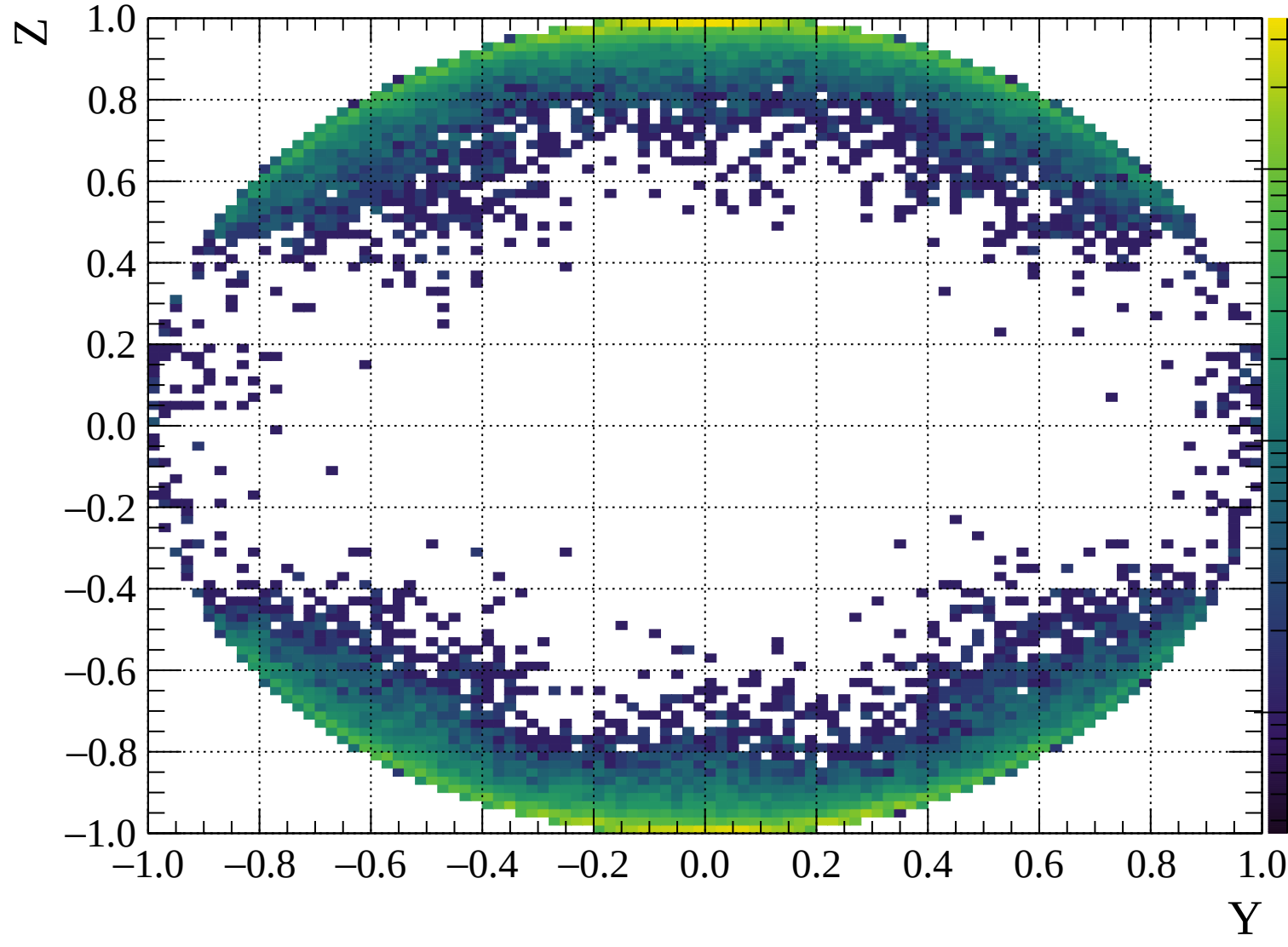


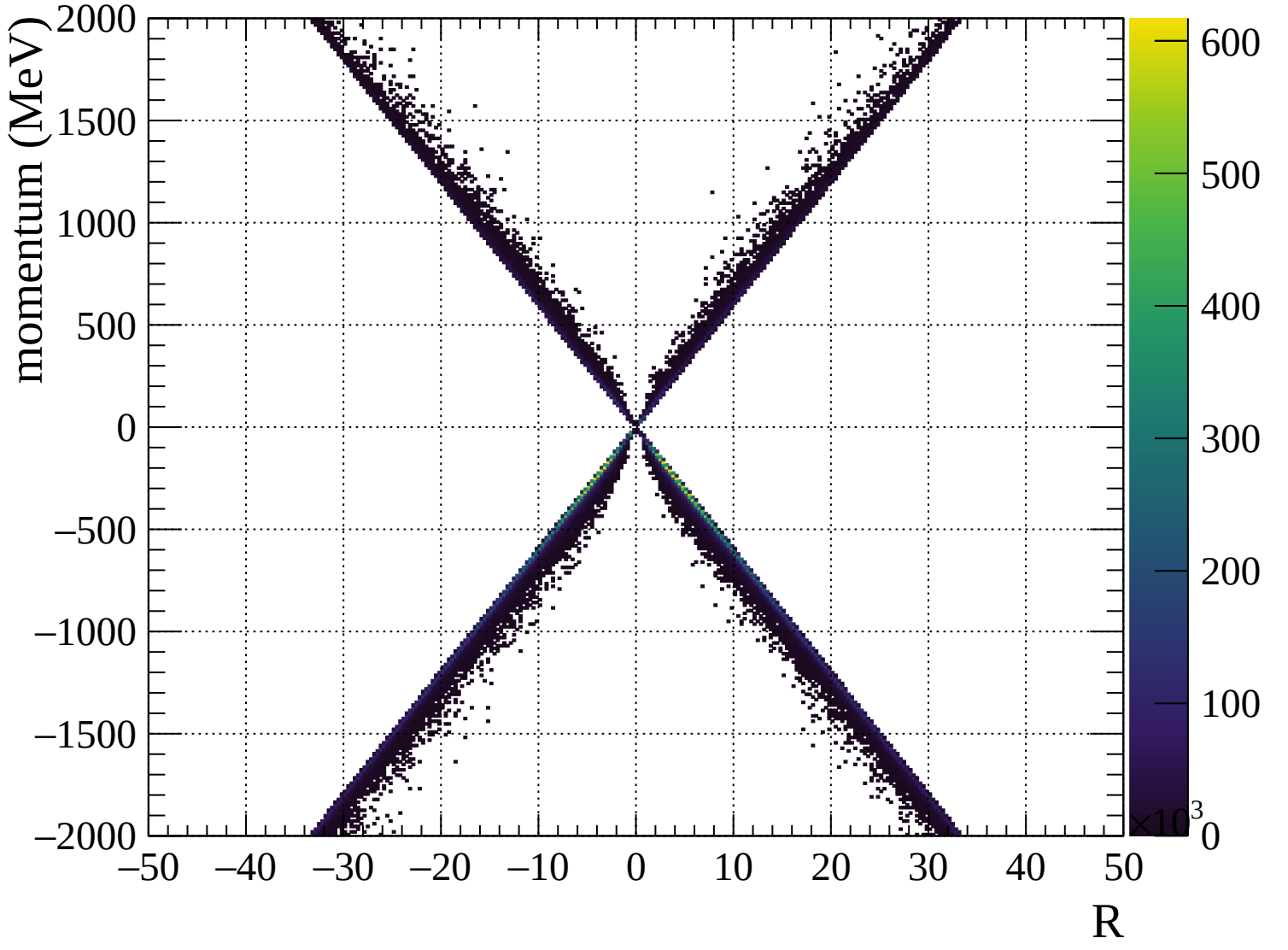


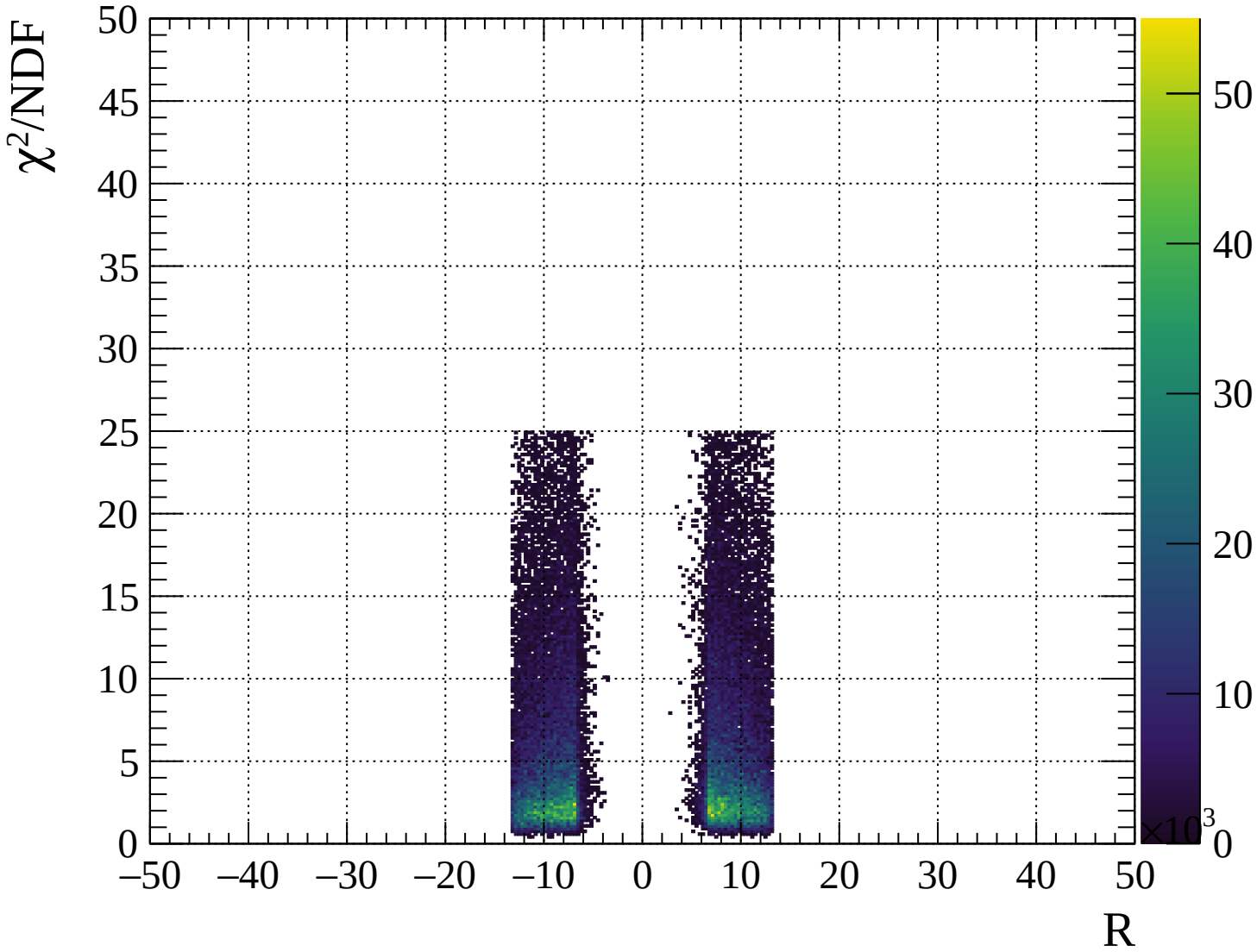




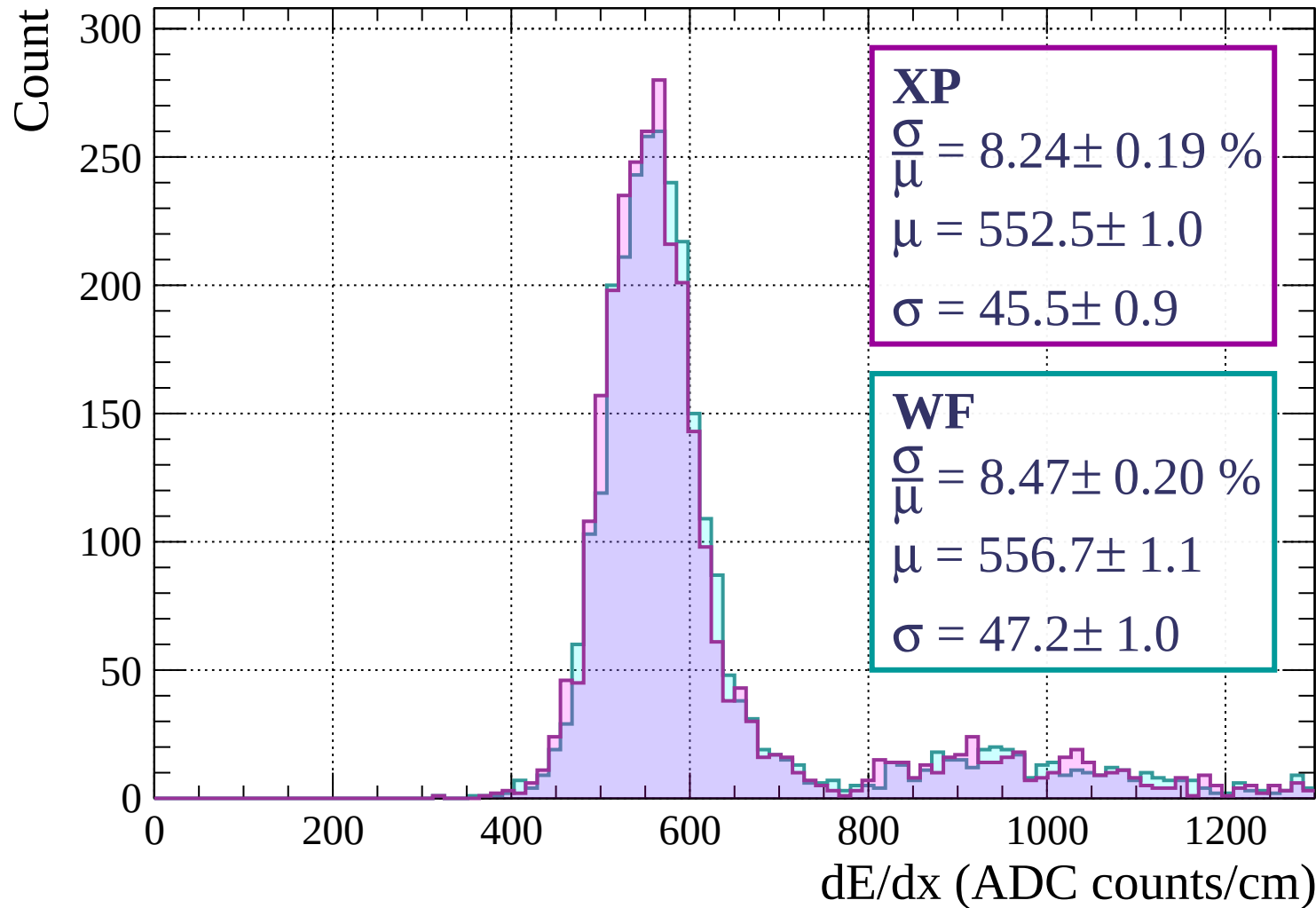






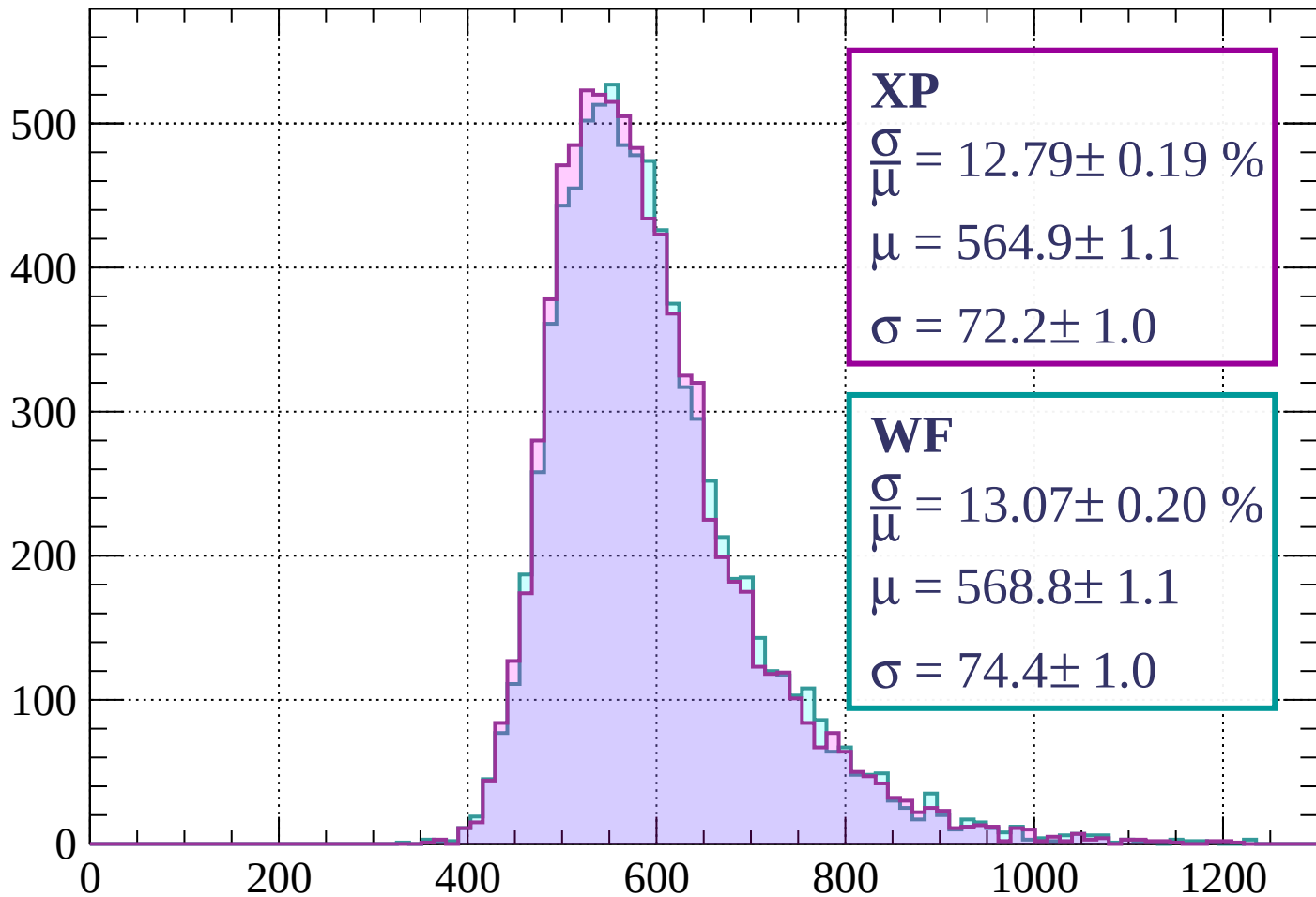


Energy loss | $0 < p < 80$



Energy loss | $80 < p < 160$

Count



XP

$$\frac{\sigma}{\mu} = 12.79 \pm 0.19 \%$$

$$\mu = 564.9 \pm 1.1$$

$$\sigma = 72.2 \pm 1.0$$

WF

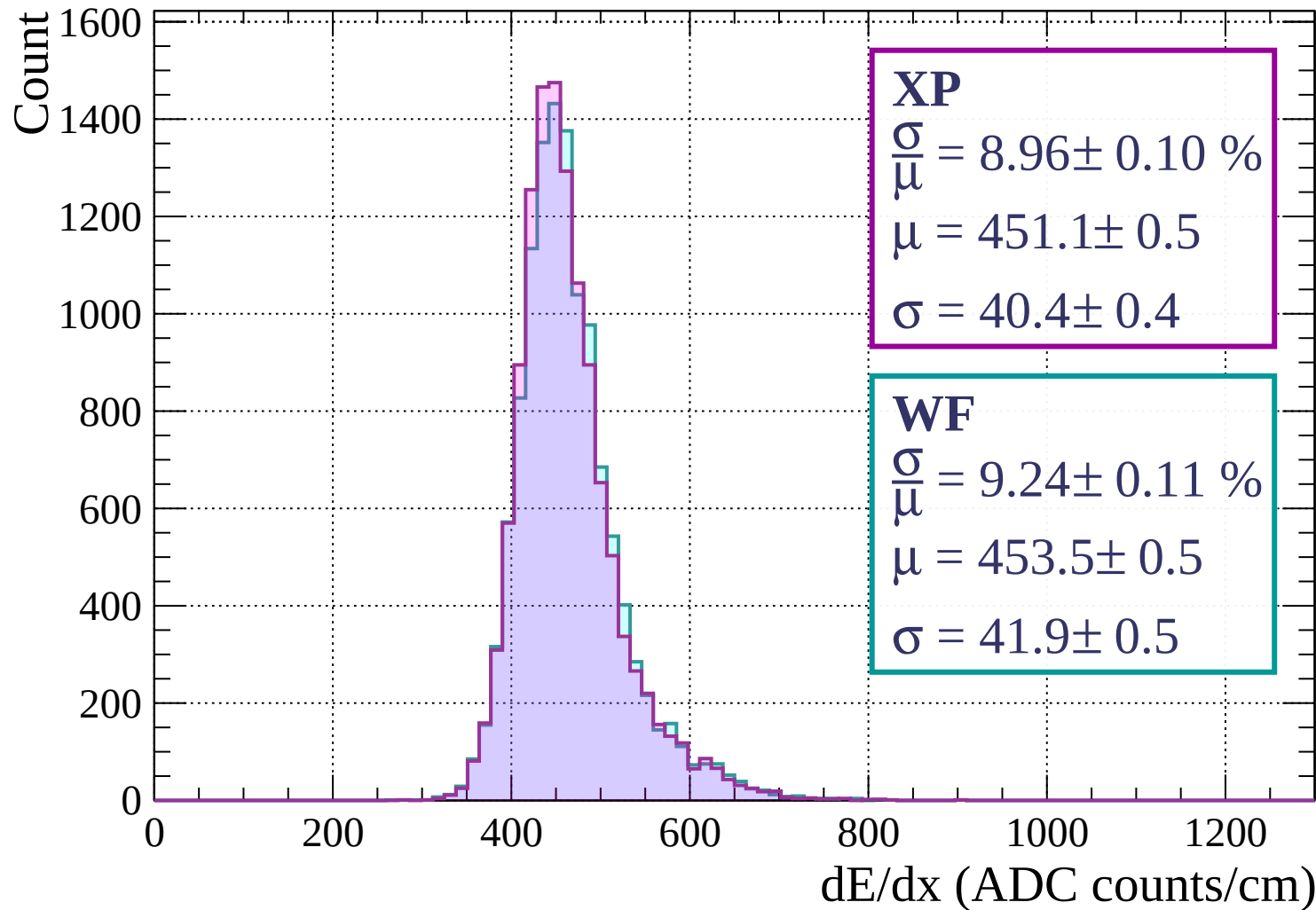
$$\frac{\sigma}{\mu} = 13.07 \pm 0.20 \%$$

$$\mu = 568.8 \pm 1.1$$

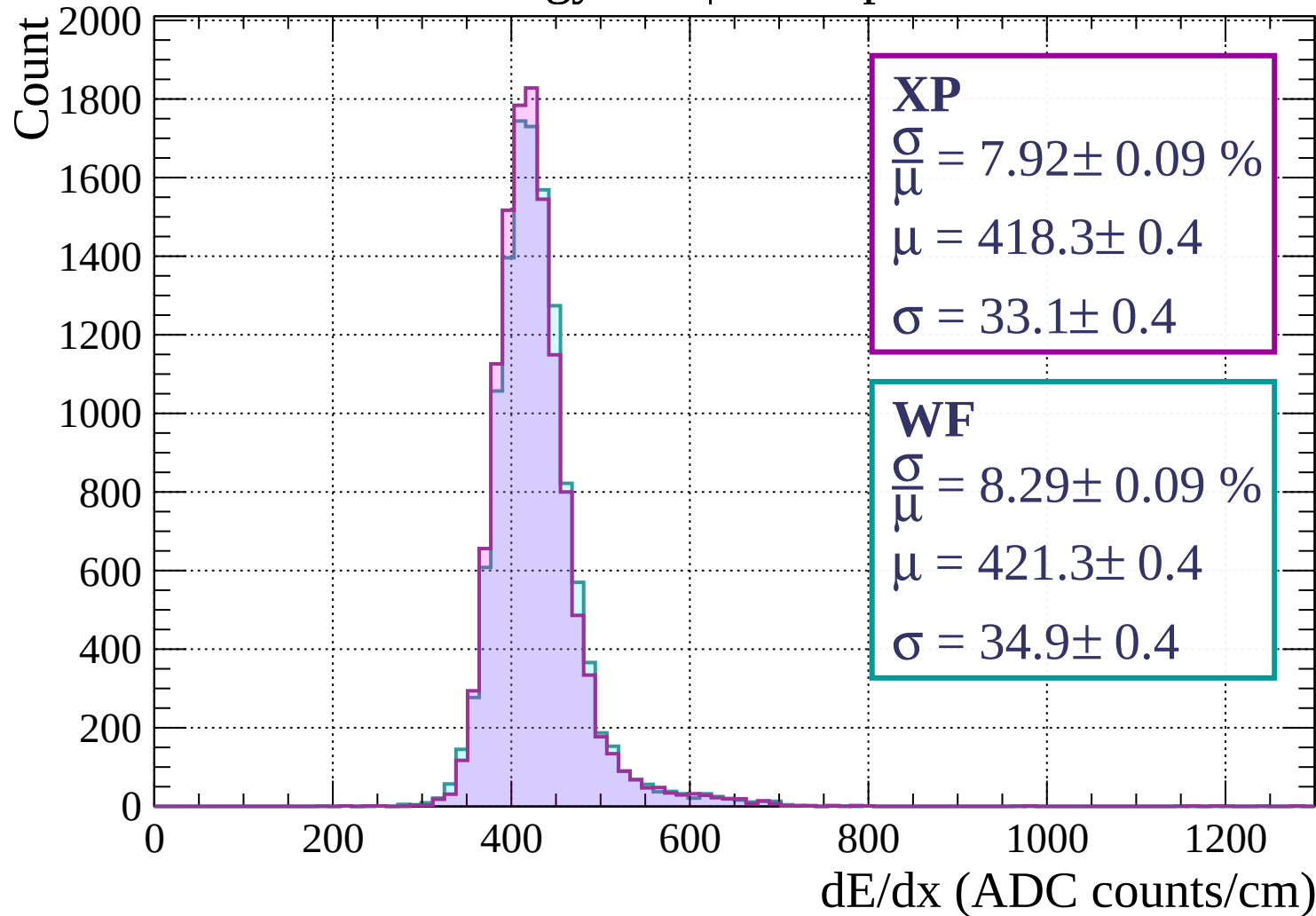
$$\sigma = 74.4 \pm 1.0$$

dE/dx (ADC counts/cm)

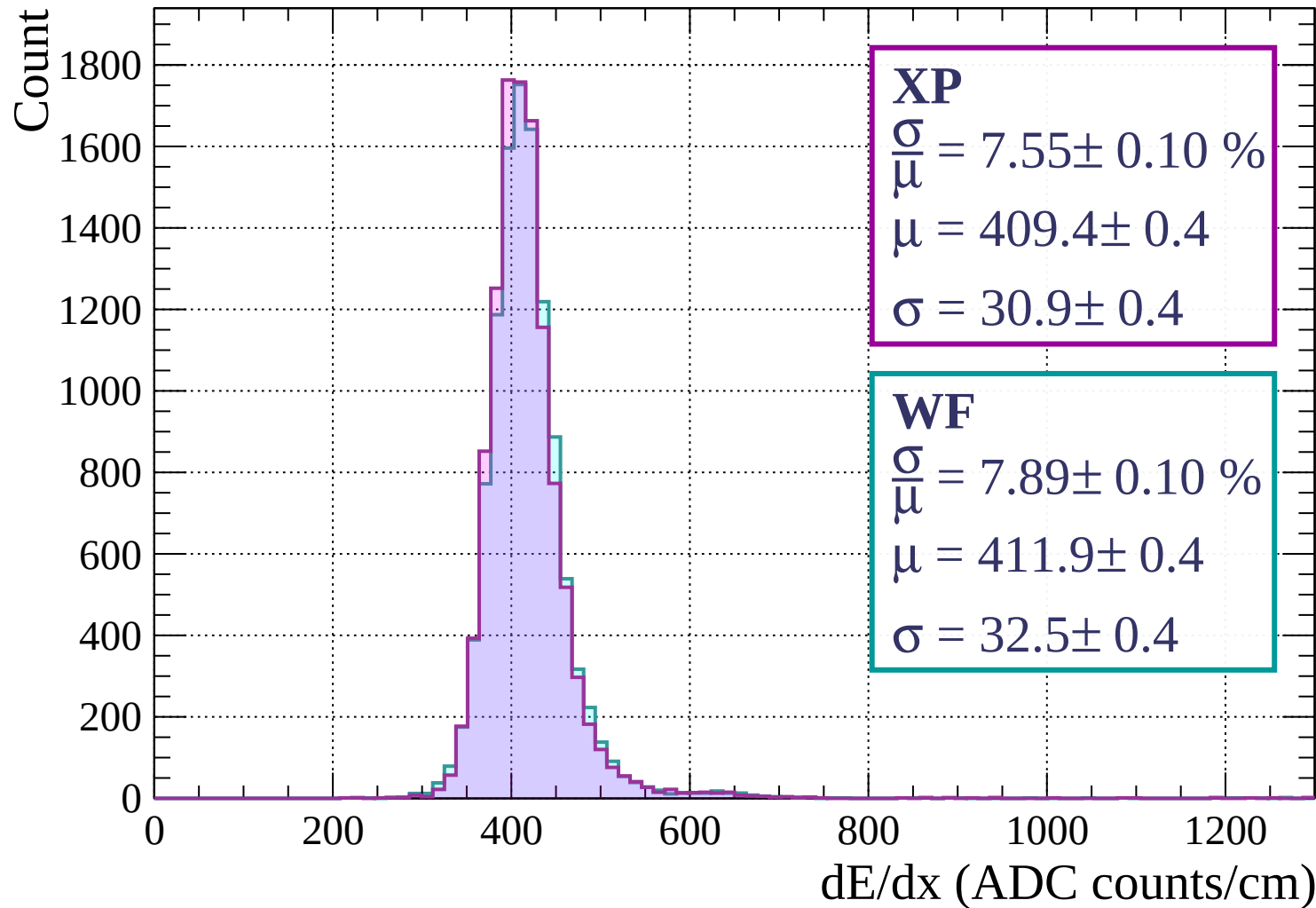
Energy loss | $160 < p < 240$



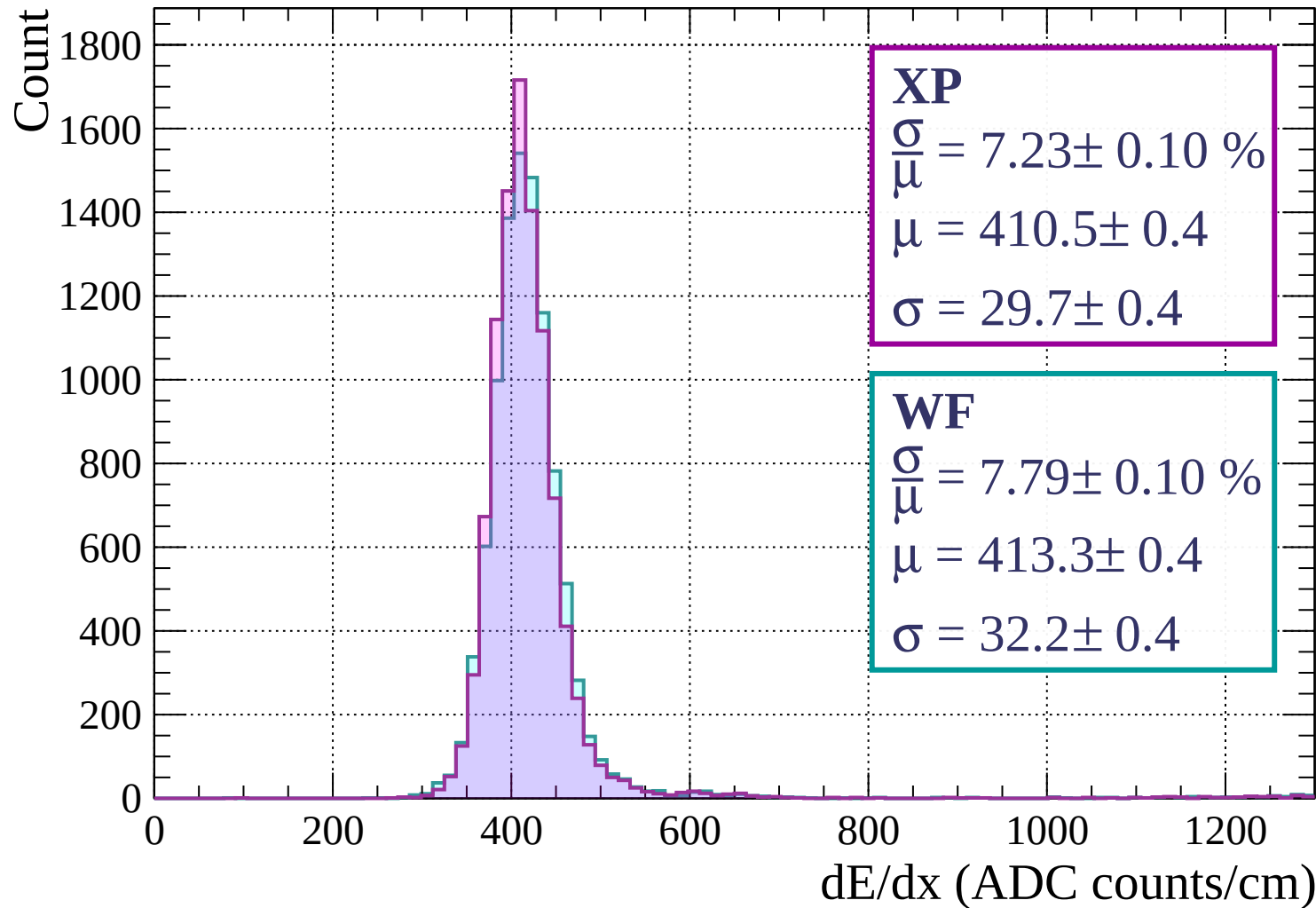
Energy loss | $240 < p < 320$



Energy loss | $320 < p < 400$



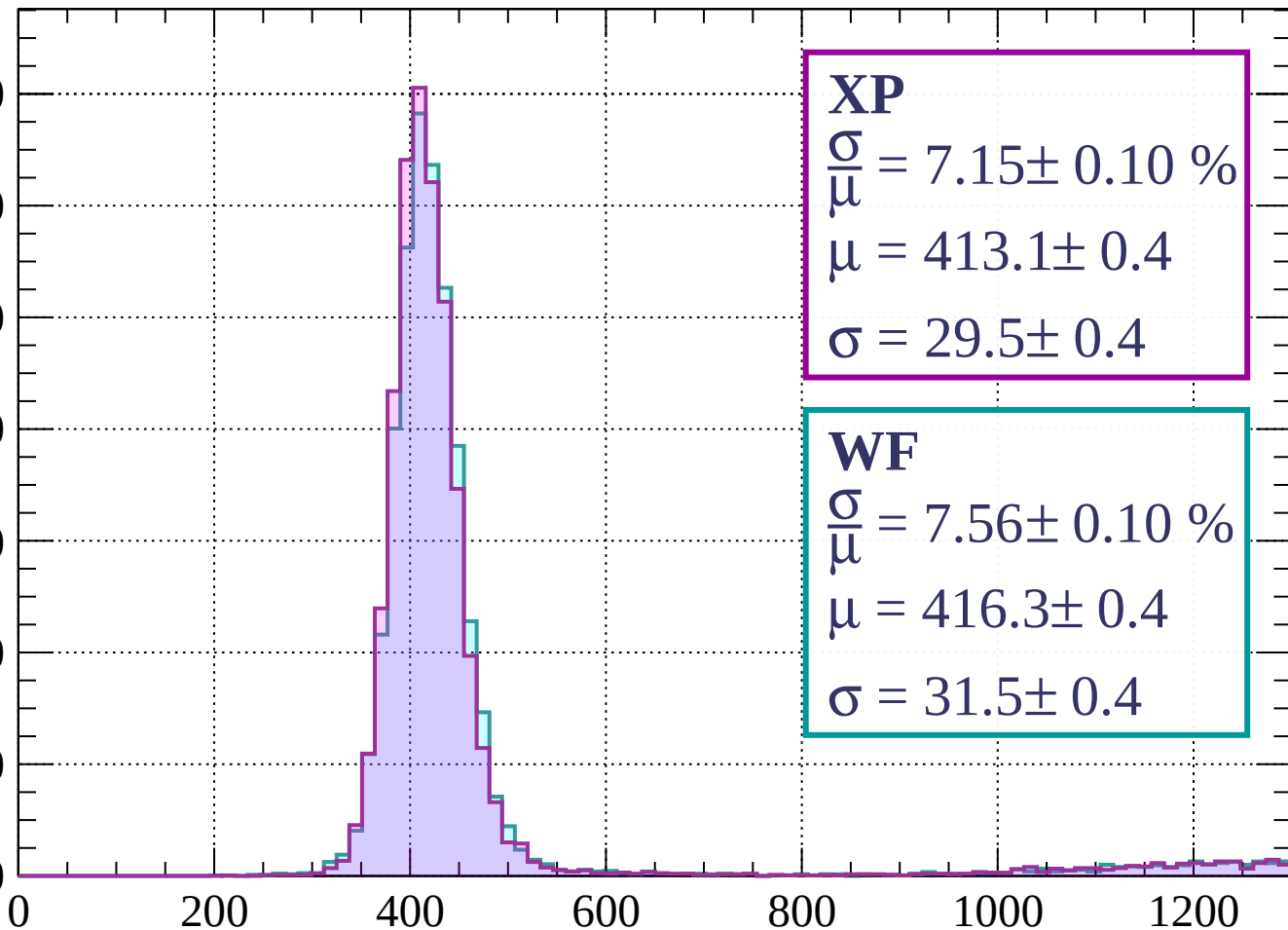
Energy loss | $400 < p < 480$



Energy loss | $480 < p < 560$

Count

1400
1200
1000
800
600
400
200
0



XP

$$\frac{\sigma}{\mu} = 7.15 \pm 0.10 \%$$

$$\mu = 413.1 \pm 0.4$$

$$\sigma = 29.5 \pm 0.4$$

WF

$$\frac{\sigma}{\mu} = 7.56 \pm 0.10 \%$$

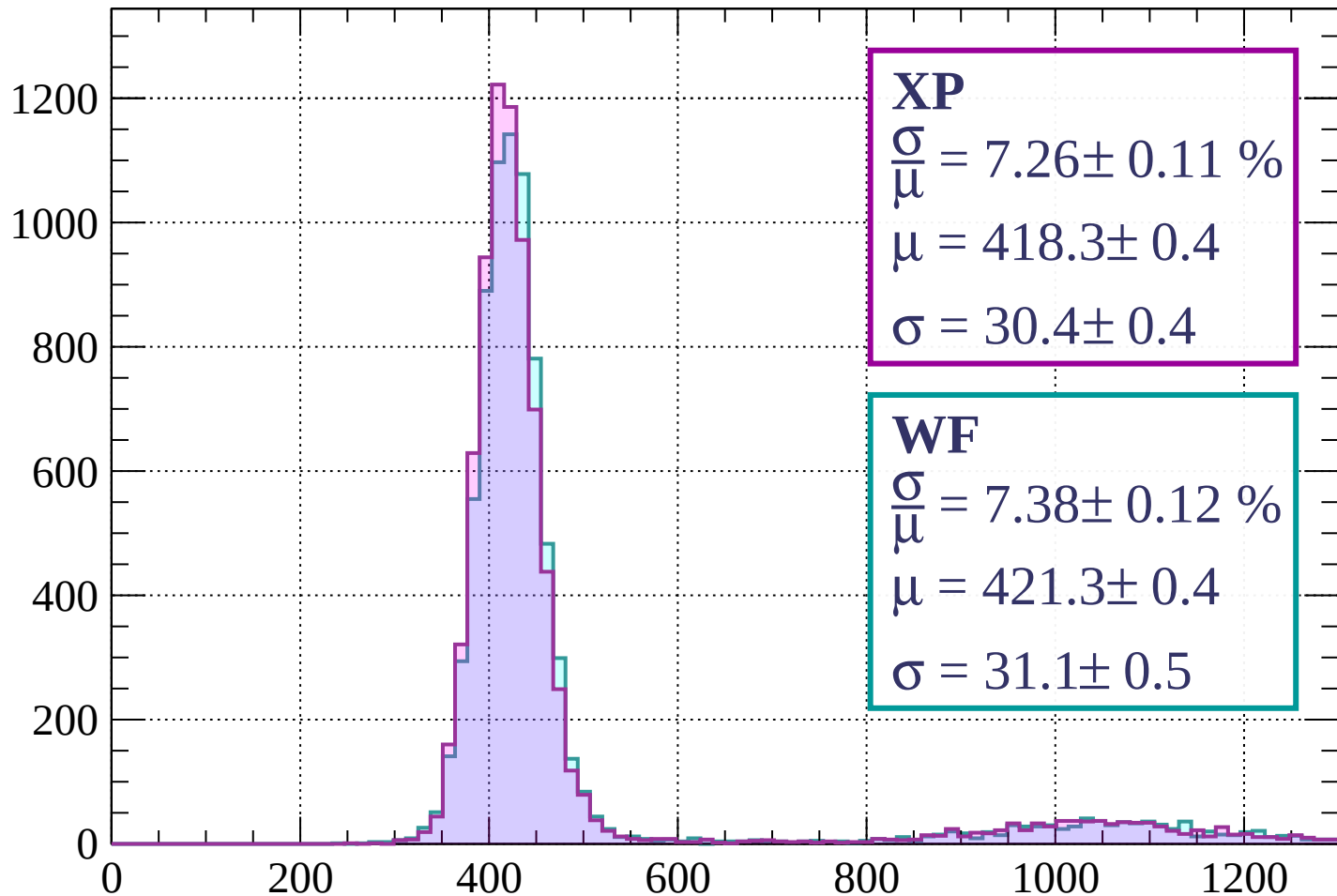
$$\mu = 416.3 \pm 0.4$$

$$\sigma = 31.5 \pm 0.4$$

dE/dx (ADC counts/cm)

Energy loss | $560 < p < 640$

Count



XP

$$\frac{\sigma}{\mu} = 7.26 \pm 0.11 \%$$

$$\mu = 418.3 \pm 0.4$$

$$\sigma = 30.4 \pm 0.4$$

WF

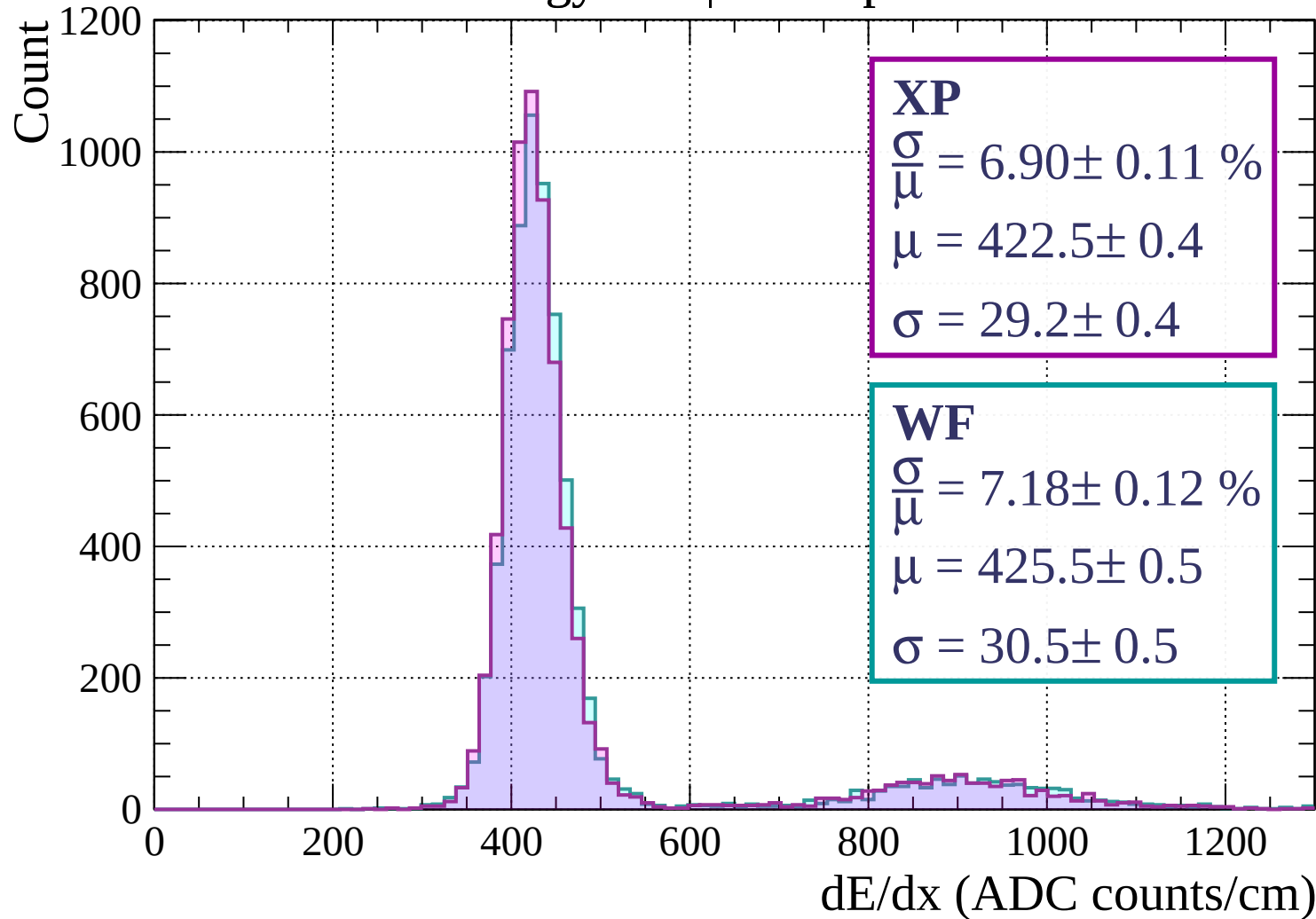
$$\frac{\sigma}{\mu} = 7.38 \pm 0.12 \%$$

$$\mu = 421.3 \pm 0.4$$

$$\sigma = 31.1 \pm 0.5$$

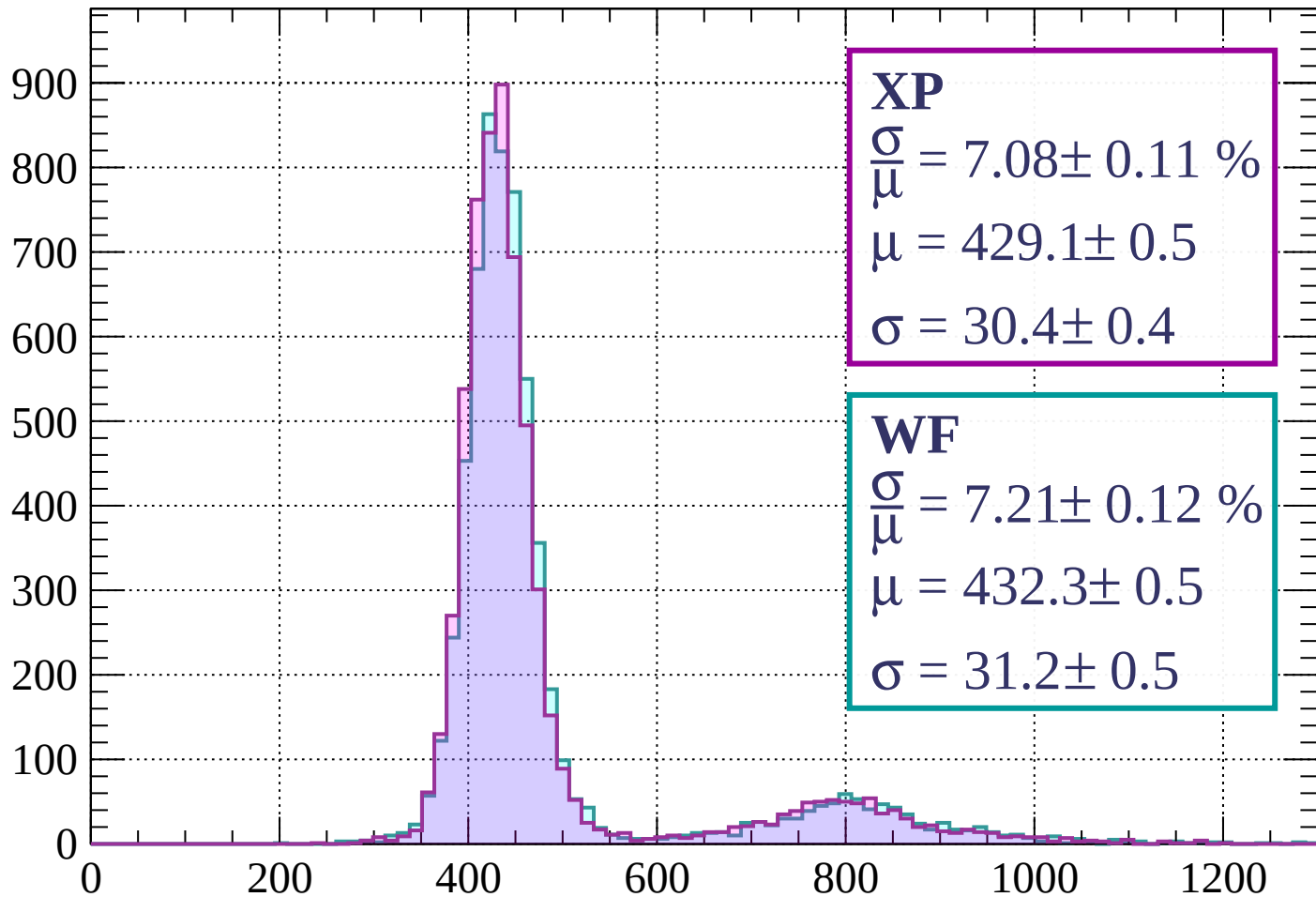
dE/dx (ADC counts/cm)

Energy loss | $640 < p < 720$



Energy loss | $720 < p < 800$

Count



XP

$$\frac{\sigma}{\mu} = 7.08 \pm 0.11 \%$$

$$\mu = 429.1 \pm 0.5$$

$$\sigma = 30.4 \pm 0.4$$

WF

$$\frac{\sigma}{\mu} = 7.21 \pm 0.12 \%$$

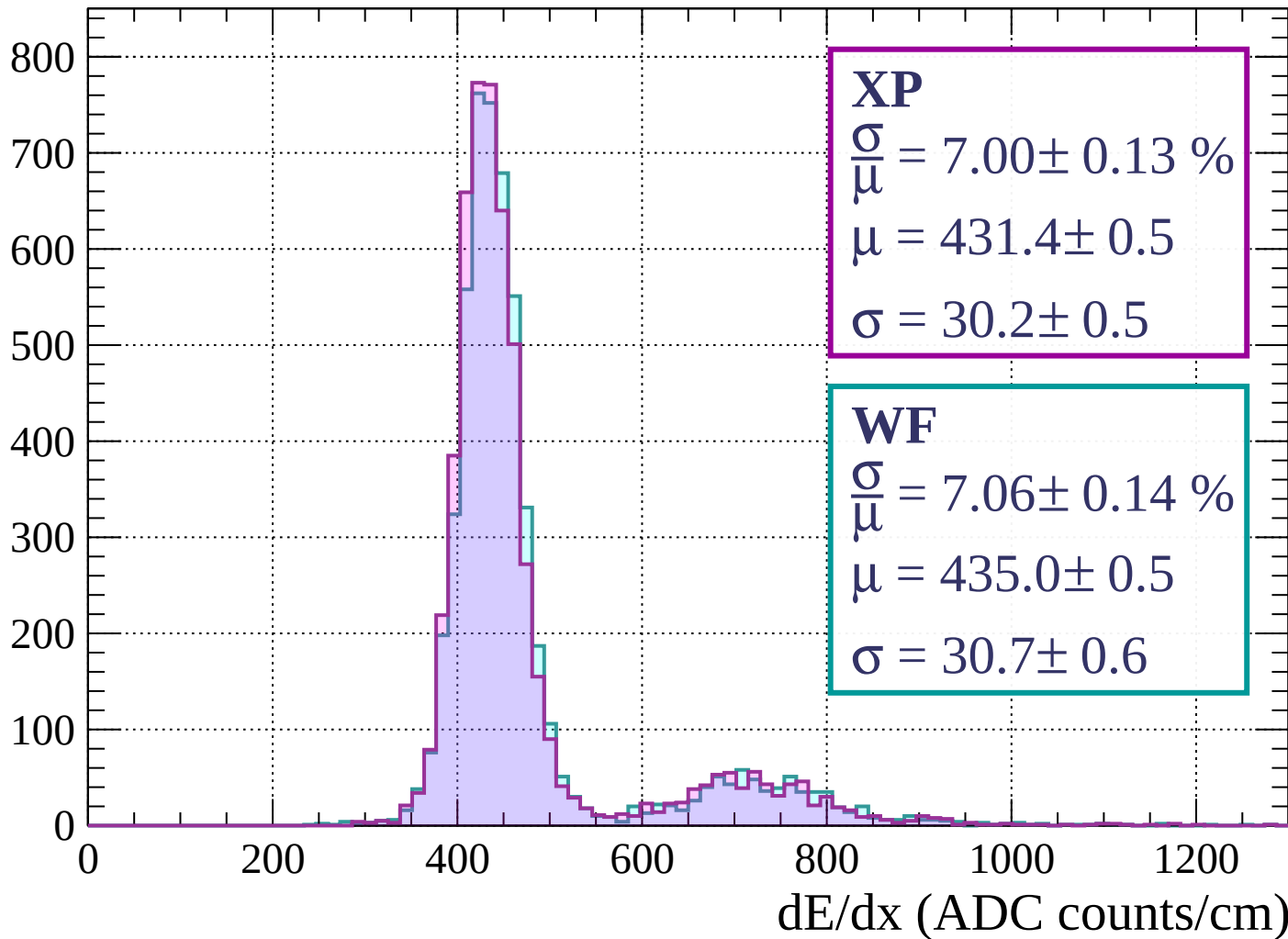
$$\mu = 432.3 \pm 0.5$$

$$\sigma = 31.2 \pm 0.5$$

dE/dx (ADC counts/cm)

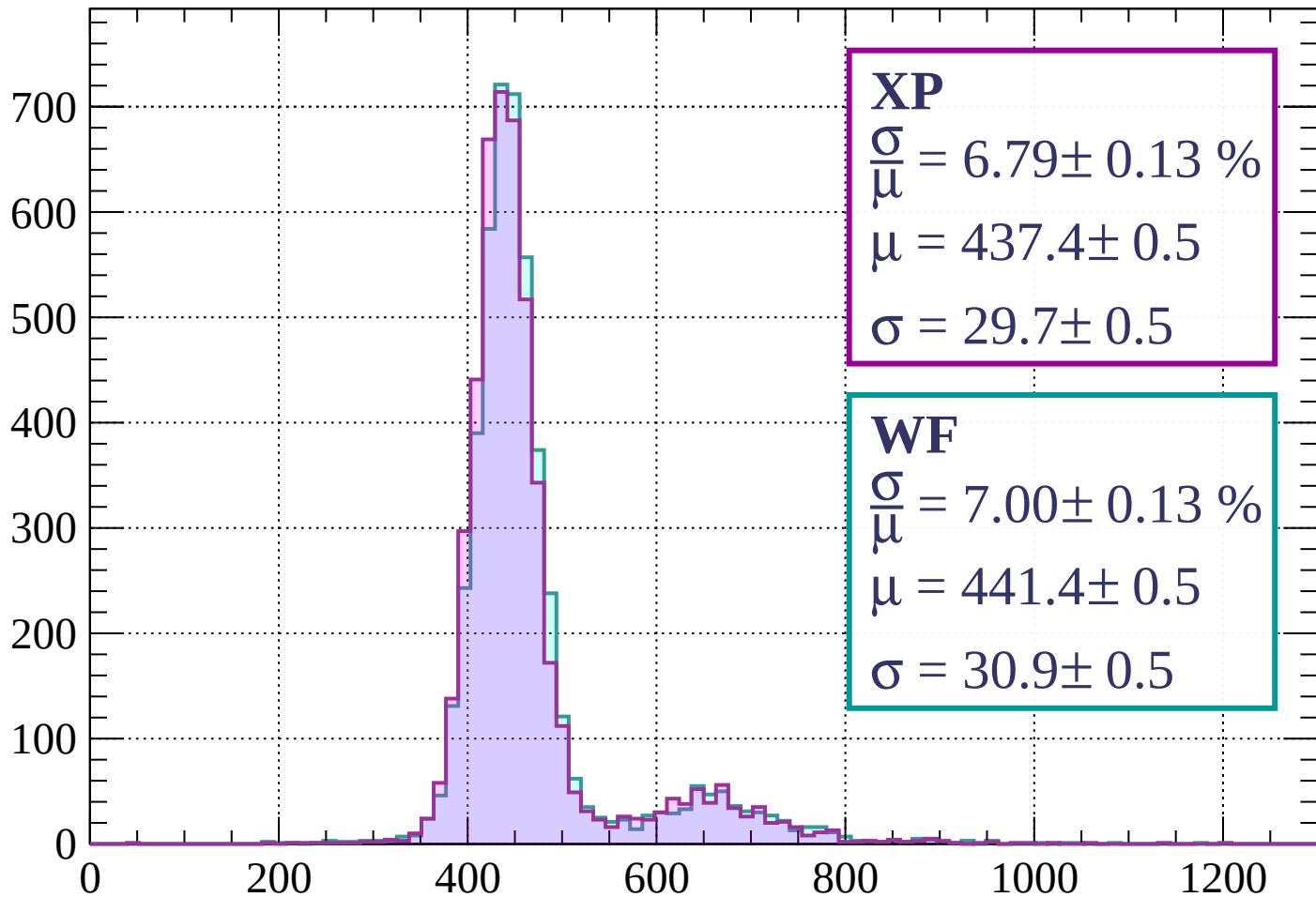
Energy loss | $800 < p < 880$

Count



Energy loss | $880 < p < 960$

Count



XP

$$\frac{\sigma}{\mu} = 6.79 \pm 0.13 \%$$

$$\mu = 437.4 \pm 0.5$$

$$\sigma = 29.7 \pm 0.5$$

WF

$$\frac{\sigma}{\mu} = 7.00 \pm 0.13 \%$$

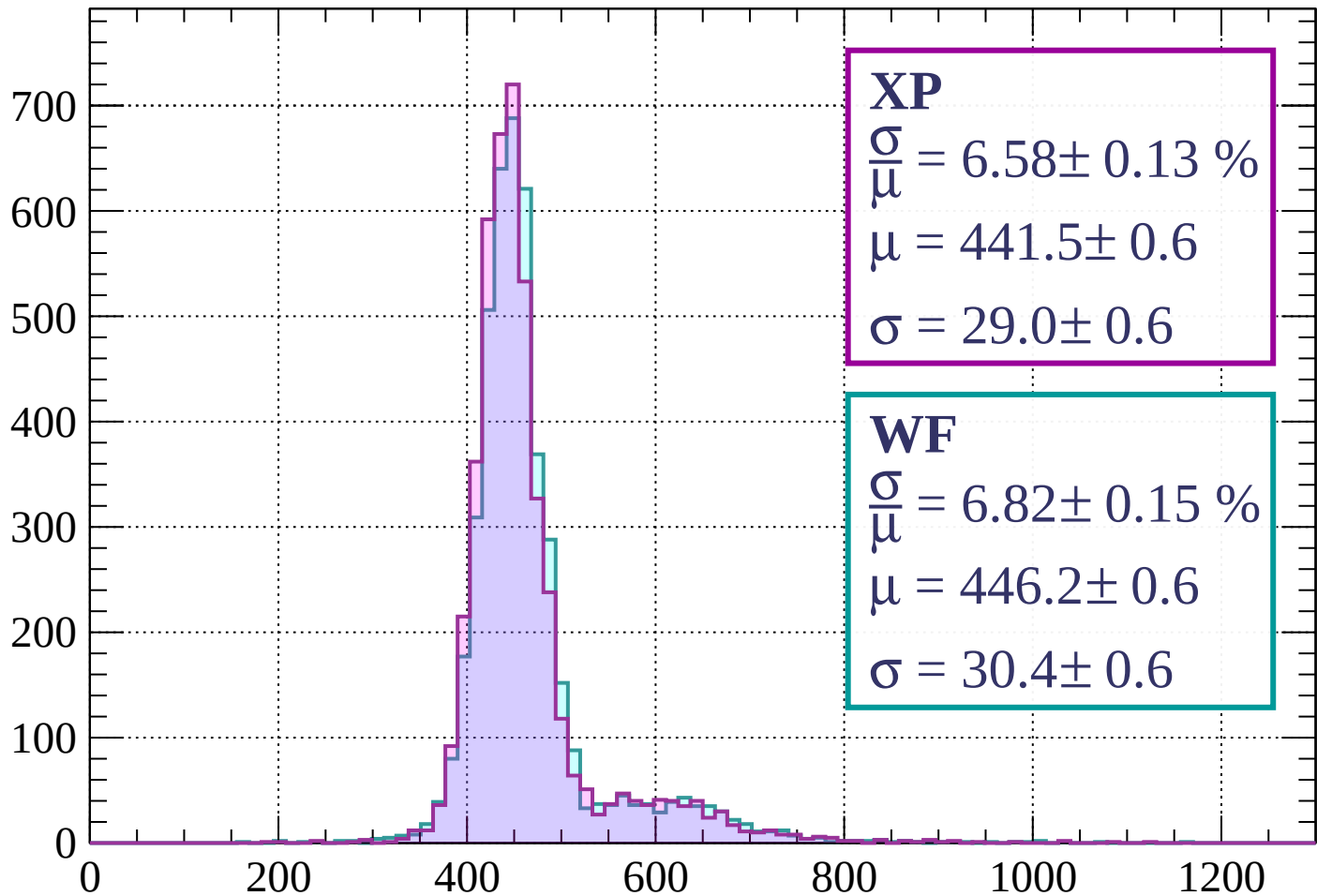
$$\mu = 441.4 \pm 0.5$$

$$\sigma = 30.9 \pm 0.5$$

dE/dx (ADC counts/cm)

Energy loss | $960 < p < 1040$

Count



XP

$$\frac{\sigma}{\mu} = 6.58 \pm 0.13 \%$$

$$\mu = 441.5 \pm 0.6$$

$$\sigma = 29.0 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 6.82 \pm 0.15 \%$$

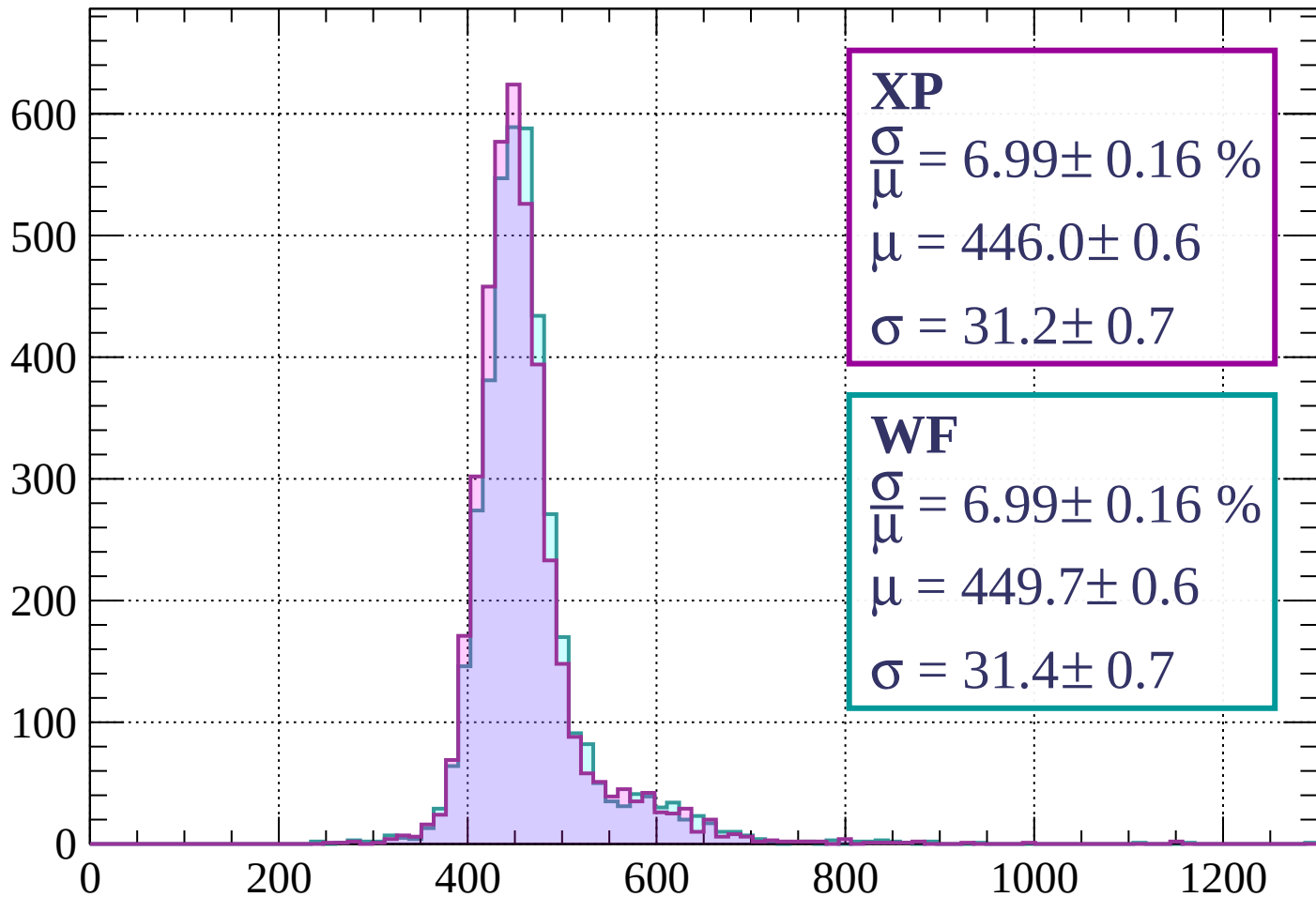
$$\mu = 446.2 \pm 0.6$$

$$\sigma = 30.4 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $1040 < p < 1120$

Count



XP

$$\frac{\sigma}{\mu} = 6.99 \pm 0.16 \%$$

$$\mu = 446.0 \pm 0.6$$

$$\sigma = 31.2 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 6.99 \pm 0.16 \%$$

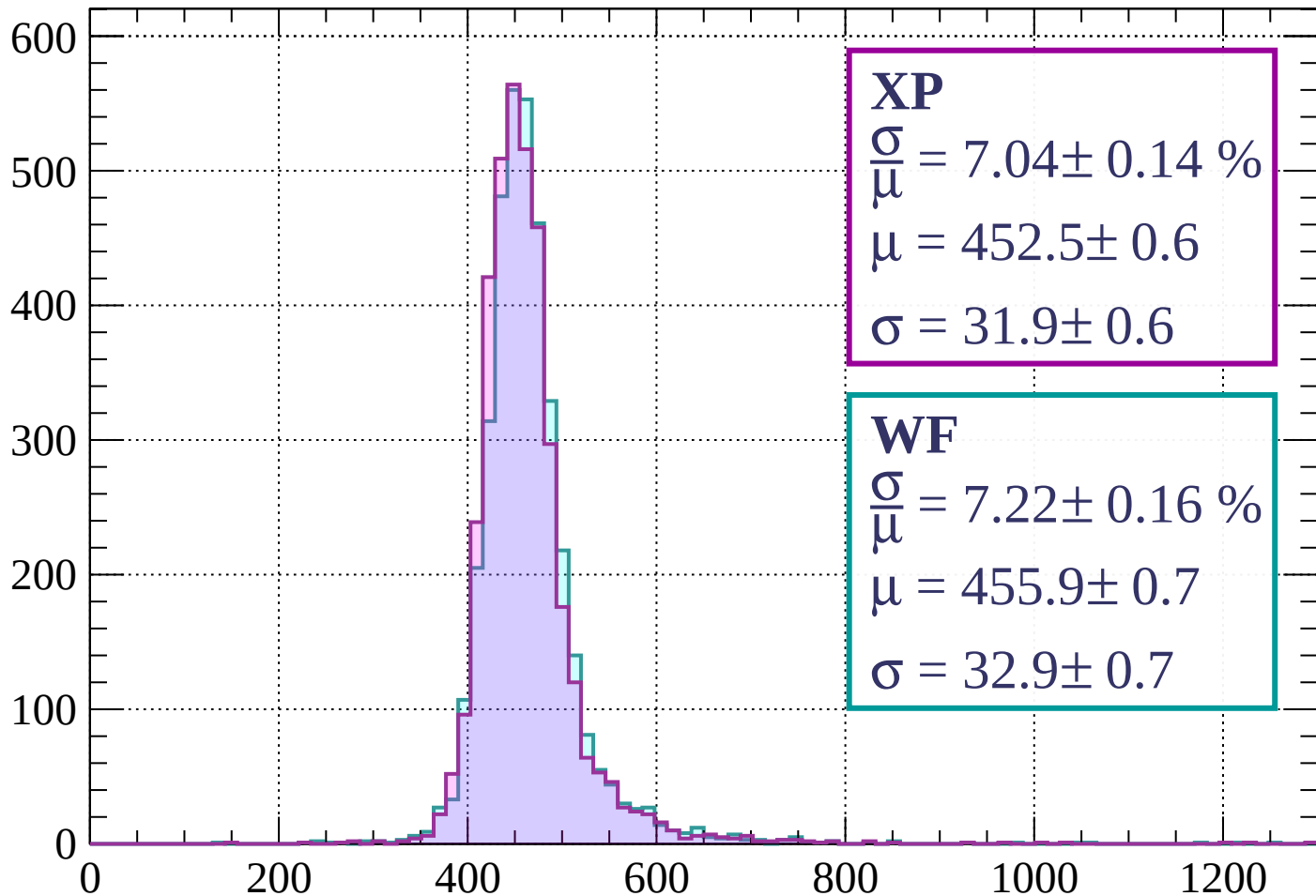
$$\mu = 449.7 \pm 0.6$$

$$\sigma = 31.4 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $1120 < p < 1200$

Count



XP

$$\frac{\sigma}{\mu} = 7.04 \pm 0.14 \%$$

$$\mu = 452.5 \pm 0.6$$

$$\sigma = 31.9 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 7.22 \pm 0.16 \%$$

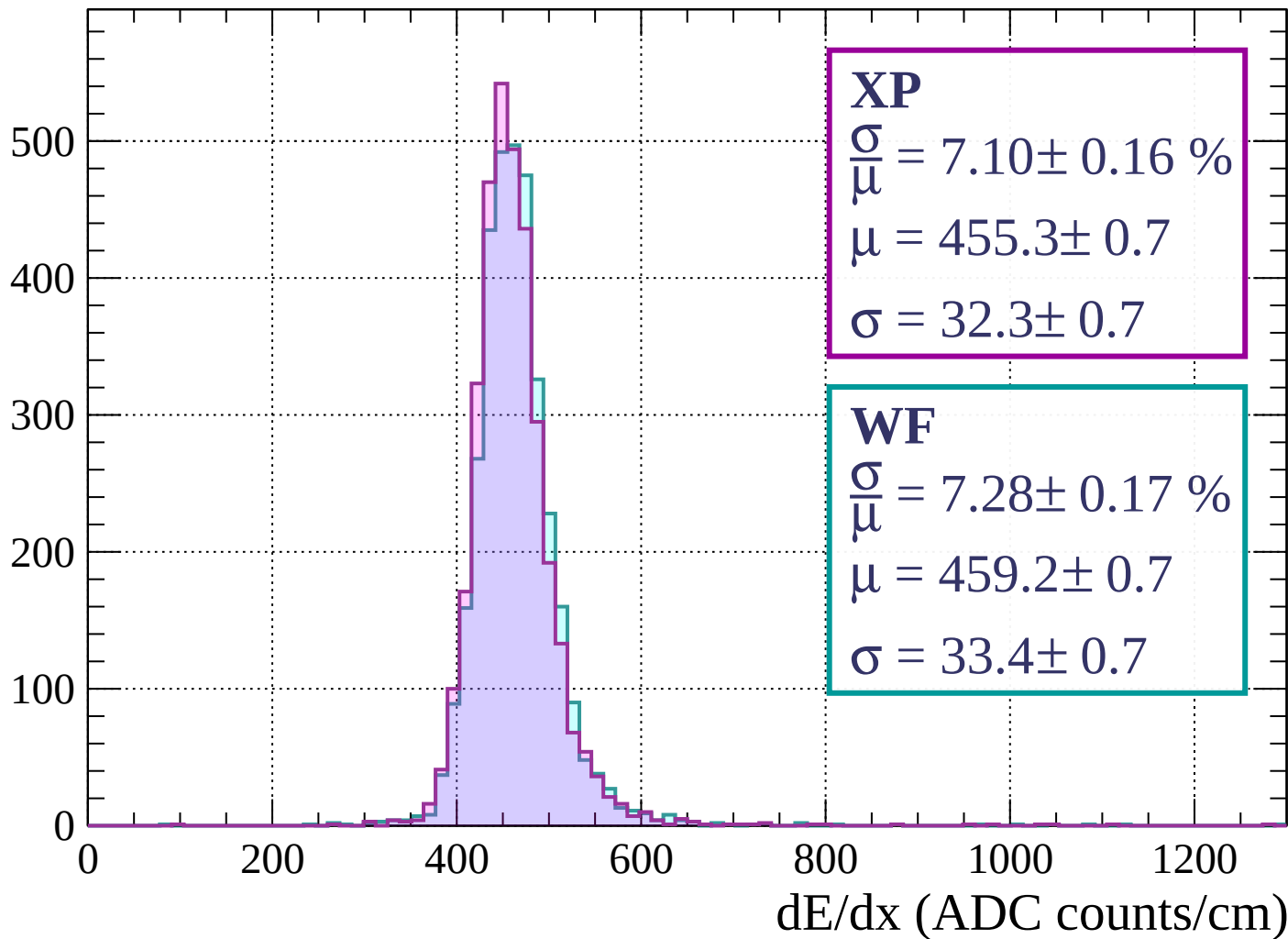
$$\mu = 455.9 \pm 0.7$$

$$\sigma = 32.9 \pm 0.7$$

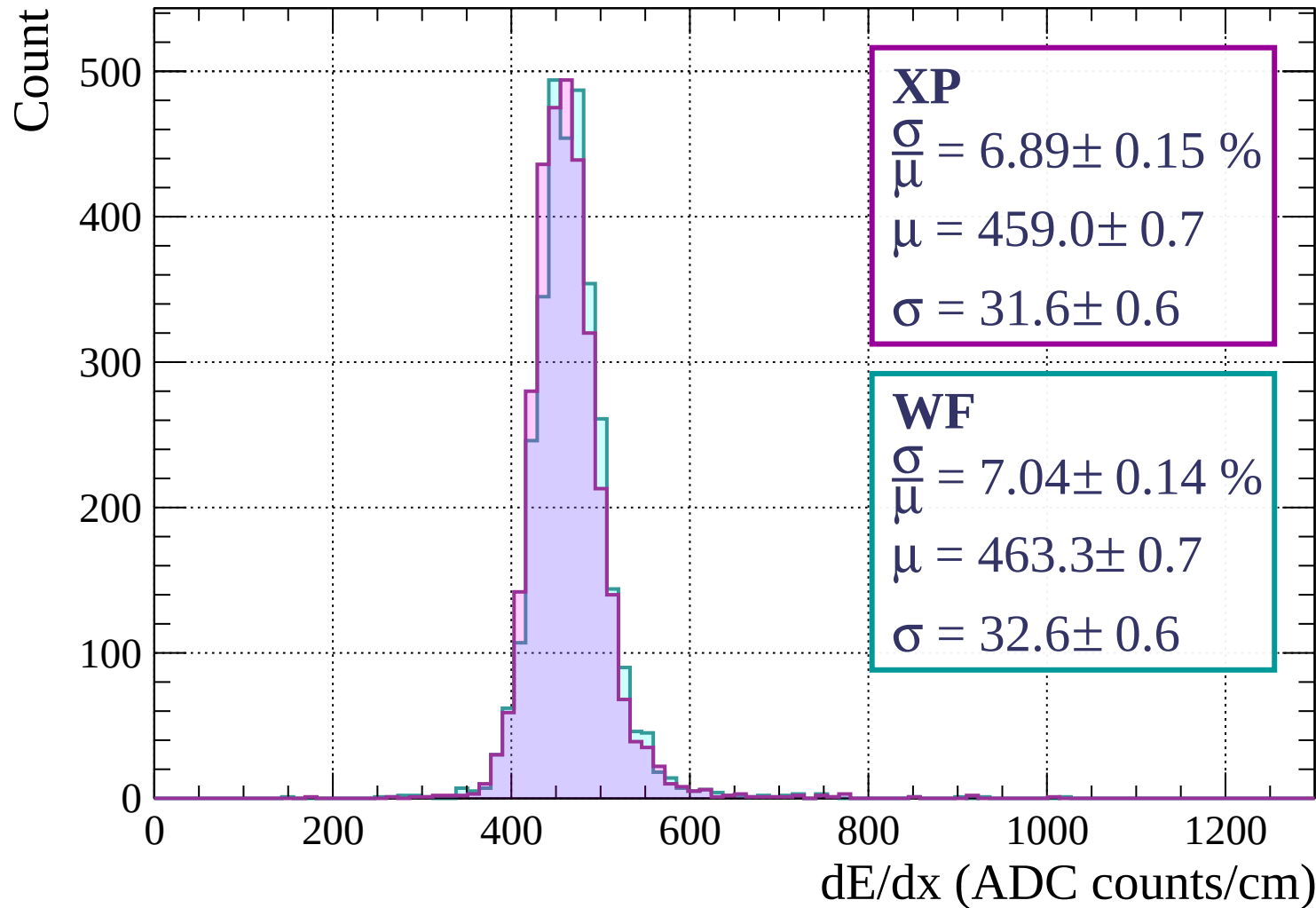
dE/dx (ADC counts/cm)

Energy loss | $1200 < p < 1280$

Count

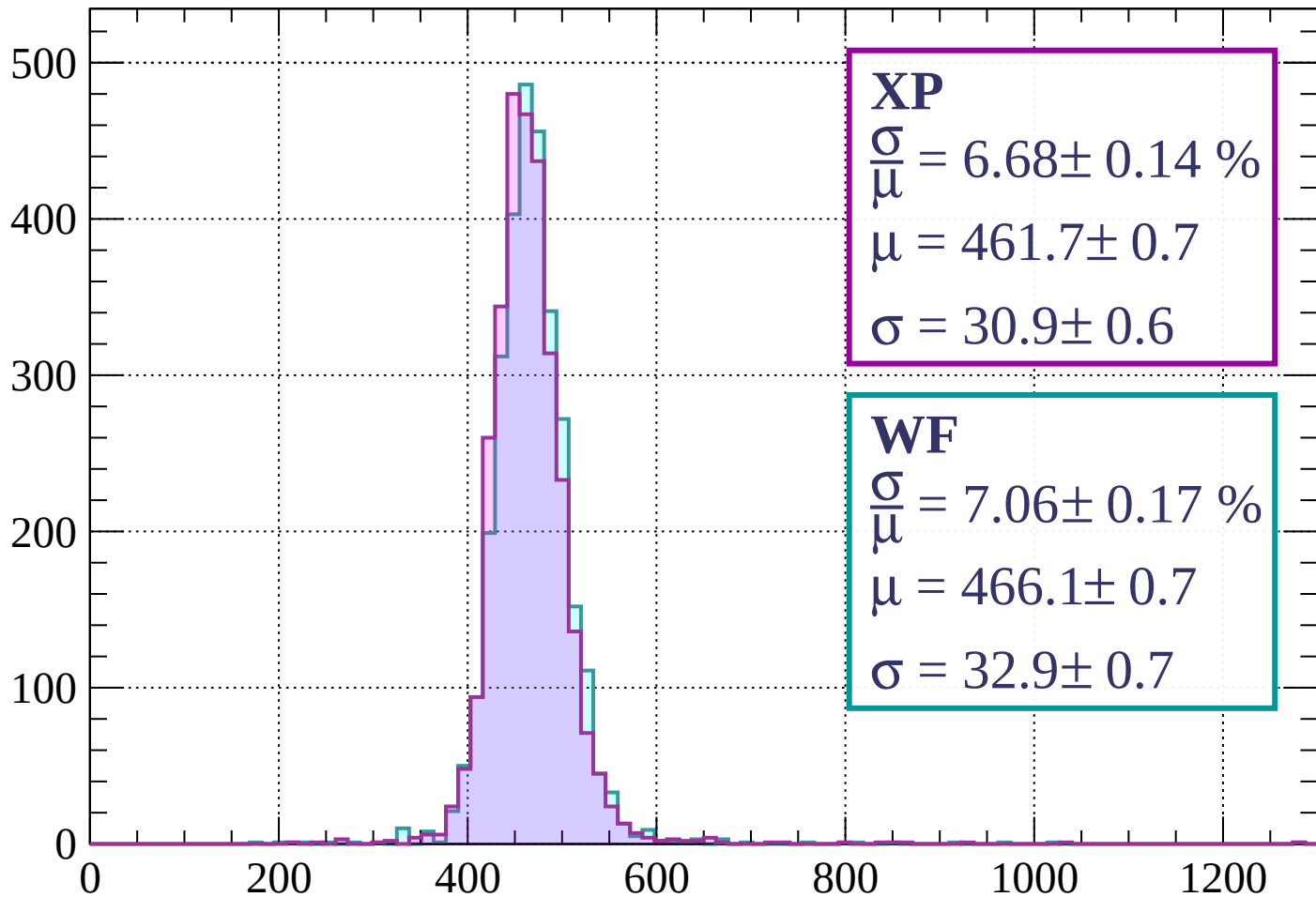


Energy loss | $1280 < p < 1360$



Energy loss | $1360 < p < 1440$

Count



XP

$$\frac{\sigma}{\mu} = 6.68 \pm 0.14 \%$$

$$\mu = 461.7 \pm 0.7$$

$$\sigma = 30.9 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 7.06 \pm 0.17 \%$$

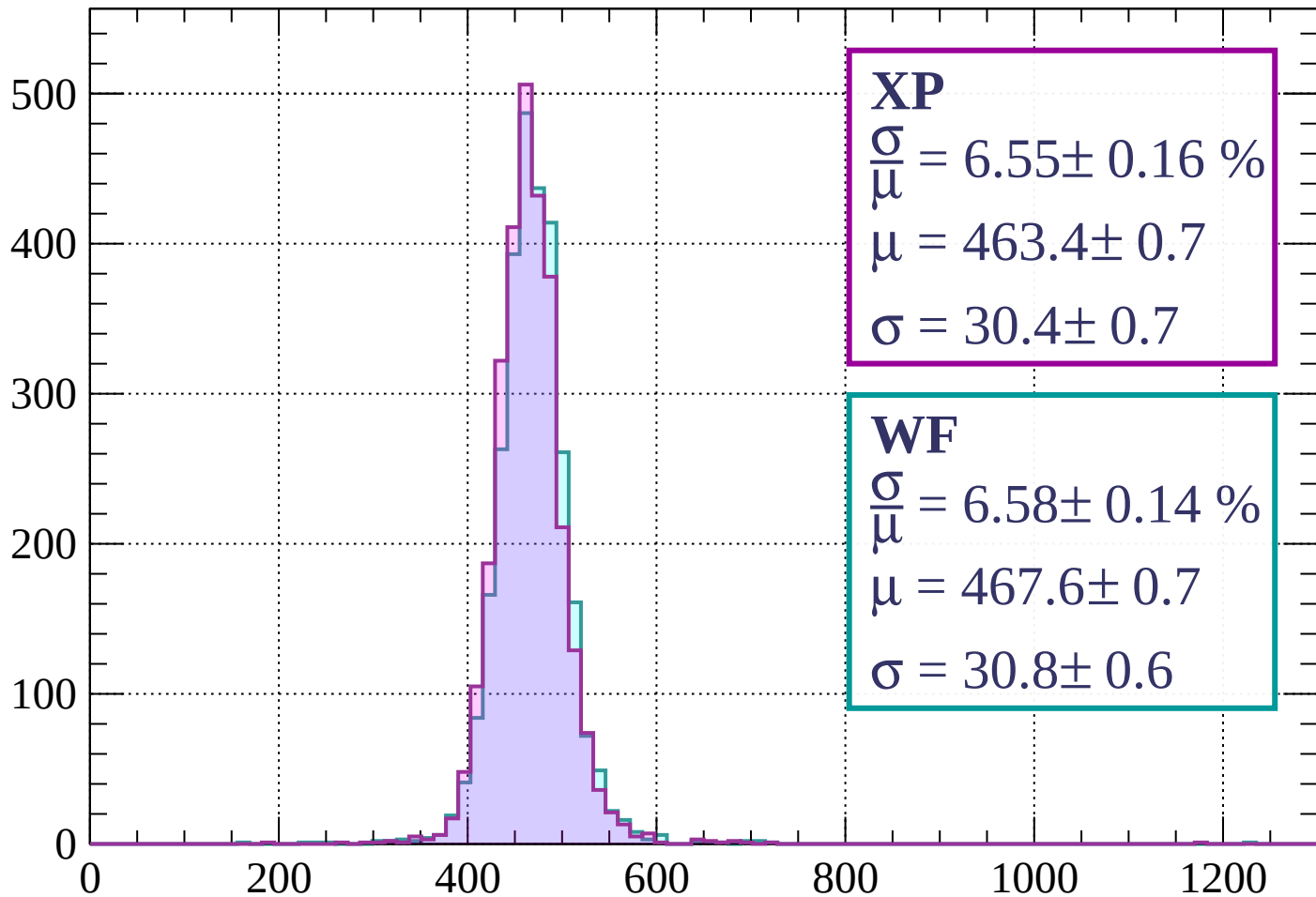
$$\mu = 466.1 \pm 0.7$$

$$\sigma = 32.9 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $1440 < p < 1520$

Count



XP

$$\frac{\sigma}{\mu} = 6.55 \pm 0.16 \%$$

$$\mu = 463.4 \pm 0.7$$

$$\sigma = 30.4 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 6.58 \pm 0.14 \%$$

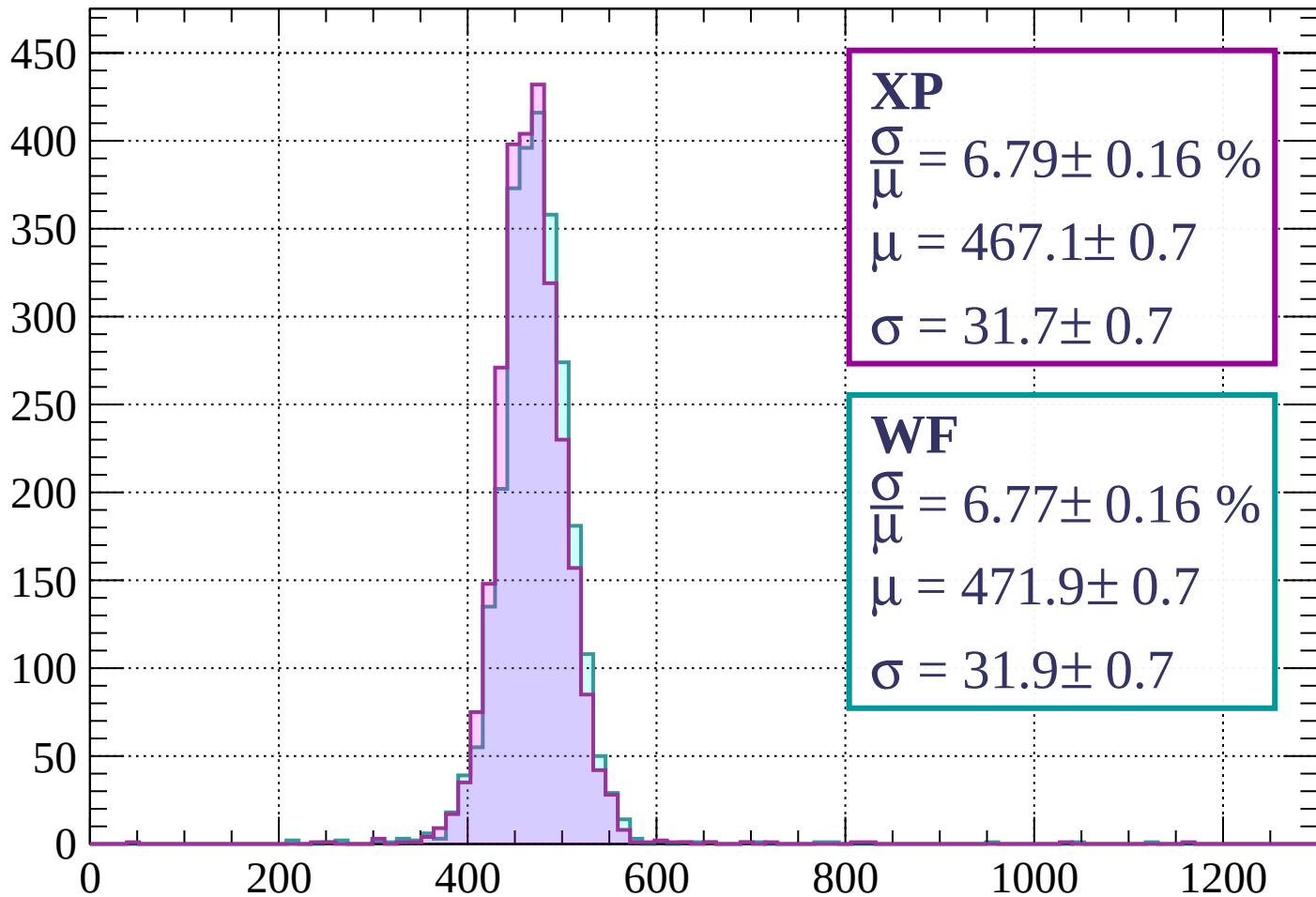
$$\mu = 467.6 \pm 0.7$$

$$\sigma = 30.8 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $1520 < p < 1600$

Count



XP

$$\frac{\sigma}{\mu} = 6.79 \pm 0.16 \%$$

$$\mu = 467.1 \pm 0.7$$

$$\sigma = 31.7 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 6.77 \pm 0.16 \%$$

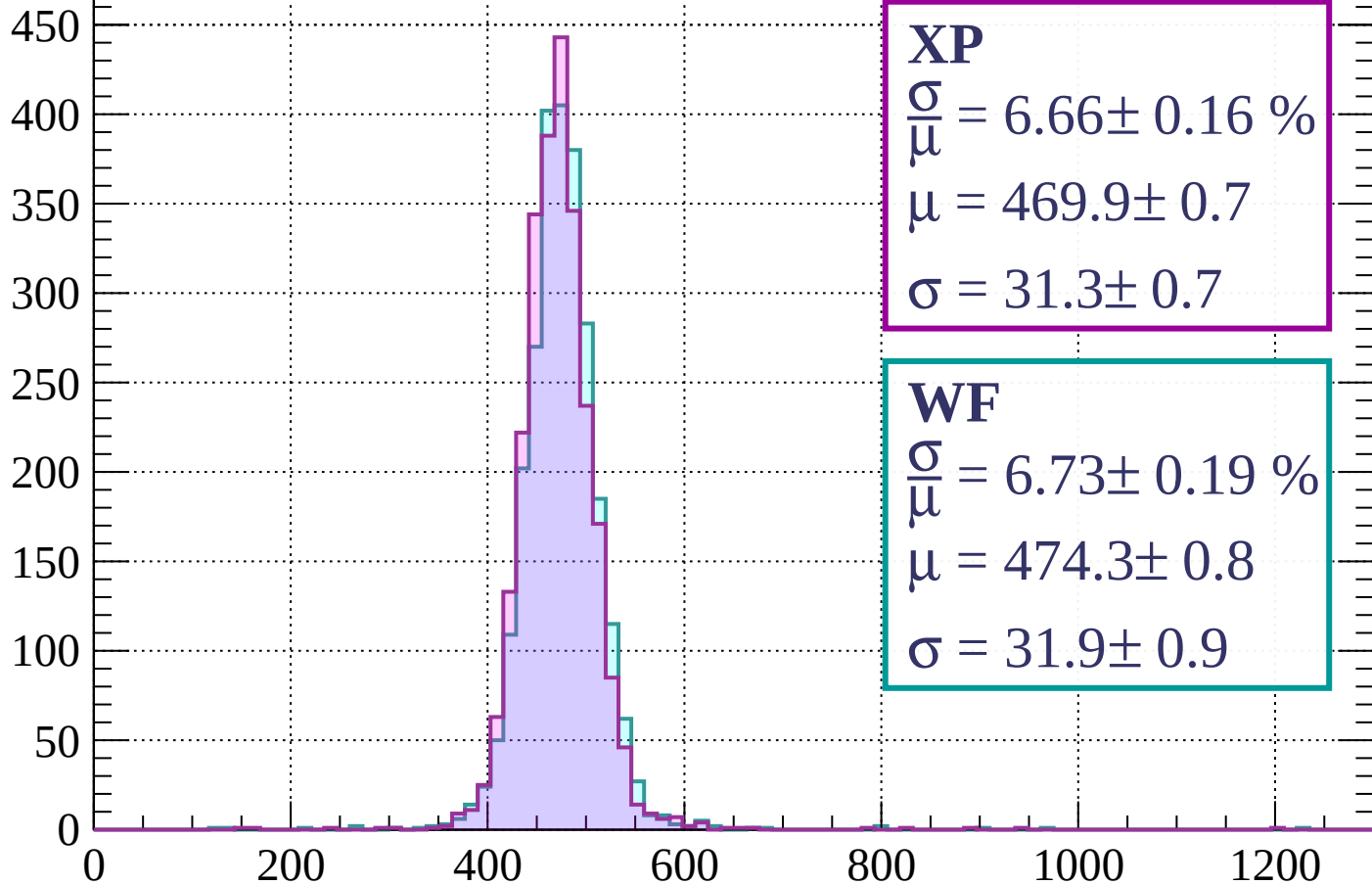
$$\mu = 471.9 \pm 0.7$$

$$\sigma = 31.9 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $1600 < p < 1680$

Count



XP

$$\frac{\sigma}{\mu} = 6.66 \pm 0.16 \%$$

$$\mu = 469.9 \pm 0.7$$

$$\sigma = 31.3 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 6.73 \pm 0.19 \%$$

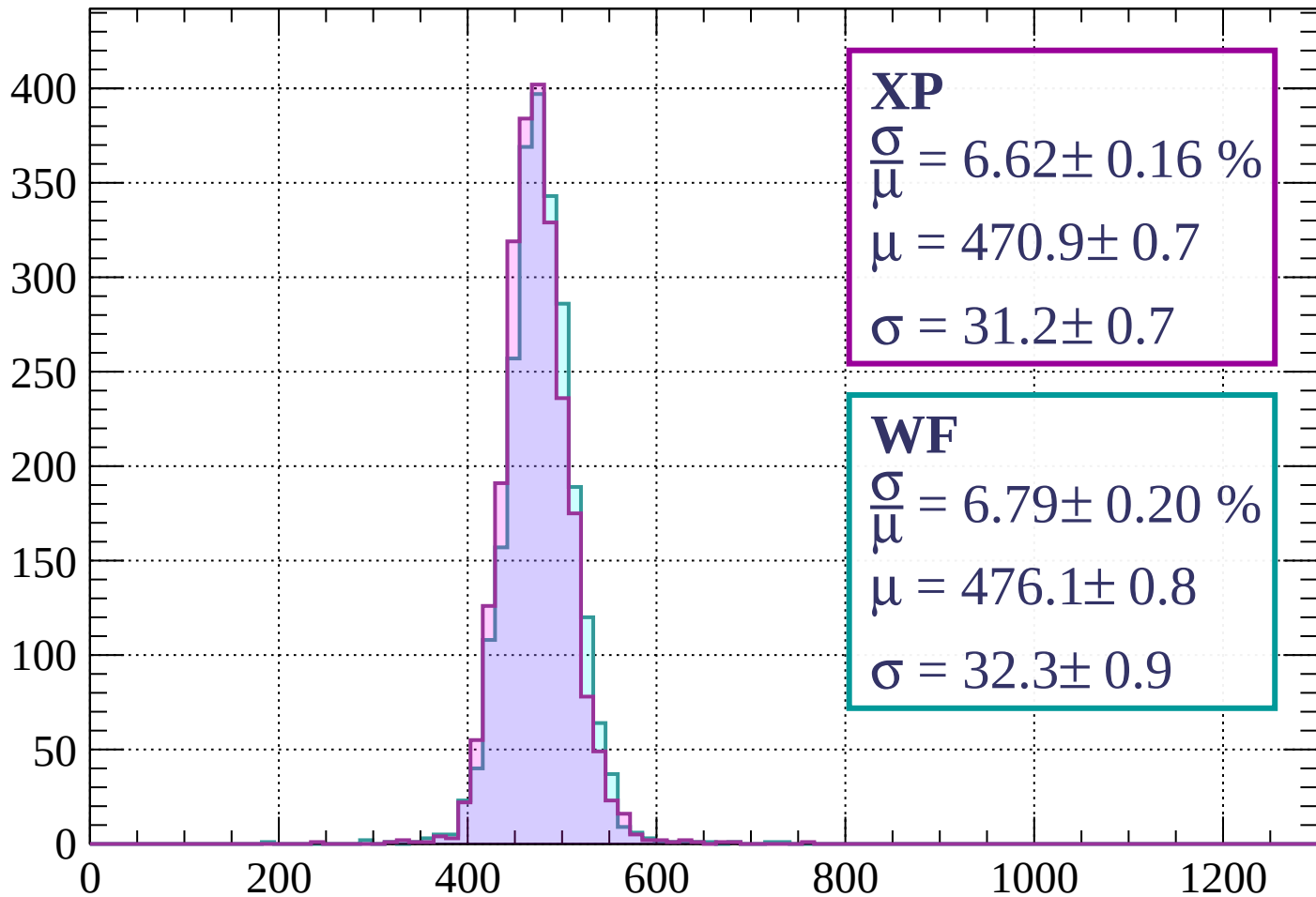
$$\mu = 474.3 \pm 0.8$$

$$\sigma = 31.9 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $1680 < p < 1760$

Count



XP

$$\frac{\sigma}{\mu} = 6.62 \pm 0.16 \%$$

$$\mu = 470.9 \pm 0.7$$

$$\sigma = 31.2 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 6.79 \pm 0.20 \%$$

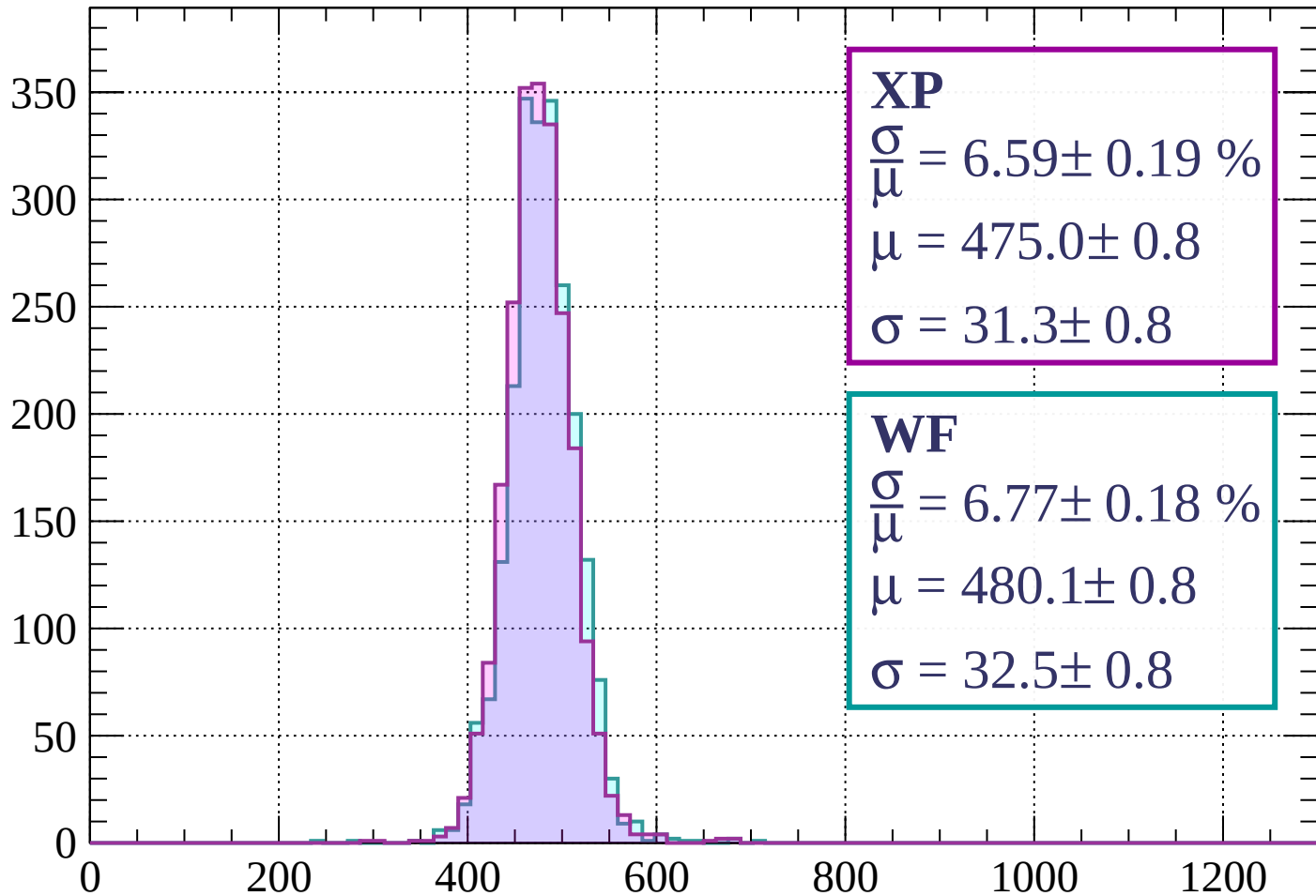
$$\mu = 476.1 \pm 0.8$$

$$\sigma = 32.3 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $1760 < p < 1840$

Count



XP

$$\frac{\sigma}{\mu} = 6.59 \pm 0.19 \%$$

$$\mu = 475.0 \pm 0.8$$

$$\sigma = 31.3 \pm 0.8$$

WF

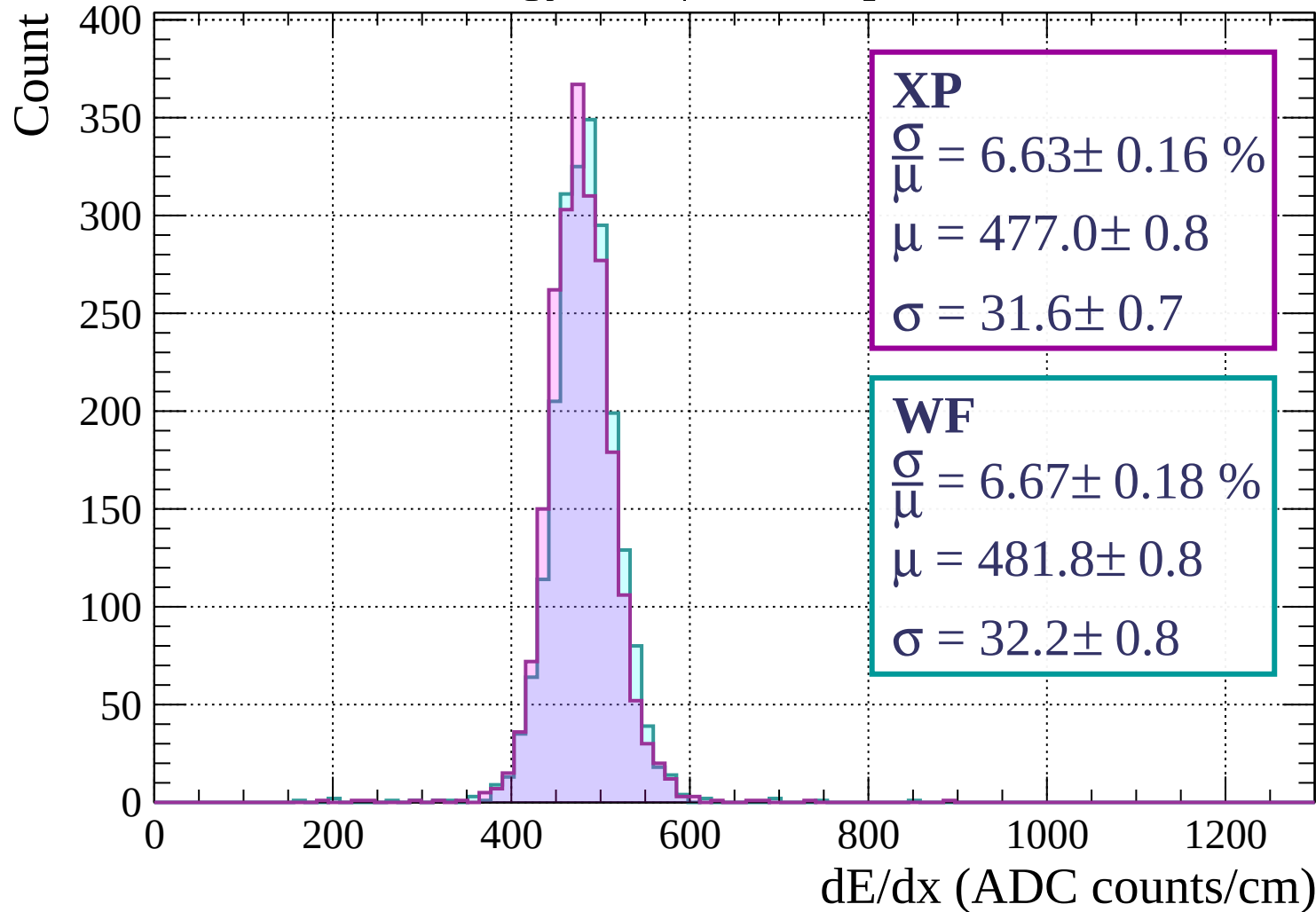
$$\frac{\sigma}{\mu} = 6.77 \pm 0.18 \%$$

$$\mu = 480.1 \pm 0.8$$

$$\sigma = 32.5 \pm 0.8$$

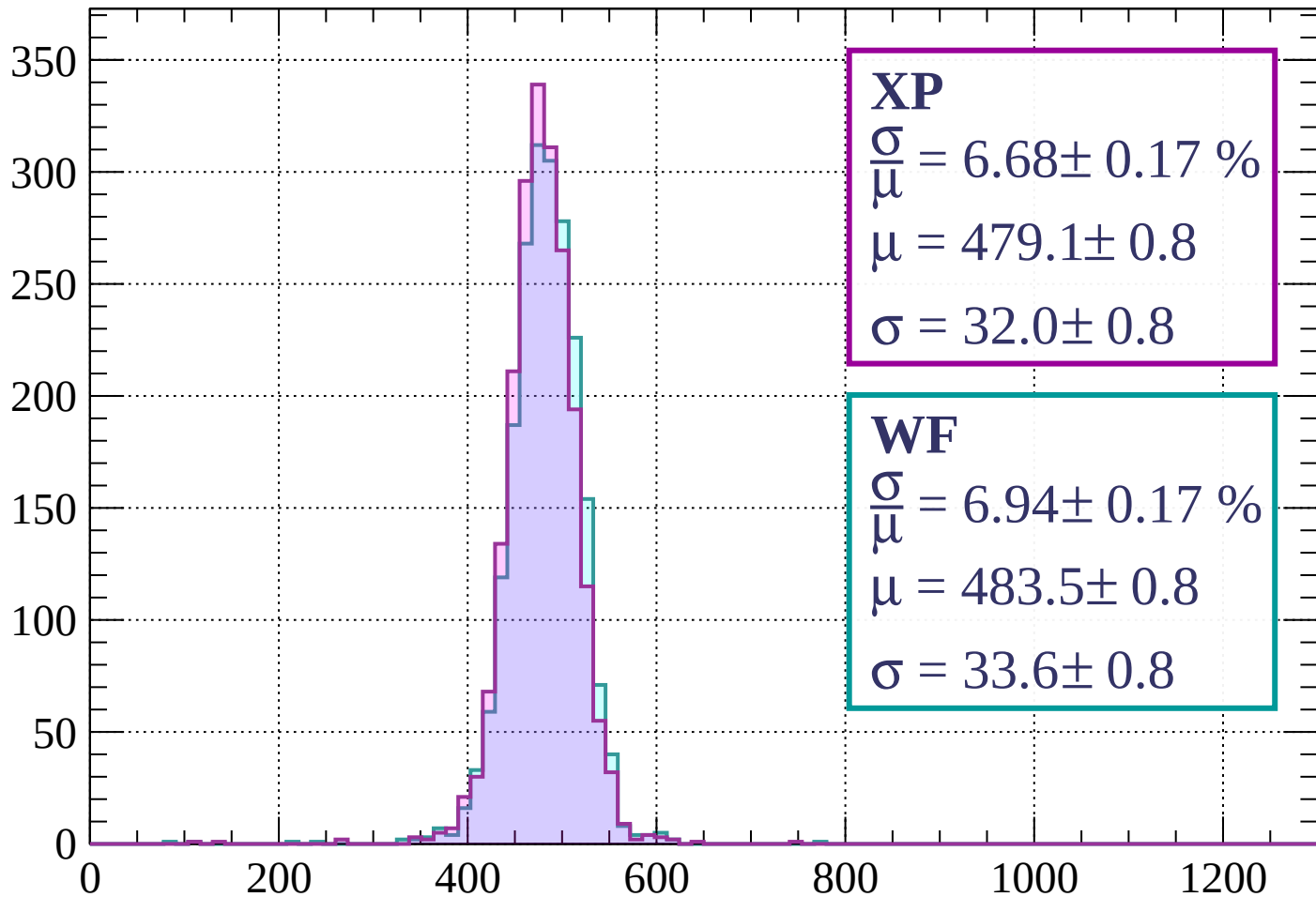
dE/dx (ADC counts/cm)

Energy loss | $1840 < p < 1920$



Energy loss | $1920 < p < 2000$

Count



XP

$$\frac{\sigma}{\mu} = 6.68 \pm 0.17 \%$$

$$\mu = 479.1 \pm 0.8$$

$$\sigma = 32.0 \pm 0.8$$

WF

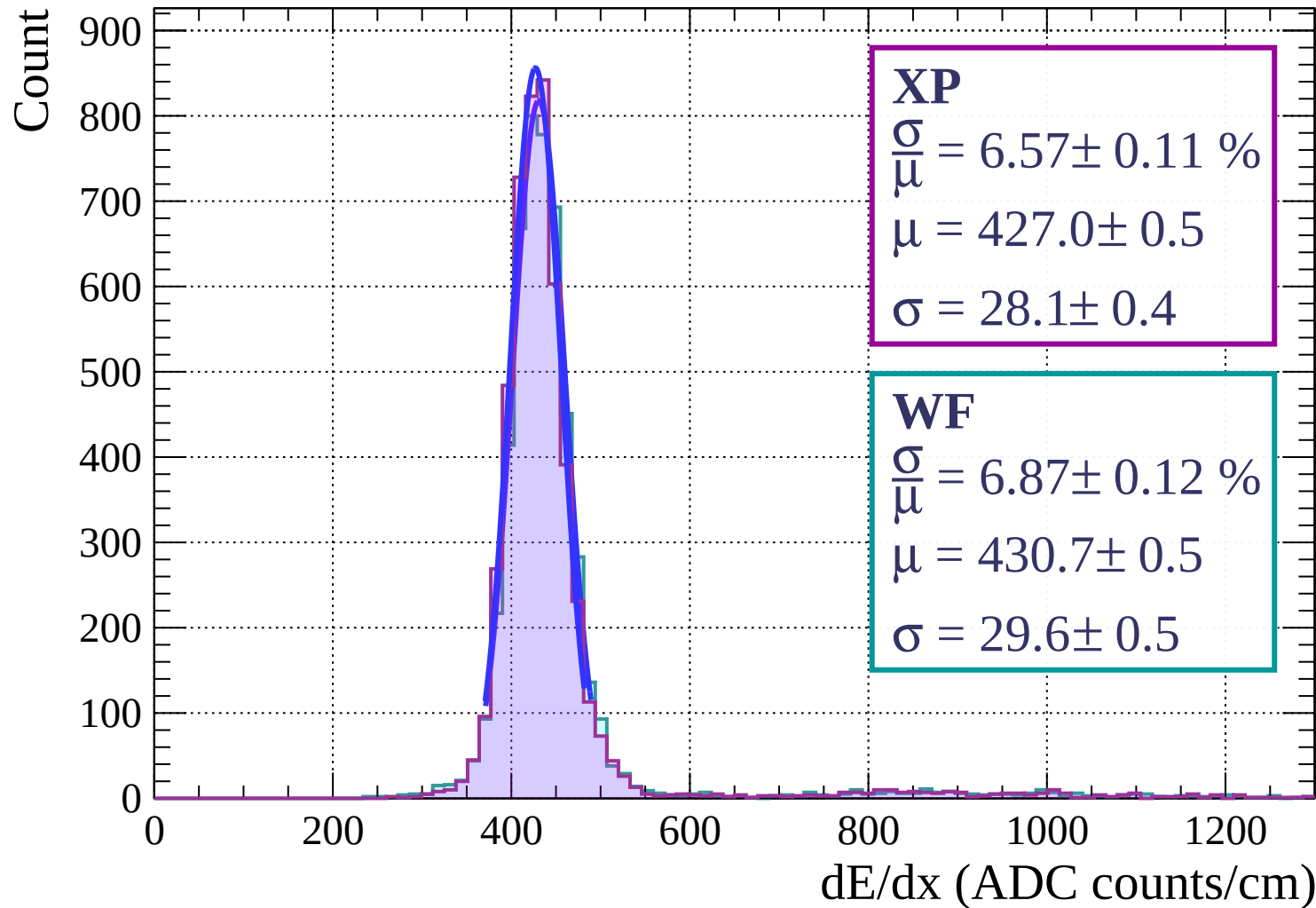
$$\frac{\sigma}{\mu} = 6.94 \pm 0.17 \%$$

$$\mu = 483.5 \pm 0.8$$

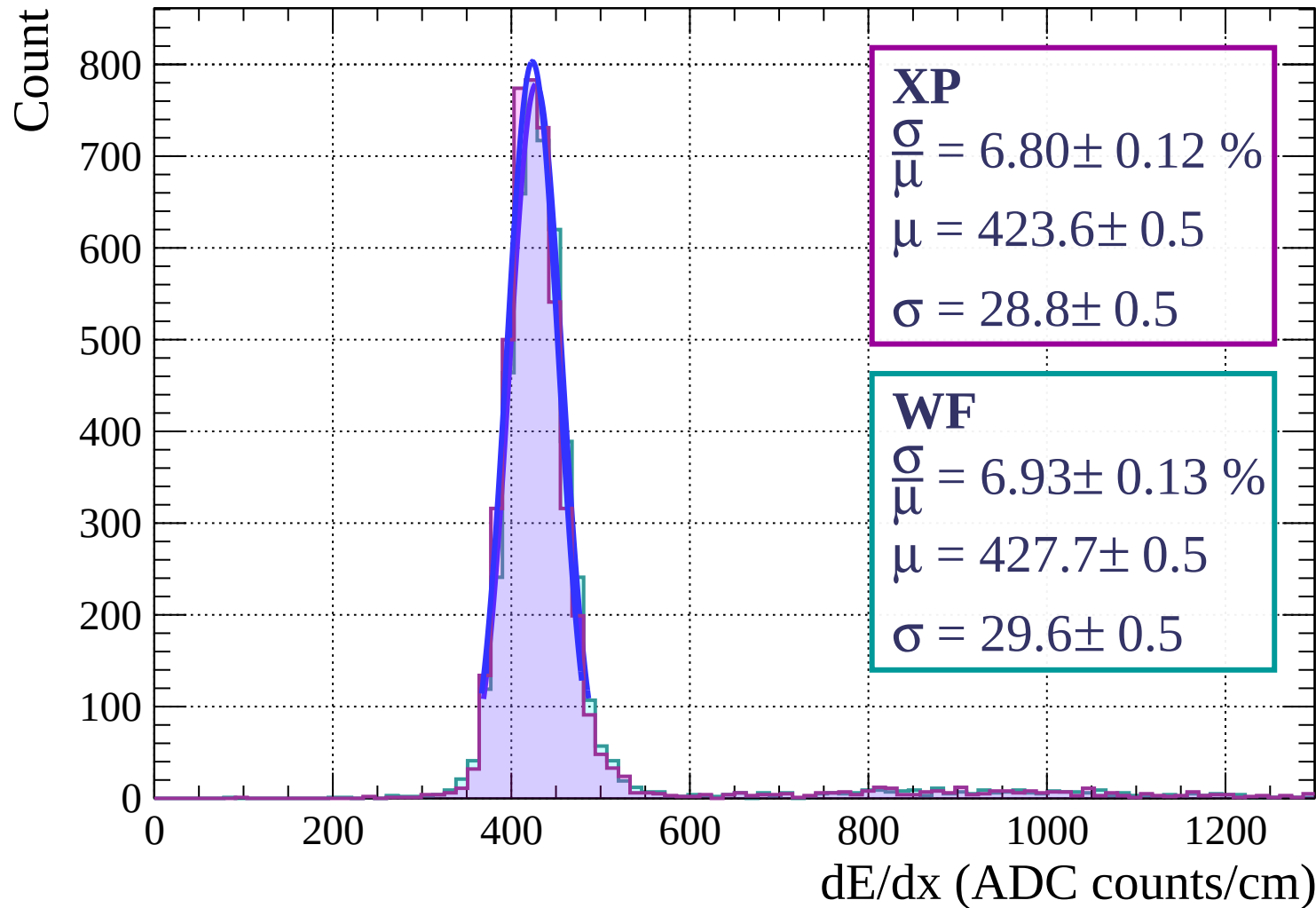
$$\sigma = 33.6 \pm 0.8$$

dE/dx (ADC counts/cm)

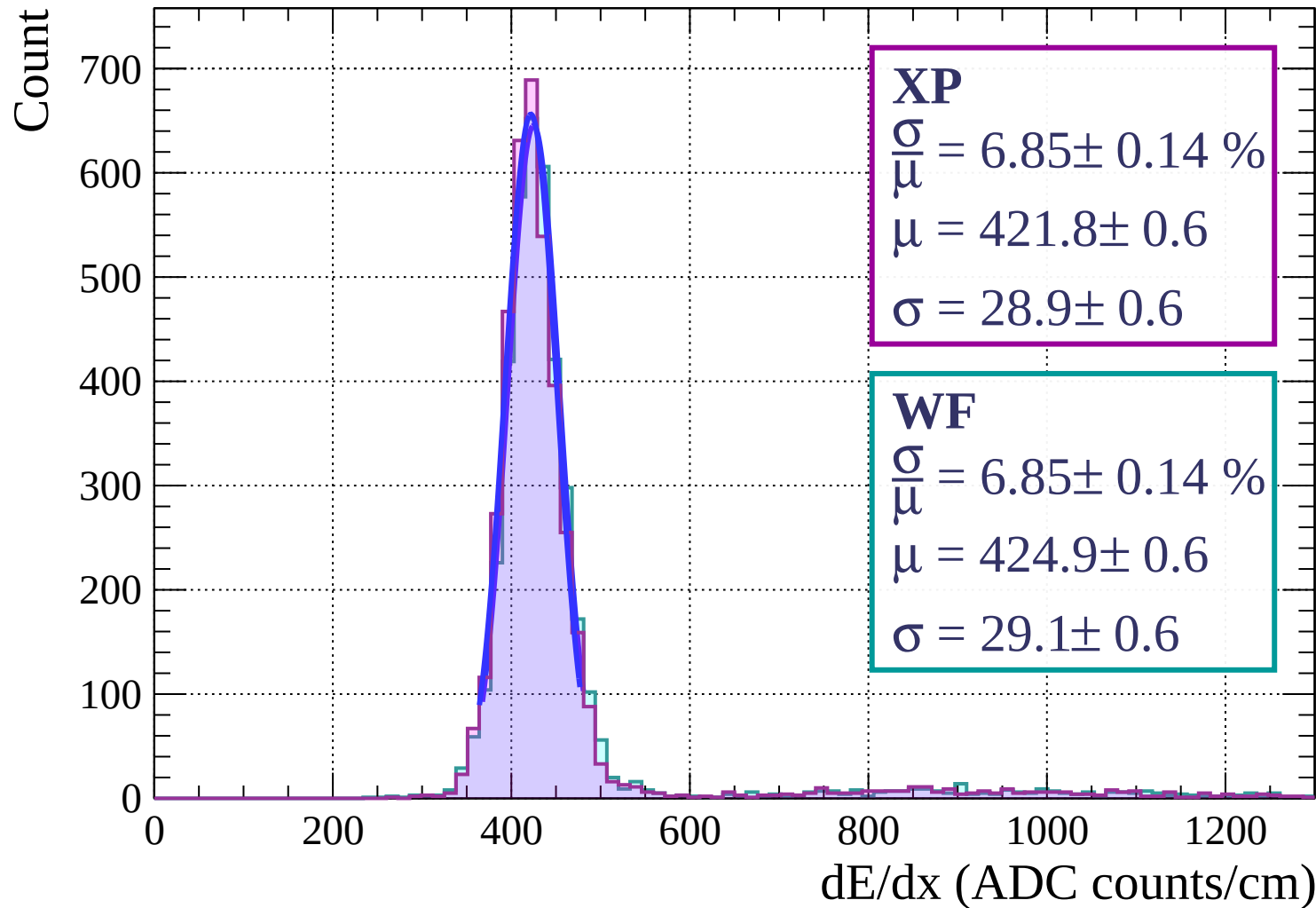
Energy loss | $0 < \phi < 3$



Energy loss | $3 < \phi < 6$

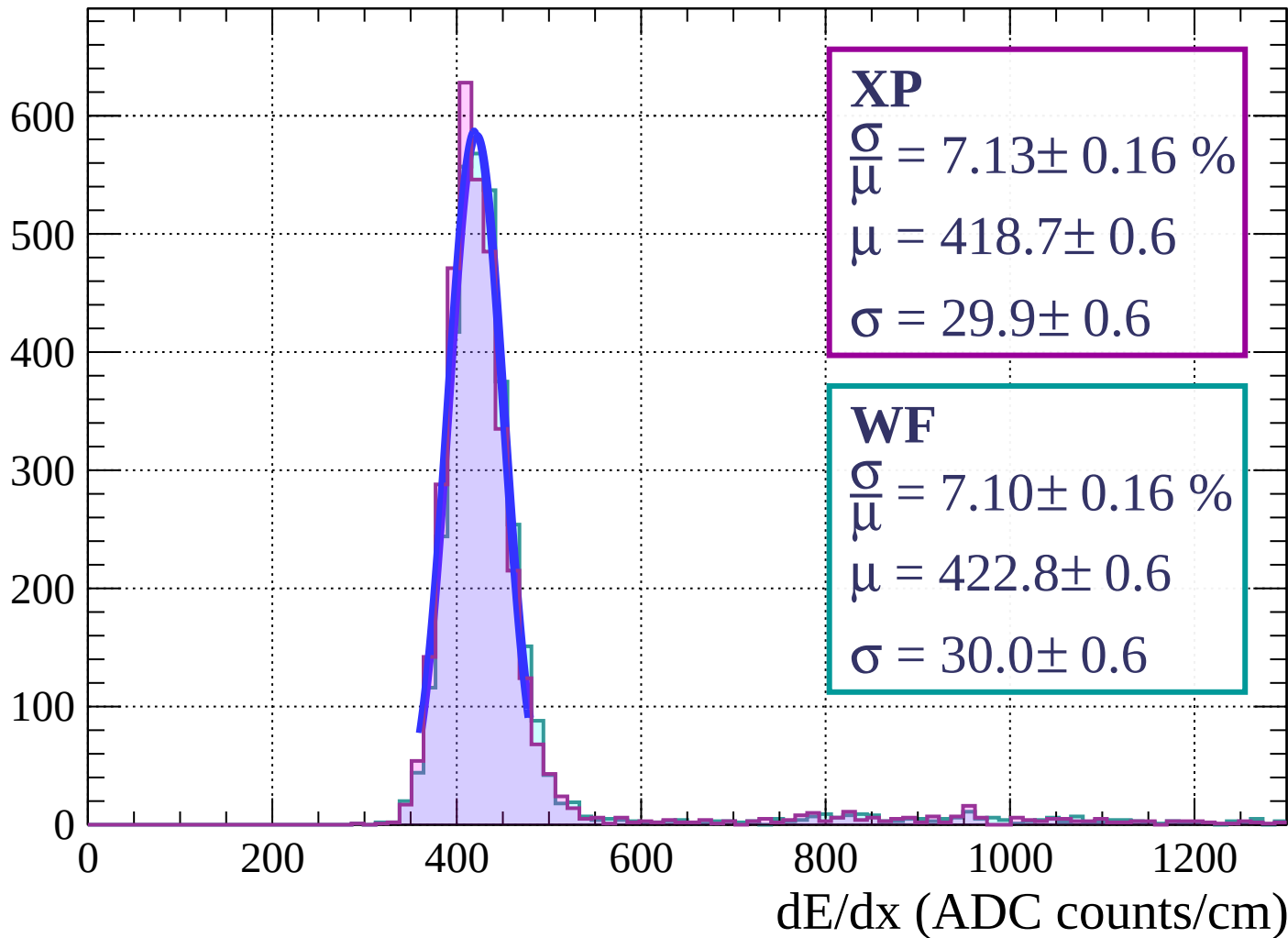


Energy loss | $6 < \phi < 9$



Energy loss | $9 < \phi < 12$

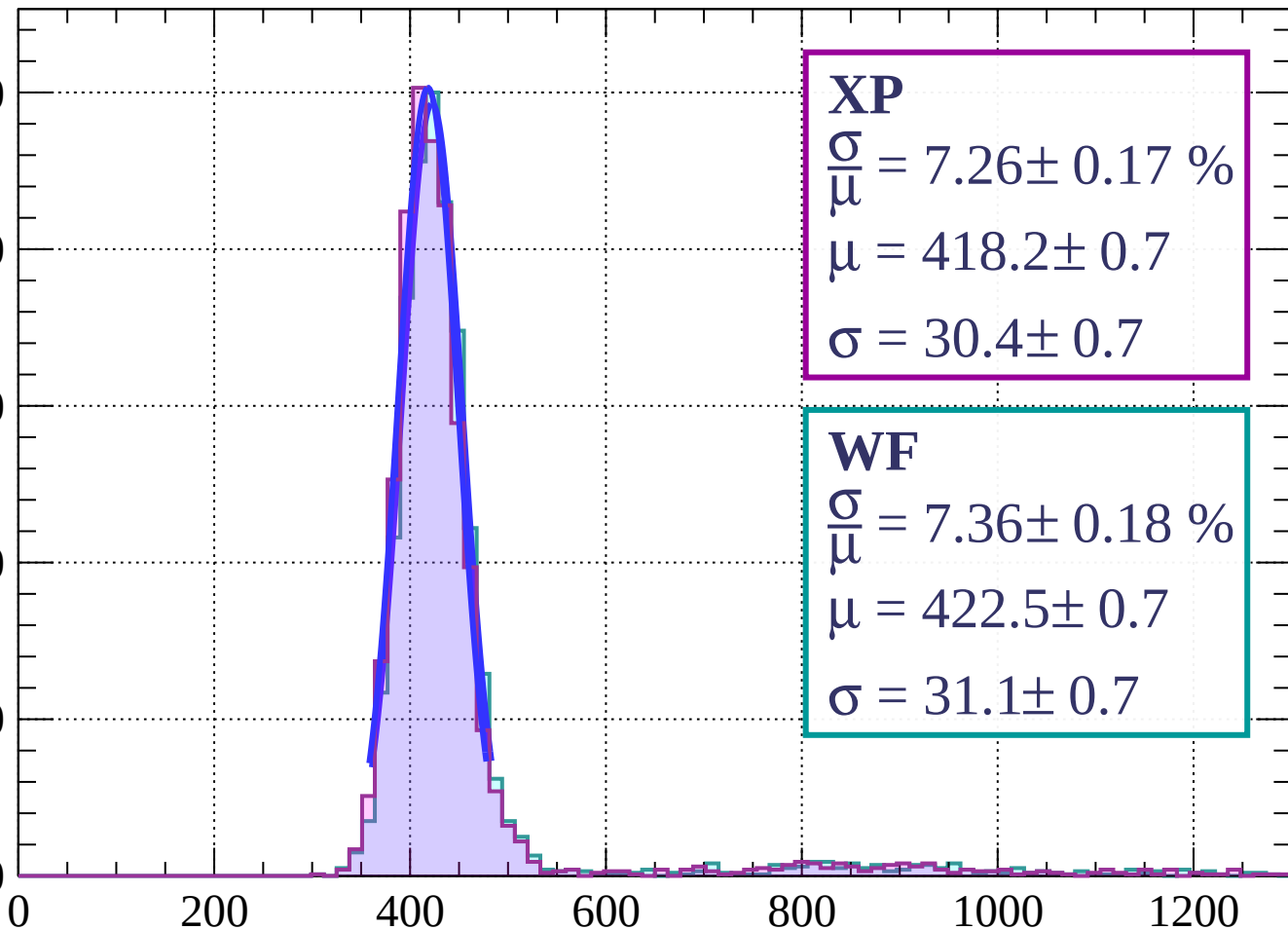
Count



Energy loss | $12 < \phi < 15$

Count

500
400
300
200
100
0



XP

$$\frac{\sigma}{\mu} = 7.26 \pm 0.17 \%$$

$$\mu = 418.2 \pm 0.7$$

$$\sigma = 30.4 \pm 0.7$$

WF

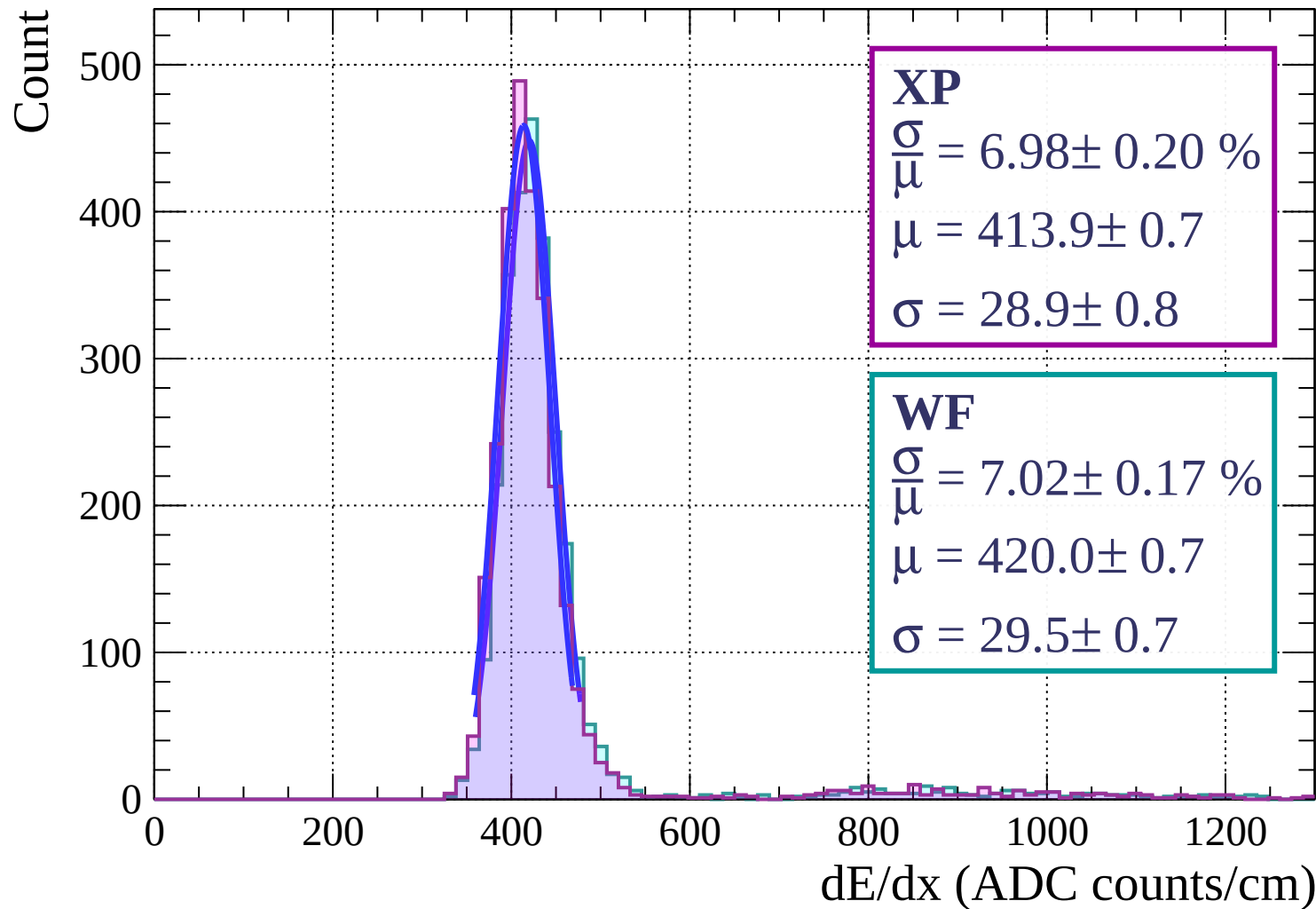
$$\frac{\sigma}{\mu} = 7.36 \pm 0.18 \%$$

$$\mu = 422.5 \pm 0.7$$

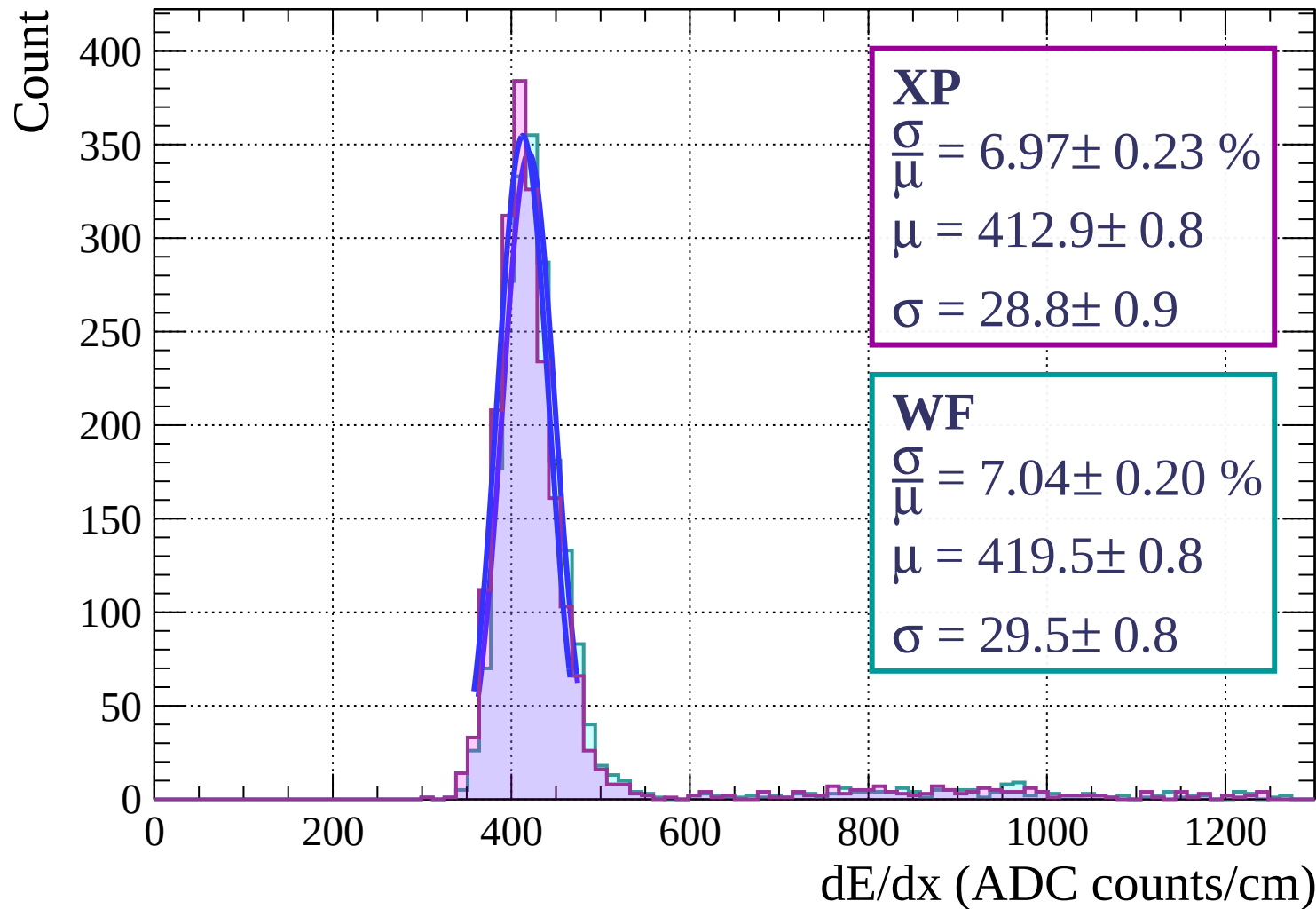
$$\sigma = 31.1 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $15 < \phi < 18$

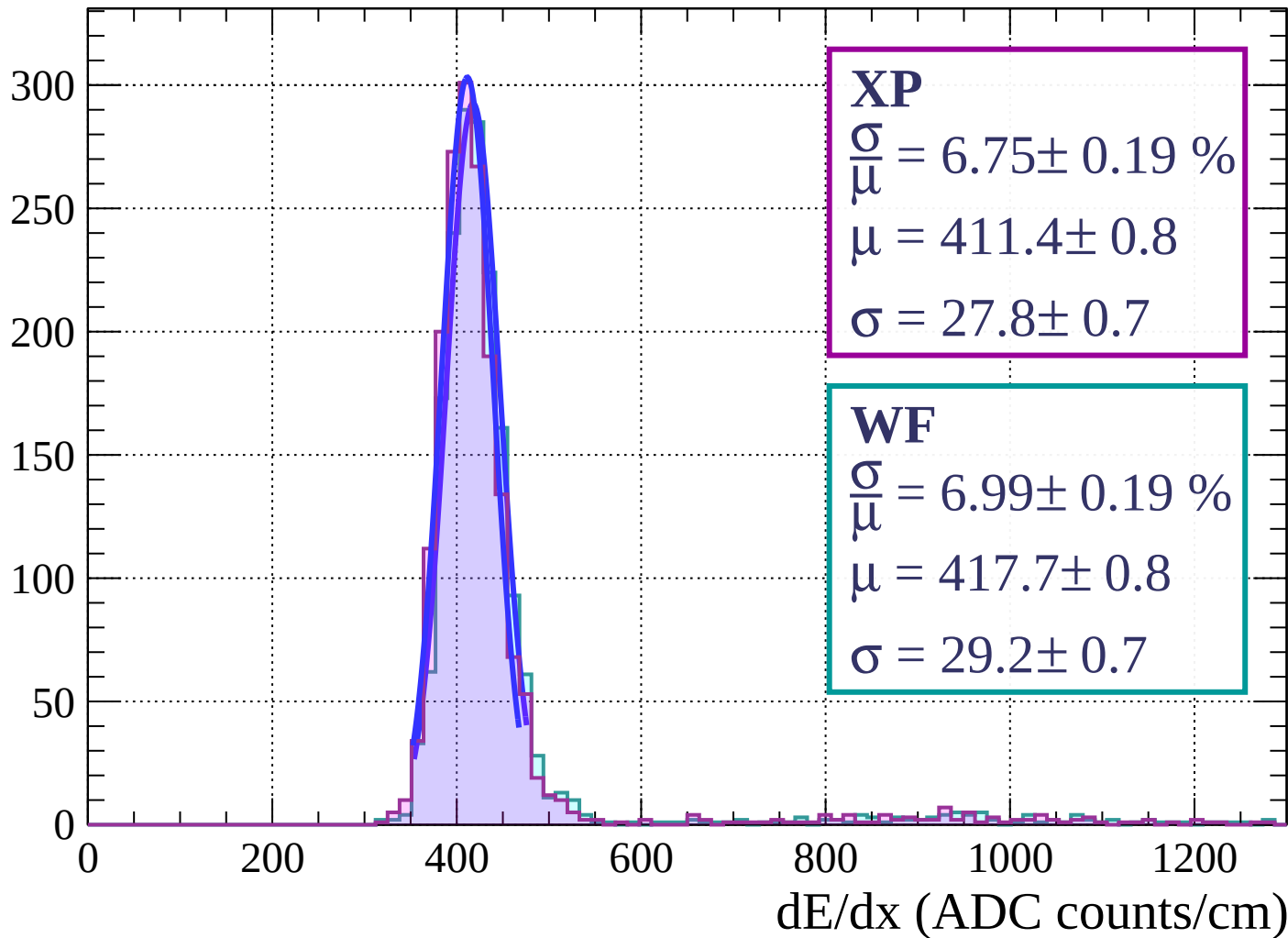


Energy loss | $18 < \phi < 21$



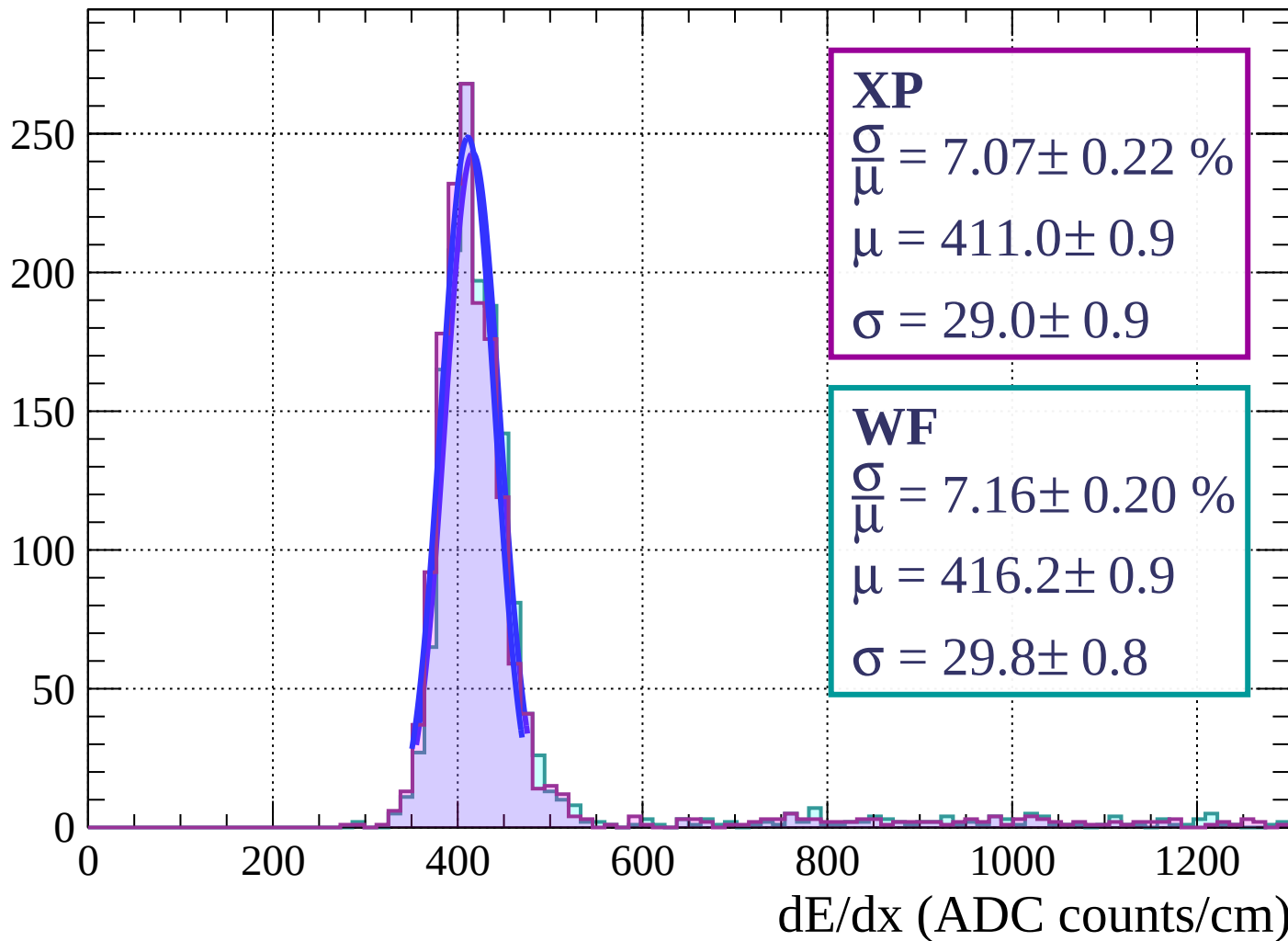
Energy loss | $21 < \phi < 24$

Count



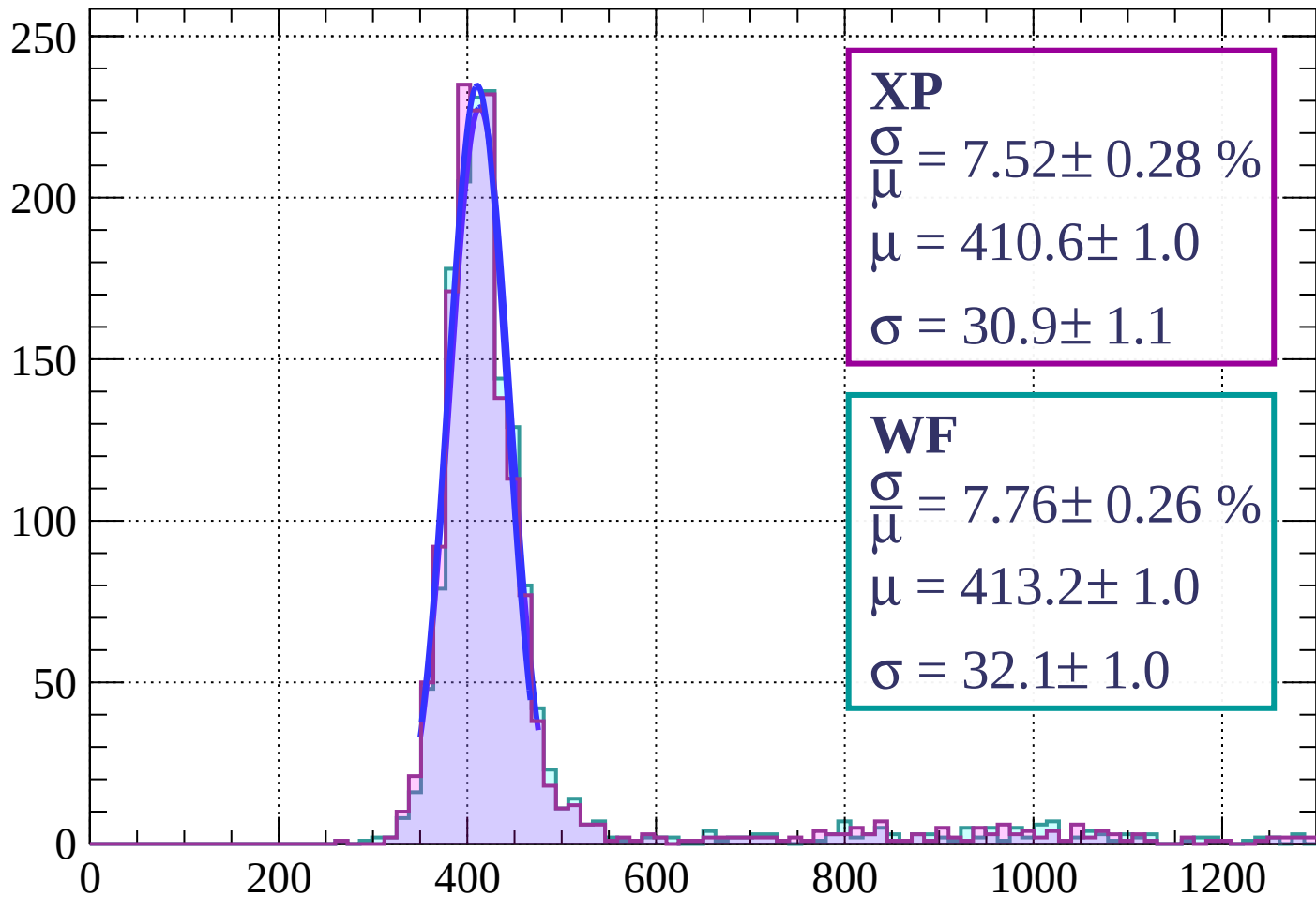
Energy loss | $24 < \phi < 27$

Count



Energy loss | $27 < \phi < 30$

Count



XP

$$\frac{\sigma}{\mu} = 7.52 \pm 0.28 \%$$

$$\mu = 410.6 \pm 1.0$$

$$\sigma = 30.9 \pm 1.1$$

WF

$$\frac{\sigma}{\mu} = 7.76 \pm 0.26 \%$$

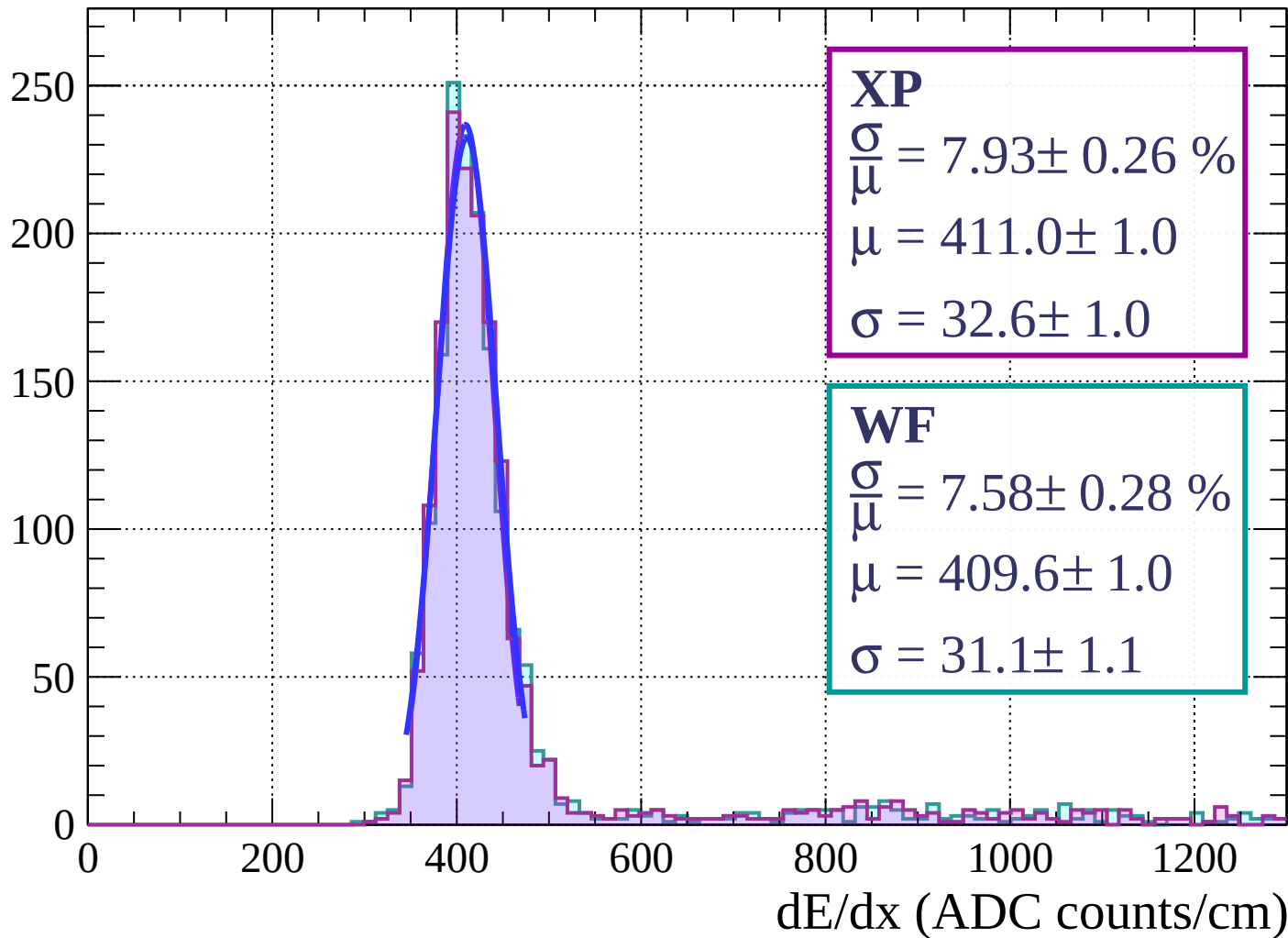
$$\mu = 413.2 \pm 1.0$$

$$\sigma = 32.1 \pm 1.0$$

dE/dx (ADC counts/cm)

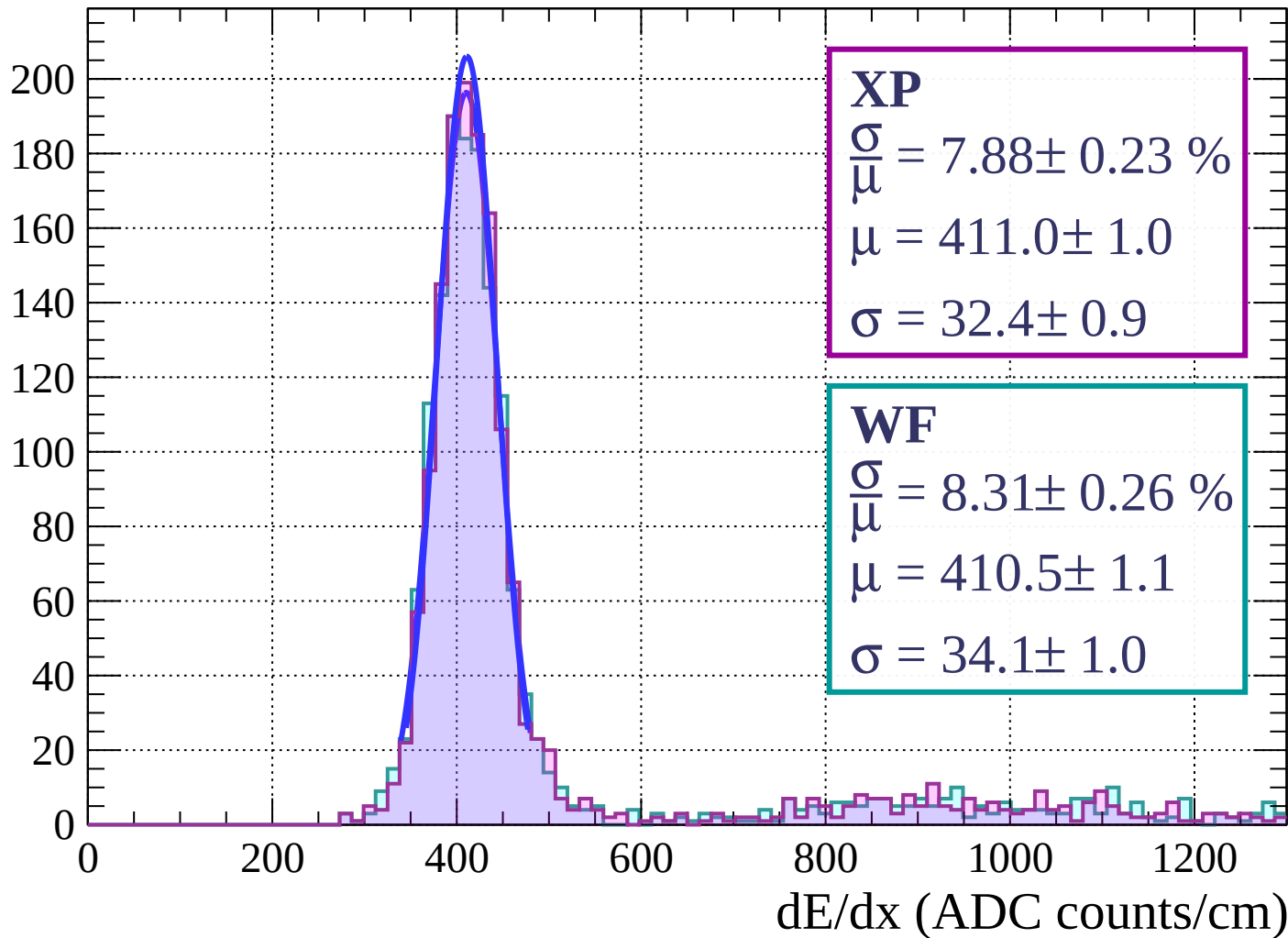
Energy loss | $30 < \phi < 33$

Count



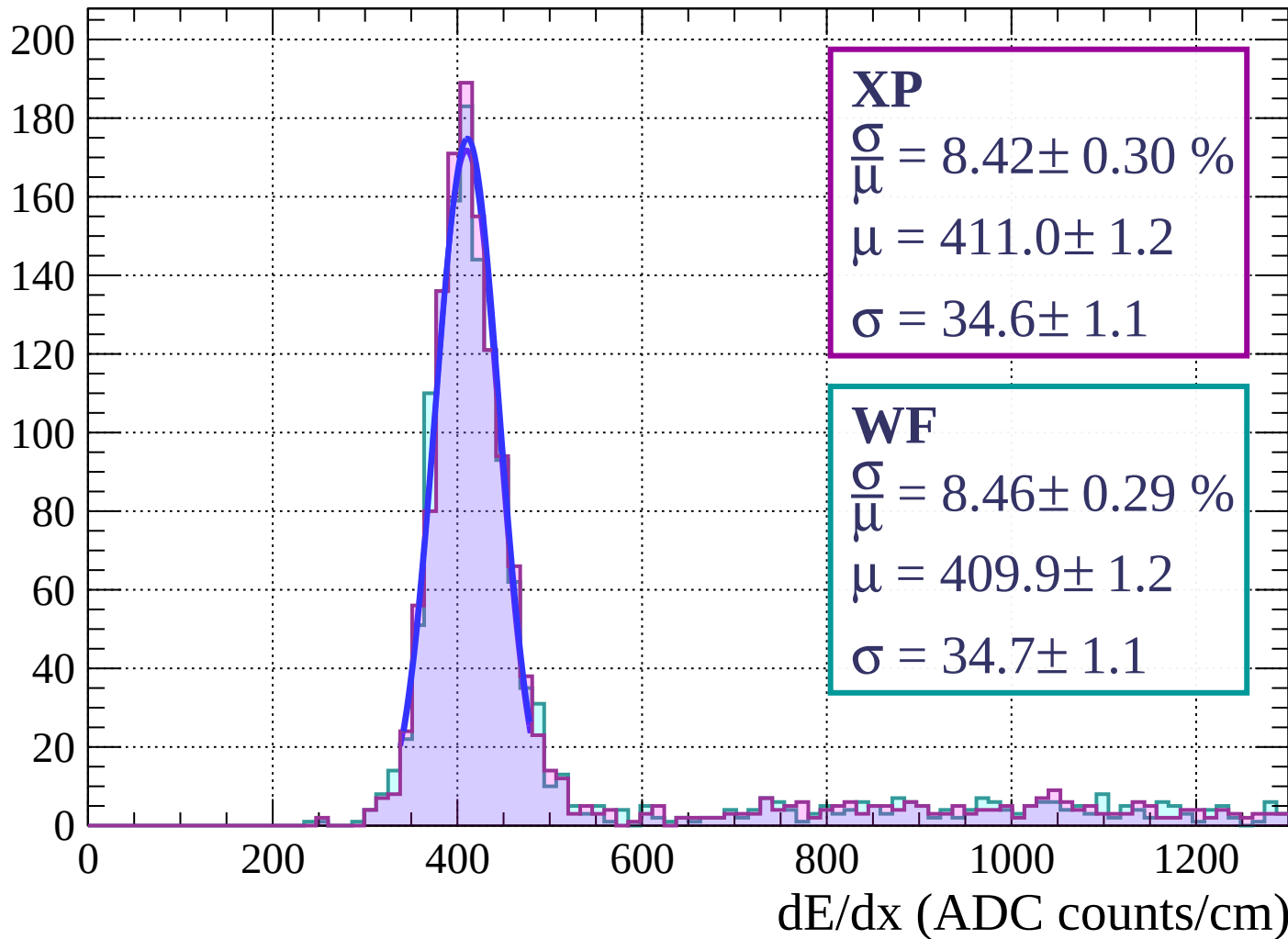
Energy loss | $33 < \phi < 36$

Count



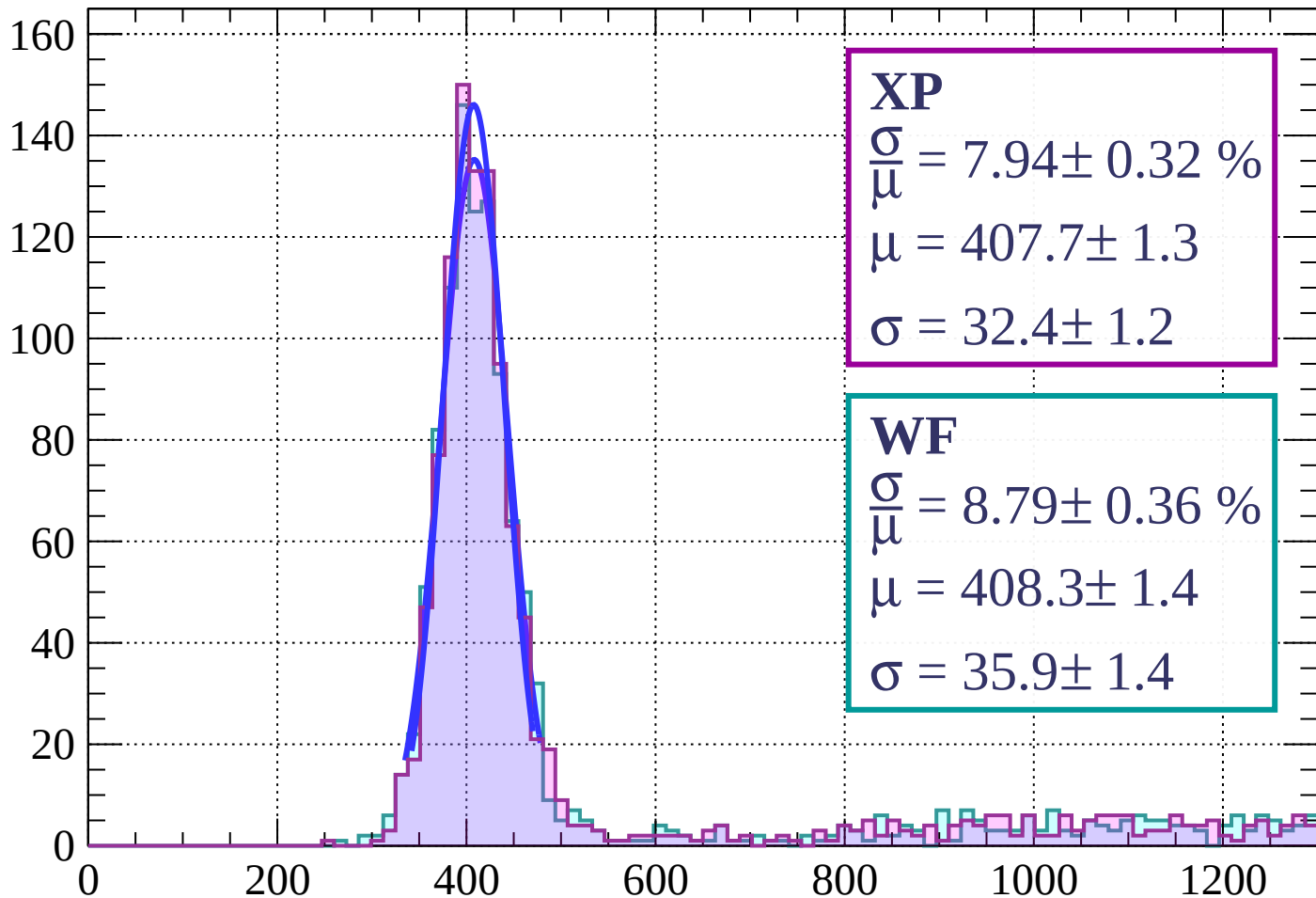
Energy loss | $36 < \phi < 39$

Count



Energy loss | $39 < \phi < 42$

Count



XP

$$\frac{\sigma}{\mu} = 7.94 \pm 0.32 \%$$

$$\mu = 407.7 \pm 1.3$$

$$\sigma = 32.4 \pm 1.2$$

WF

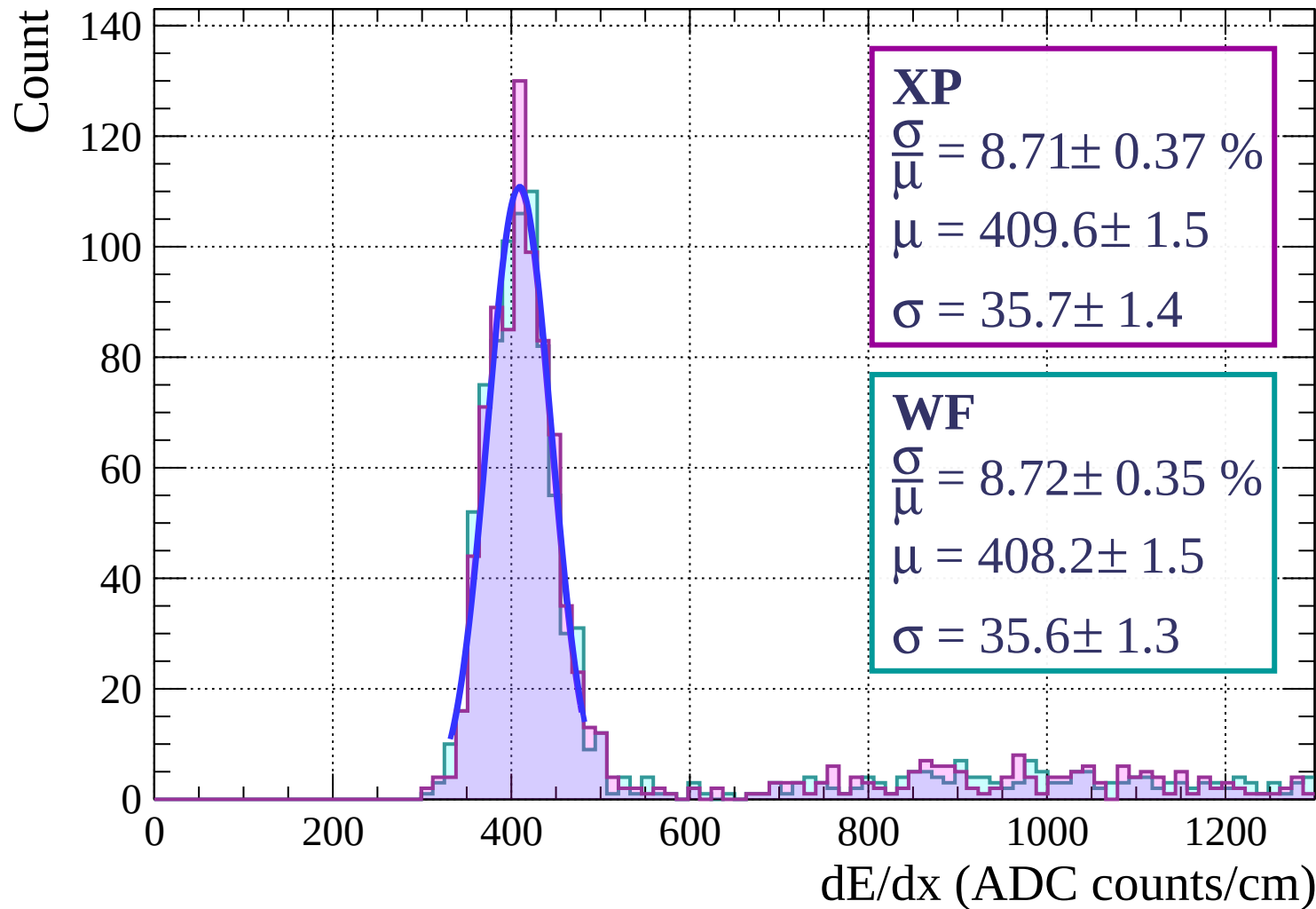
$$\frac{\sigma}{\mu} = 8.79 \pm 0.36 \%$$

$$\mu = 408.3 \pm 1.4$$

$$\sigma = 35.9 \pm 1.4$$

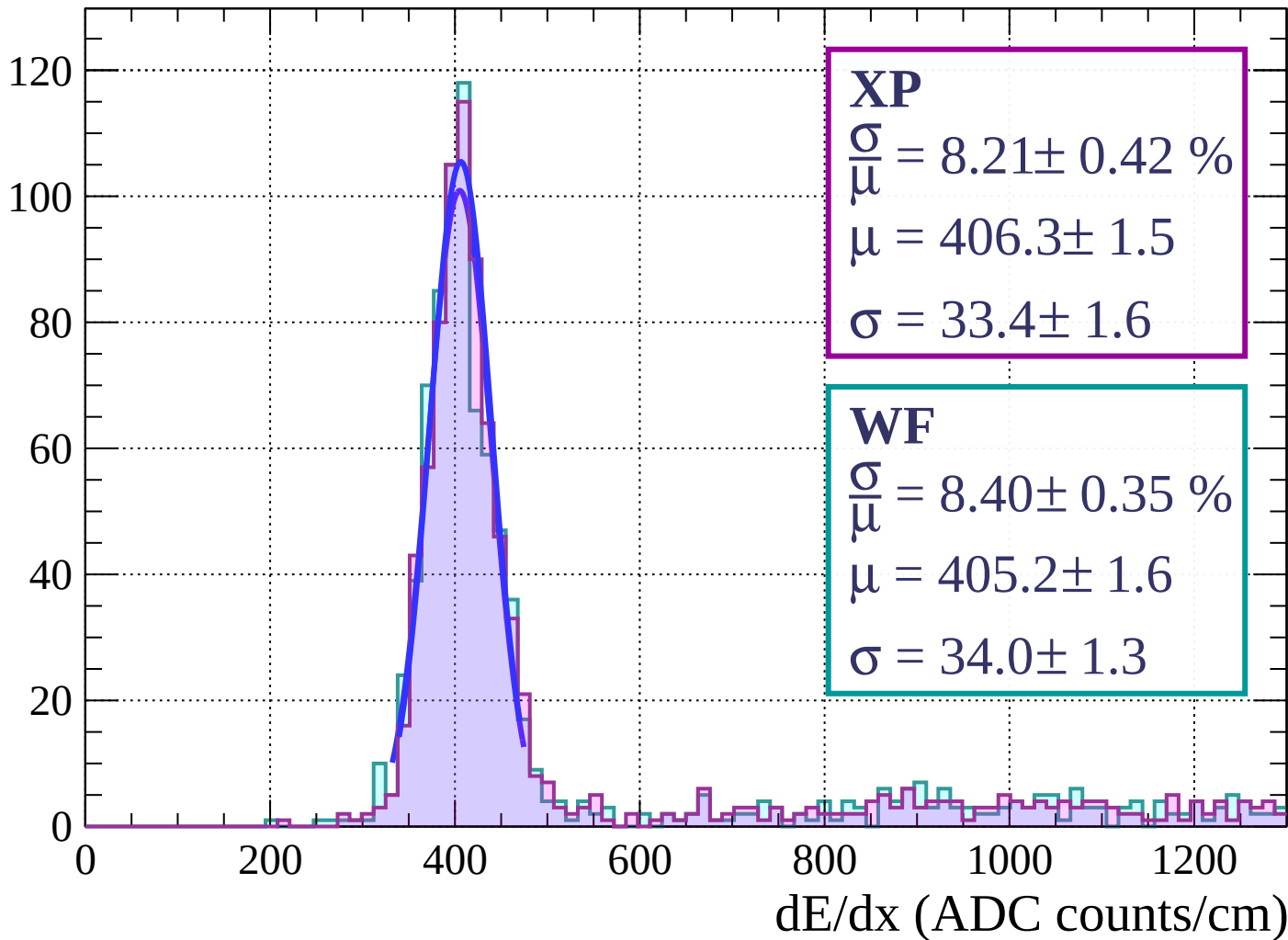
dE/dx (ADC counts/cm)

Energy loss | $42 < \phi < 45$



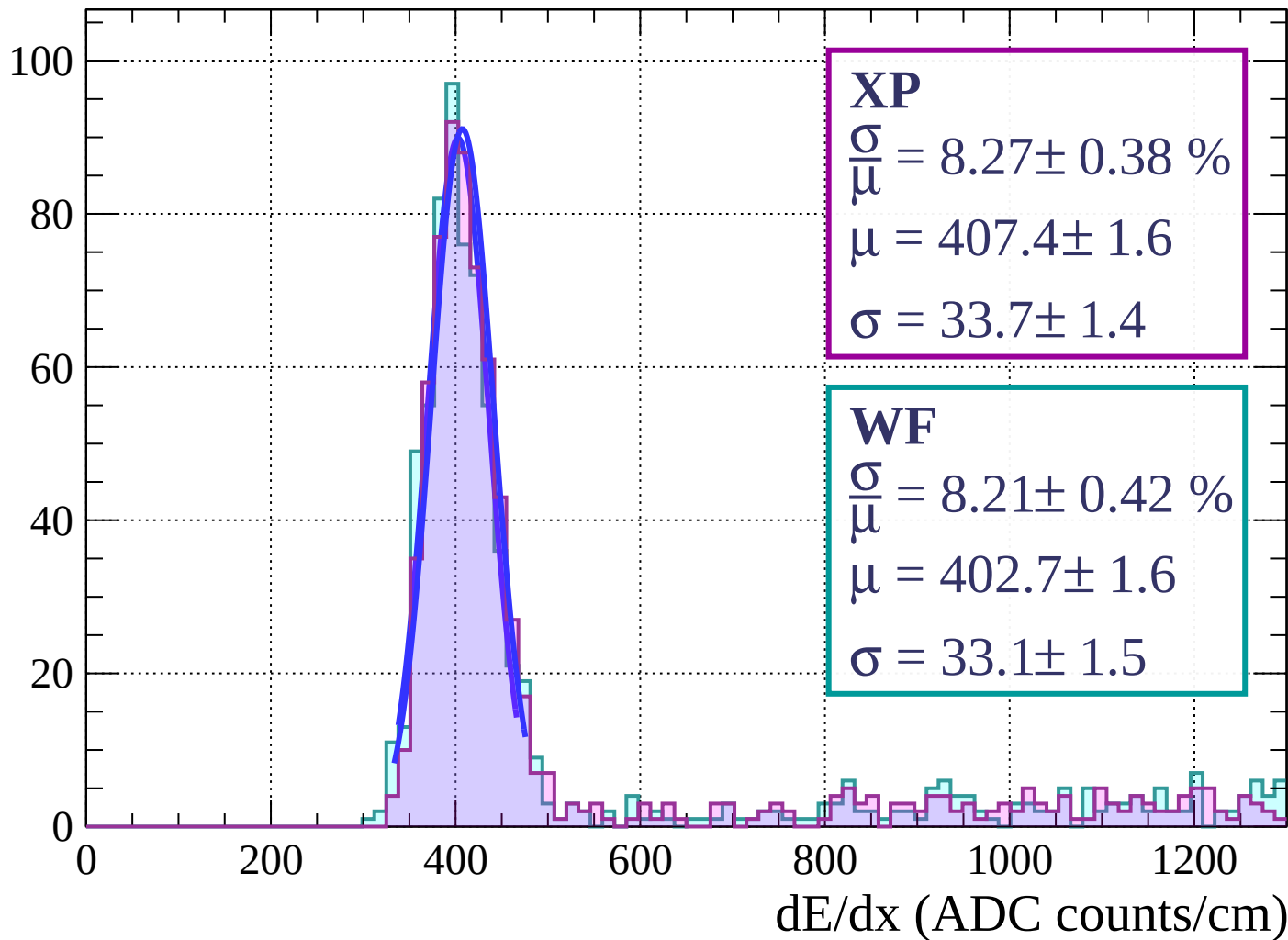
Energy loss | $45 < \phi < 48$

Count



Energy loss | $48 < \phi < 51$

Count



Energy loss | $51 < \phi < 54$

Count

XP

$$\frac{\sigma}{\mu} = 8.89 \pm 0.56 \%$$

$$\mu = 405.6 \pm 2.0$$

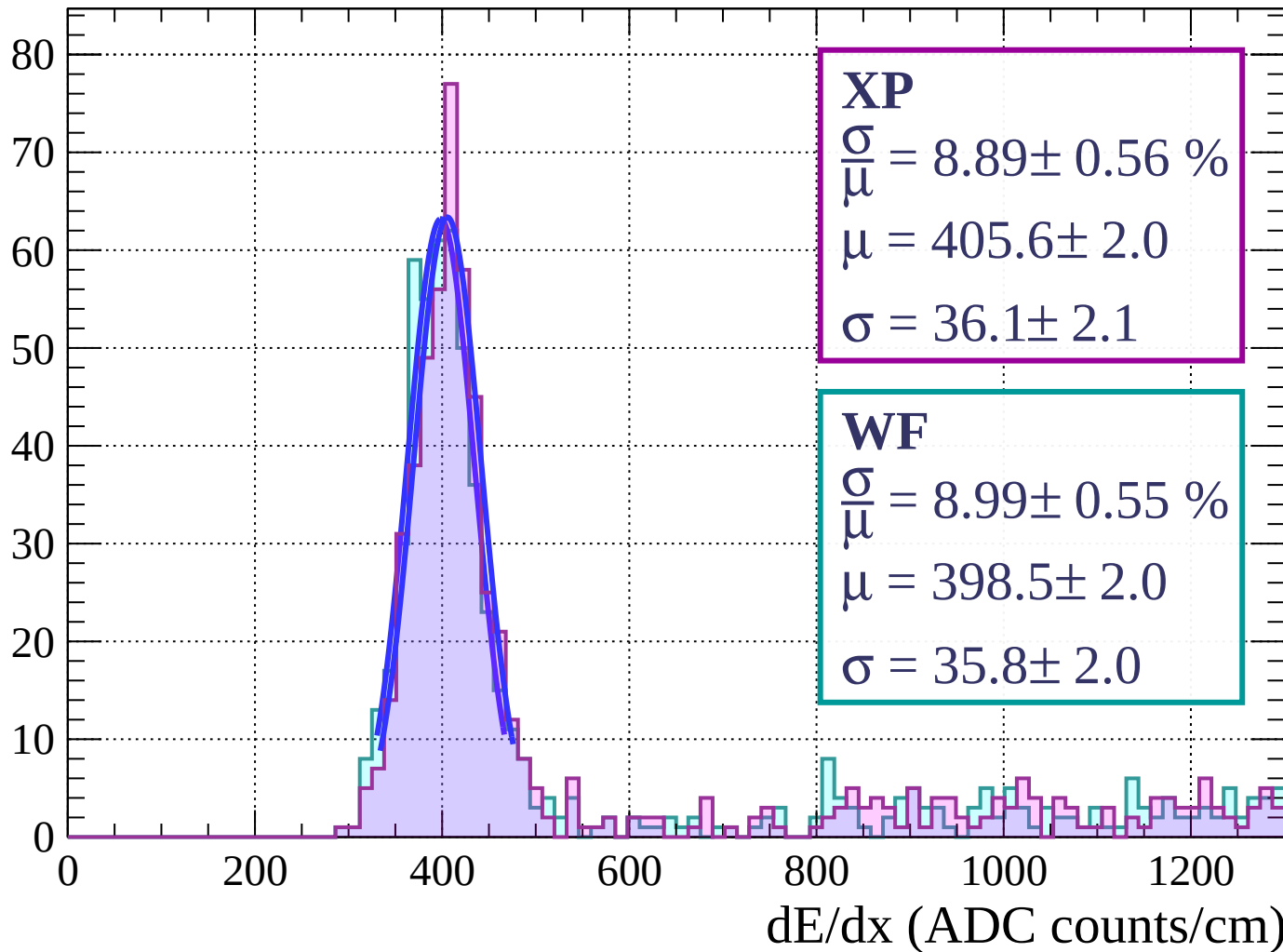
$$\sigma = 36.1 \pm 2.1$$

WF

$$\frac{\sigma}{\mu} = 8.99 \pm 0.55 \%$$

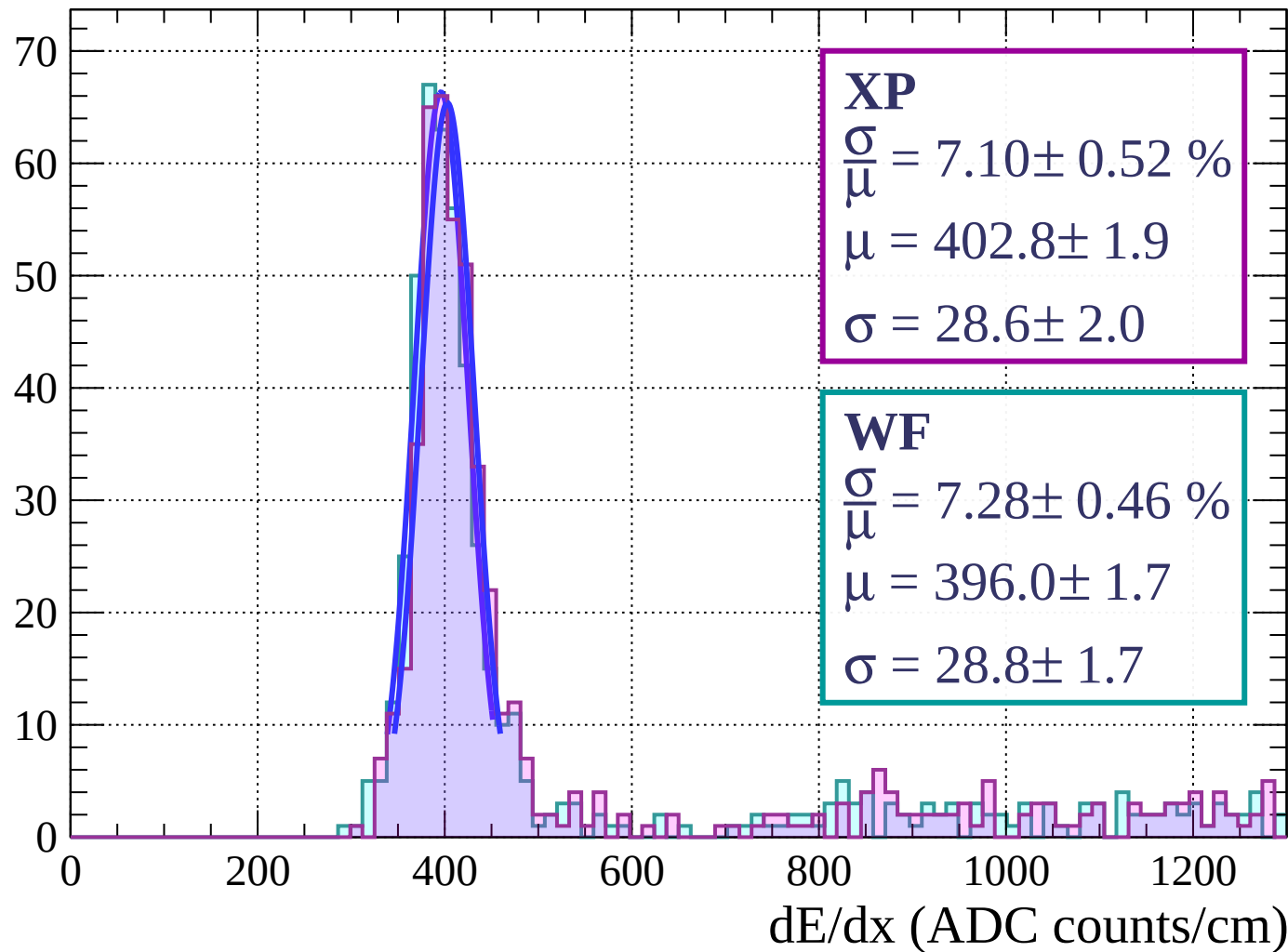
$$\mu = 398.5 \pm 2.0$$

$$\sigma = 35.8 \pm 2.0$$



Energy loss | $54 < \phi < 57$

Count



Energy loss | $57 < \phi < 60$

Count

XP

$$\frac{\sigma}{\mu} = 8.42 \pm 0.60 \%$$

$$\mu = 410.2 \pm 2.6$$

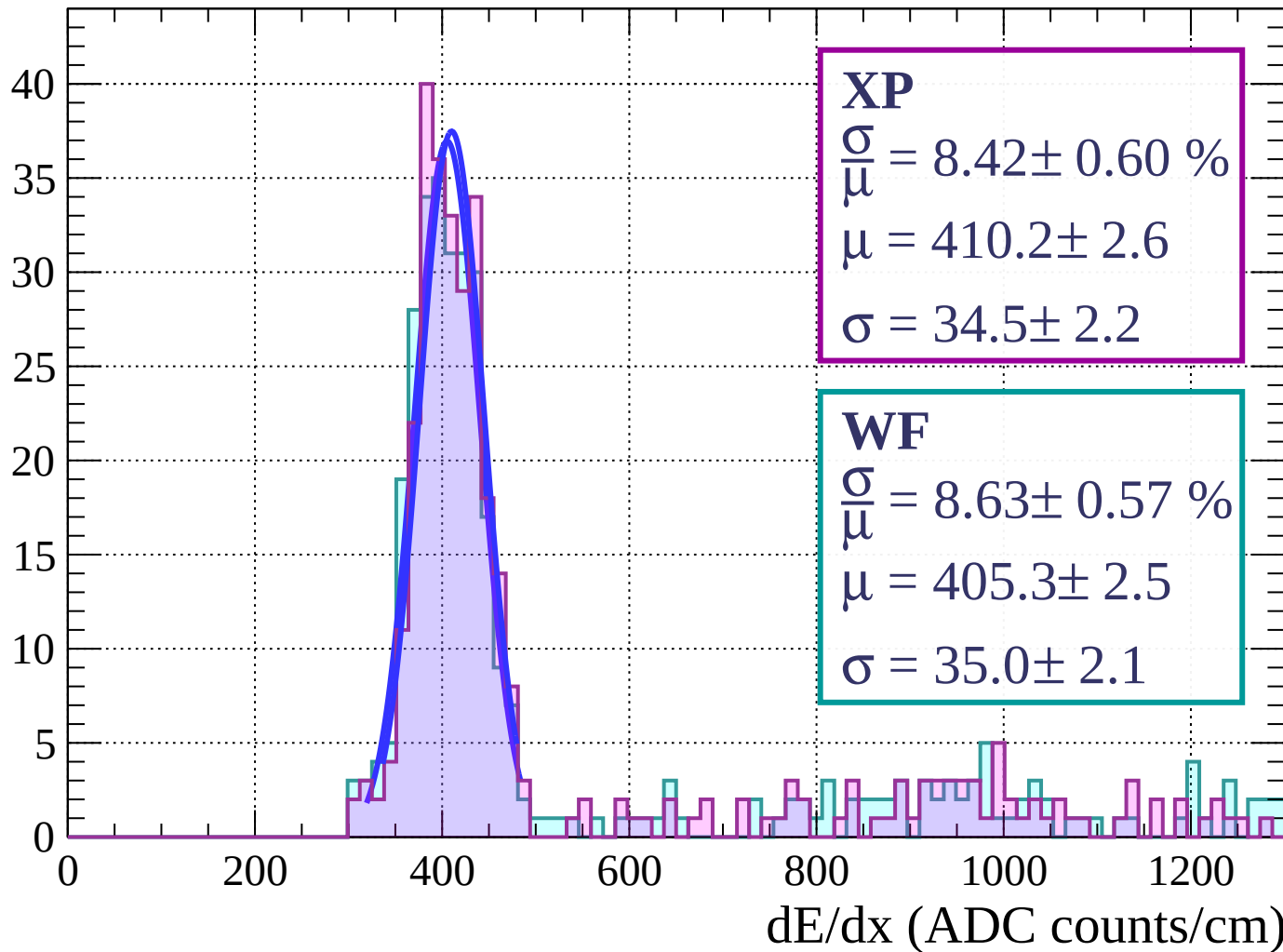
$$\sigma = 34.5 \pm 2.2$$

WF

$$\frac{\sigma}{\mu} = 8.63 \pm 0.57 \%$$

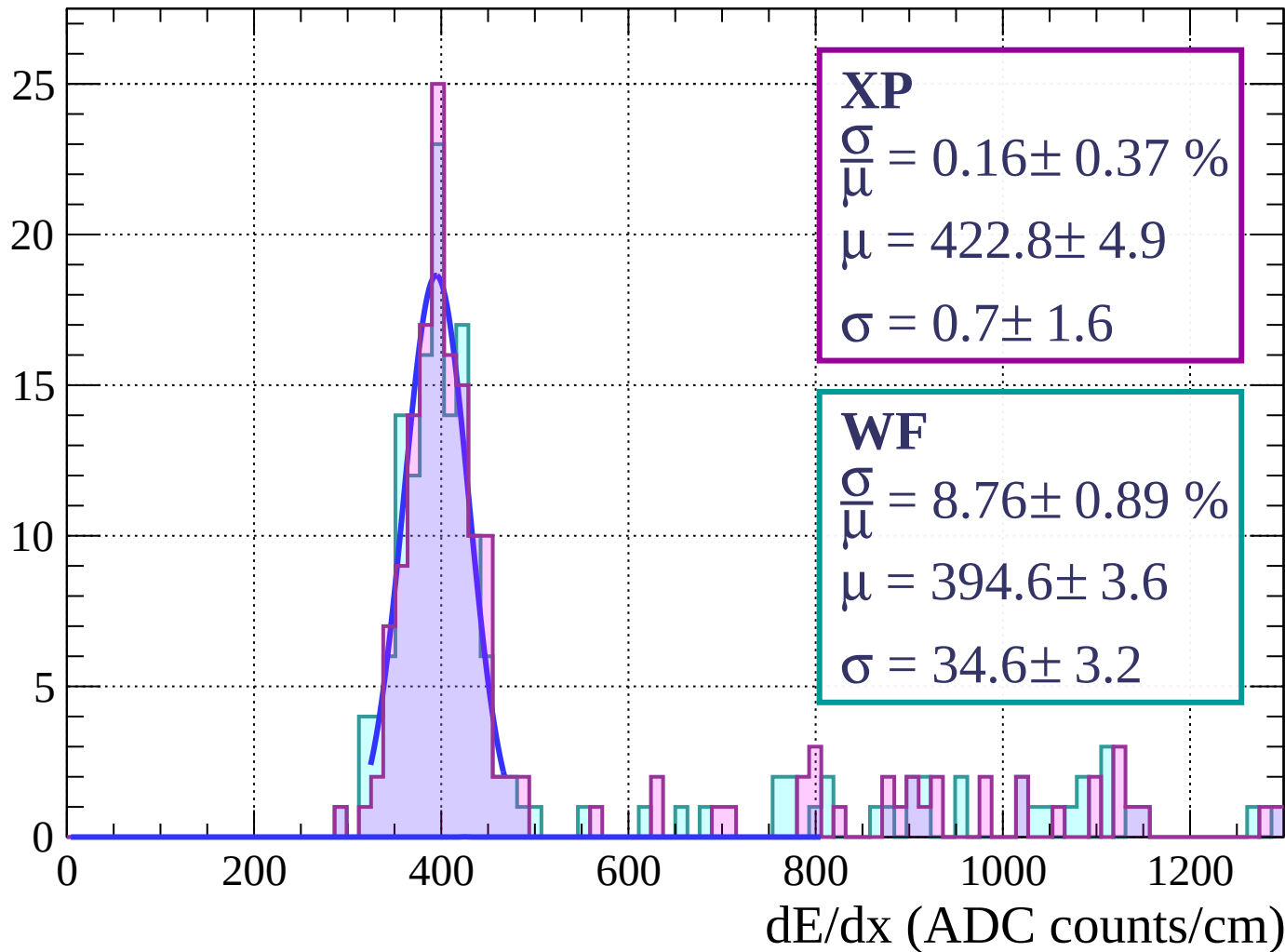
$$\mu = 405.3 \pm 2.5$$

$$\sigma = 35.0 \pm 2.1$$



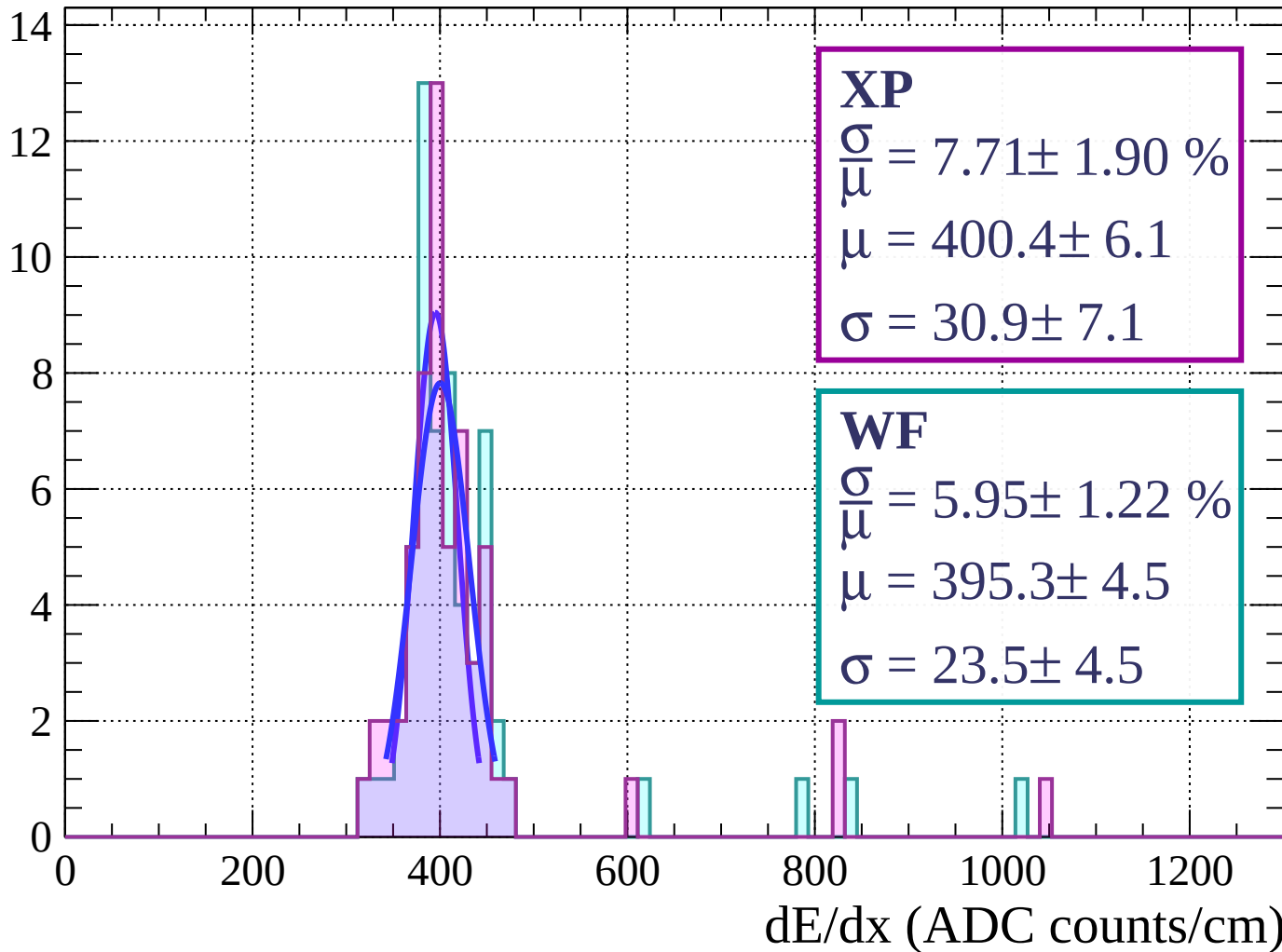
Energy loss | $60 < \phi < 63$

Count



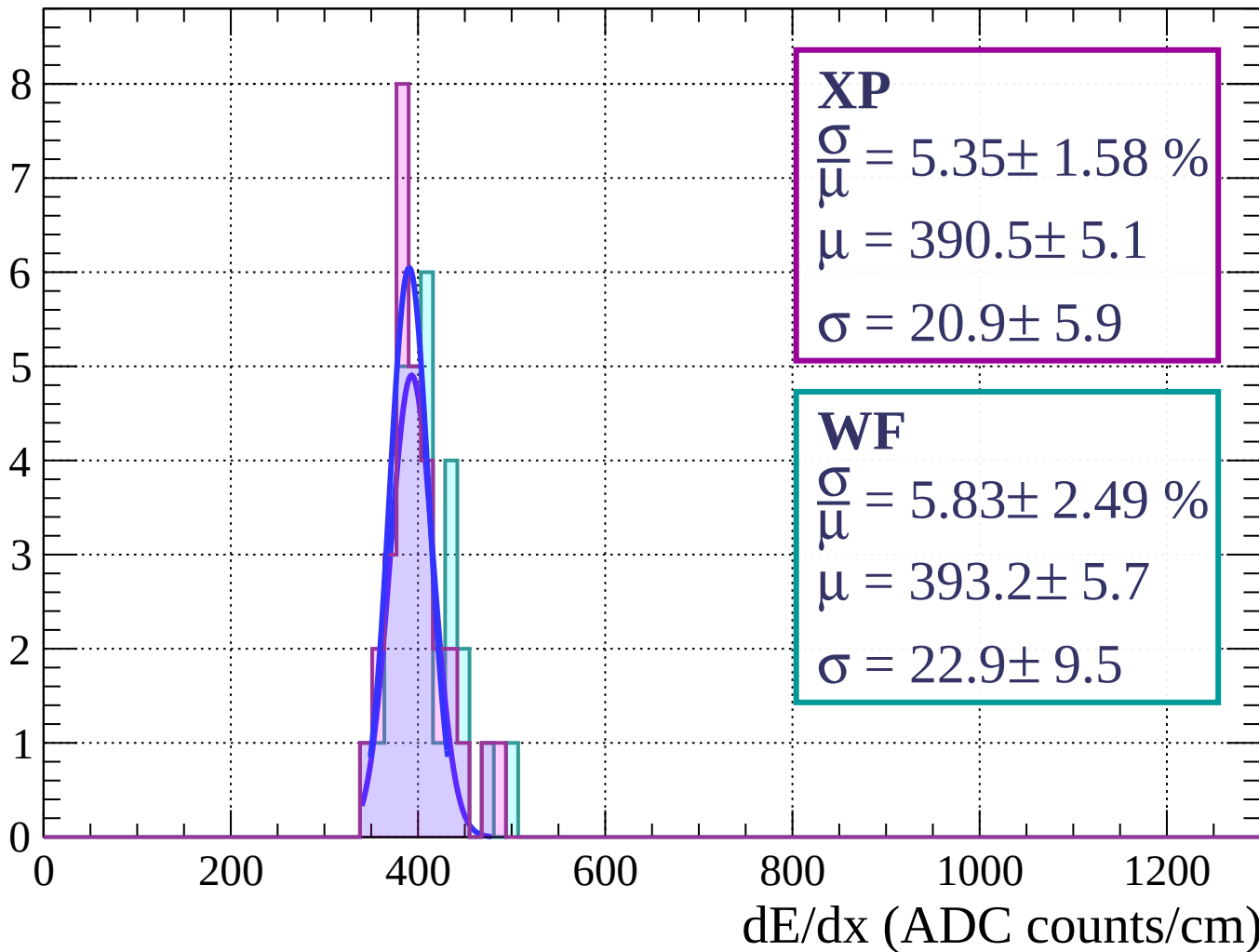
Energy loss | $63 < \phi < 66$

Count



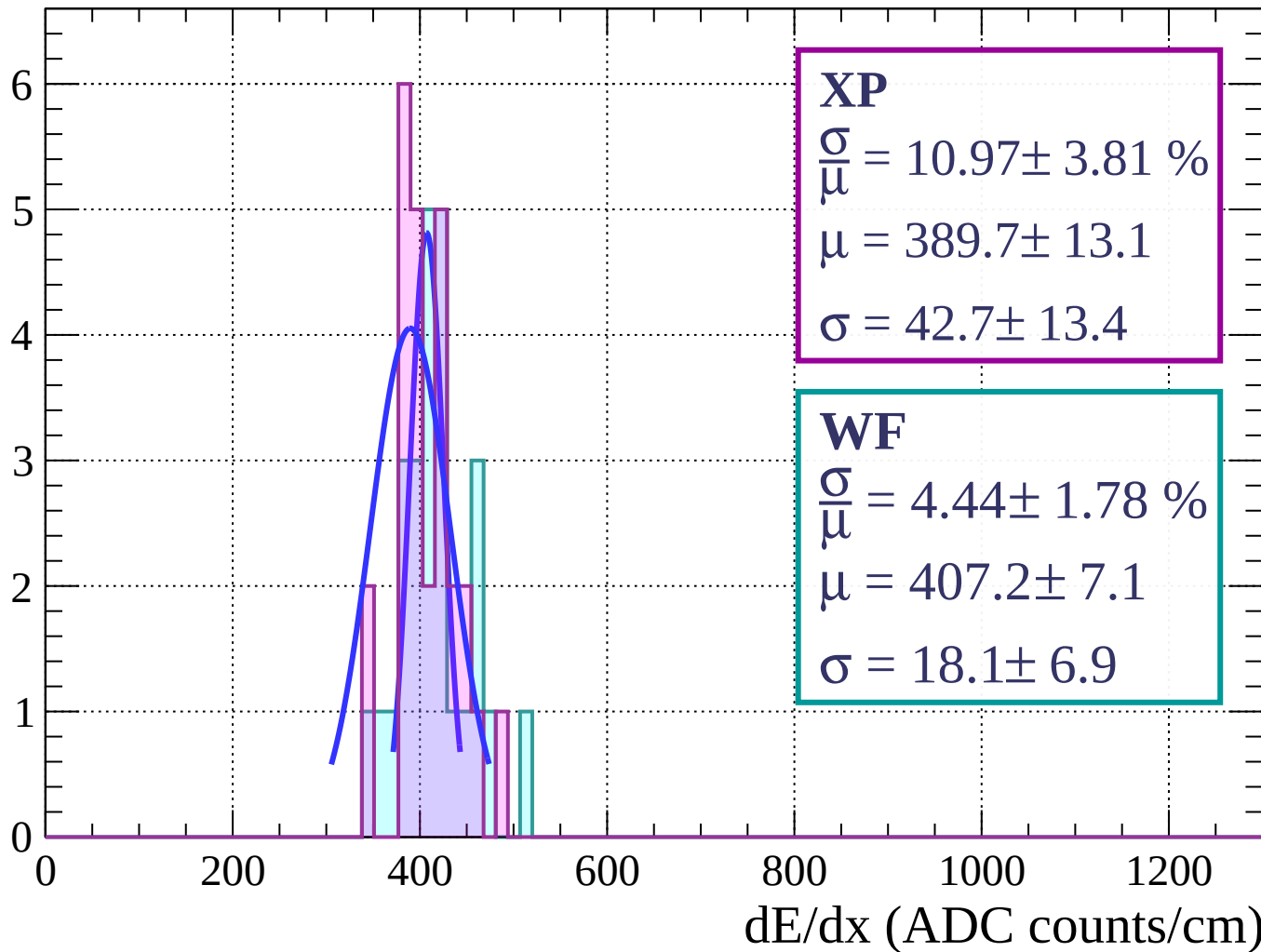
Energy loss | $66 < \phi < 69$

Count



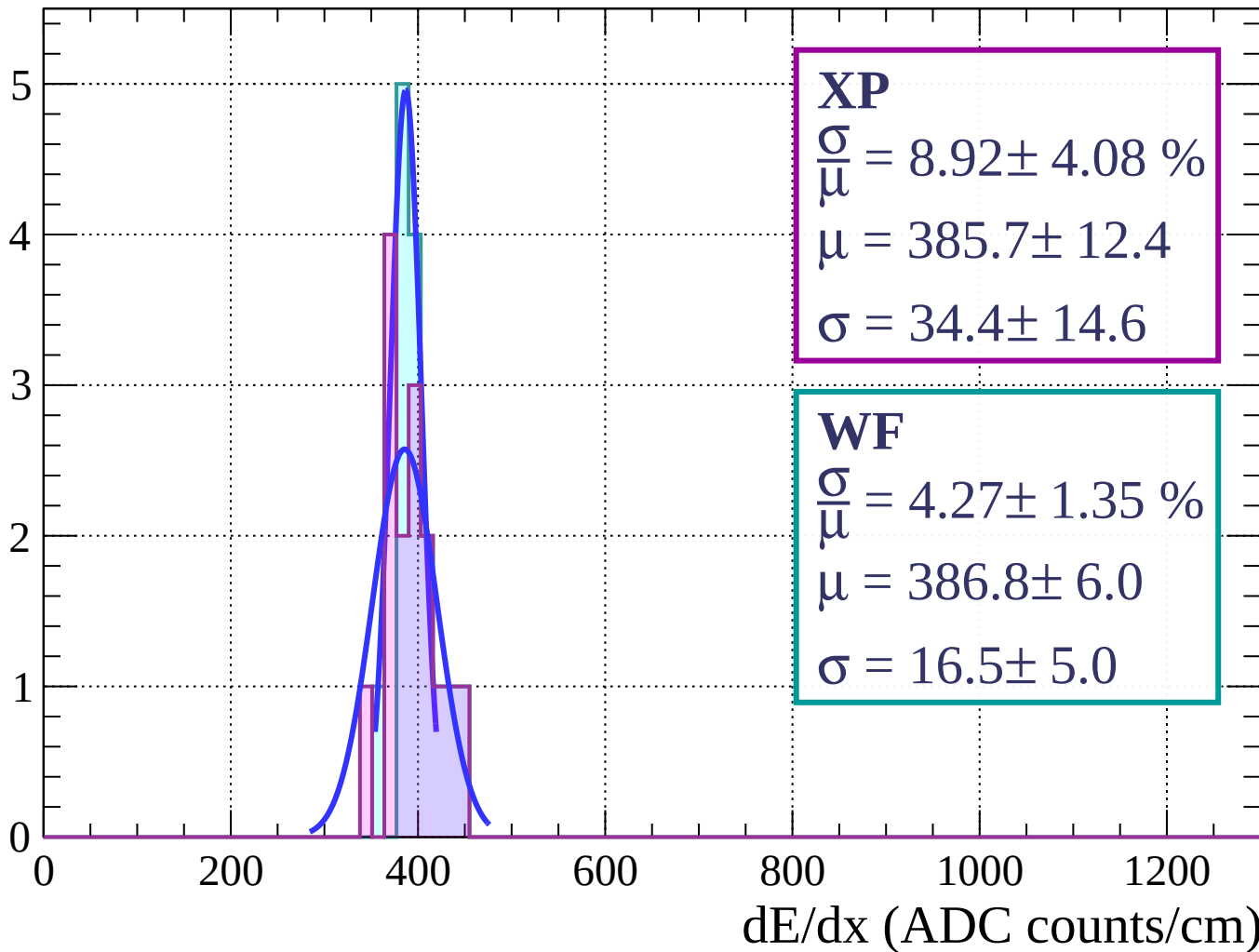
Energy loss | $69 < \phi < 72$

Count



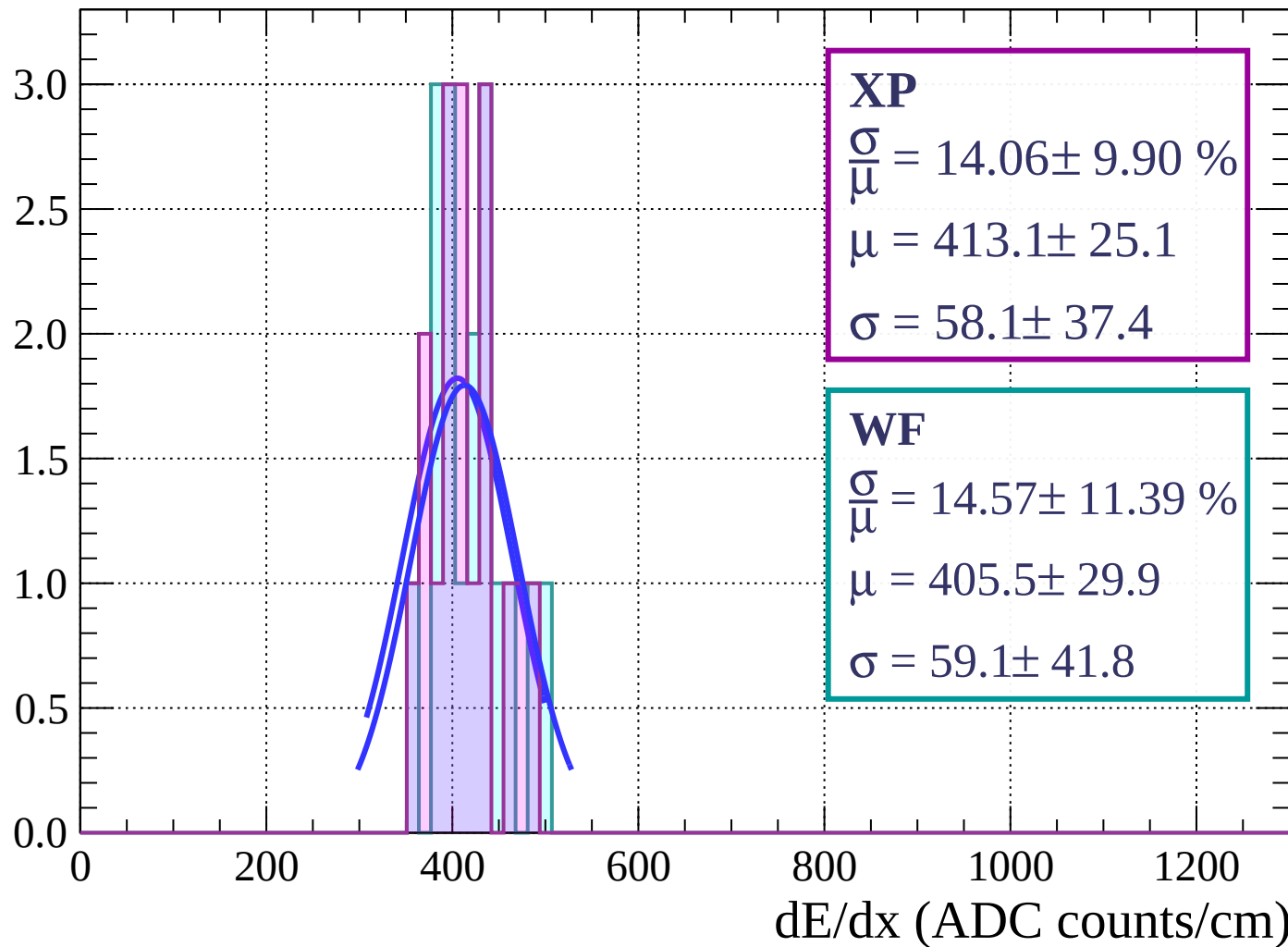
Energy loss | $72 < \phi < 75$

Count



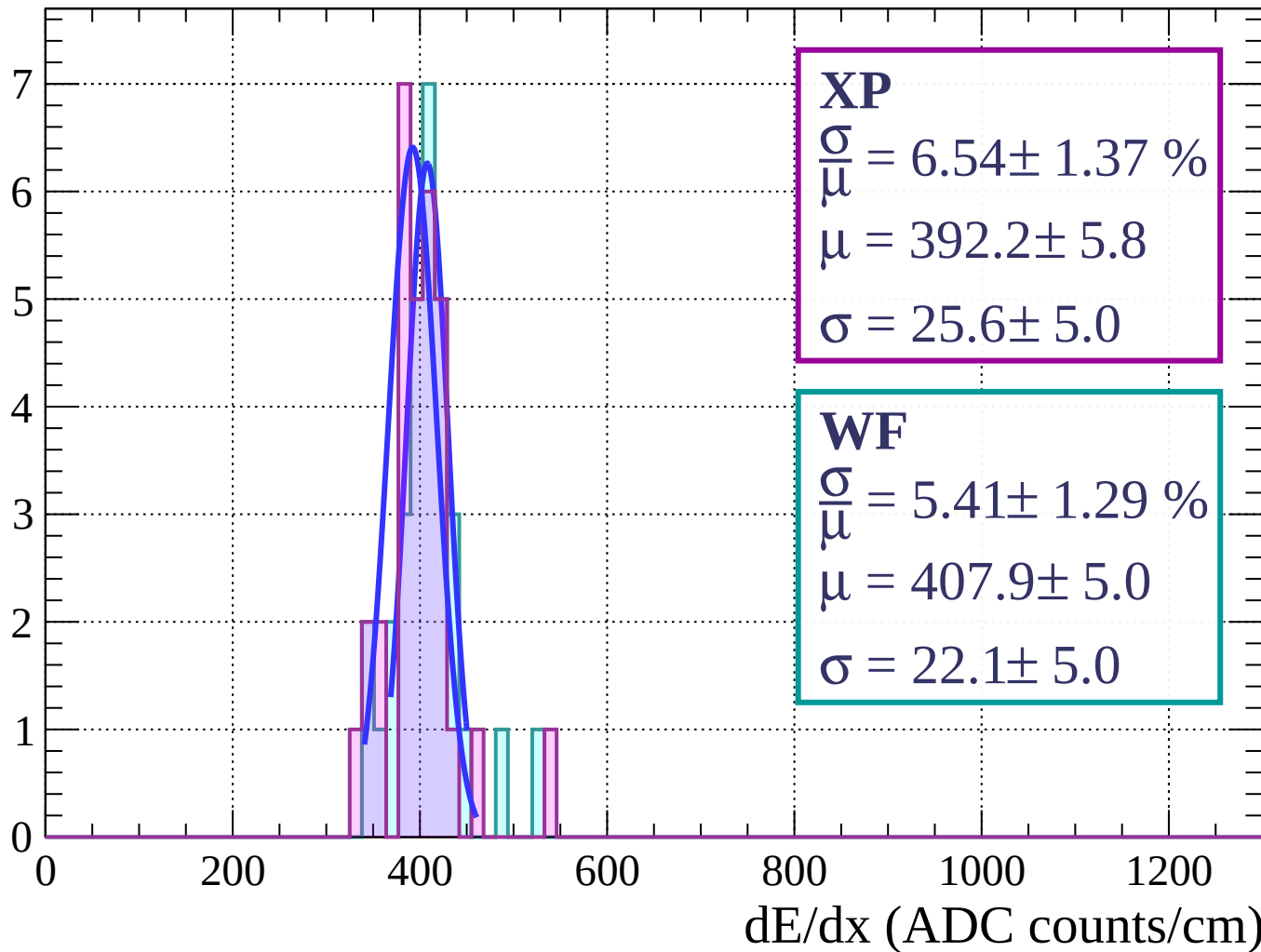
Energy loss | $75 < \phi < 78$

Count



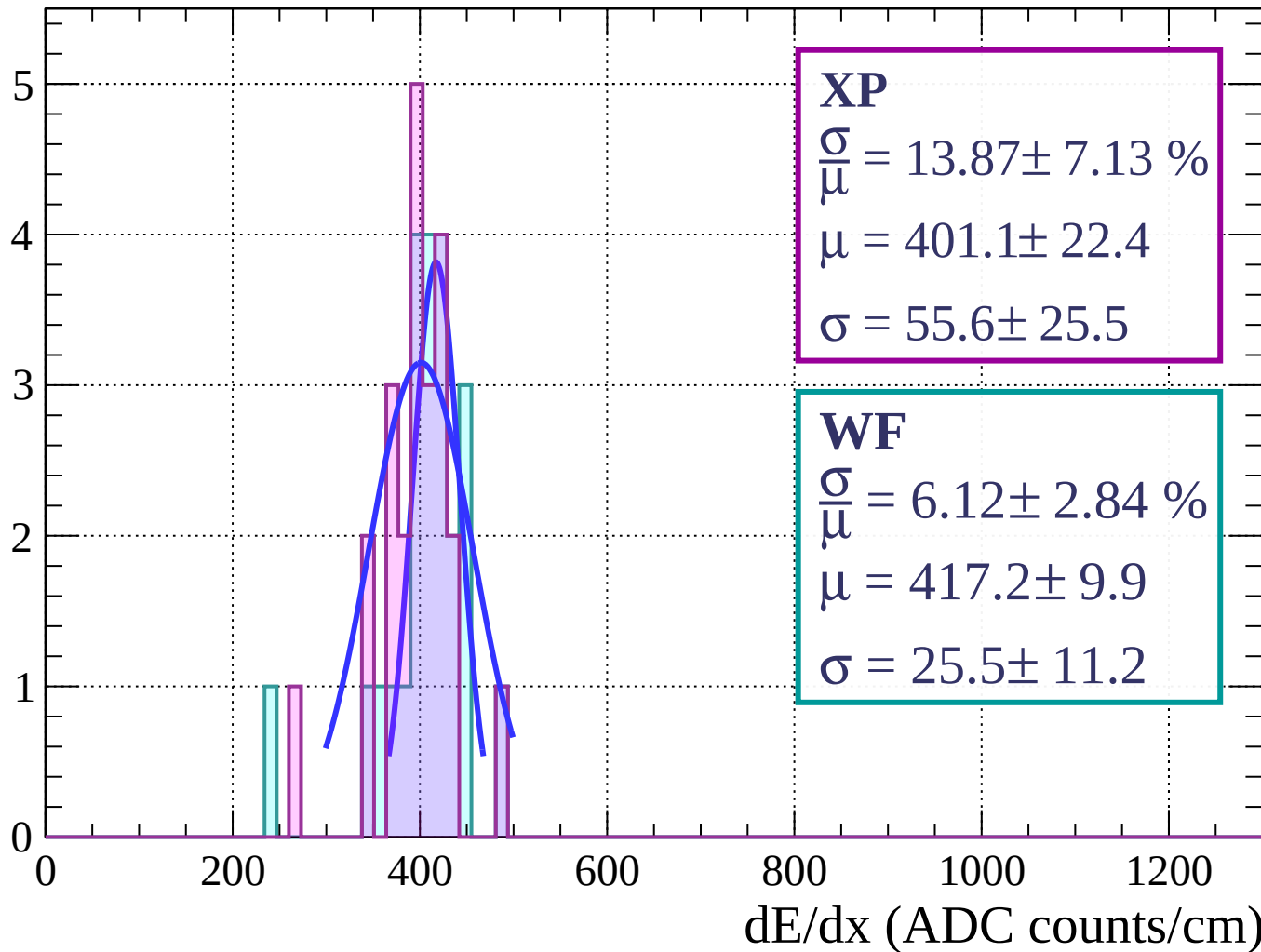
Energy loss | $78 < \phi < 81$

Count



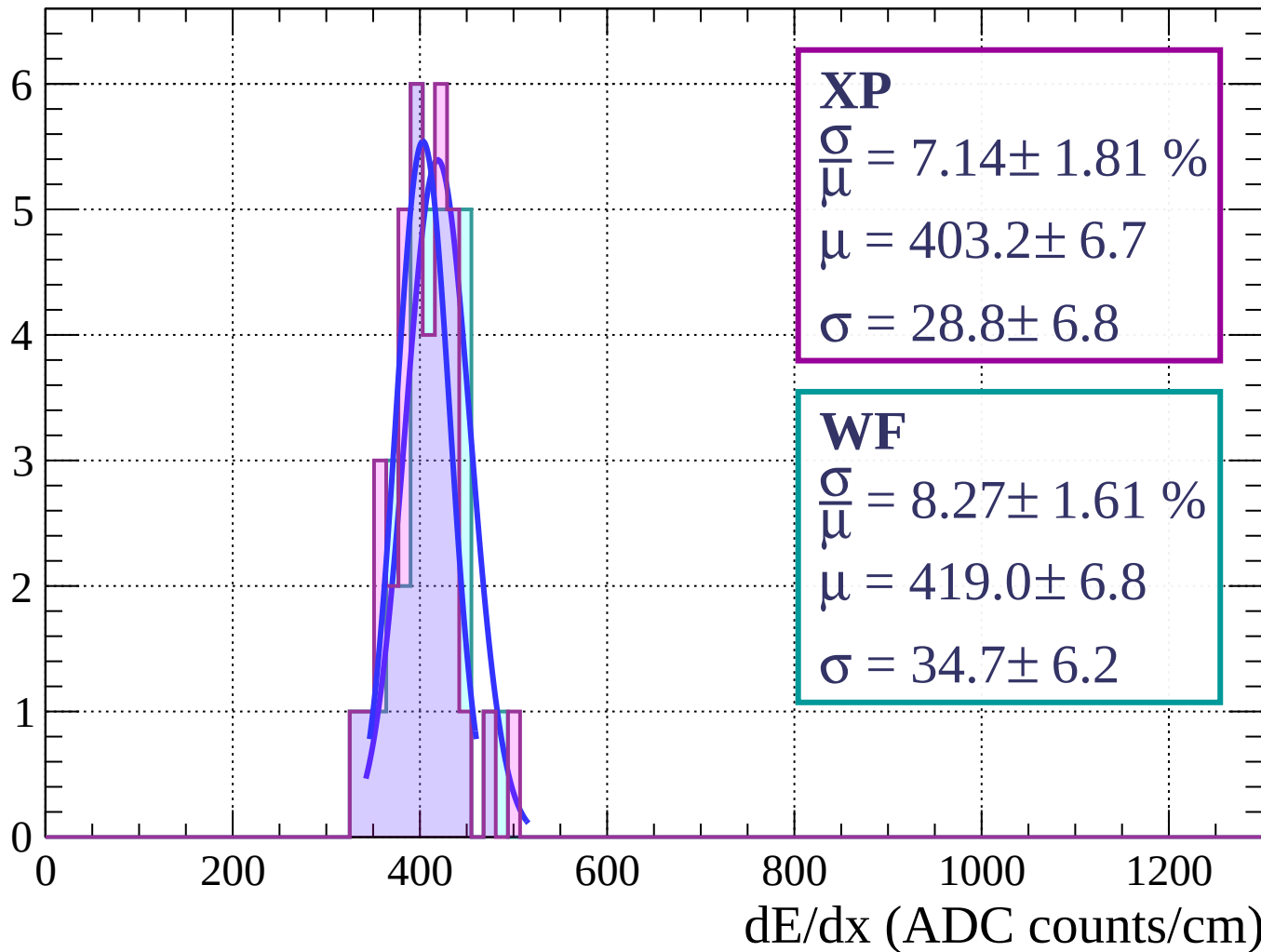
Energy loss | $81 < \phi < 84$

Count



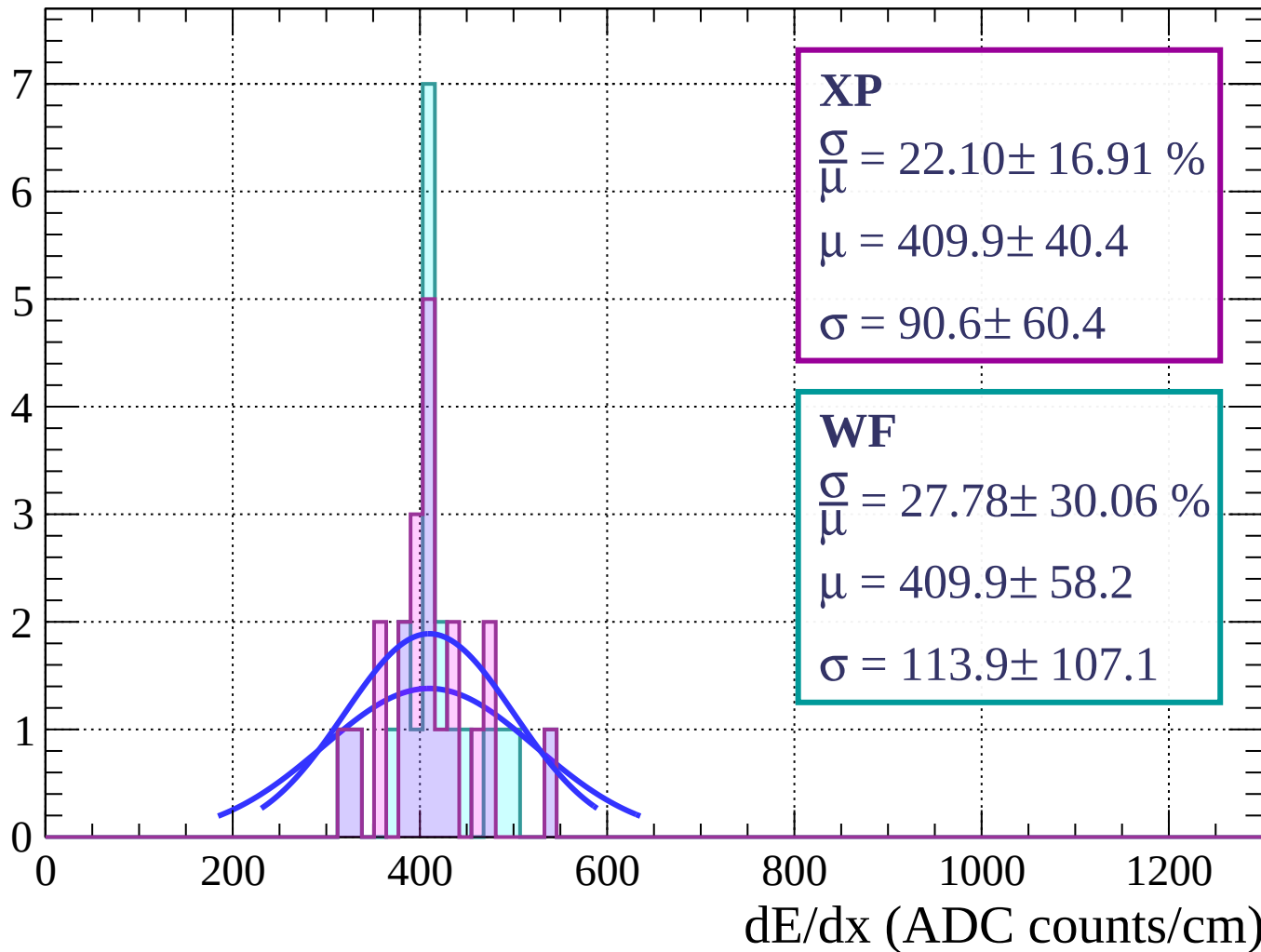
Energy loss | $84 < \phi < 87$

Count

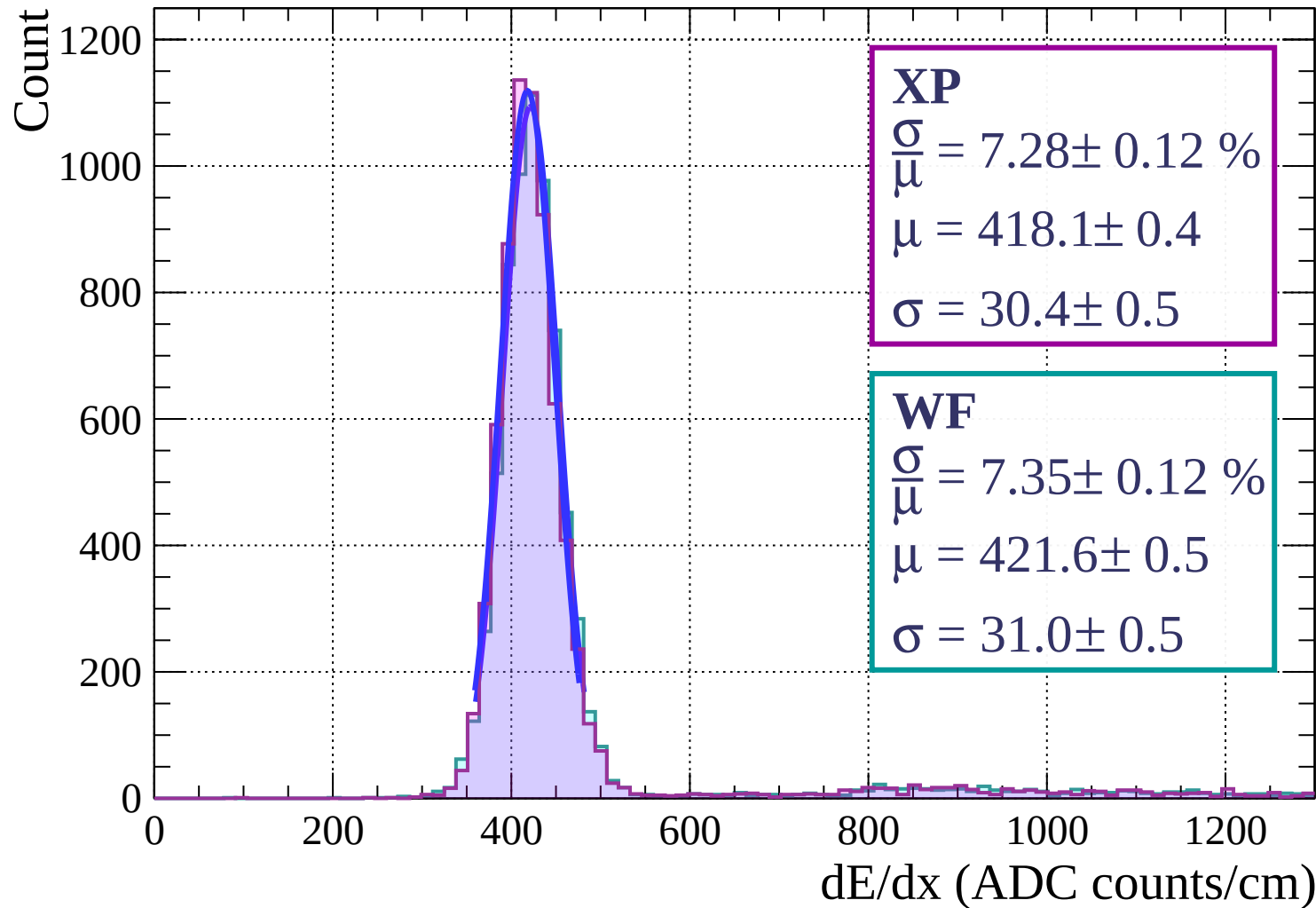


Energy loss | $87 < \phi < 90$

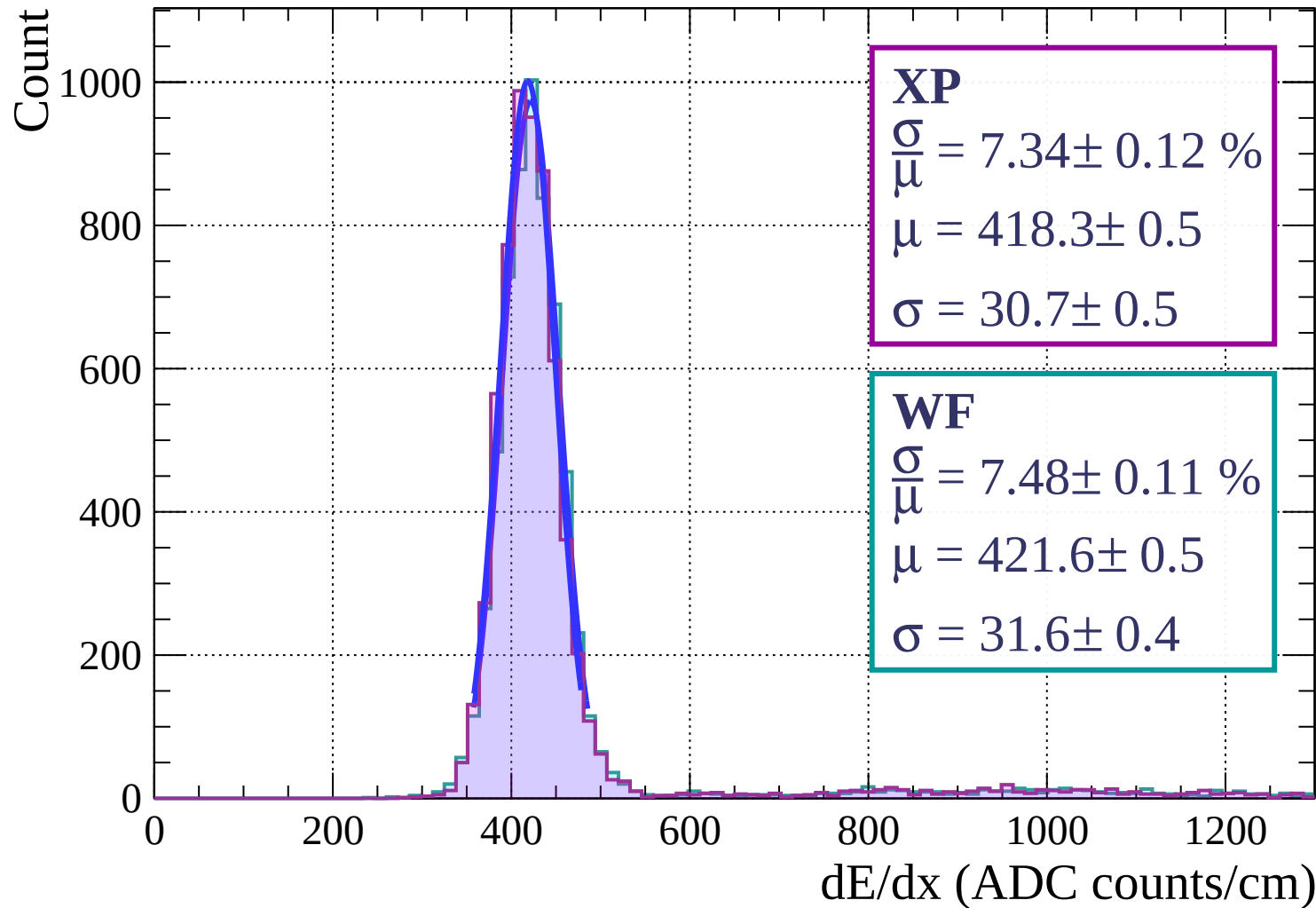
Count



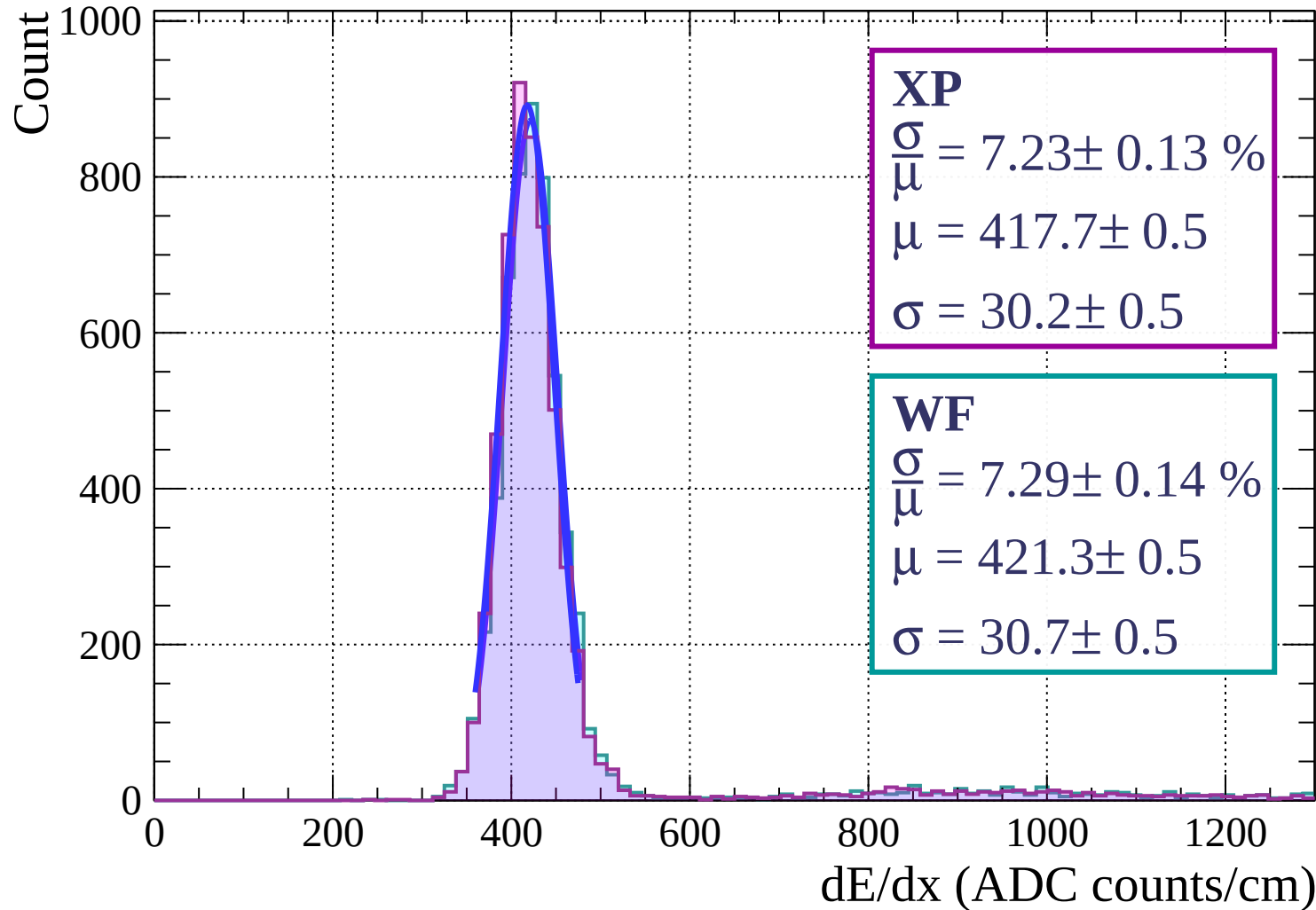
Energy loss | $0 < \theta < 3$



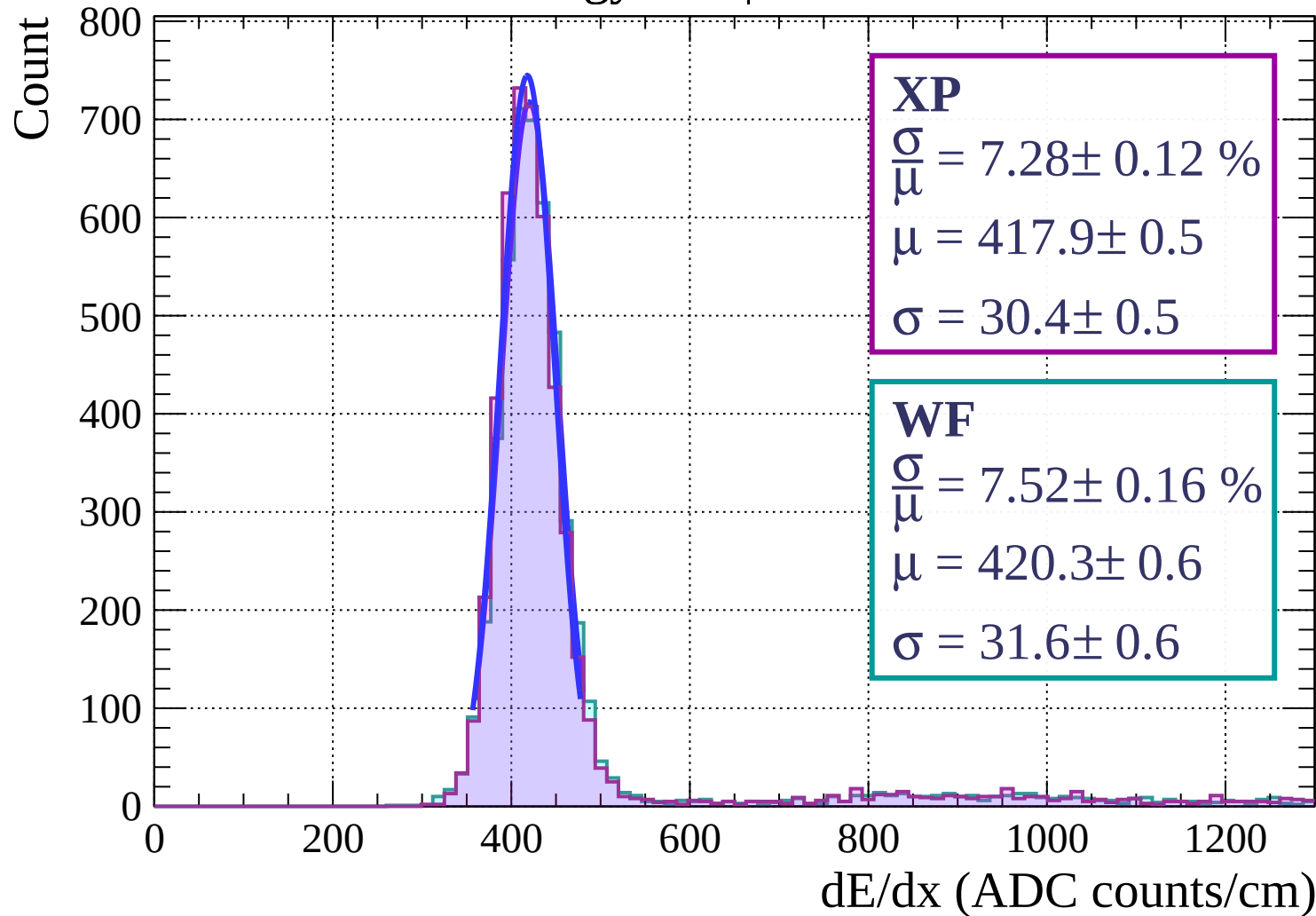
Energy loss | $3 < \theta < 6$



Energy loss | $6 < \theta < 9$

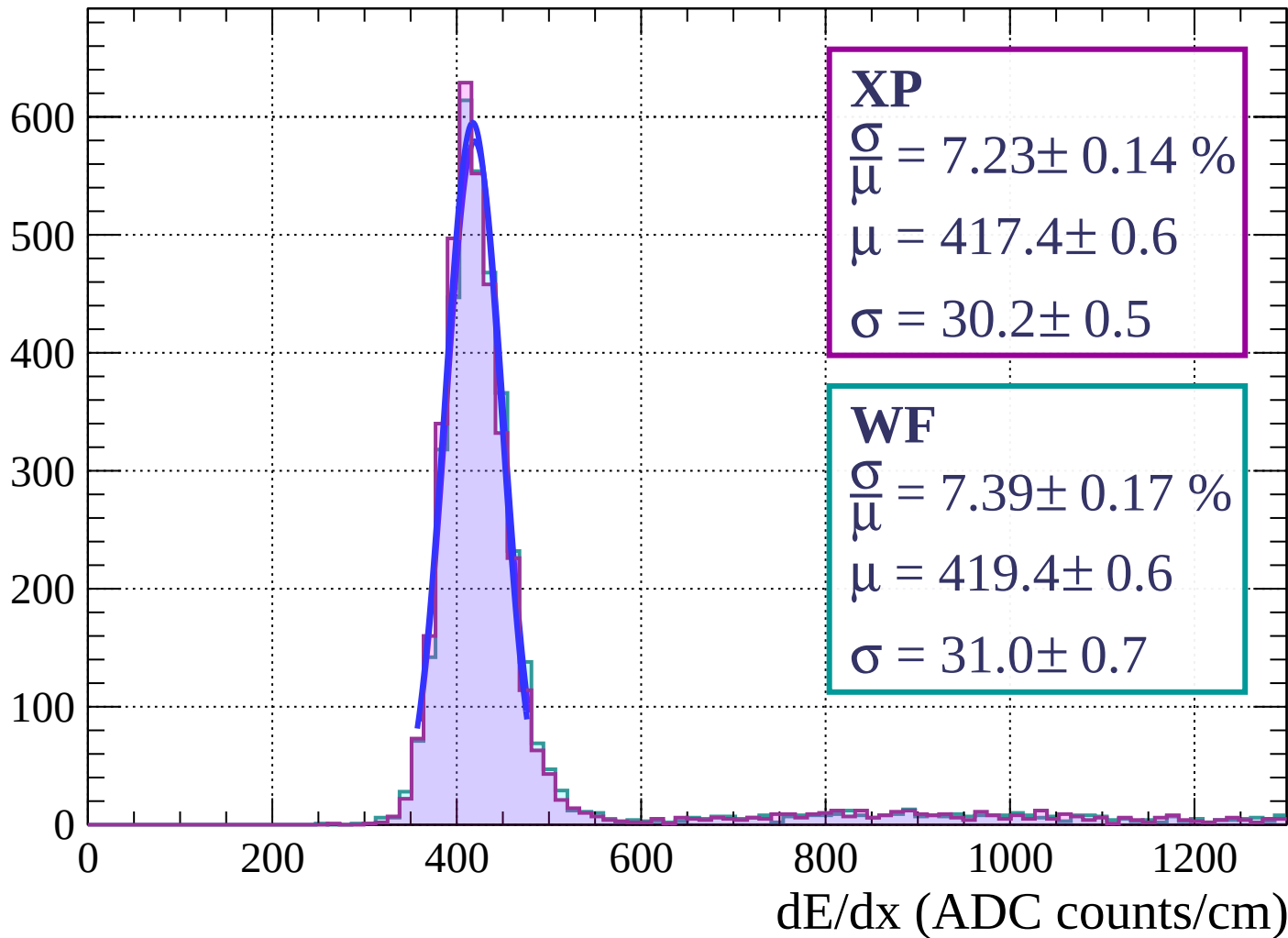


Energy loss | $9 < \theta < 12$



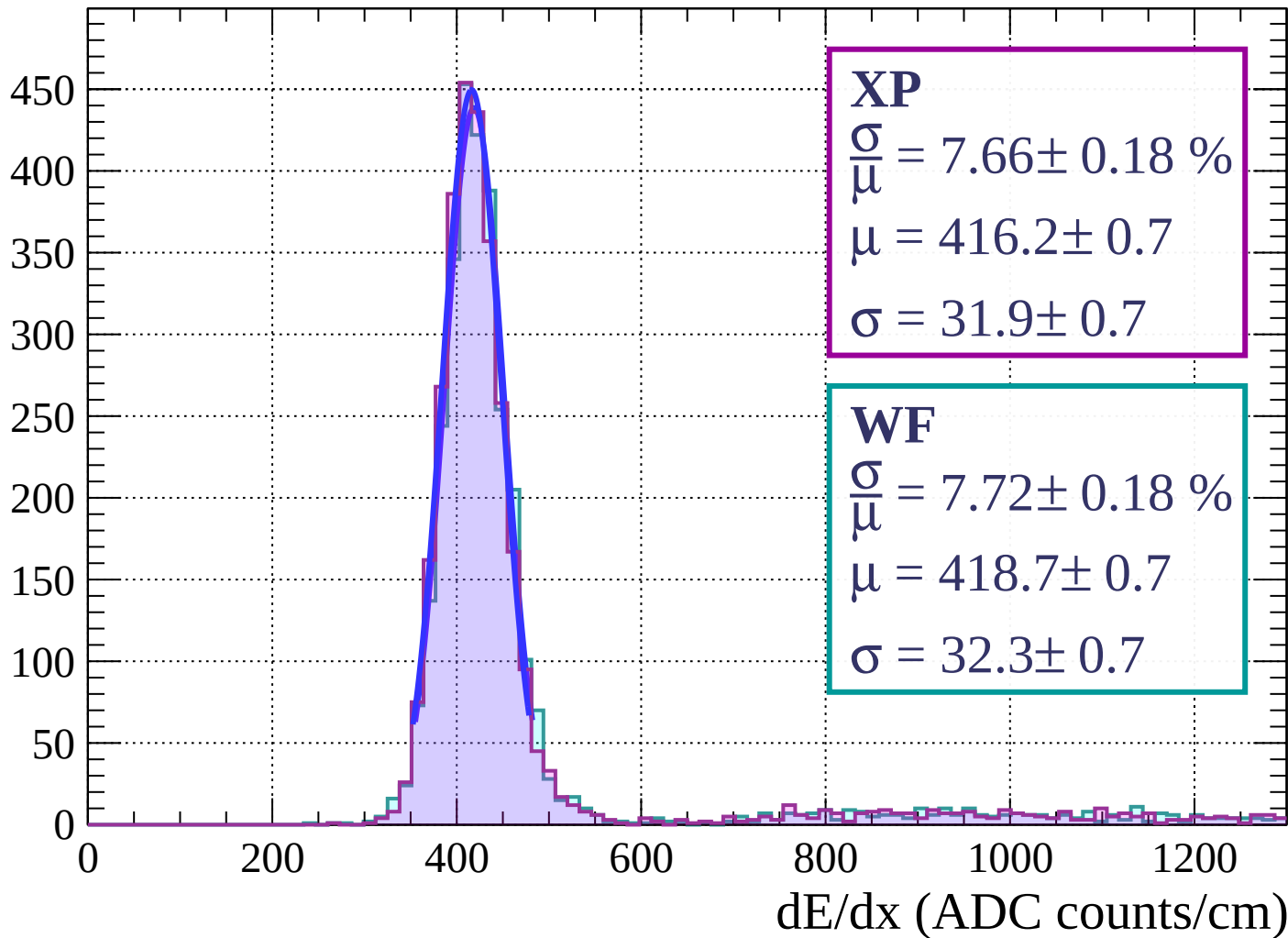
Energy loss | $12 < \theta < 15$

Count



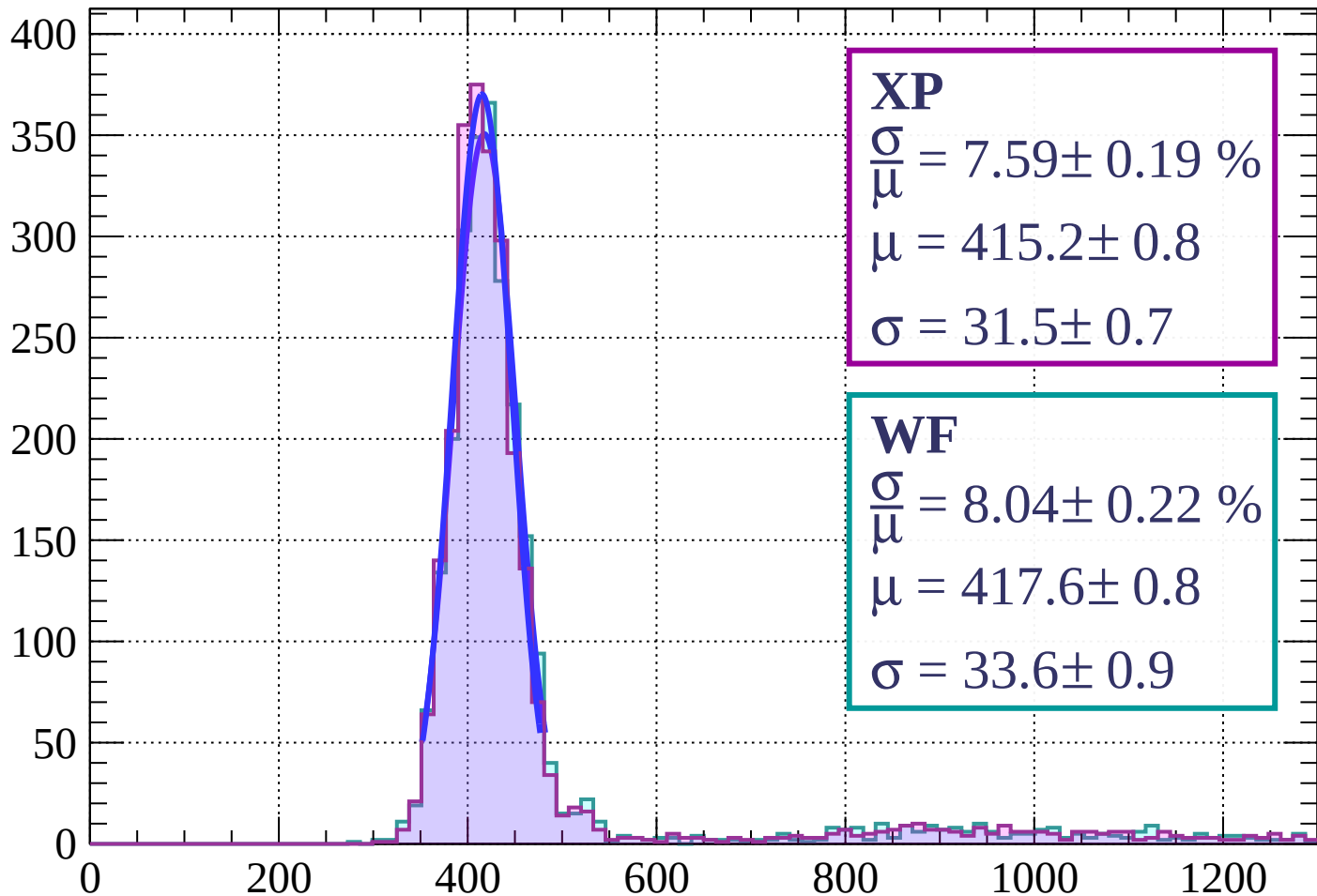
Energy loss | $15 < \theta < 18$

Count



Energy loss | $18 < \theta < 21$

Count



XP

$$\frac{\sigma}{\mu} = 7.59 \pm 0.19 \%$$

$$\mu = 415.2 \pm 0.8$$

$$\sigma = 31.5 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 8.04 \pm 0.22 \%$$

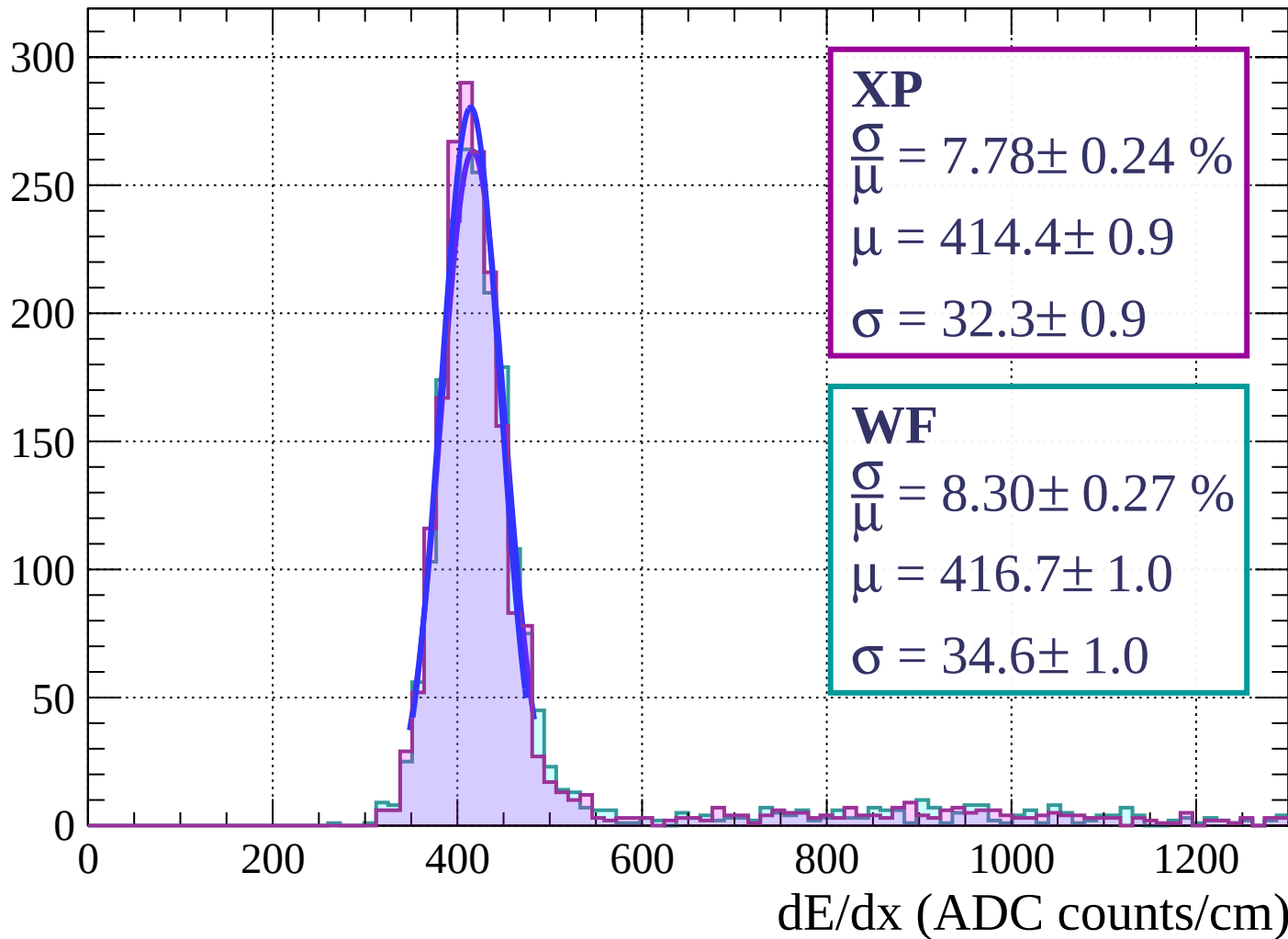
$$\mu = 417.6 \pm 0.8$$

$$\sigma = 33.6 \pm 0.9$$

dE/dx (ADC counts/cm)

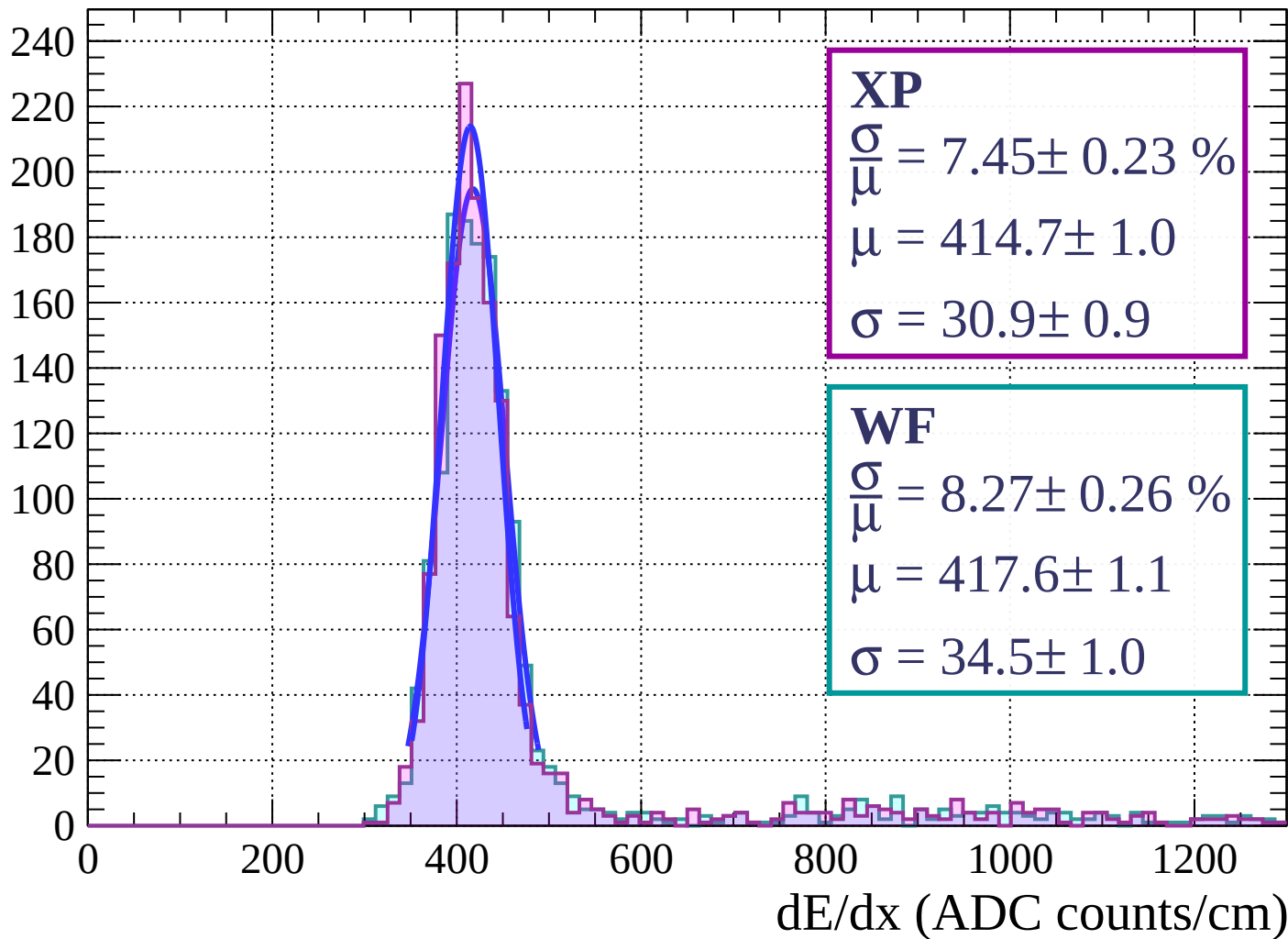
Energy loss | $21 < \theta < 24$

Count



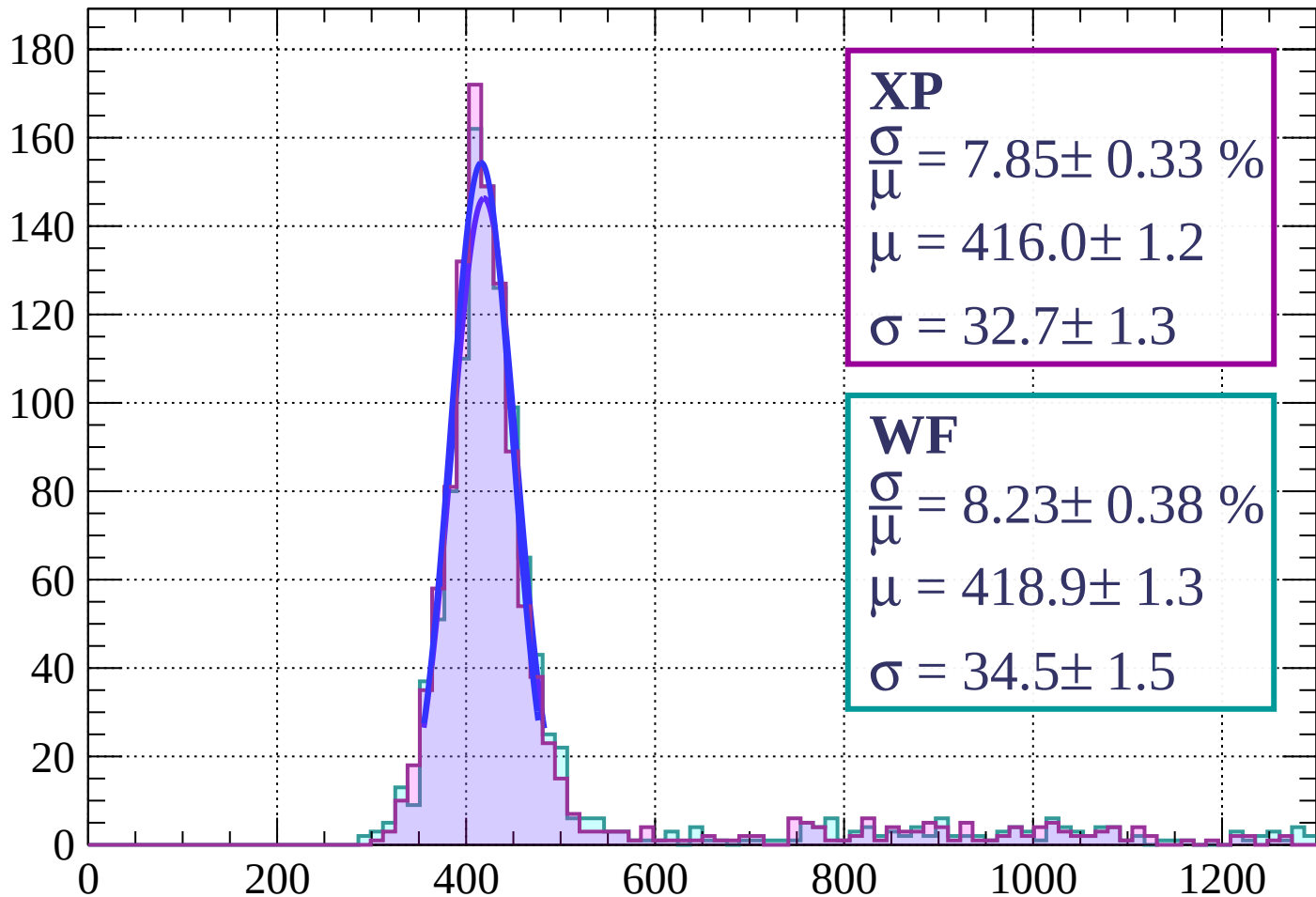
Energy loss | $24 < \theta < 27$

Count



Energy loss | $27 < \theta < 30$

Count



XP

$$\frac{\sigma}{\mu} = 7.85 \pm 0.33 \%$$

$$\mu = 416.0 \pm 1.2$$

$$\sigma = 32.7 \pm 1.3$$

WF

$$\frac{\sigma}{\mu} = 8.23 \pm 0.38 \%$$

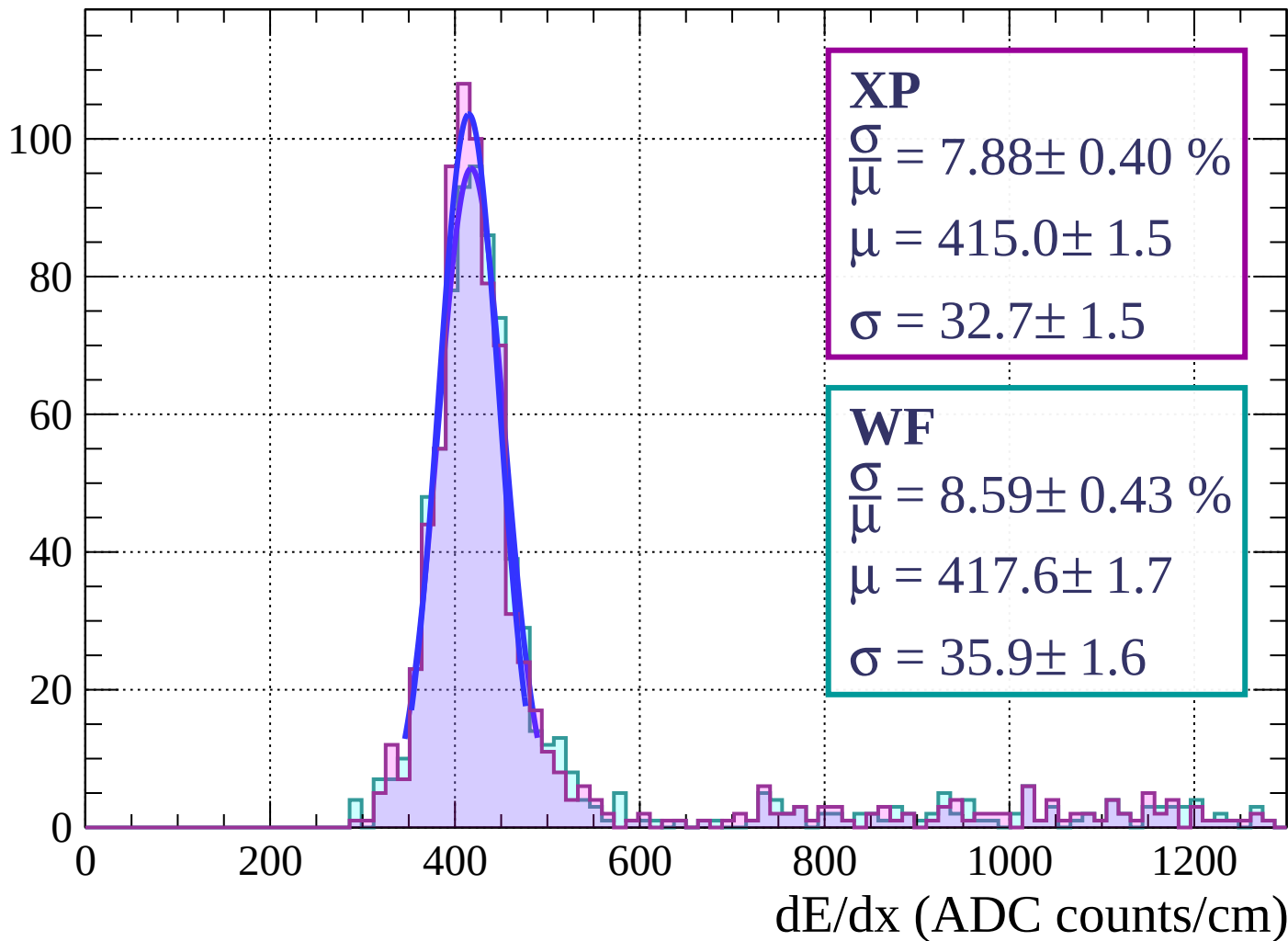
$$\mu = 418.9 \pm 1.3$$

$$\sigma = 34.5 \pm 1.5$$

dE/dx (ADC counts/cm)

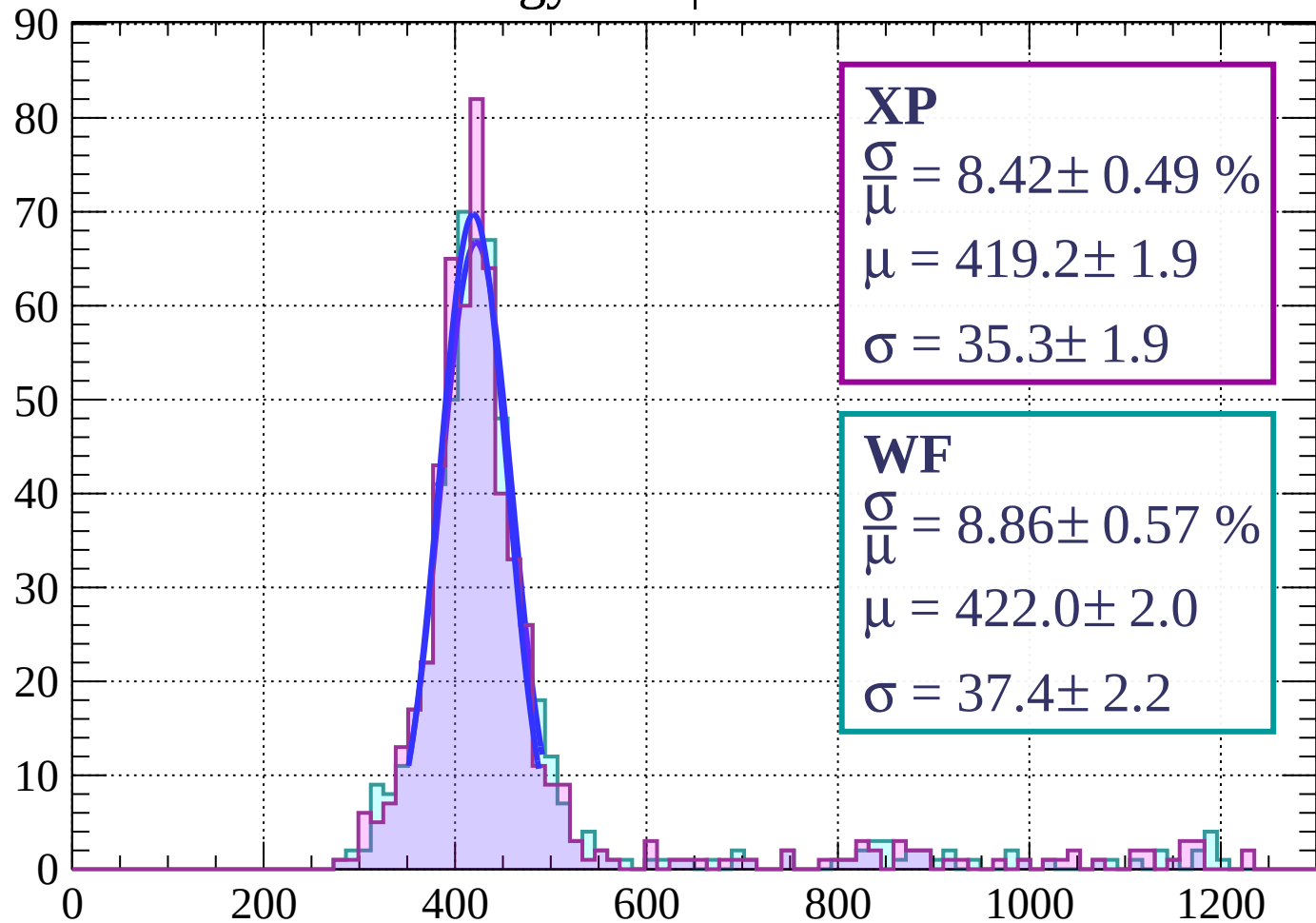
Energy loss | $30 < \theta < 33$

Count



Energy loss | $33 < \theta < 36$

Count



XP

$$\frac{\sigma}{\mu} = 8.42 \pm 0.49 \%$$

$$\mu = 419.2 \pm 1.9$$

$$\sigma = 35.3 \pm 1.9$$

WF

$$\frac{\sigma}{\mu} = 8.86 \pm 0.57 \%$$

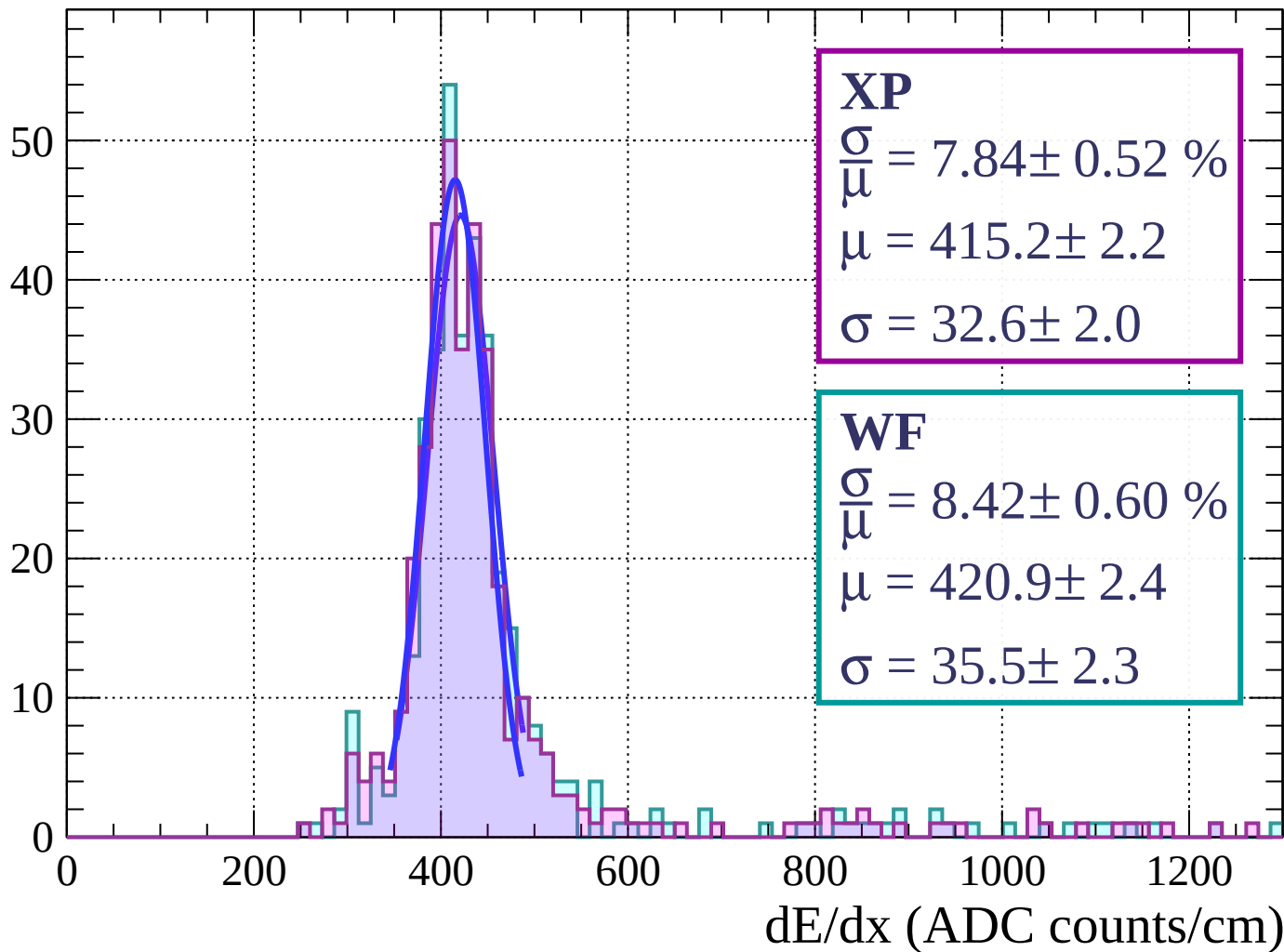
$$\mu = 422.0 \pm 2.0$$

$$\sigma = 37.4 \pm 2.2$$

dE/dx (ADC counts/cm)

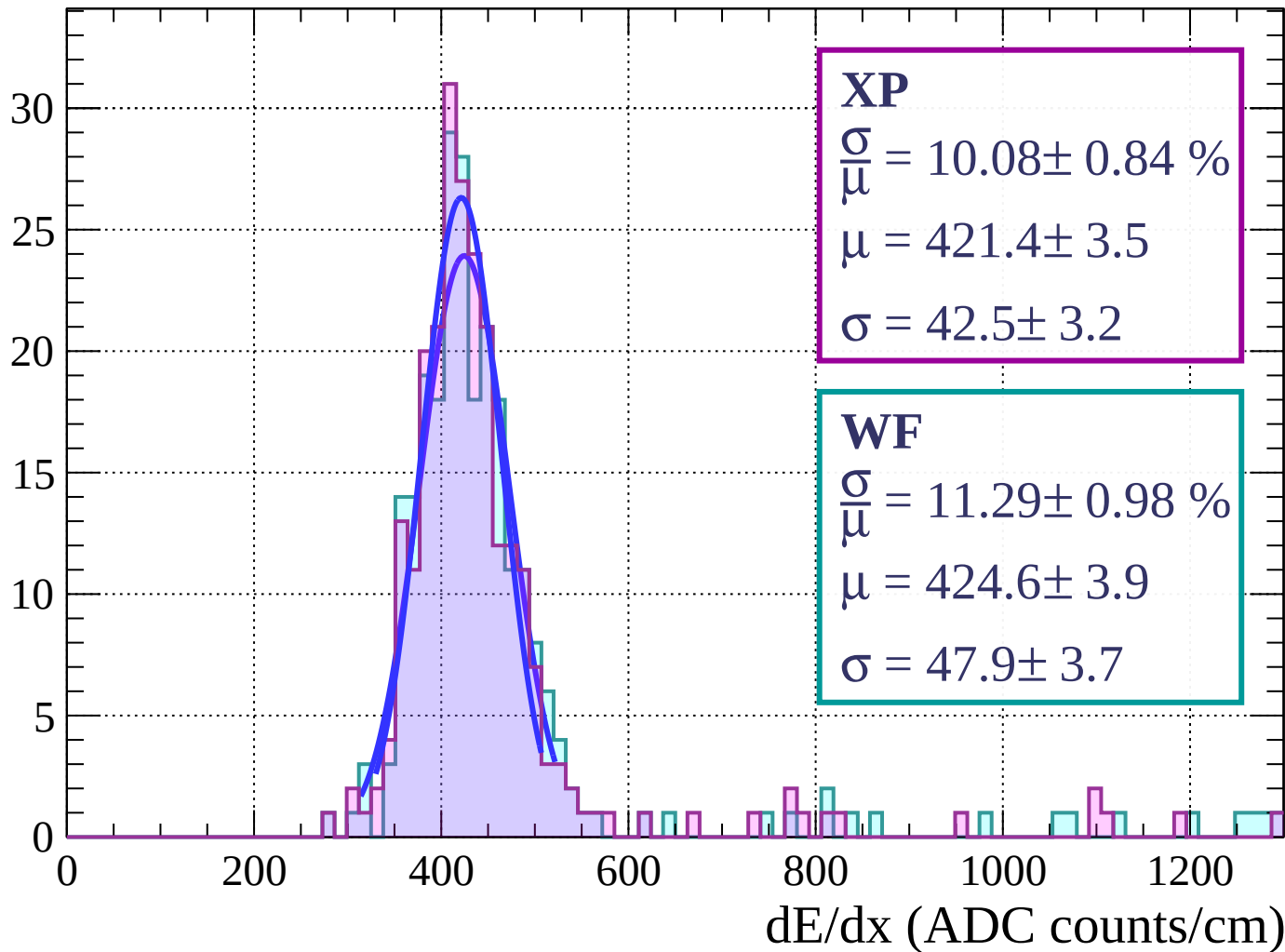
Energy loss | $36 < \theta < 39$

Count



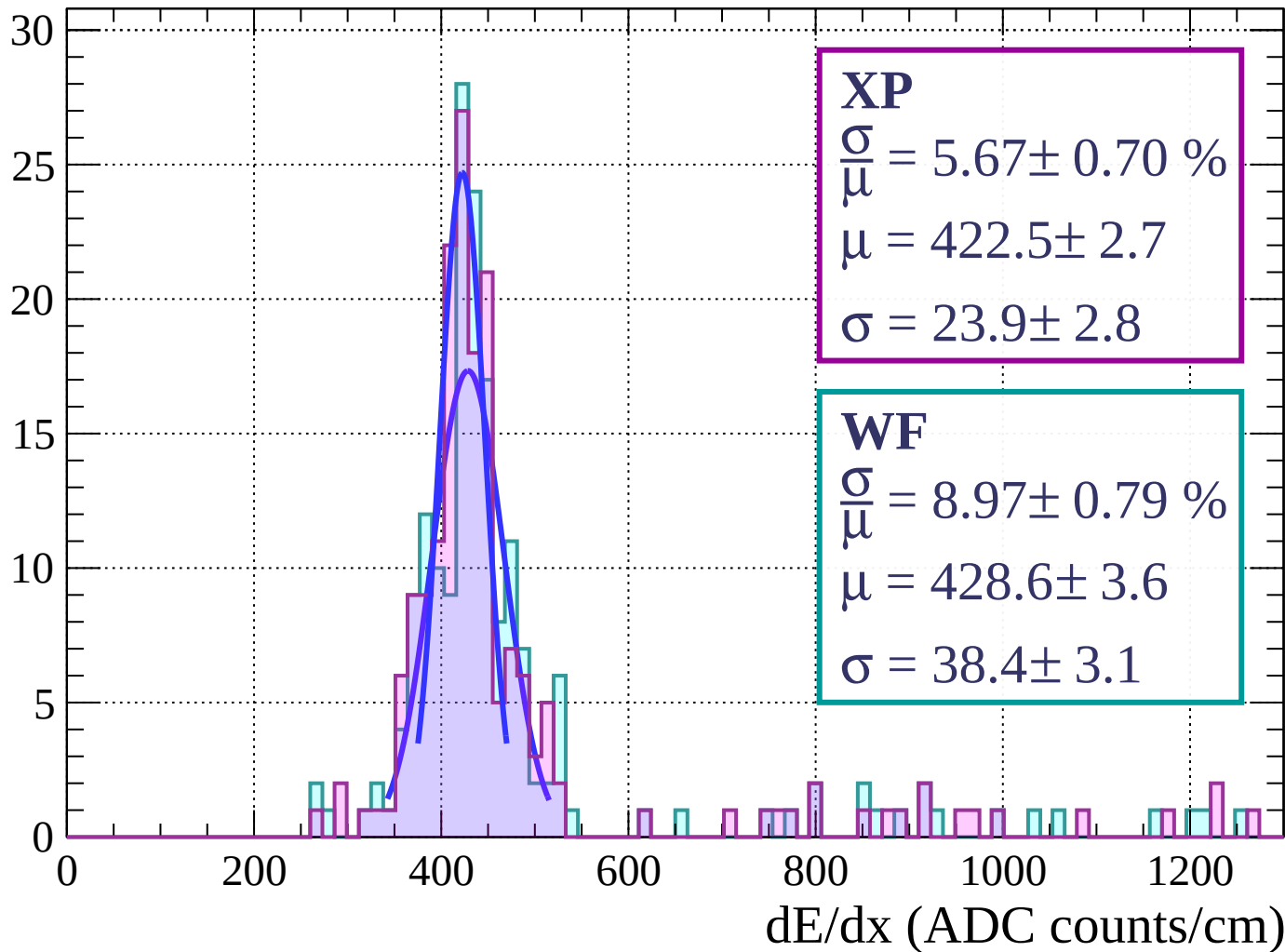
Energy loss | $39 < \theta < 42$

Count



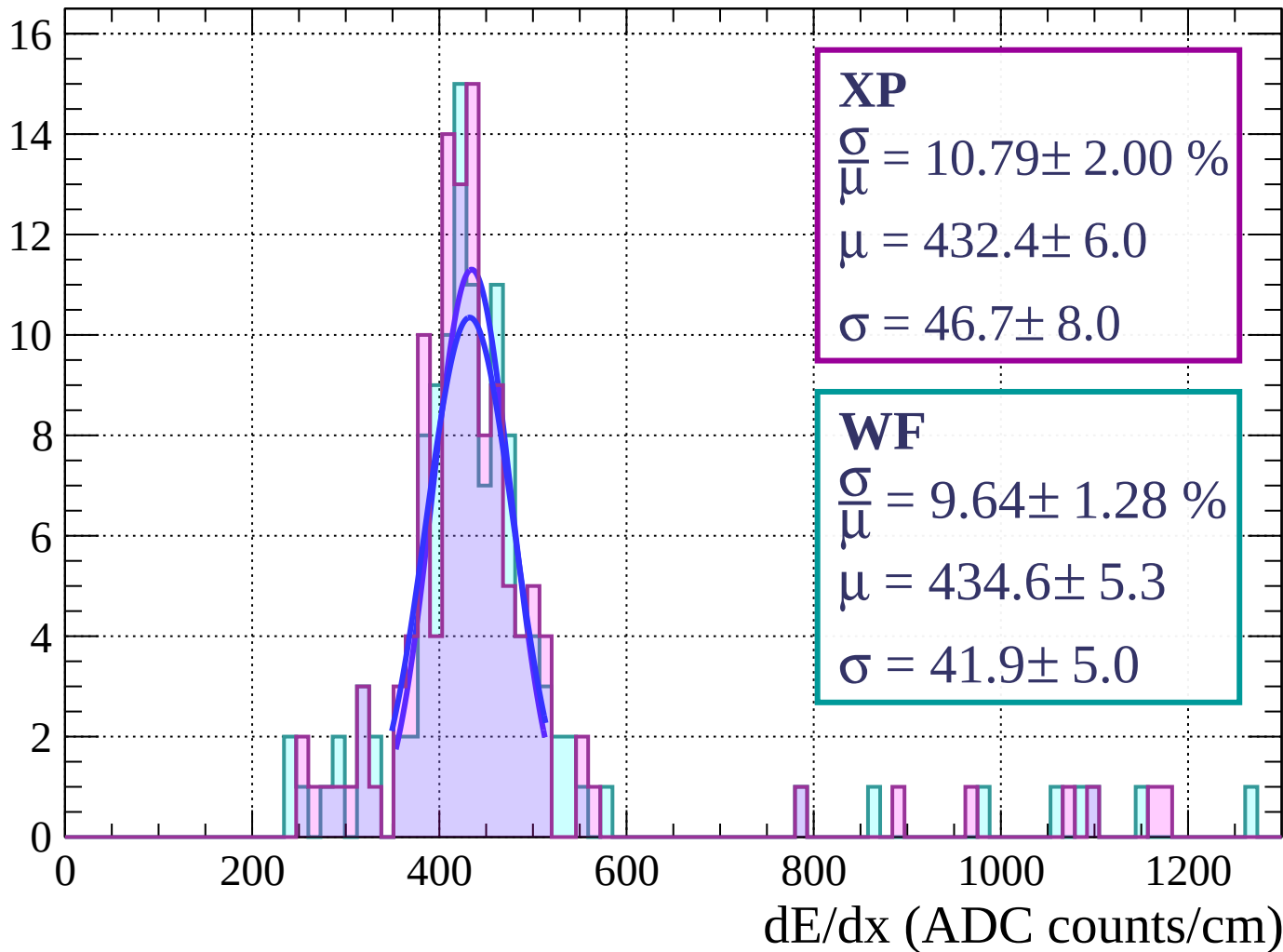
Energy loss | $42 < \theta < 45$

Count



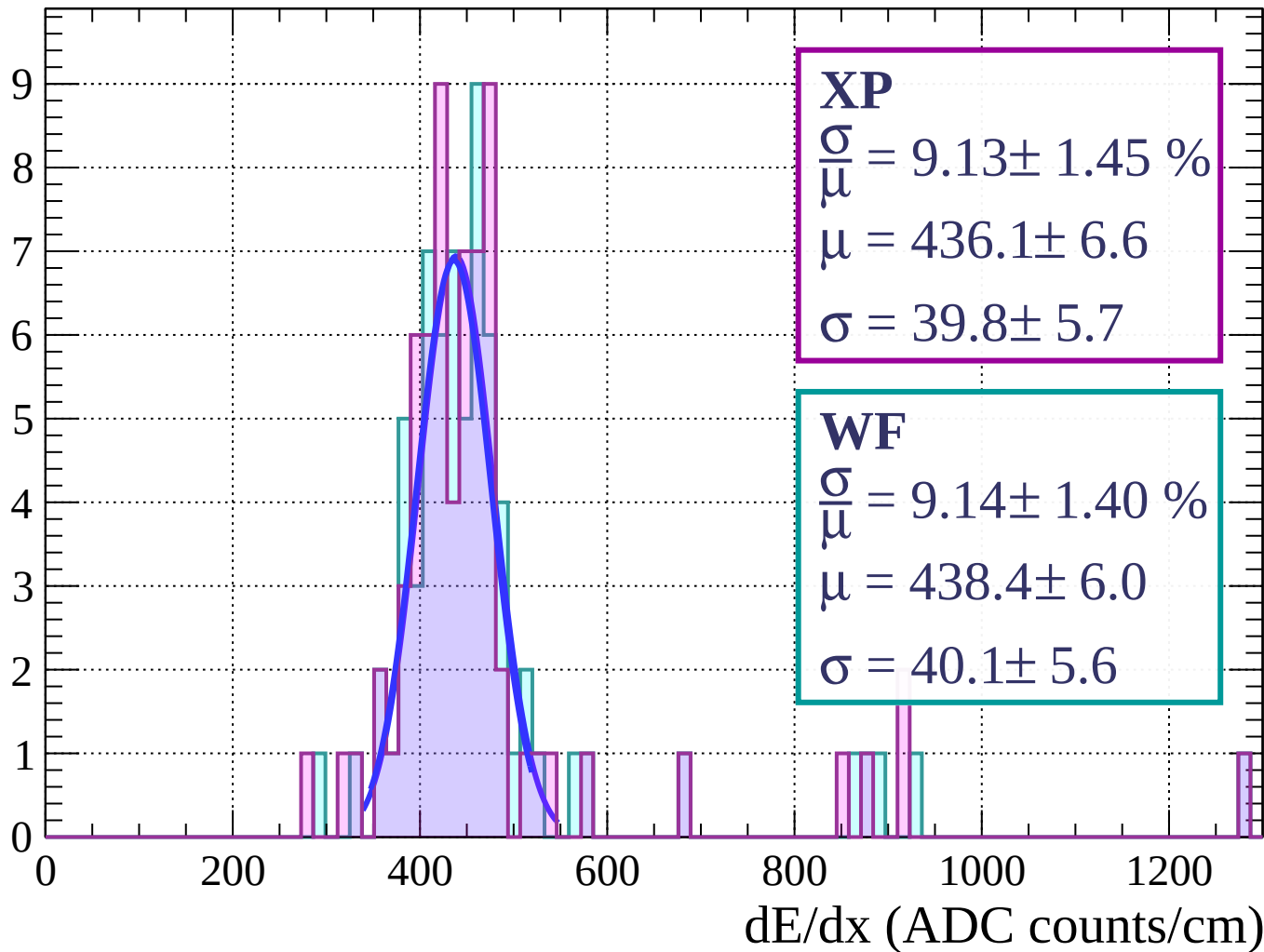
Energy loss | $45 < \theta < 48$

Count



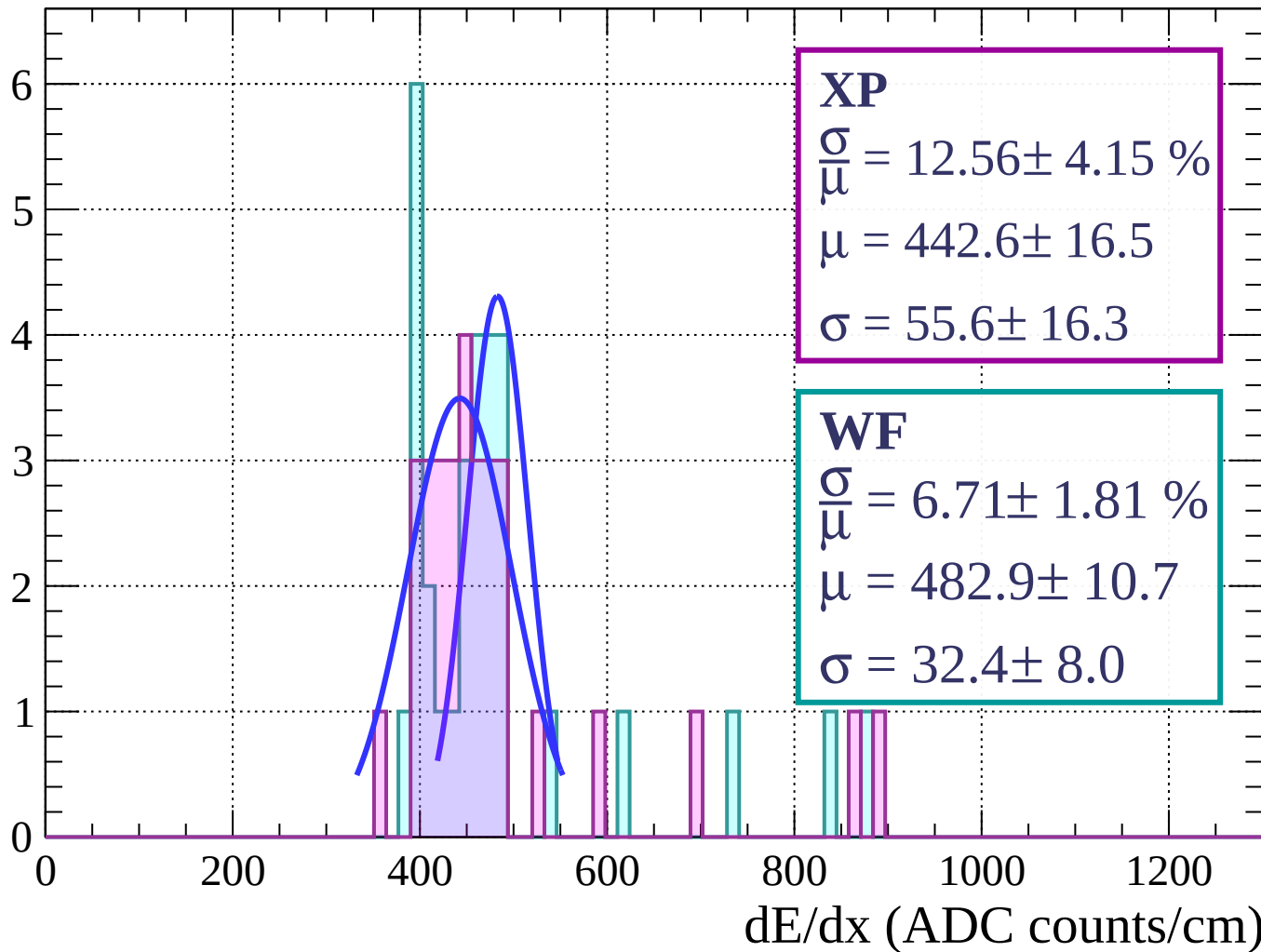
Energy loss | $48 < \theta < 51$

Count



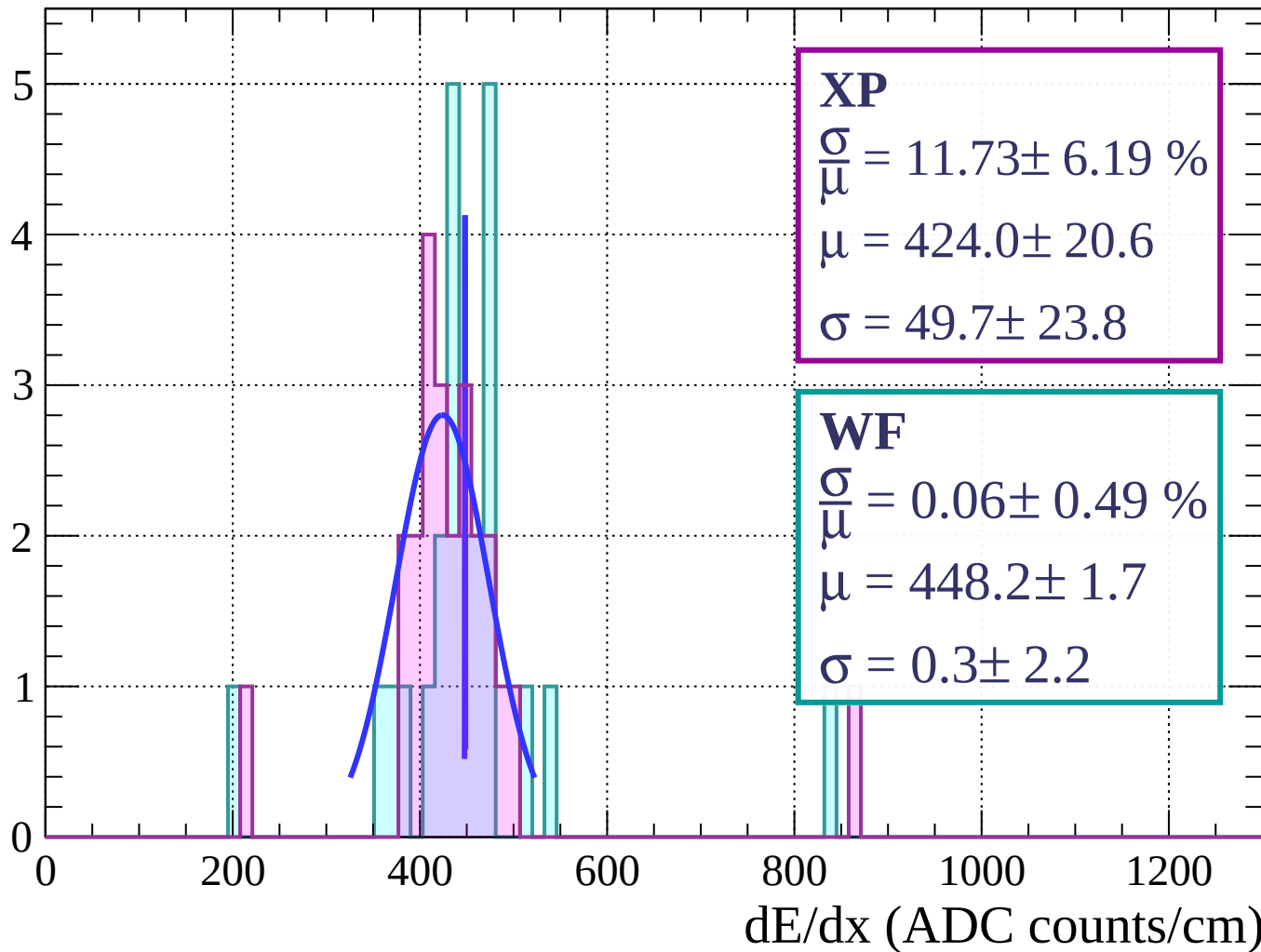
Energy loss | $51 < \theta < 54$

Count



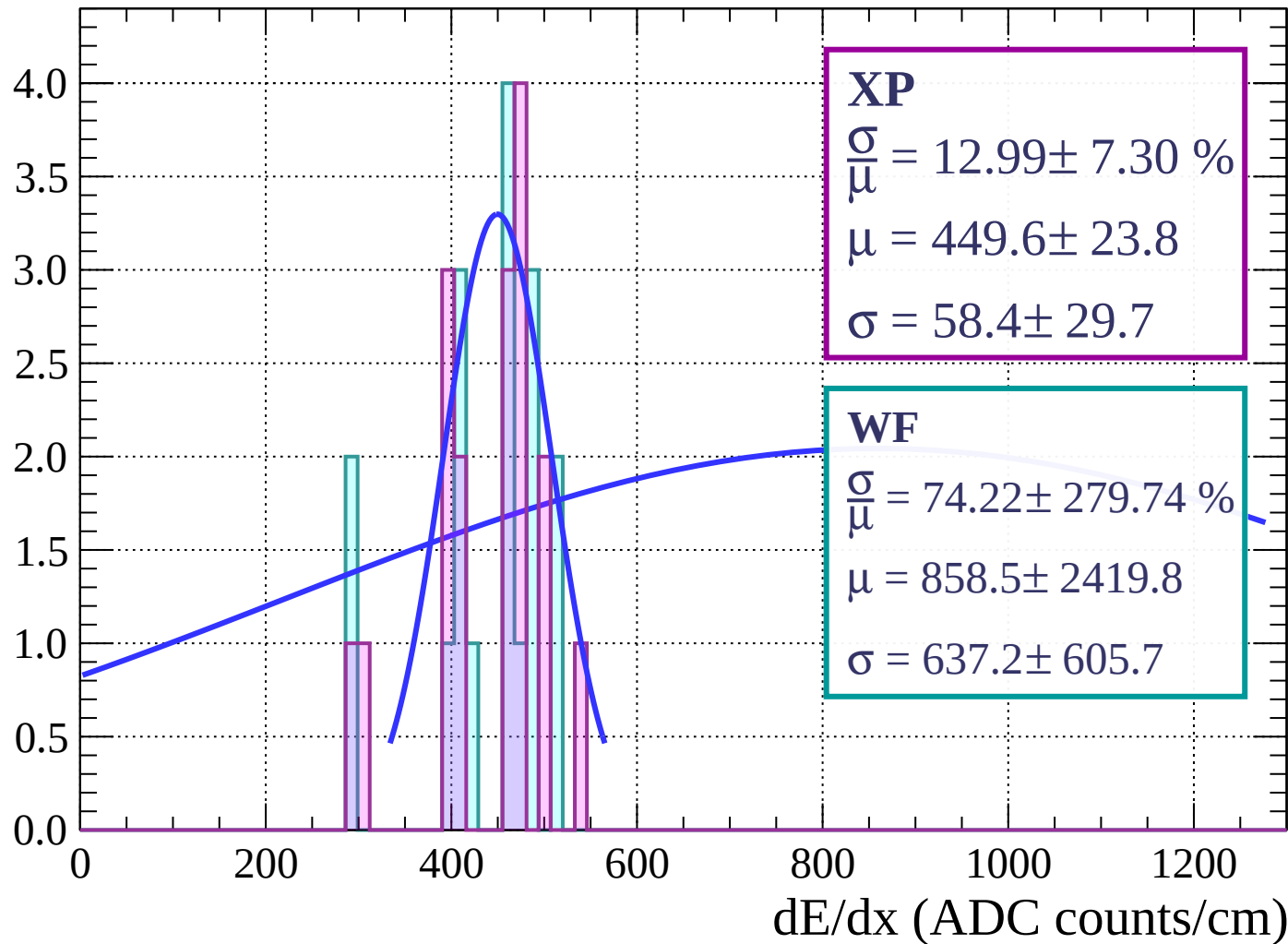
Energy loss | $54 < \theta < 57$

Count



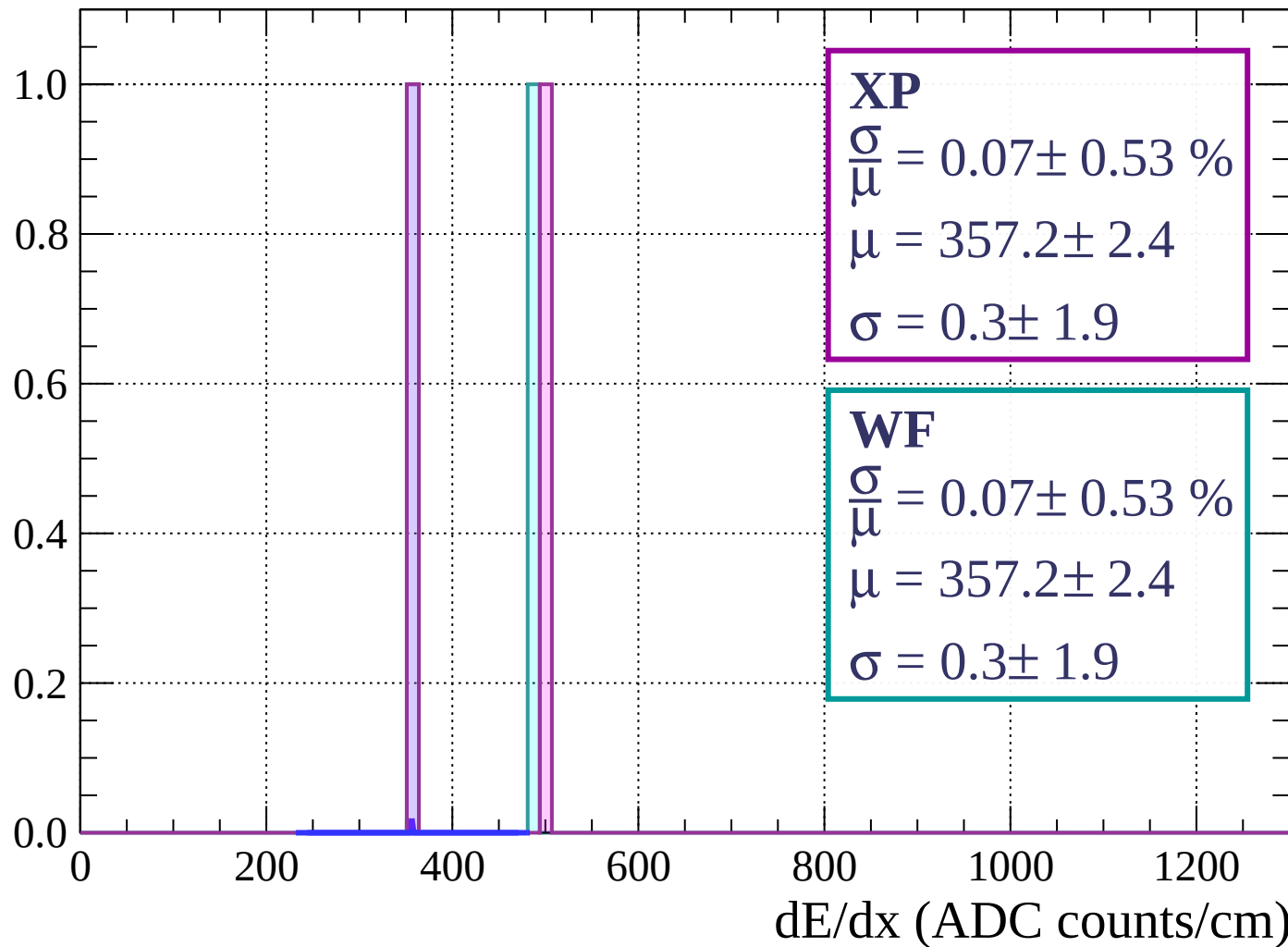
Energy loss | $57 < \theta < 60$

Count



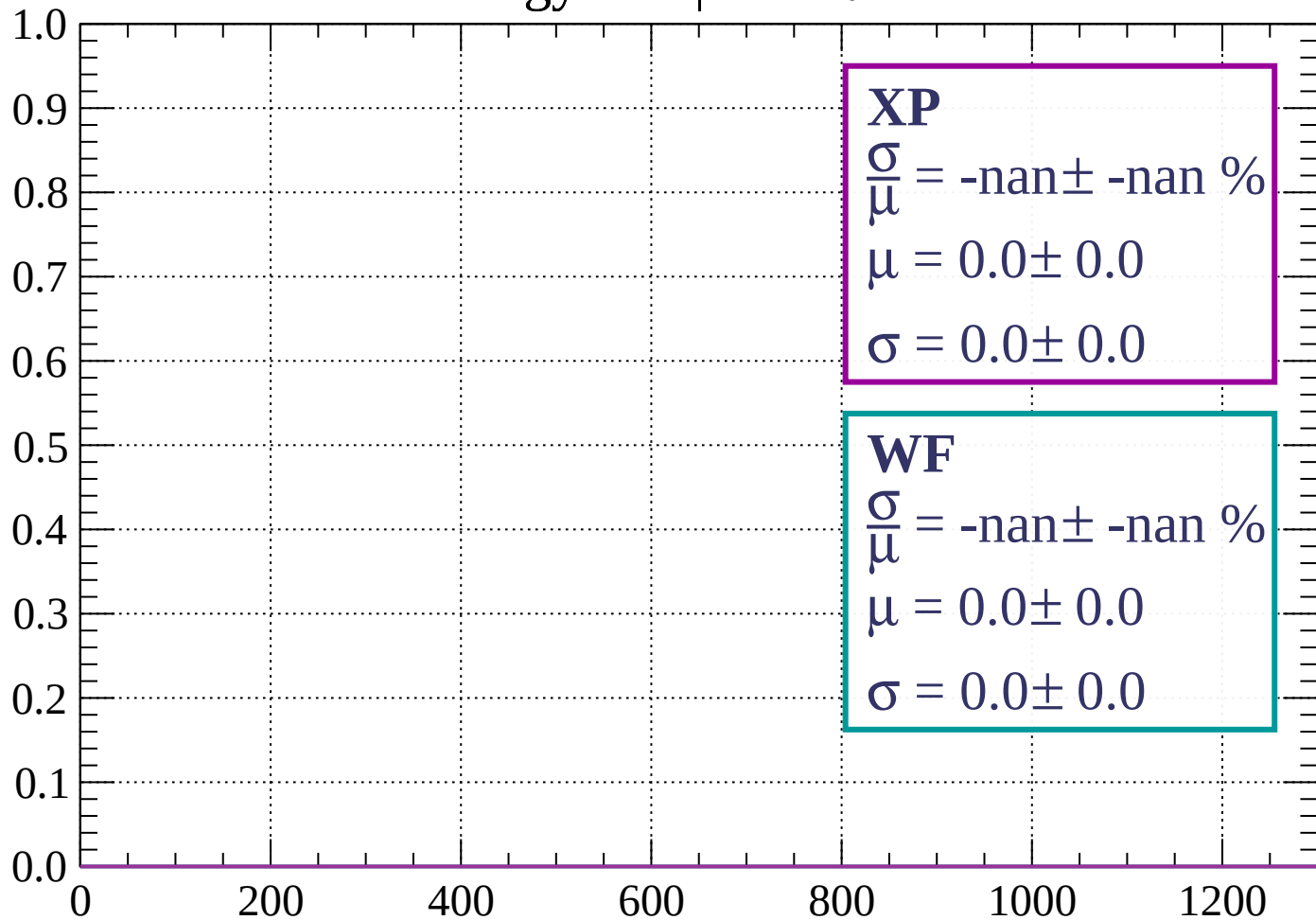
Energy loss | $60 < \theta < 63$

Count



Energy loss | $63 < \theta < 66$

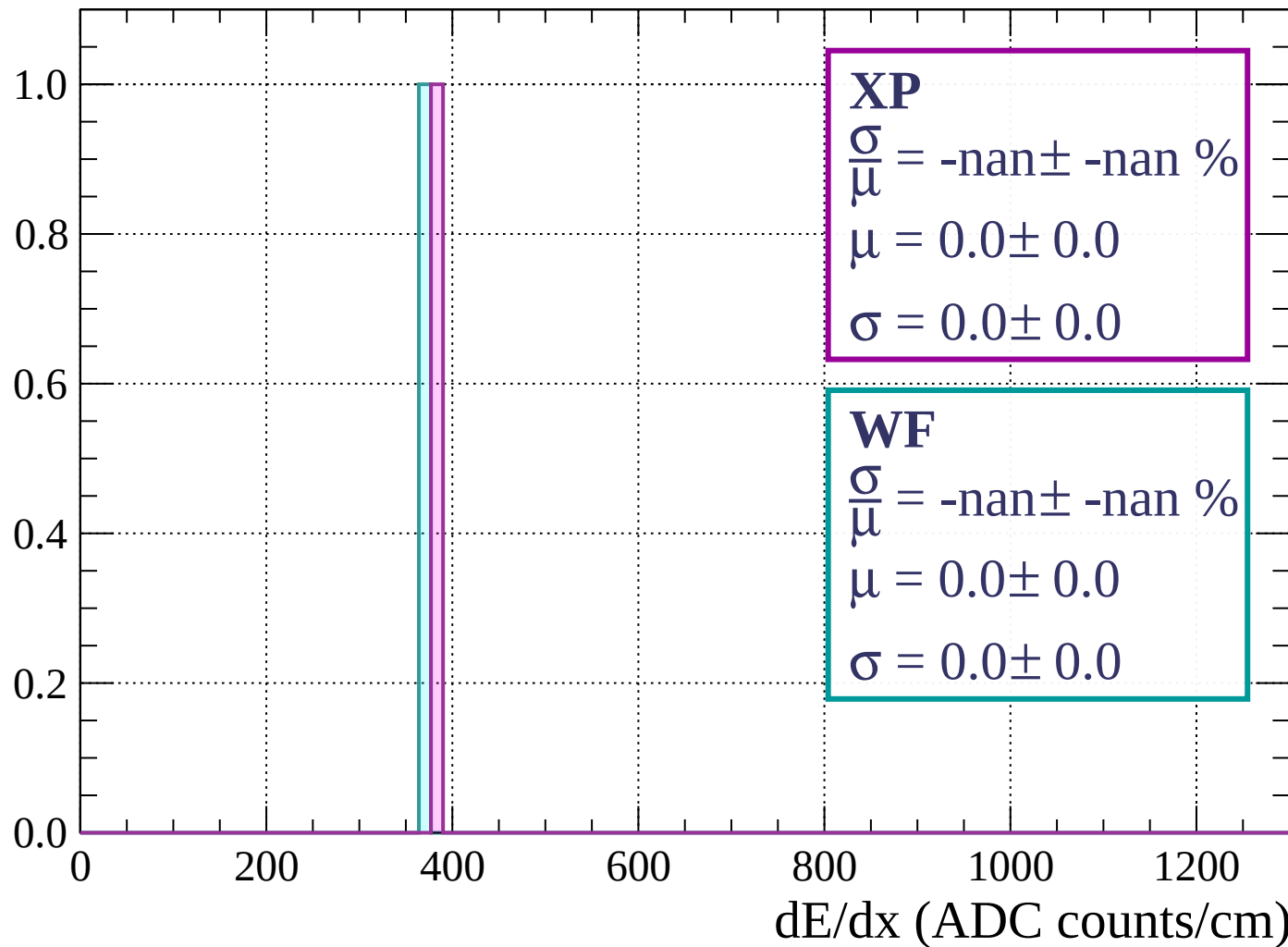
Count



dE/dx (ADC counts/cm)

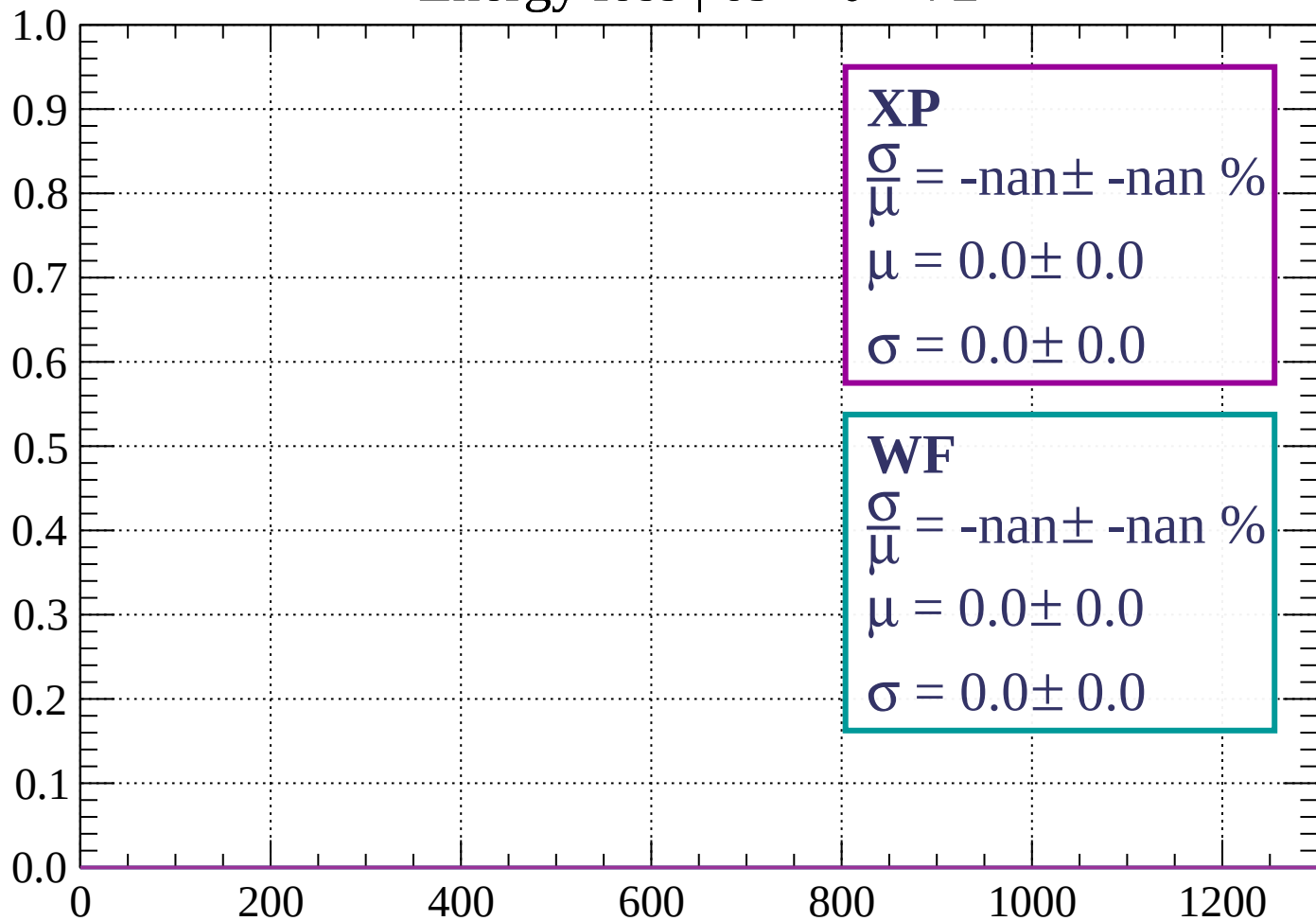
Energy loss | $66 < \theta < 69$

Count



Energy loss | $69 < \theta < 72$

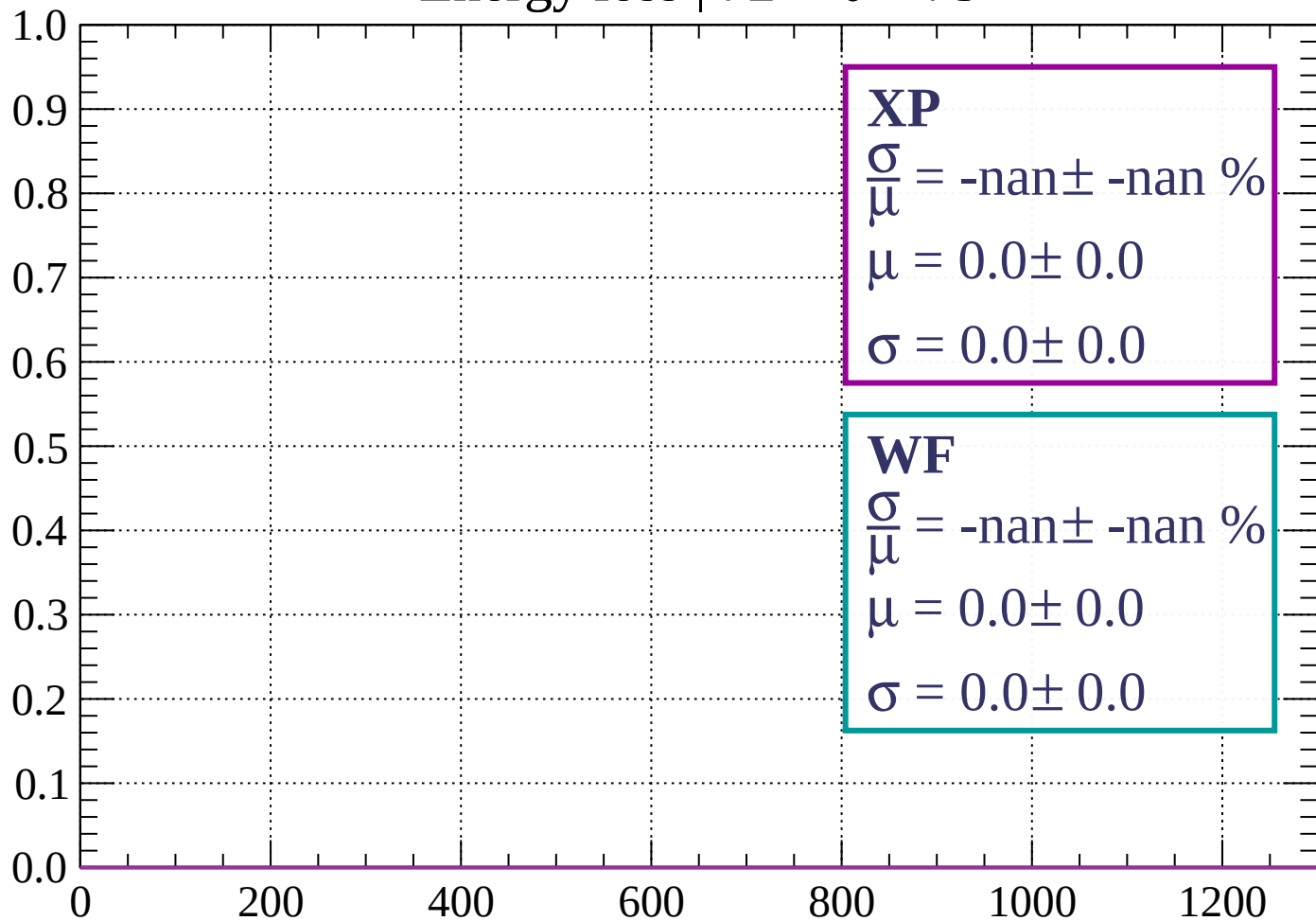
Count



dE/dx (ADC counts/cm)

Energy loss | $72 < \theta < 75$

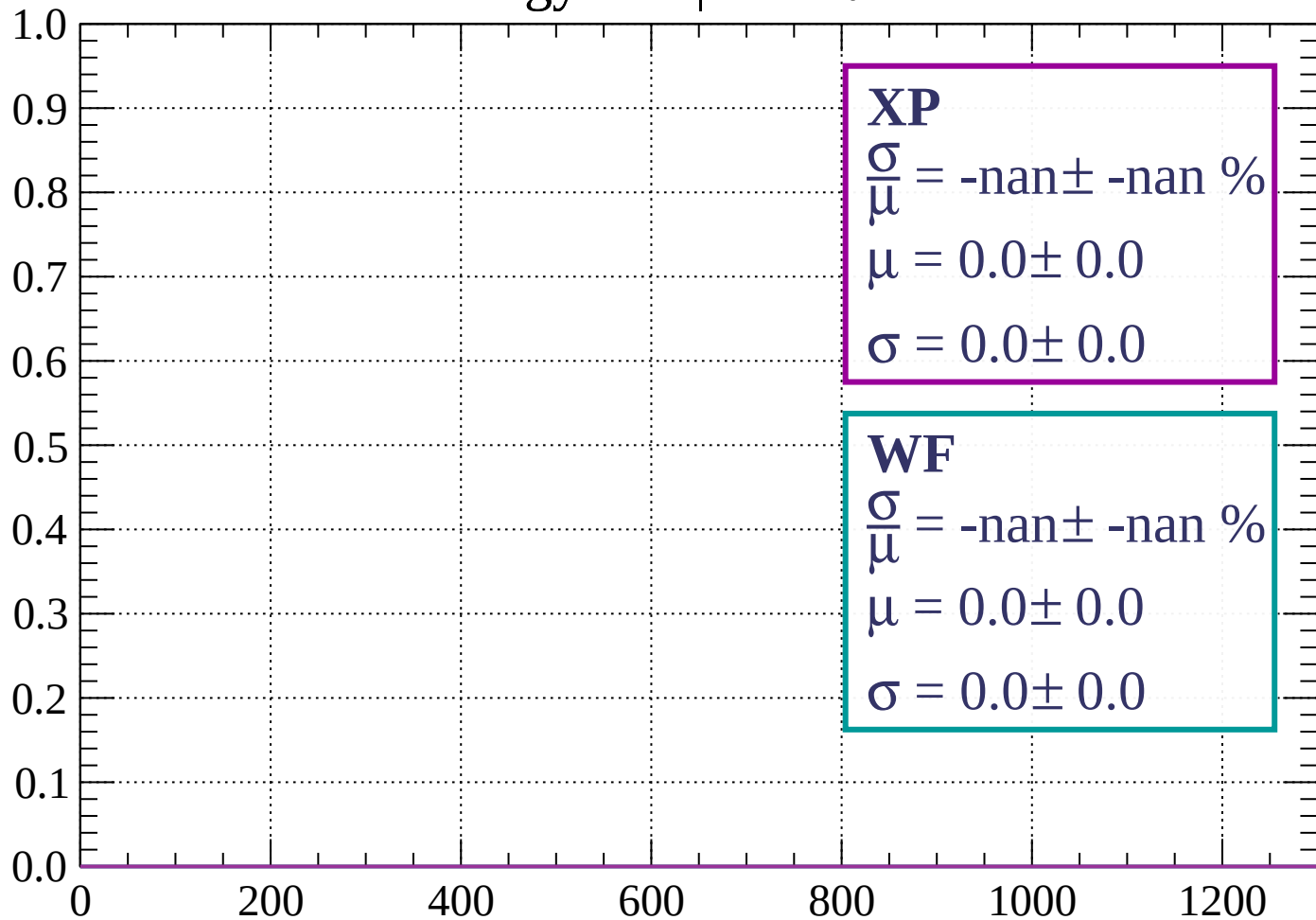
Count



dE/dx (ADC counts/cm)

Energy loss | $75 < \theta < 78$

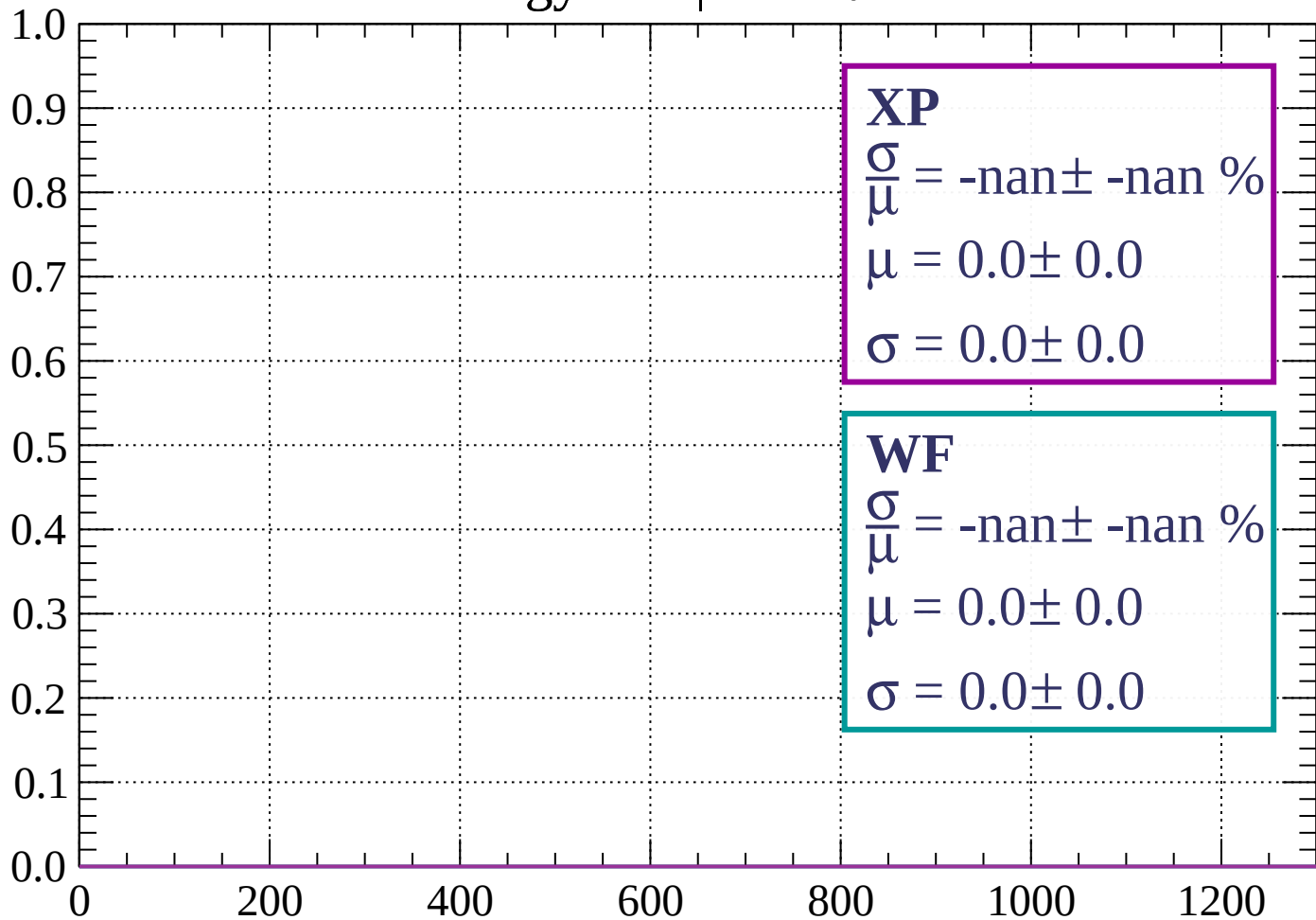
Count



dE/dx (ADC counts/cm)

Energy loss | $78 < \theta < 81$

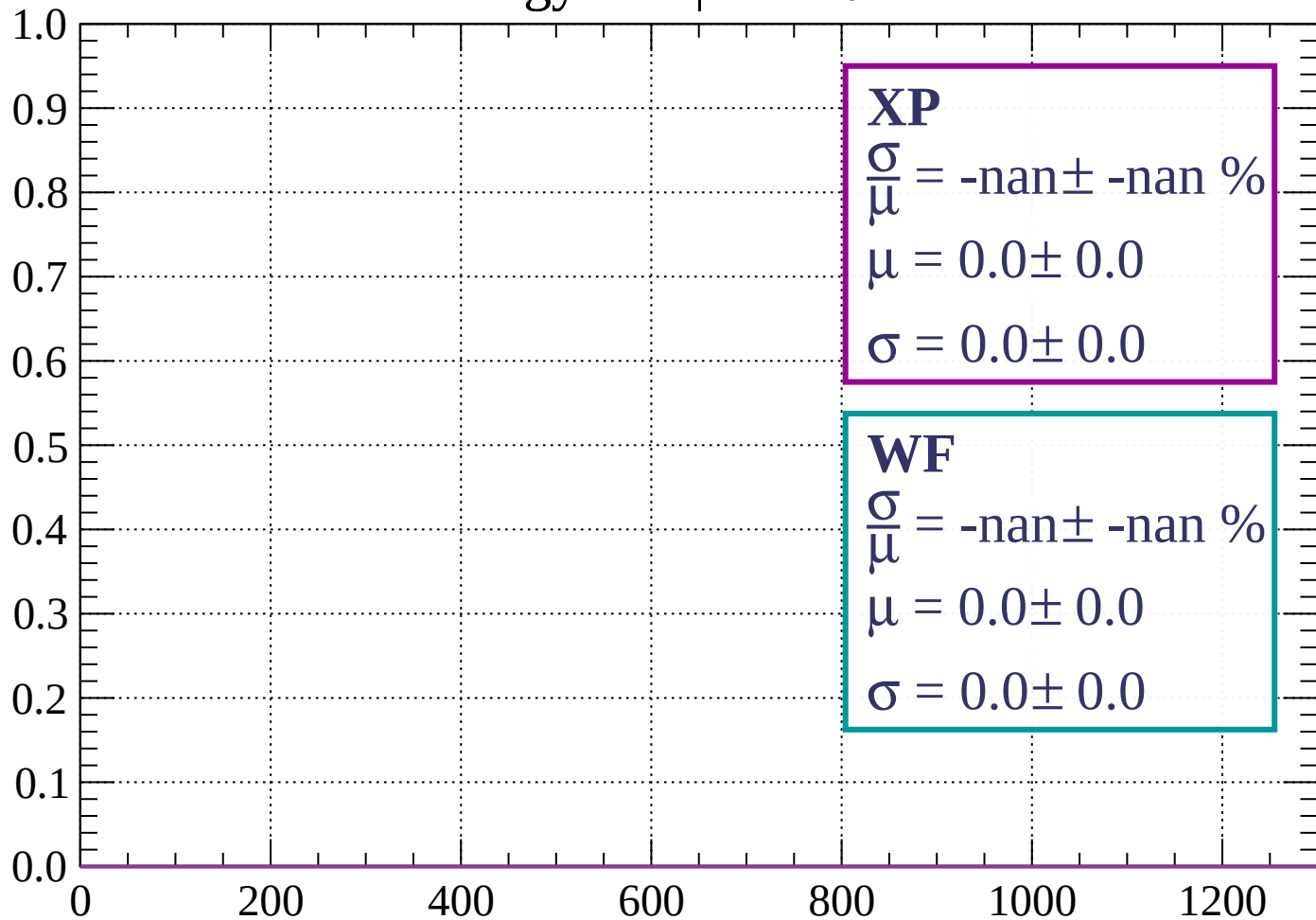
Count



dE/dx (ADC counts/cm)

Energy loss | $81 < \theta < 84$

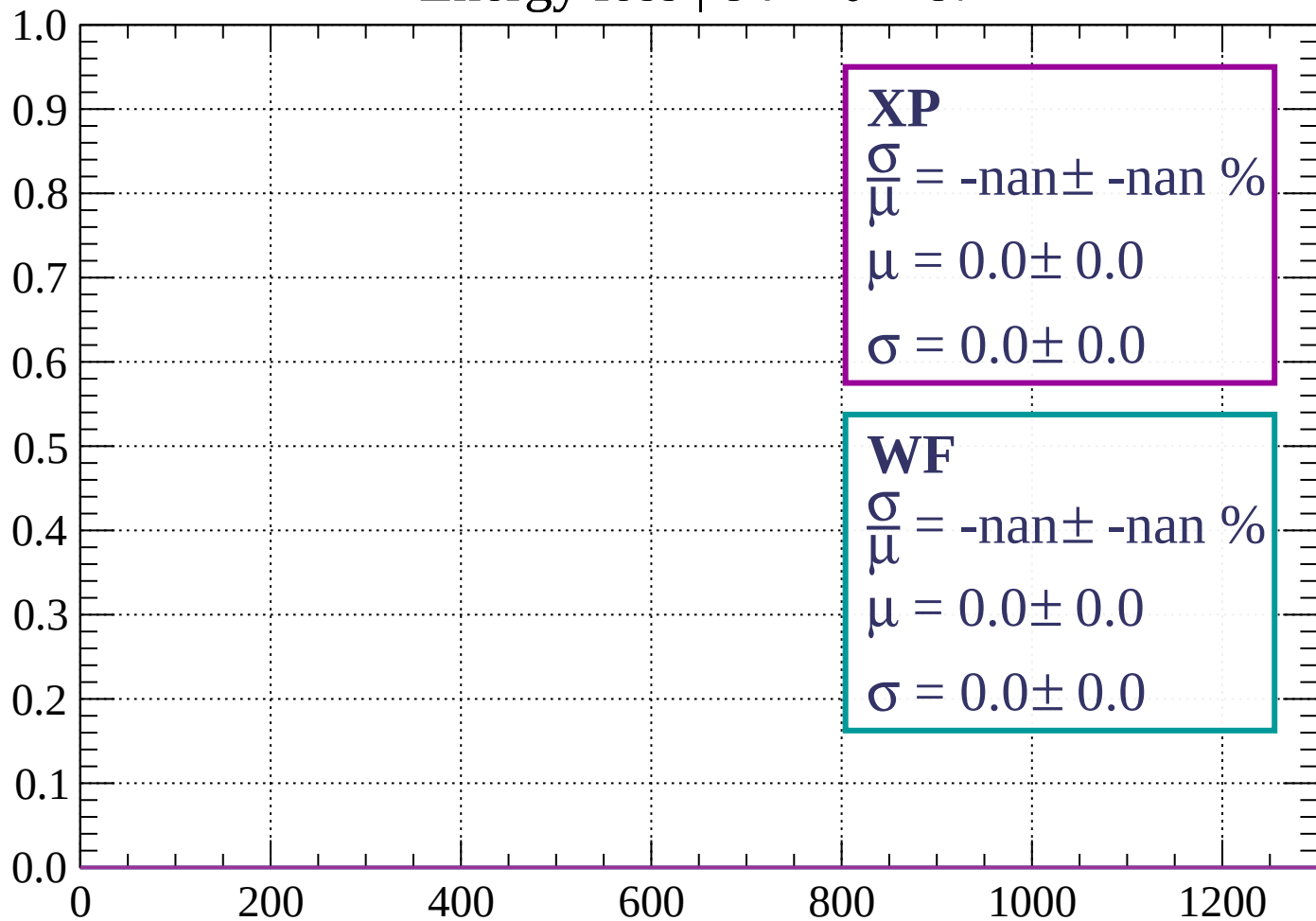
Count



dE/dx (ADC counts/cm)

Energy loss | $84 < \theta < 87$

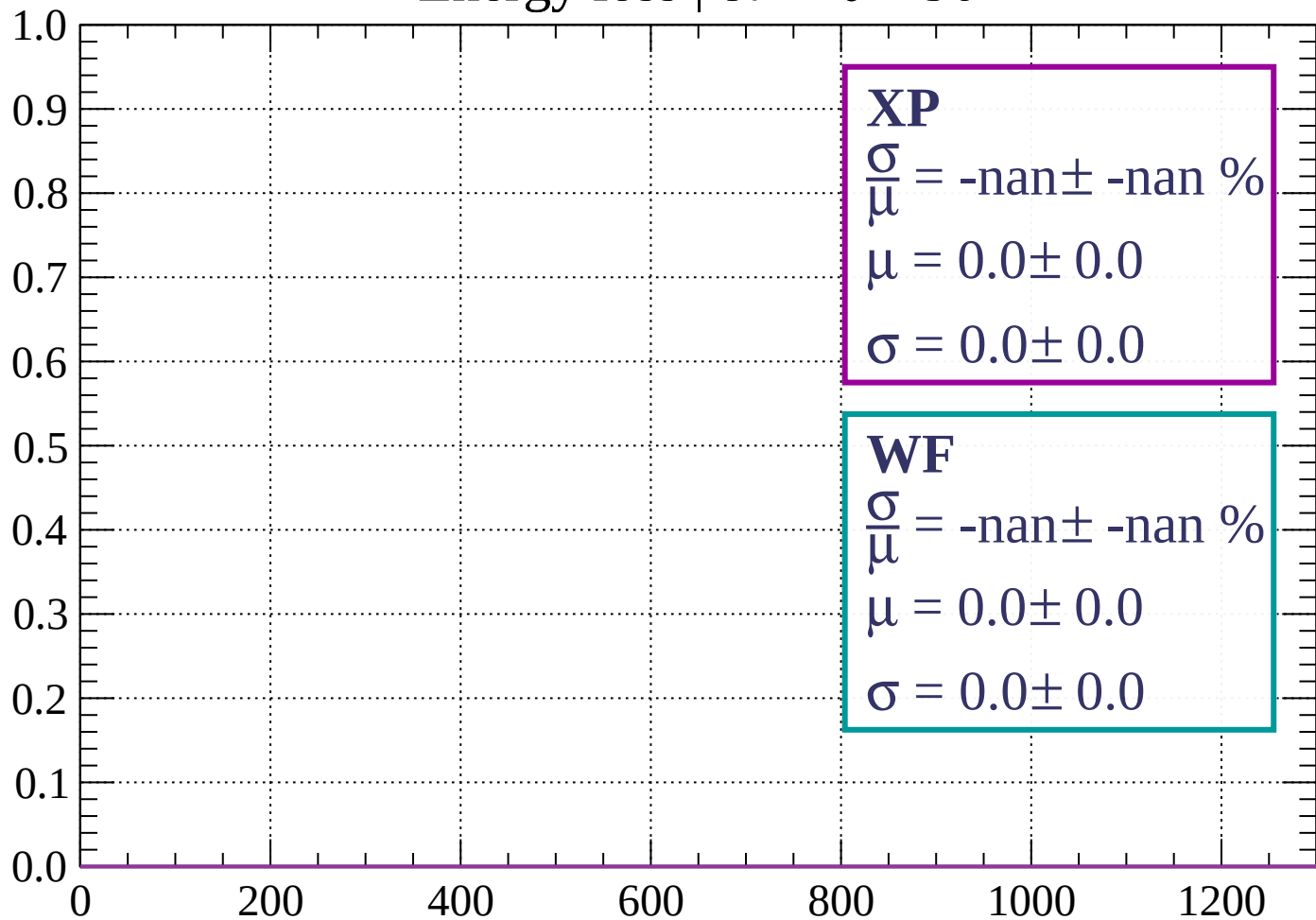
Count



dE/dx (ADC counts/cm)

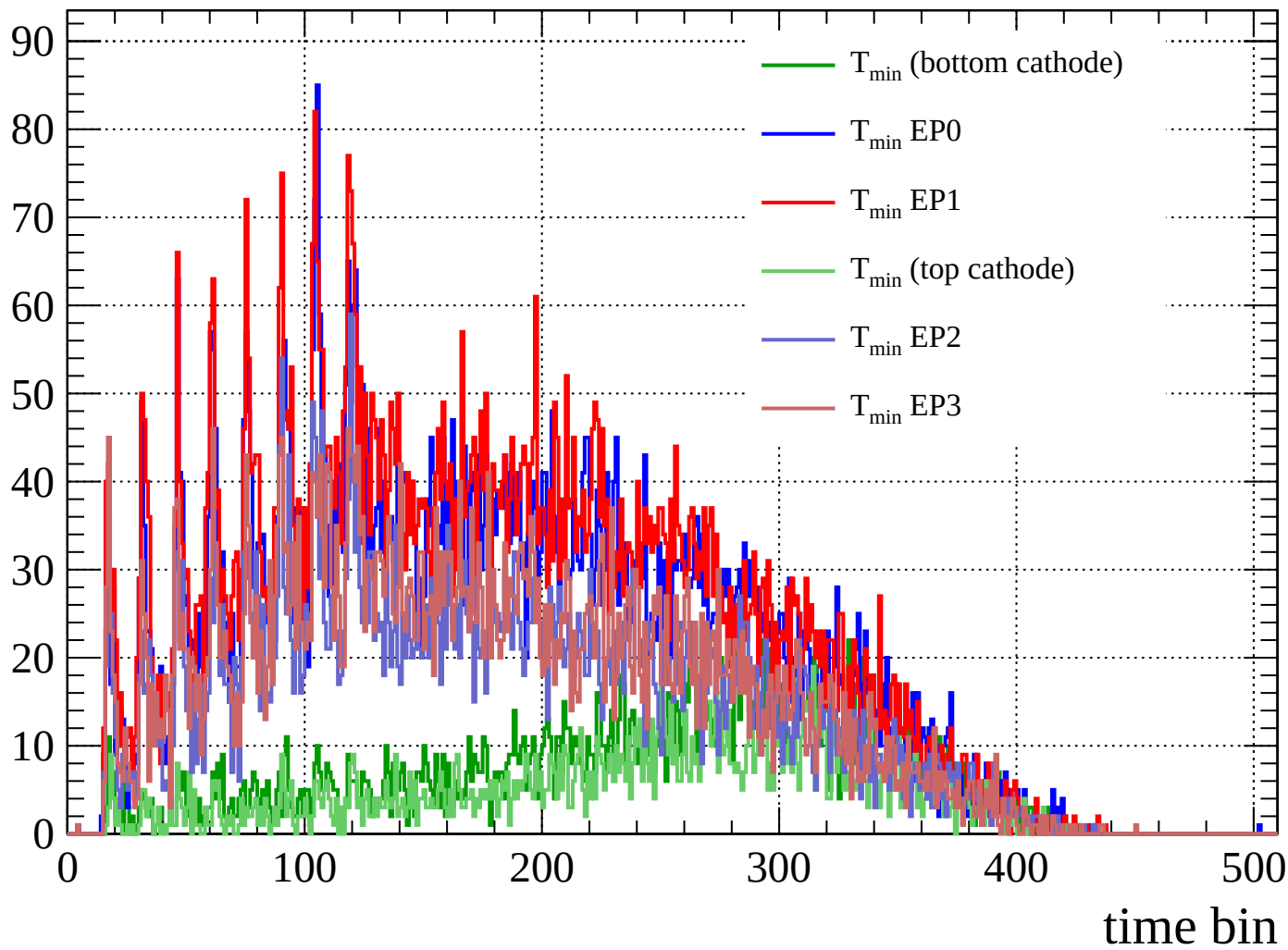
Energy loss | $87 < \theta < 90$

Count



dE/dx (ADC counts/cm)

Count



Count

